CHIRAL KOSZUL DUALITY

Ordered Fusion, Associative Chiral Fields, Bar–Cobar Geometry, and Quantum Defects

A geometric theory of doubly noncommutative chiral observables

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A chiral collision is supported on a diagonal of a complex curve. Ordered fusion is supported on an oriented interval transverse to the curve. Their common algebraic carrier is a doubly associative factorisation object. Its bar–cobar and Morita theorems hold on explicit complete loci; planar motion and stable-curve geometry supply exchange and sewing; the calculated examples begin with free algebras and culminate in Yangian, Hall, and modular structures.

Prelude: five geometric motions

Let X be a smooth complex algebraic curve. Place protected observables in a product chart

$$U \times I \subset X^{\mathrm{an}} \times \mathbb{R},$$

where U is a disc and I is an oriented interval. The local experiment has five elementary motions.

carrier	motion	algebraic structure
partial diagonal in X^I	holomorphic collision	chiral distribution
oriented interval	ordered fusion	E_1-multiplication
transverse plane	exchange	braid transport or E_2-structure
circle or ribbon surface	closure of an interval	Hochschild and cyclic sewing
stable algebraic curve	nodal degeneration	conformal-block sewing

The diagonal records singular support. The interval records temporal order in the protected topological direction. The plane records paths exchanging defects. The circle records traces. The stable curve records modular sewing. Each carrier has its own compactification, boundary maps, and deformation complex.

Two associative structures occur before exchange. A *transversely ordered chiral algebra* is an E_1-algebra for the pointwise coefficient product in factorisable D-modules. The pointwise E_1-multiplication models ordered composition on $X \times \mathbb{R}$. An *associative chiral algebra* is an E_1-algebra for chiral convolution, equivalently an algebra over a non-symmetric associative chiral operad. Its formal-disc realization is a rational complete nonlocal vertex algebra. A doubly associative chiral algebra carries both products and a mixed law on the common-decomposition sector. A braided enhancement supplies exchange through a second transverse direction or a flat meromorphic connection.

The common language is geometric. Singular products are morphisms of D-modules supported on diagonals. Ordered products are factorisation operations on intervals. Higher identities arise from codimension-two faces of compactified parameter spaces. The bar construction integrates ordered fusion over an interval. Chiral Chevalley chains integrate collision data over the Ran space. Ribbon and Deligne–Mumford sewings arise only after their edge-contraction maps have been constructed. Every scalar invariant is evaluated from one of these objects by a specified trace, period, index, or monodromy functional.

Abstract

Let $\mathrm{FactD}(X)$ be a stable presentable category of factorisable right D-modules on a smooth complex curve X. Its finite-set model carries two products. The componentwise coefficient tensor $\otimes_!$ preserves the support set, and chiral convolution $\star_{\mathrm{ch}}$ forms disjoint unions of supports. Hadamard and Cauchy products of species supply canonical inclusion and projection maps between their mixed composites. Their composite is the idempotent onto the aligned-decomposition sector. Factorisation locality is precisely the family of multiplication squares supported on this sector.

A transversely ordered chiral algebra is

$$A \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathrm{FactD}(X), \otimes_!).$$

Its interval bar is the derived boundary-to-boundary amplitude

$$\mathrm{Bar}^{\perp}_X(A) = \mathbf{1} \otimes^{\mathbb{L}}_A \mathbf{1} \simeq \int_{([0,1],\partial[0,1])} A.$$

The simplicial faces multiply adjacent transverse factors. The full chiral coefficient remains present in every bar term. An associative chiral algebra is

$$C \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathrm{DMod}(\mathrm{Ran}\,X), \star_{\mathrm{ch}}),$$

or an algebra over the local operad $\mathsf{Ass}^{\mathrm{ch}}_X$. Its operadic bar contracts chiral composition trees. Chiral Chevalley chains form a third construction, governed by the chiral Lie operation. A mixed operad $\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$ records the two associative products and their aligned interchange cells; its Boardman–Vogt resolution produces the transverse, chiral, and mixed homotopies.

Internal associative bar–cobar reconstruction holds for connected positive-weight objects and for complete inverse limits satisfying tensor-continuity and Mittag–Leffler hypotheses. Francis–Gaitsgory chiral Koszul duality belongs separately to the pro-nilpotent chiral-convolution category. The augmentation boundary has endomorphism algebra

$$A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1}),$$

and the double centralizer is the derived augmentation completion under the stated compactness hypotheses. The two-sided twisting transform carries the diagonal bimodule to the diagonal bicomodule and converts derived tensor product into derived cotensor product. Hochschild and coHochschild theories thereby form a Koszul–Morita square. Relative bars treat quiver idempotents; curved duals treat nonhomogeneous relations such as the Weyl commutator.

The complete diagonal singularity is encoded by the pole filtration

$$\mathcal{O}_{X^2}(D) \subset \mathcal{O}_{X^2}(2D) \subset \cdots \subset \mathcal{O}_{X^2}(*D),$$

whose $(m+1)$-st normal symbol carries the mode $a_{(m)}b$. Partition complexes encode collision incidence; Fulton–MacPherson screens encode the relative geometry of a collision; the stratification spectral sequence keeps these two layers distinct. On the formal disc, skew-symmetric chiral operations give vertex algebras and the Borcherds identity; ordered associative chiral operations give rational complete nonlocal vertex algebras and weak associativity. The pole filtration on the chiral operation operad has the classical Poisson-vertex operad as its associated graded.

A sequence of exact examples develops the theory from free and square-zero algebras through the quantum and Jordan planes, relative quiver bars, enveloping algebras, Zamolodchikov–Faddeev algebras, RTT Yangians, the A_2 Hall algebra, and necklace Hamiltonians. The ordered bar of $U(\mathfrak{g})$ is the Chevalley–Eilenberg complex of $\mathfrak{g}$. The legacy Motzkin and Riordan series are retained as planar enumerations and assigned to their actual combinatorial complexes. Yang–Baxter confluence, PBW bases, Hall–Serre relations, and stable trace brackets are proved by explicit reductions.

Exchange arises from planar motion or meromorphic transport. KZ flatness follows from the Arnold relation and the infinitesimal braid relations. A rigid meromorphic tensor category with a faithful tensor fibre functor reconstructs a coquasitriangular coordinate bialgebra; additional PBW, generation, and faithfulness hypotheses recognize a completed Yangian. A square-zero BRST derivation produces a bar–BRST bicomplex, and its convergent spectral sequence compares ordered bar constructions with quantum Hamiltonian reduction.

Cyclic transverse sewing and Deligne–Mumford chiral sewing carry independent genus variables. A bicoloured graph transform with explicit edge contractions satisfies

$$d\Theta + \frac{1}{2}[\Theta,\Theta]_X + \frac{1}{2}[\Theta,\Theta]_\perp + \hbar_X\Delta_X\Theta + \hbar_\perp\Delta_\perp\Theta = 0.$$

A tensor-product model realizes the two colours and their mixed Cartesian boundary squares. Strongly rational vertex operator algebras supply modular-functor sewing through their conformal blocks, while a cyclic dg algebra supplies ribbon sewing. Determinant curvature, BV anomaly, BRST anomaly, framing anomaly, and curved Koszul curvature remain classes in their native complexes; comparison morphisms relate them in concrete quantum field theories.

For a positively root-graded Lie superalgebra, the Euler supercharacter of the ordered bar is

$$\chi_\Gamma \operatorname{Bar}^\perp U(\mathfrak{n}_+) = \prod_{\alpha>0}(1-e^{-\alpha})^{\operatorname{sdim}\mathfrak{g}_\alpha}.$$

Borcherds products arise when a geometric or representation-theoretic construction identifies the graded root multiplicities and fixes the normalization. Higher-dimensional chiral operations and chiral quantizations enter through precise external theorems and explicit comparison maps.

The theorem spine

I	The Ran coefficient has pointwise and chiral products, together with a canonical retract onto aligned decompositions.
II	Product-disk factorisation on $X^{\mathrm{an}} \times \mathbb{R}$, under algebraisation and descent hypotheses, yields a transversely ordered chiral algebra.
III	Associative chiral multiplication is an independent operadic structure; its formal-disc realization is a nonlocal vertex algebra, and braided locality supplies the quantum vertex enhancement.
IV	The mixed operad $\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$ and its cofibrant resolution generate every transverse, chiral, and mixed coherence from coloured collision forests.
V	The transverse interval bar, the associative chiral operadic bar, and chiral Chevalley chains are three typed constructions with explicit comparison maps.
VI	Bar–cobar reconstruction holds on connected positive-weight and complete nilpotent loci; double centralization produces derived augmentation completion.
VII	Relative bases, augmentations, dualisability, curvature, and completion are structural data in every Koszul and Morita theorem.
VIII	Exchange, centres, BRST reduction, ribbon sewing, and modular sewing are functorial extensions generated by additional geometric carriers.
IX	Every numerical example is the homology, trace, period, monodromy, index, or Euler character of a displayed mathematical object.
X	The no-go constitution records the exact boundaries separating residues, screens, bars, centres, genera, anomalies, and quantum groups.

Notation and object types

We work over a field $\Bbbk$ of characteristic zero, except in the finite-field Hall calculations. Tensor products, limits, colimits, and mapping objects are derived. Every completion is displayed.

Symbol	**Meaning**
$\otimes_!$	componentwise coefficient tensor; transverse ordered fusion uses this product.
$\star_{\mathrm{ch}}$	chiral convolution by disjoint union; associative chiral multiplication and Ran-space chiral Koszul duality use this product.
$\mathrm{Bar}^{\perp}$	associative bar of the transverse E_1-multiplication.
$\mathrm{Bar}^{\mathrm{ch}}_{\mathrm{Ass}}$	operadic bar of associative chiral multiplication.
$\mathrm{Bar}^{\mathrm{ch}}_{\mathrm{Lie}}$	chiral Chevalley chains of a chiral Lie algebra.
$\mathbf{Z}(A)$	internal Hochschild cochains $\mathrm{RHom}_{A^e}(A, A)$, carrying an E_2-structure.
$g_X, g_\perp$	Deligne–Mumford chiral genus and transverse ribbon genus.
$\Delta_X, \Delta_\perp$	loop contractions associated to the two graph colours.

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Part I

The invariant object and its two associative directions

Chapter 1

Five carriers, three noncommutativities, one type discipline

1.1 The local experiment

Let X be a smooth complex algebraic curve, let $U \subset X^{\mathrm{an}}$ be a disc, and let $I \subset \mathbb{R}$ be an oriented interval. A pair of observables in $U \times I$ admits two independent short-distance limits. Collision in U approaches a partial diagonal and produces a diagonal-supported distribution with its complete normal filtration. Adjacency in I produces ordered associative fusion. A transverse plane supplies exchange paths, a circle supplies traces, and a degenerating algebraic curve supplies clutching maps.

The five geometric carriers are

$$\text{diagonal}, \qquad \text{interval}, \qquad \text{plane}, \qquad \text{circle}, \qquad \text{stable curve}.$$

Their algebraic structures are a chiral operation, an E_1-multiplication, braid transport, Hochschild/cyclic trace, and modular sewing. Comparison among these structures is expressed by maps induced from geometric operations on their carriers.

Theorem 1.1 (Oriented-corner identity)**.** *Let C be an oriented compact manifold with corners. Let (E, d_E) be a cochain complex. For every oriented codimension-one face F, let $\rho_F : E \to E[1]$ be a degree-one operator such that*

$$d_E \rho_F + \rho_F d_E = 0.$$

Assume that $\rho_F^2 = 0$, and that a composite $\rho_G \rho_F$ is supported on $F \cap G$. Put

$$D = d_E + \sum_F \rho_F.$$

For every codimension-two stratum S, the S-component of D^2 is

$$(D^2)_S = \sum_{S \subset F \cap G} \epsilon(S; F, G)\, \rho_G \rho_F,$$

where $\epsilon(S; F, G)$ is the corner-orientation sign. If the target decomposes as the direct sum of its stratum-supported components, then $D^2 = 0$ is equivalent to the vanishing of every displayed corner sum.

Proof. The square of d_E vanishes. The chain-map identity cancels the mixed terms, and the face-square hypothesis removes the diagonal terms. The remaining composites are indexed by ordered pairs of distinct incident faces. Grouping them by their codimension-two intersection gives the displayed sum. Opposite corner orientations supply the signs. When stratum-supported components form a direct sum, the vanishing of D^2 is equivalent to the vanishing of each component. $\square$

Associativity, chiral Jacobi and chiral associativity, Yang–Baxter compatibility, the BV master equation, and modular sewing arise by decorating different compactified parameter spaces with different coefficient maps. The oriented-corner identity supplies their common sign mechanism.

1.2 The Ran coefficient and its two products

Let $\mathrm{Fin}^{\mathrm{surj}}_{\neq\varnothing}$ denote nonempty finite sets and surjections. A surjection $\alpha : I \twoheadrightarrow J$ determines a partial diagonal

$$\Delta_\alpha : X^J \longrightarrow X^I, \qquad (x_j)_{j\in J} \longmapsto (x_{\alpha(i)})_{i\in I}.$$

In the finite-set presentation used here, a right D-module on $\operatorname{Ran} X$ is a coherent family $F_I \in \mathrm{DMod}(X^I)$ with extraordinary pullback identifications along these diagonals. The unital completion adjoins the empty set.

For a partition $\pi = \{B_1, \dots, B_r\}$ of I, let $U_\pi \subset X^I$ be the locus where points in distinct blocks are disjoint, and write $j_\pi : U_\pi \hookrightarrow X^I$.

Definition 1.2 (Factorisable coefficient)**.** A factorisable right D-module is a Ran family F equipped with coherent equivalences

$$j_\pi^! F_I \simeq j_\pi^! \Big(\boxtimes_{B\in\pi} F_B\Big)$$

for every partition, compatible with refinement, permutation, and diagonal transition maps. The chosen stable presentable enhancement is denoted $\mathrm{FactD}(X)$.

The Ran coefficient carries two monoidal products.

The pointwise coefficient product is

$$(F \otimes_! G)_I = F_I \otimes^! G_I.$$

Its support cardinality is unchanged. Endomorphisms of a fixed boundary condition compose with this product.

Chiral convolution is the Day convolution for disjoint union:

$$(F \star_{\mathrm{ch}} G)_I = \bigoplus_{I=I_1\sqcup I_2} (j_{I_1,I_2})_* j^!_{I_1,I_2}(F_{I_1} \boxtimes G_{I_2}).$$

Chiral convolution adds support cardinalities and supplies the monoidal product for Ran-space chiral Chevalley theory.

Proposition 1.3 (Cardinality pro-nilpotence)**.** *Let $\mathcal{R}_{\geq r}$ be Ran objects whose components vanish in cardinalities strictly smaller than r. Then*

$$\mathcal{R}_{\geq r} \star_{\mathrm{ch}} \mathcal{R}_{\geq s} \subseteq \mathcal{R}_{\geq r+s}.$$

Consequently, reduced chiral convolution is nilpotent modulo cardinality $> N$, and its inverse limit is pro-nilpotent.

Proof. Every summand is indexed by a disjoint union of the supporting finite sets, so cardinalities add. An $(N+1)$-fold product of reduced objects has support cardinality at least $N+1$ and vanishes in the N-th quotient. □

The pointwise product has cardinality degree zero. Chiral convolution therefore carries the pro-nilpotent cardinality filtration used in Ran-space chiral Koszul duality, while the transverse bar uses its augmentation or weight filtration.

1.3 Three forms of noncommutativity

Definition 1.4 (Transversely ordered chiral algebra)**.** A transversely ordered chiral algebra is an augmented algebra

$$A \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathrm{FactD}(X), \otimes_!).$$

The factorisable coefficient carries collision on X. The internal E_1-multiplication $m_\perp : A \otimes_! A \to A$ carries order along a transverse line.

Definition 1.5 (Associative chiral algebra)**.** An associative chiral algebra is an augmented E_1-algebra for chiral convolution,

$$C \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathcal{R}_X, \star_{\mathrm{ch}}),$$

or, in a local geometric presentation, an algebra over the non-symmetric associative chiral operad $\mathsf{Ass}^{\mathrm{ch}}_X$. Its multiplication composes ordered chiral insertions. Formal-coordinate realizations generally satisfy weak associativity rather than ordinary locality.

Definition 1.6 (Braided or exchange enhancement)**.** An exchange enhancement is an E_2-lift, a flat meromorphic transport on labelled configuration spaces, or equivalent braided categorical data. The enhancement supplies paths that exchange two ordered factors, and its geometric carrier is two-dimensional.

The three structures distinguish three forms of noncommutativity:

1. noncommutativity of an ordered word in the transverse direction;
2. noncommutativity or nonlocality of chiral fields under formal-disc composition;
3. nontrivial braid exchange after a second transverse direction or meromorphic connection is supplied.

The three structures occur independently; compatible pairs and triples are algebras over the corresponding coloured operads.

Definition 1.7 (The aligned mixed operad)**.** The operad $\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$ is generated by the transverse associative operad $\mathsf{Ass}_\perp$, the associative chiral operad $\mathsf{Ass}^{\mathrm{ch}}_X$, and one mixed square for each aligned ordered decomposition. Its mixed square identifies the two composites

$$m_X \circ (m_\perp \star_{\mathrm{ch}} m_\perp) \circ \iota \quad \text{and} \quad m_\perp \circ (m_X \otimes_! m_X),$$

where ι is the aligned-decomposition inclusion of Theorem 2.1. In a braided realization, the second composite includes the exchange of the two middle factors.

Definition 1.8 (Doubly associative chiral algebra)**.** A doubly associative chiral algebra is an algebra over $\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$. Equivalently, it is a factorisation object A equipped with coherent products

$$m_\perp : A \otimes_! A \to A, \qquad m_X : A \star_{\mathrm{ch}} A \to A,$$

together with the aligned mixed squares and all their operadic coherences.

On an aligned window the projector is the identity, and $\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$ restricts to the ordinary Boardman–Vogt tensor product of the two associative operads. On the full Ran category the projector remains part of the mixed operation.

No-go theorem 1: one associative direction does not generate the other

The forgetful functors from $\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$-algebras to transverse E_1-algebras and to associative chiral algebras admit no natural section on all factorisable coefficients. At the level of algebraic theories, $\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$ contains two independent binary generators $m_\perp$ and m_X; a term formed from one generator cannot define the other in the free algebra. The aligned mixed cells of $\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$ supply the additional geometric datum that compares the two products.

1.4 Boundary and defect origin

Let $\mathcal{C}$ be a $\mathrm{FactD}(X)$-linear factorisation category and $b \in \mathcal{C}$ a factorisation-compatible boundary condition. The internal endomorphism object

$$A_b = \mathrm{RHom}_{\mathcal{C}}(b, b)$$

composes in an order and varies factorisably in X.

Theorem 1.9 (Factorisation Morita chart)**.** *Assume that b restricts to a compact dualisable generator on a factorising basis, that these local generator equivalences commute with restriction, and that both sides satisfy Weiss descent. Then A_b is a transversely ordered chiral algebra and*

$$\mathrm{RHom}_{\mathcal{C}}(b, -) : \mathcal{C} \xrightarrow{\sim} \mathrm{Mod}_{A_b}(\mathrm{FactD}(X))$$

is an equivalence of factorisation categories.

Proof. Composition gives the internal E_1-structure. Factorisation of b on disjoint discs makes composition a morphism of coefficient systems. The ordinary compact-generator Morita theorem applies on each basis element. Compatibility on overlaps and Weiss descent glue the local equivalences. □

The algebra A_b is a Morita chart for the factorisation category. A change of compact generator changes the chart and preserves the category. Bulk comparisons are therefore formulated at the categorical or centre level.

1.5 The complete diagonal coefficient

Let $D = \Delta(X) \subset X^2$. The meromorphic extension $\mathcal{O}_{X^2}(*D)$ has the increasing pole filtration

$$P_m\mathcal{O}_{X^2}(*D) = \mathcal{O}_{X^2}((m+1)D), \qquad m \geq 0.$$

For every $m \geq 0$, the divisor exact sequence gives

$$0 \longrightarrow \mathcal{O}_{X^2}(mD) \longrightarrow \mathcal{O}_{X^2}((m+1)D) \longrightarrow N_{D/X^2}^{\otimes(m+1)} \longrightarrow 0.$$

Because $N_{\Delta/X^2} \simeq T_X$, a local coordinate z identifies the symbol of $(z-w)^{-m-1}$ with the $(m+1)$-st normal tensor. Coordinate changes mix these symbols upper triangularly. The filtered diagonal distribution is intrinsic, while the individual modes are its coordinate components.

Proposition 1.10 (Complete-coefficient principle)**.** *A chiral operation consists of the diagonal extension together with its structure morphism. The residue is its first normal symbol; higher modes are higher normal symbols; delta derivatives and differential operators are the dual distributional realization. A bounded-pole hypothesis truncates this filtration to a finite jet object.*

Proof. The divisor exact sequence identifies the quotient of pole order $m+1$ by pole order m with $N_{D/X^2}^{\otimes(m+1)}$. Local Laurent expansion identifies its coefficient with the mode multiplying $(z-w)^{-m-1}$. Formal coordinate changes preserve the filtration and act upper triangularly on its graded splittings. Duality with distributions identifies the same graded pieces with derivatives of the delta distribution. Truncation at pole order N is exactly the quotient by $P_{>N}$. $\square$

No-go theorem 2: no universal logarithmic extractor

A natural linear retraction from logarithmic one-forms to all finite principal parts does not exist. The residue is invariant under formal coordinate change, whereas the coefficient of u^{-m-1} for $m \geq 1$ mixes with lower coefficients under $u \mapsto u + a_2u^2 + \cdots$. Naturality under the formal-diffeomorphism group therefore preserves the first normal symbol and requires the complete filtered normal object for the higher symbols. A coordinate, connection, or splitting supplies the additional jet data when one is chosen.

Chapter 2

Aligned decompositions and the exact locality law

2.1 Linear species

Let $\mathcal{V}$ be an additive stable symmetric monoidal category with finite direct sums. A linear species is a functor $F : \mathrm{Fin}^{\simeq} \to \mathcal{V}$. The category of linear species carries the Hadamard product

$$(F \boxtimes_H G)(I) = F(I) \otimes G(I)$$

and the Cauchy product

$$(F \boxtimes_C G)(I) = \bigoplus_{I = I_1 \sqcup I_2} F(I_1) \otimes G(I_2),$$

where the sum is over ordered decompositions.

The component of

$$(F_1 \boxtimes_C F_2) \boxtimes_H (G_1 \boxtimes_C G_2)$$

is indexed by a pair (d, e) of ordered decompositions of I, while the component of

$$(F_1 \boxtimes_H G_1) \boxtimes_C (F_2 \boxtimes_H G_2)$$

is indexed by one decomposition d. The one-decomposition composite is exactly the diagonal sector (d, d) inside the two-decomposition composite.

Theorem 2.1 (Aligned-decomposition splitting)**.** *Canonical natural transformations*

$$\begin{aligned}\iota :&(F_1 \boxtimes_H G_1) \boxtimes_C (F_2 \boxtimes_H G_2) \longrightarrow (F_1 \boxtimes_C F_2) \boxtimes_H (G_1 \boxtimes_C G_2),\\ \pi :&(F_1 \boxtimes_C F_2) \boxtimes_H (G_1 \boxtimes_C G_2) \longrightarrow (F_1 \boxtimes_H G_1) \boxtimes_C (F_2 \boxtimes_H G_2),\end{aligned}$$

with

$$\pi\iota = \mathrm{id}, \qquad e := \iota\pi, \qquad e^2 = e.$$

The image of e is the common-decomposition summand. The maps ι and π are compatible with associativity, units, and the symmetry of $\boxtimes_H$.

Proof. At arity I, include the summand indexed by d into the summand indexed by (d, d); this is ι. Project the double sum onto those pairs with equal decompositions; this is π. The two composites act as the identity on every common-decomposition summand and vanish on every complementary summand. For three Cauchy factors, both coherence composites include, respectively project onto, triples with one common ordered decomposition. Hence every coherence diagram reduces to equality of the same direct-summand map. □

No-go theorem 3: no global strong-duoidal isomorphism

Unless all independent-decomposition summands vanish, ι is not surjective and π is not injective. Therefore Hadamard and Cauchy products are not related by a universal interchange isomorphism. Any theorem using a strong duoidal law must either restrict to an aligned subcategory or retain the idempotent e.

Definition 2.2 (Aligned window)**.** A full replete subcategory of species is aligned if it is closed under both products and if the projector e acts as the identity on every iterated mixed product of its objects. On such a window, ι and π are inverse and the two products form a strong duoidal pair.

Finite support in a fixed decomposition type gives an elementary aligned window. The unrestricted Ran category retains the idempotent e as part of its mixed structure.

2.2 Sheaf-level locality

Fix an ordered decomposition $d = (I_1, I_2)$ and the corresponding disjointness inclusion $j_d : U_d \hookrightarrow X^I$. A factorisation structure gives

$$\delta_d : j_d^! A_I \xrightarrow{\sim} j_d^!(A_{I_1} \boxtimes A_{I_2}).$$

An ambient pointwise multiplication $m : A \otimes_! A \to A$ belongs to $\mathrm{FactD}(X)$ exactly when the square

$$\begin{tikzcd}
j_d^!(A_I \otimes_! A_I) \arrow[r, "j_d^! m"] \arrow[d, "\delta_d \otimes_! \delta_d"'] & j_d^! A_I \arrow[d, "\delta_d"] \\
j_d^!\big((A_{I_1} \otimes_! A_{I_1}) \boxtimes (A_{I_2} \otimes_! A_{I_2})\big) \arrow[r, "m \boxtimes m"'] & j_d^!(A_{I_1} \boxtimes A_{I_2})
\end{tikzcd}$$

commutes for every d.

Theorem 2.3 (Exact locality criterion)**.** *A pointwise E_1-multiplication on the underlying Ran object defines a transversely ordered chiral algebra precisely when all of the displayed disjoint-locus squares and their higher E_1 coherences commute. Under finite-set indexing, these equations lie in the image of the aligned projector of Theorem 2.1.*

Proof. A morphism in the category of factorisation objects is by definition a Ran morphism intertwining all factorisation equivalences. Applying this definition to every operation of the E_1-algebra gives the stated squares. The same decomposition d occurs before and after multiplication, which is exactly the common-decomposition sector. □

The locality squares are the sheaf-level bialgebra law. Their common decomposition places every mixed operation in the image of the aligned projector.

2.3 The two coproducts on the transverse bar

Once $m_\perp$ is a morphism of factorisation objects, every face and degeneracy in the simplicial bar is a morphism of factorisation objects. Geometric realization therefore retains the chiral factorisation coproduct. Independently, the normalized word complex has the deconcatenation coproduct

$$\Delta_{\mathrm{dec}}[a_1|\cdots|a_r] = \sum_{p=0}^{r}[a_1|\cdots|a_p] \otimes [a_{p+1}|\cdots|a_r].$$

The factorisation coproduct separates support on the curve. The deconcatenation coproduct cuts the transverse word. Their aligned interchange makes the two coalgebra structures commute.

2.4 The geometric meaning of the projector

The aligned projector selects a single ordered decomposition of the physical experiment and uses it in both monoidal directions. Four consequences follow.

1. Locality is expressed on the summands indexed by one common decomposition.
2. The transverse bar inherits factorisation because every simplicial face preserves that decomposition.
3. Iterated transverse and chiral bars acquire a comparison after the mixed operad supplies its aligned cells.
4. A strong duoidal formula is recovered on subcategories where the projector acts as the identity.

The idempotent is the algebraic record of one common geometric cut.

Chapter 3

Rectangular recognition and its exact hypotheses

3.1 Product-disk factorisation

Let Disk_X be a factorising basis of sufficiently small analytic discs in X^{an}, and let $\mathrm{Disk}_1^{\mathrm{ord}}$ be the little oriented intervals operad. The product basis consists of rectangles $U \times I$. A multimorphism is a family of pairwise disjoint rectangular embeddings, ordered in the interval coordinate.

Definition 3.1 (Rectangular holomorphic–topological factorisation algebra)**.** A rectangular factorisation algebra $\mathcal{O}$ on $X^{\mathrm{an}} \times \mathbb{R}$ satisfies:

1. Weiss descent on the product-disk basis;
2. local constancy under isotopy in the interval variable;
3. a holomorphic algebraisation in the X-variable by a factorisable right D-module;
4. compatibility of that algebraisation with the finite colimits and tensor products used to form the structure maps.

Recognition theorem 3.2 (Rectangular recognition)**.** *Let* $\mathrm{Fact}^{\mathrm{rect}}_{\mathrm{hol,lc}}(X \times \mathbb{R})$ *denote rectangular factorisation algebras satisfying Weiss descent, local constancy in the interval variable, holomorphic algebraisation by factorisable right* D*-modules, and compatibility of algebraisation with the tensor products and finite colimits defining the structure maps. Restriction to a standard interval induces an equivalence*

$$\mathrm{Fact}^{\mathrm{rect}}_{\mathrm{hol,lc}}(X \times \mathbb{R}) \simeq \mathrm{Alg}_{\mathsf{E}_1}(\mathrm{FactD}(X), \otimes_!).$$

The internal multiplication is fusion of ordered subintervals; every operation is a morphism of chiral coefficient systems.

Proof. A product-disk algebra is an algebra over the Boardman–Vogt tensor product $\mathrm{Disk}_X \otimes \mathrm{Disk}_1^{\mathrm{ord}}$. The operadic exponential law identifies its algebras with E_1-algebras internal to Disk_X-algebras. Local constancy identifies all sufficiently small intervals through a contractible space of choices, and ordered embeddings produce the E_1-operations. Conversely, an internal E_1-algebra evaluates a family of rectangles by external factorisation in X, followed by ordered multiplication. The tensor-product relation is exactly the locality square. Weiss descent and the algebraisation hypothesis extend the equivalence from the basis to the asserted categories. □

Remark 3.3 (Scope of rectangular recognition). The domain of the equivalence is the rectangular algebraic subcategory displayed in the theorem. Analytic theories with irregular singularities, non-algebraic monodromy, discontinuous completions, or uncontrolled infinite products enter only after a separate algebraisation functor satisfying the four defining conditions has been constructed.

3.2 Block-Ran recognition

An ordered block profile is a finite ordered list $\mathbf{I} = (I_1, \ldots, I_r)$ of finite chiral label sets. Its carrier is $X^{I_1} \times \cdots \times X^{I_r}$. Chiral refinements occur inside a block; transverse fusion joins adjacent blocks. Let BRan_X be the Grothendieck construction of the simplicial associative indexing category acting on the Ran diagram.

Theorem 3.4 (Block-Ran recognition)**.** *Transversely ordered chiral algebras are equivalent to coCartesian sections over* BRan_X *that are Segal in the block direction and factorising inside each chiral block. Marking a left endpoint, right endpoint, or both endpoints gives left modules, right modules, and bimodules. Gluing marked intervals computes relative tensor products.*

Proof. The simplicial nerve of an internal E_1-algebra has r-simplices $A^{\otimes_! r}$ and satisfies the Segal condition. The active map $[2] \to [1]$ recovers multiplication, $[0] \to [1]$ recovers the unit, and the simplicial identities recover all coherence. Factorisation inside each block says these maps live in $\mathrm{FactD}(X)$. Straightening and unstraightening provide inverse constructions. The marked statements are the corresponding coloured operads. □

3.3 Opposites and three reversals

Reflection of the interval, $t \mapsto -t$, gives $A^{\mathrm{op},\perp}$ and reverses the transverse multiplication. Reversal of an exchange path inverts braid transport. Verdier duality acts on the D-module coefficient. A comparison among these three involutions is additional duality data in a rigid model.

No-go theorem 4: order does not imply exchange

The ordered chamber $\mathrm{Conf}_n^<(\mathbb{R})$ is contractible and contains no path exchanging two labelled points. Hence an E_1-product determines associativity but no braid generator and no R-matrix. Exchange requires a second transverse direction, a meromorphic connection, or braided categorical input.

Chapter 4

Associative chiral algebras and nonlocal vertex fields

4.1 The associative chiral direction

The transverse theory places an ordered product on a factorisable chiral coefficient. Associative chiral composition is a second operation: it composes ordered singular fields on the curve. The operad $\mathsf{Ass}_X^{\mathrm{ch}}$ carries this additional structure.

Let $\mathsf{Ass}_X^{\mathrm{ch}}(I)$ denote the complex of ordered chiral operations with inputs labelled by I. Geometrically, an operation is defined on the complement of collision diagonals, carries an ordering of the input flags, and extends as an allowed distribution across a specified diagonal. Operadic substitution is composition along nested partial diagonals, with the order of flags retained. Forgetting the order and taking the signed commutator maps to the usual chiral Lie operad. Adjoining a braiding satisfying the equations of the quantum vertex enhancement produces the braided operadic extension.

Definition 4.1 (Associative chiral multiplication)**.** An associative chiral multiplication on a coefficient C is an algebra structure over $\mathsf{Ass}_X^{\mathrm{ch}}$. In arity two it is a distributional map

$$\mu_X : j_* j^*(C \boxtimes C) \longrightarrow \Delta_! C$$

with ordered inputs. In arity three, the two nested collision composites agree in a strict $\mathsf{Ass}_X^{\mathrm{ch}}$-algebra; a cofibrant resolution replaces the equality by a specified associative homotopy and its higher coherences.

A Beilinson–Drinfeld chiral Lie algebra has a binary operation that is skew with respect to the transposition of the two input factors and satisfies the chiral Jacobi identity. An associative chiral multiplication retains the two input orders separately. Whenever transposition acts on the chosen singular extension, the signed commutator of the two ordered products defines a chiral Lie operation.

4.2 Formal-disc realization

Let V be a complete linearly topologized vector space, let $\mathbf{1} \in V$, and let

$$Y(-, z) : V \longrightarrow \mathrm{Hom}_{\mathrm{cont}}(V, V((z))), \qquad a \longmapsto Y(a, z) = \sum_{n \in \mathbb{Z}} a_{(n)} z^{-n-1}.$$

Definition 4.2 (Rational complete nonlocal vertex algebra)**.** A rational complete nonlocal vertex algebra consists of $(V, Y, \mathbf{1}, D)$ with the following properties.

1. $Y(\mathbf{1}, z) = \mathrm{id}_V$, and $Y(a, z)\mathbf{1} \in V[[z]]$ has constant term a.
2. $[D, Y(a, z)] = \partial_z Y(a, z)$, and $D\mathbf{1} = 0$.
3. For every $a, b, c \in V$, an integer $N \geq 0$ satisfies

$$(z_0 + z_2)^N Y(a, z_0 + z_2)Y(b, z_2)c = (z_0 + z_2)^N Y(Y(a, z_0)b, z_2)c.$$

4. Every continuous matrix coefficient of an iterated field is an expansion of a rational function on the ordered formal configuration space, with finite pole order along its partial diagonals; the expansions attached to nested configuration domains agree under restriction.

The fourth condition is the rationality and descent datum that turns formal fields into a distributional chiral operation.

Weak associativity is the three-point operadic composition law after clearing the diagonal denominator. Rationality glues its formal expansions over the ordered configuration domains.

Definition 4.3 (Quantum vertex enhancement)**.** Assume that V is topologically free over $\Bbbk[[\hbar]]$. A quantum vertex enhancement of Theorem 4.2 consists of an invertible braiding

$$\mathcal{S}(z) : V \widehat{\otimes} V \longrightarrow V \widehat{\otimes} V \widehat{\otimes} \Bbbk((z))[[\hbar]], \qquad \mathcal{S}(z) \equiv \mathrm{id} \pmod{\hbar},$$

satisfying

$$\begin{aligned}
&\mathcal{S}_{21}(-z)\mathcal{S}(z) = \mathrm{id}, \\
&\mathcal{S}_{12}(z_1)\mathcal{S}_{13}(z_1 + z_2)\mathcal{S}_{23}(z_2) = \mathcal{S}_{23}(z_2)\mathcal{S}_{13}(z_1 + z_2)\mathcal{S}_{12}(z_1), \\
&[D \otimes 1, \mathcal{S}(z)] = -\frac{d}{dz}\mathcal{S}(z).
\end{aligned}$$

For every $a, b, c \in V$ and $M \geq 0$, braided locality requires an $N \geq 0$ such that

$$\begin{aligned}
&(z - w)^N Y(z)(1 \otimes Y(w))\mathcal{S}^{12}(z - w)(a \otimes b \otimes c) \\
&\qquad = (z - w)^N Y(w)(1 \otimes Y(z))(b \otimes a \otimes c) \pmod{\hbar^M}.
\end{aligned}$$

Together with the weak associativity of Theorem 4.2, these identities define the $\hbar$-adic quantum vertex structure of Etingof–Kazhdan and its structural formulation in De Sole et al. [18], Li [38]. All tensor products are completed in the $\hbar$-adic and given linear topologies.

Recognition theorem 4.4 (Formal-disc recognition)**.** *Choose a formal coordinate on the disc. Formal-coordinate expansion defines an equivalence between:*

1. *unital translation-equivariant, pole-complete algebras over the ordered associative chiral operad $\mathsf{Ass}^{\mathrm{ch}}_X$; and*
2. *rational complete nonlocal vertex algebras in the sense of Theorem 4.2.*

Under this equivalence, the arity-three operadic composition is weak associativity, the unit is the vacuum, and infinitesimal translation is D. A braiding satisfying the equations of the quantum vertex enhancement gives the corresponding $\hbar$-adic quantum vertex algebra structure [18, 38]. Restriction to skew-symmetric chiral operations recovers the Beilinson–Drinfeld/formal-variable equivalence for vertex algebras, and the pole associated graded is the classical operad governing Poisson vertex algebras [3, 34].

Proof. Expand an ordered chiral distribution along the formal coordinate diagonals. Lower truncation gives $Y(a,z)b \in V((z))$; the unit and translation action give the vacuum, creation, and derivative identities. Operadic substitution on the two nested three-point domains yields two expansions of one rational distribution, and clearing its common diagonal denominator gives weak associativity. Higher operadic substitutions give the compatible rationality data of Theorem 4.2.

Conversely, rational matrix coefficients define continuous distributions on every ordered formal configuration space. Compatibility of their expansions supplies restriction to nested domains and hence operadic substitution. Weak associativity identifies the two three-point substitutions, while the vacuum and translation identities provide the nullary and unary operations. The constructions preserve all arities and are inverse. The quantum statement is the same comparison with the exchange operator inserted between the two ordered expansions. □

No-go theorem 5: a transverse E_1-algebra is not a quantum vertex algebra

The datum of an object in $\mathrm{Alg}_{\mathsf{E}_1}(\mathrm{FactD}(X), \otimes_!)$ consists of a factorisable coefficient and a transverse multiplication. The state–field map $Y(a,z)b$ and the exchange operator $\mathcal{S}(z)$ are additional operations with independent axioms. Hence the transverse algebra alone determines neither a nonlocal vertex structure nor a quantum vertex structure; the associative chiral coefficient supplies Y, and the braided enhancement supplies $\mathcal{S}$.

4.3 The genuinely doubly noncommutative object

On the formal disc, an algebra over $\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$ consists of a nonlocal vertex operation Y_X, a transverse associative product $m_\perp$, and the aligned identity

$$Y_X\big(m_\perp(a,b), z\big)m_\perp(c,d) = m_\perp(Y_X(a,z)c,\, Y_X(b,z)d)$$

whenever the four inputs are represented by the same ordered decomposition. A braided model inserts $\mathcal{S}$ between the middle factors before the right-hand product. The cofibrant mixed operad replaces this equality by a distinguished homotopy and adds all higher compatibility cells. The field product, the transverse word, and the exchange operator therefore provide three independent sources of noncommutativity.

4.4 Separated-factor construction

Let C_X be a rational associative chiral algebra whose underlying factorisable coefficient is a commutative algebra for $\otimes_!$, and assume that the pointwise and chiral products satisfy the aligned mixed law. Let B be an augmented associative algebra. The tensor product

$$A = B \otimes C_X$$

has transverse multiplication $m_B \otimes m^{\mathrm{pt}}_{C_X}$, associative chiral multiplication $1_B \otimes m^{\mathrm{ch}}_{C_X}$, and the tensor-factorwise mixed law. The tensor product $B \otimes C_X$ gives strict doubly associative examples. The free object on arbitrary generators is instead $\mathrm{Free}_{\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}}(M)$; its two-coloured tree filtration records the interactions between the two products. A confluent spectral exchange quotient gives the Zamolodchikov–Faddeev models of Chapter 16.

Chapter 5

Homotopy-coherent generators and relations

5.1 Derived generators and relations

A derived presentation records generators, relation maps, chosen null-homotopies, and the higher homotopies among those choices. Its quotient is the homotopy pushout of the corresponding free-algebra diagram.

Fix a combinatorial symmetric monoidal model category $\mathcal{M}$ whose localization is the chosen stable symmetric monoidal ∞-category. Assume that $\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$, its module colours, and their Boardman–Vogt resolutions are admissible, well pointed, and Σ-cofibrant in $\mathcal{M}$. Let

$$\mathcal{P} = W\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$$

be a cofibrant resolution of the aligned mixed operad.

Definition 5.1 (Derived presentation)**.** Let V be a collection of generators, R a collection of relation cells, and $\partial_0, \partial_1 : \mathrm{Free}_{\mathcal{P}}(R) \rightrightarrows \mathrm{Free}_{\mathcal{P}}(V)$ the two sides of each relation. The derived algebra presented by (V, R) is the homotopy coequalizer

$$A \simeq \mathrm{hcoeq}\big(\mathrm{Free}_{\mathcal{P}}(R) \rightrightarrows \mathrm{Free}_{\mathcal{P}}(V)\big),$$

equivalently the corresponding homotopy pushout after adjoining relation disks.

A cofibrant replacement of this homotopy coequalizer is the generators-and-relations object needed for derived mapping spaces. A map $A \to B$ is a choice of images of generators, null-homotopies of relations, and all higher compatibility data encoded by $\mathcal{P}$.

5.2 Two-coloured levelled forests

A cell of $\mathcal{P}$ is represented by a levelled forest whose internal edges are coloured $\perp$ or X. A transverse edge resolves associative composition along the ordered line. A chiral edge resolves nested collision. For a forest T, set

$$\mathrm{or}(T) = \det \Bbbk E_{\perp}(T) \otimes \det \Bbbk E_X(T).$$

The cellular complex is

$$\mathsf{Word}_{\mathcal{P}}(I) = \bigoplus_{T} \mathrm{Dec}(T) \otimes \mathrm{or}(T)$$

with

$$d = d_{\mathrm{dec}} + d_{\perp} + d_X.$$

Theorem 5.2 (Mixed boundary identity)**.** *With the product orientation,*

$$d_{\perp}^2 = d_X^2 = 0, \qquad d_{\perp} d_X + d_X d_{\perp} = 0,$$

and d_{dec} supercommutes with both edge differentials. Hence $d^2 = 0$.

Proof. A codimension-two cell has two contracted edges. If the colours agree, the two orders give the opposite boundary orientations of the same cellular operad. If the colours differ, both orders give the same decorated forest but exchange the two one-dimensional determinant factors, producing a minus sign. Naturality of the decoration differential gives the remaining anticommutators. ☐

When vertices are decorated by operations, $d_{\perp}^2 = 0$ is the Stasheff system, $d_X^2 = 0$ is associative chiral coherence, and the mixed anticommutator is the entire hierarchy of locality/interchange homotopies.

5.3 Rectification and the ∞-category

Theorem 5.3 (Cofibrant comparison)**.** *Assume:*

1. *$\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$ is admissible in $\mathcal{M}$;*
2. *$W\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$ is cofibrant and the augmentation $W\mathsf{Mix}^{\mathrm{ch}}_{X,\perp} \to \mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$ is an aritywise weak equivalence;*
3. *change of operad along this augmentation is a Quillen equivalence.*

Then strict algebras over the target operad and homotopy-coherent algebras over $\mathcal{P}$ present equivalent ∞-categories. Derived presentations computed in either model represent the same mapping functor.

Proof. The hypotheses make $\mathcal{P} \to \mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$ a cofibrant resolution and identify the localized algebra categories by change of operad. Homotopy pushouts between cofibrant diagrams compute colimits in the localized ∞-category. ☐

The model category supplies explicit cofibrant representatives. The localized mapping space is the invariant mapping object.

5.4 Low arity

At arity two there are two primitive binary operations $m_{\perp}$ and m_X. At arity three, the codimension-one faces produce

$$m_{\perp}(m_{\perp}(a,b),c) - m_{\perp}(a, m_{\perp}(b,c)),$$
$$m_X(m_X(a,b),c) - m_X(a, m_X(b,c)),$$

and the mixed square compares

$$m_X(m_\perp(a,b), m_\perp(c,d)) \quad \text{with} \quad m_\perp(m_X(a,c), m_X(b,d))$$

on a common decomposition. In a braided model, the middle factors are transported by the exchange operator before the second expression is evaluated. At arity four the pentagons, chiral nested-collision cells, and mixed prisms appear automatically. The arity-four coherences are the cellular boundaries of the cofibrant mixed operad.

5.5 Homotopy transfer

Given a deformation retract

$$(H, d_H) \underset{p}{\overset{i}{\rightleftarrows}} (V, d_V), \qquad ip - \mathrm{id}_V = d_V h + h d_V,$$

evaluate a decorated forest by placing original operations at vertices and h on internal edges. The mixed boundary identity proves the Maurer–Cartan equation for the transferred two-direction structure. The same formula is the algebraic skeleton of perturbative Feynman rules: interaction vertices decorate vertices, propagators decorate edges, and boundary restrictions of compactified graph configurations implement edge contraction.

No-go theorem 6: incidence is not screen geometry

The nerve of the incidence poset of collision strata can be contractible even when the geometric screen has nonzero cohomology. For three points on the affine complex line, the projectivized screen is $\mathbb{P}^1 \setminus \{0, 1, \infty\}$, whose first cohomology has rank two, whereas the incidence fibre with terminal corolla is contractible. Therefore a poset resolution cannot replace screen chains unless a comparison theorem proves that the omitted rows are acyclic.

Part II

Bar, cobar, Morita, and homotopy-coherent presentations

Chapter 6

The transverse interval bar

6.1 The simplicial object

Let

$$A \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathrm{FactD}(X), \otimes_!), \qquad \varepsilon : A \to \mathbf{1}, \qquad \overline{A} = \mathrm{fib}(\varepsilon).$$

The two-sided simplicial bar is

$$B^{\perp}_r(A) = \mathbf{1} \otimes_! A^{\otimes_! r} \otimes_! \mathbf{1}, \qquad r \geq 0.$$

Interior faces multiply adjacent factors, outer faces use the augmentation, and degeneracies insert the unit. Its geometric realization is

$$\mathrm{Bar}^{\perp}_X(A) = \left| B^{\perp}_{\bullet}(A) \right| \simeq \mathbf{1} \otimes^{\mathbb{L}}_A \mathbf{1}.$$

Every simplicial map belongs to $\mathrm{FactD}(X)$ by the exact locality criterion; hence the entire bar remains a chiral coefficient system.

6.2 Normalized words and signs

Write $s\overline{A}$ for cohomological suspension. Normalization gives the tensor coalgebra

$$T^c(s\overline{A}) = \bigoplus_{r \geq 0} (s\overline{A})^{\otimes_! r}.$$

For homogeneous $a_i \in \overline{A}$, abbreviate $[a_1|\cdots|a_r] = sa_1 \otimes \cdots \otimes sa_r$. We fix

$$b_2(sa \otimes sb) = (-1)^{|a|} s(ab).$$

The multiplication coderivation is therefore

$$b_m[a_1|\cdots|a_r] = \sum_{i=1}^{r-1} (-1)^{\sum_{j<i}(|a_j|+1)+|a_i|} [a_1|\cdots|a_i a_{i+1}|\cdots|a_r].$$

The internal differential is the tensor differential on suspended letters. For an A_∞-algebra, add the coderivations induced by every m_k.

Proposition 6.1 (Bar identity)**.** *The total bar differential $d_B = d_{\mathrm{int}} + \sum_{k \geq 2} b_{m_k}$ squares* *to zero exactly when the* A_∞*-relations hold. In the strict ungraded case, the square on a three-letter word is the suspended associator*

$$b_m^2[a|b|c] = \pm[(ab)c - a(bc)].$$

Proof. A coderivation of the cofree tensor coalgebra is determined by its projection to cogenerators. The projection of d_B^2 in arity n is precisely the signed sum of all substitutions $m_r(1^{\otimes i} \otimes m_s \otimes 1^{\otimes j})$ with $r + s = n + 1$, which is the n-th Stasheff identity. □

The deconcatenation coproduct

$$\Delta_{\mathrm{dec}}[a_1|\cdots|a_r] = \sum_{p=0}^{r}[a_1|\cdots|a_p] \otimes [a_{p+1}|\cdots|a_r]$$

makes $\mathrm{Bar}_X^{\perp}(A)$ a conilpotent coassociative coalgebra in $\mathrm{FactD}(X)$. The bar differential is a coderivation because every multiplication contracts a consecutive subword.

6.3 Interval factorisation homology

Theorem 6.2 (Bar as interval amplitude)**.** *Let the endpoints of* $[0,1]$ *carry the augmentation module. Then*

$$\mathrm{Bar}_X^{\perp}(A) \simeq \mathbf{1} \otimes_A^{\mathbb{L}} \mathbf{1} \simeq \int_{([0,1],\partial[0,1])} A.$$

More generally, for a right module R *and left module* L*,*

$$\mathrm{Bar}(R, A, L) \simeq R \otimes_A^{\mathbb{L}} L \simeq \int_{([0,1],\{0,1\})} (A; R, L).$$

Proof. Cover the interval by an ordered chain of small subintervals and endpoint collars. The factorisation Cech diagram is the two-sided simplicial bar. Excision identifies its colimit with factorisation homology and with the derived relative tensor product. Refinement of the cover is cofinal. □

A bar word is an ordered sequence of intermediate boundary operations. The bar differential fuses adjacent operations, and the factorisable chiral coefficient accompanies every letter and multiplication map. Bar degree measures transverse interval length.

No-go theorem 7: collision residue cannot replace the bar differential

The transverse bar differential has source $(s\overline{A})^{\otimes_! r}$ and target $(s\overline{A})^{\otimes_!(r-1)}$, and it is induced by $m_\perp$. A collision residue changes the normal degree of a distribution on X^I and is induced by restriction to a diagonal. Their source and target gradings differ. A bicomplex can carry both maps as anticommuting differentials; each differential retains its own geometric degree and coefficient map.

6.4 Associahedra and the monadic bar

The simplicial bar presents the derived tensor product. The Stasheff associahedron resolves the multiplication used to evaluate it. If A is an algebra over a cofibrant A_∞-operad, the operadic bar has a cellular filtration with terms

$$\bigoplus_{r\geq 0} C_*(K_r) \otimes (s\overline{A})^{\otimes_! r}.$$

Rectification gives a zigzag of quasi-isomorphisms to the simplicial bar of a strict model. The simplex resolves the derived tensor product, while the associahedron resolves the multiplication used in its faces.

6.5 Completion

The direct-sum bar is conilpotent. In formal or filtered field theory one often needs

$$\widehat{\mathrm{Bar}}^{\perp}(A) = \prod_{r\geq 0} (s\overline{A})^{\otimes_! r}.$$

Let $F^N A$ be a complete decreasing filtration such that every m_k raises total filtration by at least $k-1$.

Proposition 6.3 (Completed-bar convergence)**.** *Assume that each quotient $A/F^N A$ has finite bar length in a fixed total weight and that the cohomology tower satisfies Mittag–Leffler. Then*

$$H^*(\widehat{\mathrm{Bar}}^{\perp}(A)) \cong \varprojlim_N H^*(\mathrm{Bar}^{\perp}(A/F^N A)).$$

The bar differential is continuous and the completion is separated.

Proof. The completed bar is the inverse limit of the finite quotient bars. The Mittag–Leffler hypothesis kills $\varprojlim^1$, and the filtration-growth assumption makes every coderivation continuous. $\square$

Chapter 7

The associative chiral bar and the theorem of two bars

7.1 Operadic chiral bar

Let C be an augmented algebra over $\mathsf{Ass}_X^{\mathrm{ch}}$. The operadic bar is a conilpotent coalgebra over the bar cooperad $B\mathsf{Ass}_X^{\mathrm{ch}}$:

$$\mathrm{Bar}_{\mathsf{Ass}}^{\mathrm{ch}}(C) = \bigoplus_T B\mathsf{Ass}_X^{\mathrm{ch}}(T) \otimes_{\mathrm{Aut}\, T} (s\overline{C})^{\otimes\, \mathrm{in}(T)}.$$

The index T runs over ordered chiral composition trees. The differential has three sources:

1. the internal differential of C;
2. the cooperadic differential of the chiral resolution, including screen chains when present;
3. contraction of an internal chiral edge by the associative chiral multiplication.

The entire principal-part distribution remains attached to each vertex and edge. Its normal symbols supply residues, higher modes, and differential-operator components.

On the formal disc, this bar resolves iterated nonlocal fields. Its length is the number of chiral composition vertices, and its relations are weak associativity together with all higher homotopies. When an exchange operator is present, the chiral trees are braided and the differential uses the corresponding transport.

7.2 Chiral Chevalley chains are a third construction

A chiral Lie algebra $\mathfrak{g}$ has Chevalley chains

$$\mathrm{CE}_*^{\mathrm{ch}}(\mathfrak{g}) = \big(\mathrm{Cofree}_{\mathsf{Com}^c}^{\mathrm{ch}}(s\mathfrak{g}), d_{\mathrm{int}} + d_{[-,-]_{\mathrm{ch}}}\big).$$

Francis–Gaitsgory chiral Koszul duality uses this chiral Chevalley construction. Enveloping and commutator morphisms compare it with the associative chiral bar, while a monoidal realization compares either chiral construction with the transverse interval bar.

Theorem 7.1 (The theorem of two associative bars)**.** *Let A be a doubly associative chiral algebra in the sense of Theorem 1.8. Then:*

1. $\mathrm{Bar}^{\perp}(A)$ *is a conilpotent coalgebra for transverse deconcatenation and remains an as-*

sociative chiral coefficient whenever the mixed locality maps are compatible with the bar faces;

2. $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A)$ *is a conilpotent chiral cooperadic object and remains a transverse* E_1*-algebra whenever the same mixed data are compatible with chiral edge contraction;*
3. *the iterated objects* $\mathrm{Bar}^{\perp}\,\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A)$ *and* $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}\,\mathrm{Bar}^{\perp}(A)$ *are linked by a canonical comparison on the aligned-decomposition sector;*
4. *if the mixed operad is a distributive law whose comparison map of composite operads is an aritywise equivalence, and if both bar realizations preserve geometric realizations of the resulting bisimplicial objects, the comparison is an equivalence and the total differential is*

$$d_{\mathrm{tot}} = d_{\mathrm{int}} + d_{\perp} + d_X, \qquad d_{\perp} d_X + d_X d_{\perp} = 0.$$

Proof. The first two statements apply each bar functor internally to the algebra category for the other operation. The mixed cells make every face map a morphism for the remaining structure. The aligned projector supplies the comparison between the two orders of common decomposition. Under a distributive law, the two-sided bar construction is the geometric realization of one bisimplicial/cooperadic object; Fubini for homotopy colimits gives the equivalence. The anticommutator is the mixed edge identity of Theorem 5.2. □

No-go theorem 8: no unnamed comparison of bars

The expressions $\mathrm{Bar}^{\perp}(A)$ and $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A)$ belong to bar constructions for different monoidal products and carry different geometric gradings. An equivalence between them consists of a strong monoidal realization, a distributive law, and the resulting chain map. Equality of Euler characteristics supplies a numerical consequence of such a map, rather than the map itself.

7.3 Universal enveloping comparison

Suppose $\mathfrak{g}$ is a chiral Lie algebra and $U^{\mathrm{ch}}(\mathfrak{g})$ is an associative chiral enveloping algebra satisfying PBW. The universal twisting map induces

$$\mathrm{CE}^{\mathrm{ch}}_{*}(\mathfrak{g}) \longrightarrow \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(U^{\mathrm{ch}}(\mathfrak{g})).$$

Under the chiral PBW and connectivity hypotheses, the associated graded map is the ordinary Koszul comparison between symmetric coalgebras and the bar of a tensor quotient, hence a quasi-isomorphism. The enveloping comparison thereby identifies the Lie-type and associative-chiral chain models.

For a transversely ordered enveloping algebra $U(\mathfrak{g})$ in the coefficient category, the analogous comparison is

$$\mathrm{CE}_{*}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{Bar}^{\perp}(U(\mathfrak{g})),$$

which is the classical Cartan–Eilenberg theorem internal to the coefficient category. The two statements share a proof pattern but have different multiplications.

Chapter 8

Bar–cobar reconstruction and the completion locus

8.1 Cobar

Let C be a coaugmented conilpotent coassociative coalgebra in a stable symmetric monoidal category $\mathcal{V}$, and write $\overline{C}$ for its coaugmentation coideal. The cobar construction is

$$\mathrm{Cobar}(C) = \big(T(s^{-1}\overline{C}), d_C + d_\Delta\big),$$

where d_Δ is the derivation induced by the reduced coproduct. The universal twisting morphisms give the adjunction

$$\mathrm{Bar} : \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathcal{V}) \rightleftarrows \mathrm{Coalg}^{\mathrm{coaug}}_{B\mathsf{E}_1}(\mathcal{V}) : \mathrm{Cobar}\,.$$

The notation on the coalgebra side includes the divided-power or norm data required by the ambient category; in ordinary characteristic-zero chain complexes it reduces to the familiar conilpotent coassociative coalgebra.

8.2 Nilcompletion and conilcompletion

An augmented algebra has an operadic nilpotent tower; its inverse limit is the nilcompletion $A^\wedge_{\mathrm{nil}}$. A coalgebra over the bar cooperad has a conilpotent tower; its inverse limit is the conilcompletion $C^\wedge_{\mathrm{conil}}$.

External theorem 8.1 (Nilcomplete operadic bar–cobar duality)**.** *Let $\mathcal{V}$ be a stable presentable symmetric monoidal ∞-category whose tensor product preserves colimits separately, and let $\mathcal{O}$ be a reduced operad. The operadic bar–cobar adjunction restricts to an equivalence between nilcomplete $\mathcal{O}$-algebras and conilcomplete divided-power coalgebras over the bar cooperad $B\mathcal{O}$. The nilcomplete and conilcomplete subcategories are the maximal full subcategories on which the unit and counit are equivalences [32].*

External theorem 8.2 (Point-set comparison of Koszul adjunctions)**.** *In differential graded categories, the classical operadic bar–cobar adjunction and the completed point-set bar–cobar adjunction fit into a commutative inclusion–restriction square. Passing to underlying ∞-categories gives the corresponding Koszul adjunction and identifies its relation to the constructions of Lurie, Francis–Gaitsgory, and Heuts. Thus a chain model and its higher-categorical* *completion are related by explicit derived functors rather than by an implicit identification [33].*

Corollary 8.3 (Internal associative Koszul duality)**.** *Let* $(\mathcal{V}, \otimes, \mathbf{1})$ *satisfy the hypotheses of Theorem 8.1, and let*

$$\mathcal{V} = \mathcal{V}_{\geq 0} \supset \mathcal{V}_{\geq 1} \supset \cdots, \qquad \mathcal{V}_{\geq r} \otimes \mathcal{V}_{\geq s} \subset \mathcal{V}_{\geq r+s}$$

be a complete separated multiplicative filtration for which tensor product commutes with completion. Then

$$\mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathcal{V})_{\mathrm{nilcomp}} \simeq \mathrm{Coalg}^{\mathrm{coaug}}_{B\mathsf{E}_1}(\mathcal{V})_{\mathrm{conilcomp}}.$$

For $(\mathrm{FactD}(X), \otimes_!)$*, a positive augmentation-weight filtration gives this equivalence on the complete positive subcategory. For chiral convolution on* $\mathrm{DMod}(\mathrm{Ran}\, X)$*, the cardinality filtration of Theorem 1.3 gives the pro-nilpotent Ran-space setting of Francis–Gaitsgory [25].*

Proof. Apply Theorem 8.1 to the reduced associative operad. The displayed multiplicative filtration identifies the nilpotent and conilpotent towers with the chosen completions. Positive augmentation weight supplies the filtration for pointwise tensor, while support cardinality supplies it for chiral convolution. □

The bar-cooperad coalgebra includes the norm or divided-power structure required by the ambient stable category. Ordinary characteristic-zero chain complexes recover conilpotent coassociative coalgebras.

8.3 Positive-weight proof

A positive weight grading gives a direct proof of reconstruction, independent of the general nilcomplete theorem.

Theorem 8.4 (Positive reconstruction)**.** *Let*

$$A = \mathbf{1} \oplus \prod_{w \geq 1} A_w, \qquad A_u \otimes_! A_v \longrightarrow A_{u+v},$$

be connected and complete, and assume that every weight has finitely many decompositions into positive weights. Then the counit

$$\mathrm{Cobar}\,\mathrm{Bar}(A) \longrightarrow A$$

is a filtered quasi-isomorphism. Dually, if a coaugmented coalgebra is conilpotent, complete in a positive weight grading, and has finitely many coproduct decompositions in every weight, then the unit $C \to \mathrm{Bar}\,\mathrm{Cobar}(C)$ *is a filtered quasi-isomorphism.*

Proof. Fix total weight w. Only finitely many subdivided words occur. Filter $\mathrm{Cobar}\,\mathrm{Bar}(A)$ by the number of surviving cuts. The associated graded of each fixed underlying word is the augmented cellular chain complex of the simplex of its subdivisions. The extra degeneracy joining the first cut contracts this complex onto the unsplit word. Thus the associated-graded counit is a quasi-isomorphism in every weight. Completeness promotes it to the filtered equivalence. The coalgebra proof uses the finite coradical filtration in each weight. □

8.4 Mittag–Leffler limits

For a complete filtered algebra $A \simeq \varprojlim A/F^N$, reconstruction is inherited from the finite quotients when the cohomology towers satisfy Mittag–Leffler and tensor product is continuous. The Milnor exact sequence identifies the additional term as $\varprojlim^1 H^{*-1}$; the Mittag–Leffler condition makes this term vanish.

No-go theorem 9: bar–cobar without completion

In the filtered and pro-nilpotent settings used here, the canonical map $\mathrm{Cobar}\,\mathrm{Bar}(A) \to A$ identifies the source with the nilcompletion of A. The bar–cobar counit is an equivalence on the nilcomplete locus and in finite positive weights. An unqualified equality outside that locus suppresses the completion computed by the double-centralizer map.

8.5 Deformation complexes

The convolution complex

$$\mathrm{Conv}(B\mathsf{E}_1, \mathrm{End}_A)$$

controls deformations of the associative multiplication. Its Maurer–Cartan elements are homotopy-coherent products, and its gauge equivalences are operadic homotopies. In the two-direction theory, the complete deformation complex is the total convolution algebra of the bicoloured cooperad. Its differential splits into internal, transverse, chiral, and mixed components. Curvature can be inserted only after its arity and degree are specified.

Chapter 9

Koszul–Morita theory, relative bases, and the centre

9.1 The augmentation boundary

For an augmented algebra A, the unit is a left and right module. The augmentation-boundary algebra is

$$A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1}).$$

When bar terms are dualisable, or continuous internal Hom is used,

$$A^! \simeq \mathrm{RHom}_{\mathcal{V}}(\mathrm{Bar}(A), \mathbf{1}).$$

The endomorphism algebra $A^!$ is the boundary Koszul dual. Quadratic and Verdier duals map to it through the comparison morphisms of Chapter 10.

Theorem 9.1 (Double centralizer)**.** *Assume A is connective, derived augmentation completion is computed by the cobar totalization of the Amitsur complex, and $\mathbf{1}$ is compact in the completed module category. Then*

$$A^{!!} := \mathrm{RHom}_{A^!}(\mathbf{1}, \mathbf{1}) \simeq A^{\wedge}_{\varepsilon}.$$

On the nilcomplete locus this is A.

Proof. The bar resolution identifies the thick subcategory generated by $\mathbf{1}$ with perfect $A^!$-modules. The endomorphisms of the corresponding Yoneda generator form the double centralizer. The universal algebra acting on the augmentation-generated subcategory is derived completion along $\ker \varepsilon$, computed by the assumed Amitsur totalization. □

9.2 One-sided and two-sided twisting

Put $C = \mathrm{Bar}(A)$ and let $\tau : C \to A$ be the universal twisting cochain. For an A-module M, define

$$K_A(M) = C \otimes_! M$$

with the sum of the internal differential and the twisting term obtained from Δ_C, τ, and the module action. The first factor makes it a C-comodule. The reverse functor is $L_C(N) = A \otimes_! N$ with the dual twisting differential.

Theorem 9.2 (Positive-complete module–comodule duality)**.** *Let A be connected and complete in a positive weight grading, assume finite weight decompositions, and assume that tensor product is exact on the chosen module category and commutes with weight completion. Put $C = \mathrm{Bar}(A)$. The twisting functors K_A and L_C are inverse equivalences*

$$\mathrm{Mod}_A^{\mathrm{comp},+} \simeq \mathrm{Comod}_C^{\mathrm{conil},+}.$$

For two algebras A, B satisfying the same hypotheses, the two-sided twisting functor gives

$$\mathrm{Bimod}_{A,B}^{\mathrm{comp},+} \simeq \mathrm{Bicomod}_{\mathrm{Bar}(A),\mathrm{Bar}(B)}^{\mathrm{conil},+},$$

and carries the regular diagonal bimodule to the regular diagonal bicomodule.

Proof. Filter the composites by bar cuts and coradical length. In each fixed weight, the associated graded is the subdivision complex of an ordered word with an extra degeneracy, hence contractible onto the unsplit module or comodule. Finite decomposition gives convergence; completion passes from finite windows to the full object. □

For composable bimodules,

$$K_{A,D}^{\mathrm{bi}}(M \otimes_B^{\mathbb{L}} N) \simeq K_{A,B}^{\mathrm{bi}}(M) \square_{\mathrm{Bar}\,B}^{\mathbb{R}} K_{B,D}^{\mathrm{bi}}(N).$$

Thus tensor composition becomes cotensor composition. Mapping objects identify Hochschild cochains with coHochschild cochains, and Hochschild chains with coHochschild chains, in the complete categories.

9.3 Relative augmentation bases

A separable or semisimple algebra S provides the natural augmentation base for quiver algebras, multi-boundary systems, and other algebras with distinguished idempotents.

Definition 9.3 (Relative bar)**.** For an S-algebra A equipped with a retraction $A \to S$, the relative reduced bar is

$$\mathrm{Bar}_S(A) = S \otimes_A^{\mathbb{L}} S \simeq T_S^c(s\overline{A}_S)$$

with tensors and multiplication taken over S.

For a quiver, $S = \bigoplus_i \Bbbk e_i$ is the vertex-idempotent algebra. Tensoring over S automatically retains only composable paths. The relative tensor product retains precisely the composable paths and computes the augmentation boundary over the vertex-idempotent algebra.

9.4 The internal derived centre

The derived centre is

$$\mathbf{Z}(A) = \mathrm{RHom}_{A^e}(A, A), \qquad A^e = A \otimes_! A^{\mathrm{op}}.$$

The brace operations make $\mathbf{Z}(A)$ an E_2-algebra internal to the coefficient category. In the Morita $(\infty, 2)$-category of algebras and bimodules, $\mathbf{Z}(A)$ is the endomorphism E_2-algebra of the identity A-bimodule and represents coherent central actions on A [40]. Morita equivalences identify these endomorphism objects.

A physical bulk theory $\mathcal{B}$ coupled to a boundary condition produces a bulk-to-boundary-centre map

$$\mathcal{B} \longrightarrow \mathbf{Z}(A_b).$$

An open–closed theorem identifies a selected physical bulk sector with the centre when the displayed map is a quasi-isomorphism. The full bulk theory also carries its stress tensor, global state space, and anomaly theory.

No-go theorem 10: a boundary centre does not determine an arbitrary bulk

The bulk-to-centre construction has no inverse on the category of all bulk–boundary theories. Let $\mathcal{D}$ be a nontrivial decoupled bulk theory whose chosen boundary condition is the tensor unit and whose bulk-to-boundary action is null-homotopic. The product bulks $\mathcal{B}$ and $\mathcal{B} \otimes \mathcal{D}$ induce the same boundary algebra and the same map to $\mathbf{Z}(A_b)$, while their bulk observable complexes differ by the factor $\mathrm{Obs}(\mathcal{D})$. Thus $\mathcal{B} \simeq \mathbf{Z}(A_b)$ follows only from a stated open–closed quasi-isomorphism together with the locality and generation hypotheses of the physical theory.

Chapter 10

Quadratic, PBW, curved, and Verdier comparisons

10.1 Quadratic recognition

Let V be dualisable and

$$A = T(V)/(R), \qquad R \hookrightarrow V \otimes_! V.$$

The quadratic dual algebra is

$$A^{q!} = T(V^*[-1])/(R^\perp),$$

and the quadratic dual coalgebra is the largest subcoalgebra of $T^c(sV)$ whose quadratic core lies in s^2R.

Theorem 10.1 (Quadratic recognition)**.** *Assume every weight component is dualisable and the diagonal Koszul complex is exact in positive homological degree. Then the quadratic dual coalgebra includes quasi-isomorphically into* $\mathrm{Bar}(A)$*, and continuous duality gives*

$$A^! \simeq A^{q!}.$$

Proof. Filter the bar by tensor weight. Its linear strand is the quadratic Koszul complex. The exactness hypothesis collapses the spectral sequence onto the quadratic dual coalgebra. Continuous linear duality turns the coalgebra into the stated algebra. □

No-go theorem 11: quadratic dual is not automatically derived Koszul dual

The orthogonal-relations algebra $A^{q!}$ is the zeroth approximation supplied by the quadratic strand. The canonical comparison $A^{q!} \to \mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$ is a quasi-isomorphism exactly on the Koszul locus. Higher homology of the Koszul complex survives as higher derived endomorphisms and obstructs that comparison.

10.2 PBW deformations and curvature

A nonhomogeneous relation has the form

$$r - \alpha(r) - \beta(r), \qquad \alpha : R \to V, \qquad \beta : R \to \mathbf{1}.$$

The linear term becomes a differential on the quadratic dual, and the constant term becomes curvature.

Theorem 10.2 (Curved PBW duality)**.** *If the Braverman–Gaitsgory overlap conditions hold, then the filtered algebra has associated graded $T(V)/(R)$, and its dual is a curved dg algebra*

$$(A^{q!}, d_\alpha, h_\beta), \qquad d_\alpha^2 = [h_\beta, -], \qquad d_\alpha h_\beta = 0.$$

Conversely, these curved identities are equivalent to resolution of the degree-three overlap ambiguities.

Proof. Evaluate the ambiguity space $(R \otimes V) \cap (V \otimes R)$. Its filtration-two component gives the derivation condition for d_α; the filtration-one and filtration-zero components give $d_\alpha^2 = [h_\beta, -]$ and $d_\alpha h_\beta = 0$. The Rees spectral sequence then identifies the associated graded algebra. □

For the Weyl relation $yx - xy = 1$, the constant term defines the curvature element of the dual curved algebra.

10.3 Verdier duality

Verdier duality acts on holonomic factorisation D-modules and exchanges !- and $*$-extensions. When every arity component is holonomic, dualisable, and compatible with the finite push-forwards defining factorisation, Verdier duality carries the structure maps to dual coalgebra maps. A continuous perfect pairing between the bar resolution and its Verdier dual then produces a comparison with the boundary Koszul dual. Perfectness of that comparison is the self-duality theorem in a concrete example.

No-go theorem 12: four dualities are not one operation

The interval bar $\mathbf{1} \otimes^{\mathbb{L}}_A \mathbf{1}$, boundary dual $\mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$, quadratic dual $T(V^*[-1])/(R^\perp)$, and Verdier dual $\mathbb{D}A$ have different variances and hypotheses. A finite Koszul self-dual model can carry comparison maps among these four objects. Replacing the maps by equality erases the theorem.

Part III

Calculated noncommutative and doubly noncommutative models

Chapter 11

Free and square-zero ordered chiral algebras

11.1 Scalar extension and separated factors

Let $\mathbf{1}_X$ be the tensor unit of $(\mathrm{FactD}(X), \otimes_!)$, and let B be an augmented associative dg algebra. The constant coefficient extension

$$B_X = B \otimes \mathbf{1}_X$$

is a transversely ordered chiral algebra. Its transverse multiplication is induced by m_B, and its collision coefficient is the unit factorisation object. The ordered bar is the internalization of the ordinary bar:

$$\mathrm{Bar}_X^{\perp}(B_X) \simeq \mathrm{Bar}(B) \otimes \mathbf{1}_X.$$

More generally, let F be an augmented commutative algebra object for $\otimes_!$. Over a field, or whenever the Kunneth morphism is an equivalence, the external tensor product satisfies

$$\mathrm{Bar}_X^{\perp}(B \otimes F) \simeq \mathrm{Bar}(B) \otimes \mathrm{Bar}_X^{\perp}(F).$$

If F also carries associative chiral multiplication and the aligned mixed law, then $B \otimes F$ is doubly associative. Varying B while holding F fixed preserves the chiral coefficient and changes the transverse bar through the displayed Kunneth factor.

11.2 The free algebra

Let $B = T(V)$ with the standard augmentation.

Theorem 11.1 (Bar of a free algebra)**.** *The augmentation bar admits a natural deformation retract*

$$\mathrm{Bar}(T(V)) \simeq \Bbbk \oplus sV.$$

Equivalently,

$$\mathrm{Tor}_0^{T(V)}(\Bbbk, \Bbbk) = \Bbbk, \qquad \mathrm{Tor}_1^{T(V)}(\Bbbk, \Bbbk) = V, \qquad \mathrm{Tor}_r^{T(V)}(\Bbbk, \Bbbk) = 0 \quad (r \geq 2).$$

The same deformation retract holds for the constant coefficient extension $T(V) \otimes \mathbf{1}_X$.

Proof. Fix a word $v_1 \cdots v_n$ in the generators. The normalized bar subcomplex spanned by all ways of inserting bars between its letters is the augmented cellular chain complex of the $(n-1)$-simplex of cuts. If $n > 1$, the extra degeneracy that removes the first cut contracts this complex. If $n = 1$, one suspended generator remains. Summing over all words gives the retract. ☐

The free resolution identifies every higher normalized bar word with a contractible simplex of cuts. Higher numerical counts attached to the same generators therefore belong to chain states, geometric coefficients, or another differential.

11.3 The square-zero algebra

Let $B = \Bbbk \oplus M$ with $M^2 = 0$.

Proposition 11.2 (Bar of a square-zero algebra). *The multiplication part of the normalized bar differential vanishes, and*

$$\mathrm{Bar}(B) = T^c(sM)$$

with only the internal differential. If M is a d-dimensional vector space in degree zero, then

$$\dim \mathrm{Tor}_r^B(\Bbbk, \Bbbk) = d^r, \qquad P_B(t) = \frac{1}{1 - dt}.$$

Proof. Every adjacent multiplication in the augmentation ideal equals zero. The normalized bar differential therefore vanishes, and the homology is the full tensor coalgebra. ☐

Free and square-zero products on the same generator space yield, respectively, a bar concentrated in degrees zero and one and a full tensor coalgebra. Bar homology records multiplication.

11.4 A first doubly associative factor

Let C_X be a pointwise-commutative associative chiral algebra with aligned mixed law, and let $V_\perp$ be a finite-dimensional generator space. The separated-factor algebra

$$T(V_\perp) \otimes C_X$$

has the free scalar transverse multiplication on the first factor and the associative chiral multiplication on the second. The Kunneth comparison gives

$$\mathrm{Bar}^\perp(T(V_\perp) \otimes C_X) \simeq (\Bbbk \oplus sV_\perp) \otimes \mathrm{Bar}^\perp(C_X).$$

Replacing $T(V_\perp)$ by $\Bbbk \oplus V_\perp$ replaces the first factor by $T^c(sV_\perp)$. The associative chiral bar of the second factor is unchanged. The two choices give an explicit comparison of the interval and chiral bars for one fixed chiral coefficient.

Chapter 12

The quantum and Jordan planes

12.1 The quantum plane

Fix $q \in \Bbbk^\times$ and set

$$A_q = \Bbbk\langle x, y\rangle/(yx - qxy), \qquad \varepsilon(x) = \varepsilon(y) = 0.$$

Orient the relation as $yx \mapsto qxy$. Every word reduces uniquely to $x^a y^b$, so these monomials form a PBW basis and

$$H_{A_q}(z) = \frac{1}{(1-z)^2}.$$

Theorem 12.1 (Quantum-plane Koszul resolution)**.** *The complex of free left modules*

$$0 \longrightarrow A_q(-2) \xrightarrow{d_2} A_q(-1)^{\oplus 2} \xrightarrow{d_1} A_q \xrightarrow{\varepsilon} \Bbbk \longrightarrow 0,$$

with

$$d_1(a, b) = ax + by, \qquad d_2(c) = (cy, -cqx),$$

is exact. Consequently

$$\mathrm{Tor}_*^{A_q}(\Bbbk, \Bbbk) \cong \Bbbk \oplus \Bbbk\{sx, sy\} \oplus \Bbbk\{\omega_q\},$$

where $|\omega_q| = 2$, and a normalized bar representative is

$$\omega_q = [y|x] - q[x|y].$$

The Poincare polynomial is $1 + 2t + t^2$.

Proof. The identity $d_1 d_2(c) = c(yx - qxy) = 0$ proves that the maps form a complex. The Ore-extension presentation $A_q = \Bbbk[x][y;\sigma]$, $\sigma(x) = qx$, makes A_q a domain; hence d_2 is injective. The image of d_1 is the augmentation ideal. The PBW basis gives

$$H_{A_q}(z)(1 - 2z + z^2) = 1.$$

Since the complex is already exact at its two ends, the graded Euler identity forces the middle homology to vanish. Tensoring the minimal resolution with $\Bbbk$ kills both differentials. In the normalized bar, $b([y|x] - q[x|y]) = yx - qxy = 0$. □

The quadratic dual is

$$A_q^! = \Bbbk\langle \xi, \eta\rangle/(\xi^2, \eta^2, \xi\eta + q\eta\xi).$$

The ordered parameter q survives in the Yoneda product even though the ungraded Betti numbers agree with the commutative plane.

12.2 The Jordan plane

Set

$$A_J = \Bbbk\langle x, y\rangle/(yx - xy - x^2), \qquad \varepsilon(x) = \varepsilon(y) = 0.$$

The rule $yx \mapsto xy + x^2$ lowers inversion number. The leading word yx has an empty overlap set. Bergman's diamond lemma therefore gives the PBW basis $x^a y^b$.

Theorem 12.2 (Jordan-plane Koszul resolution)**.** *The complex*

$$0 \longrightarrow A_J(-2) \xrightarrow{d_2} A_J(-1)^{\oplus 2} \xrightarrow{d_1} A_J \xrightarrow{\varepsilon} \Bbbk \longrightarrow 0,$$

with

$$d_1(a, b) = ax + by, \qquad d_2(c) = (c(y - x), -cx),$$

is exact. Thus

$$P_{\mathrm{Tor}^{A_J}(\Bbbk,\Bbbk)}(t) = 1 + 2t + t^2,$$

and the degree-two bar class is represented by

$$\omega_J = [y|x] - [x|y] - [x|x].$$

Proof. The composite is $c((y-x)x-xy) = c(yx-xy-x^2) = 0$. The Ore-extension presentation $A_J = \Bbbk[x][y;\delta]$, $\delta(x) = x^2$, makes A_J a domain, so d_2 is injective. The image of d_1 is the augmentation ideal. The normal forms $x^a y^b$ give the Hilbert identity $H_{A_J}(z)(1-2z+z^2) = 1$, which forces vanishing of the middle homology. Tensoring with the augmentation module makes both differentials zero, and the bar boundary of ω_J is the defining relation. □

The quantum and Jordan planes have identical Betti numbers and distinct products, normal forms, quadratic duals, and module categories. The structure maps distinguish algebras that their Betti numbers identify.

12.3 Finite-weight bar matrices

Choose the monomial bases of normalized bar words at fixed polynomial weight. Multiplication gives exact matrices over $\mathbb{Q}(q)$ for the quantum plane and over $\mathbb{Q}$ for the Jordan plane. Specialization to $q = 2$ gives the following ranks; generic ranks agree away from the finite determinantal exceptional locus.

algebra	weight	degree	chain dimension	outgoing rank	homology dimension
A_q	1	1	2	0	2
A_q	2	1	3	0	0
A_q	2	2	4	3	1
A_q	3	1	4	0	0
A_q	3	2	12	4	0
A_q	3	3	8	8	0
A_J	1	1	2	0	2
A_J	2	1	3	0	0
A_J	2	2	4	3	1
A_J	3	1	4	0	0
A_J	3	2	12	4	0
A_J	3	3	8	8	0

All weights four and five have zero homology. Direct substitution in the differential gives $b\omega_q = b\omega_J = 0$.

12.4 Internal chiral realization

The constant coefficient extensions $A_q \otimes \mathbf{1}_X$ and $A_J \otimes \mathbf{1}_X$ are transversely ordered chiral algebras, and their ordered bars are the displayed Koszul coalgebras tensored with $\mathbf{1}_X$. Let F be an augmented pointwise-commutative factorisation algebra for which the Kunneth map is an equivalence. Then

$$\mathrm{Bar}^{\perp}(A_q \otimes F) \simeq \mathrm{Bar}(A_q) \otimes \mathrm{Bar}^{\perp}(F), \qquad \mathrm{Bar}^{\perp}(A_J \otimes F) \simeq \mathrm{Bar}(A_J) \otimes \mathrm{Bar}^{\perp}(F).$$

Every collision map of F acts on the second tensor factor. If F also carries associative chiral multiplication with the aligned mixed law, both tensor products are strict doubly associative examples.

Chapter 13

Augmentation obstructions, Weyl curvature, and matrix sectors

13.1 The Weyl obstruction

Let

$$W_1 = \Bbbk\langle x, y\rangle/(yx - xy - 1).$$

Theorem 13.1 (Augmentation criterion for the Weyl algebra)**.** *For $W_\hbar = \Bbbk[\hbar]\langle x, y\rangle/(yx - xy - \hbar)$, a unital algebra morphism $W_\hbar \to R$ to a commutative $\Bbbk[\hbar]$-algebra exists only on the fibre where the image of $\hbar$ vanishes. In particular, the scalar augmentation fibre is $\hbar = 0$, while W_1 has an empty space of scalar augmentations.*

Proof. Applying a morphism to the defining relation gives $\varepsilon(y)\varepsilon(x) - \varepsilon(x)\varepsilon(y) = \varepsilon(\hbar)$. The left side vanishes in a commutative target, so $\varepsilon(\hbar) = 0$. Conversely, the special fibre $\hbar = 0$ admits the augmentation $x, y \mapsto 0$. □

Three canonical replacements carry the Weyl relation.

1. The homogenized/Rees algebra

$$W_\hbar = \Bbbk[\hbar]\langle x, y\rangle/(yx - xy - \hbar),$$

   with $\hbar$ central, is augmented over the special fibre $\hbar = 0$. The fibre at $\hbar = 0$ is the commutative plane, and the fibre at $\hbar = 1$ is the first Weyl algebra.
2. The nonhomogeneous quadratic dual is curved. The constant term produces a curvature element h with $d^2 = [h, -]$, as in Theorem 10.2.
3. A specified module or boundary idempotent defines a relative bar.

Curved dual of the first Weyl algebra

The quadratic part is the polynomial plane, whose dual is the exterior algebra $\Lambda(\xi, \eta)$. The constant functional on the relation becomes, up to the suspension convention, the curvature

$$h = \xi\eta.$$

The homogeneous two-generator dual has zero differential. Since h is central in the graded exterior algebra, the curved identity reads $d^2 = [h, -] = 0$. The nonzero arity-zero term records the constant Weyl commutator.

13.2 The matrix obstruction

Theorem 13.2 (Scalar augmentation criterion for matrix algebras)**.** *The affine scheme of unital algebra morphisms $M_n(\Bbbk) \to \Bbbk$ is a point for $n = 1$ and is empty for $n > 1$.*

Proof. For $n = 1$, the identity map gives the unique morphism. For $n > 1$, the kernel of a unital morphism is a proper two-sided ideal. Simplicity forces the kernel to vanish, while a morphism to a commutative algebra annihilates every commutator. The nonzero commutator $[E_{12}, E_{21}] = E_{11} - E_{22}$ gives the contradiction. □

Matrix-valued fields define ordered chiral algebras relative to a module, an idempotent, or an augmented ambient algebra. Morita theory transports their boundary categories. The matrix trace supplies cyclic closure and the identity representation supplies module-valued bars.

13.3 An augmented noncommutative Frobenius algebra

Let $D = \Bbbk[\epsilon]/(\epsilon^2)$ and

$$B_n = D \times M_n(\Bbbk), \qquad \varepsilon_B(a + b\epsilon, M) = a.$$

The algebra B_n is noncommutative for $n > 1$ and carries the displayed augmentation. Give it the symmetric Frobenius trace

$$\tau_B(a + b\epsilon, M) = b + \mathrm{Tr}(M).$$

The pairing $\langle u, v\rangle = \tau_B(uv)$ is nondegenerate: the dual-number block pairs 1 with ϵ, and the matrix block has the trace pairing.

Proposition 13.3 (Bar and handle element of B_n)**.** *As an augmented algebra,*

$$\Bbbk \otimes^{\mathbb{L}}_{B_n} \Bbbk \simeq \Bbbk \otimes^{\mathbb{L}}_{D} \Bbbk \simeq T^c(s\Bbbk\epsilon).$$

Thus $\dim \mathrm{Tor}^{B_n}_r(\Bbbk, \Bbbk) = 1$ *for every* $r \geq 0$*. The Frobenius handle element is*

$$e_{B_n} = (2\epsilon, nI_n).$$

Consequently

$$\tau_B(e_{B_n}) = 2 + n^2, \qquad \tau_B(e^g_{B_n}) = n^{g+1} \quad (g \geq 2).$$

The displayed values are the exact algebraic handle contractions of the symmetric Frobenius pairing.

Proof. Modules over a direct product decompose. The augmentation module is $(\Bbbk, 0)$, so its derived tensor product over $D \times M_n$ is the derived tensor product over D. Since the augmentation ideal of D is one-dimensional square-zero, Theorem 11.2 gives the tensor coalgebra. For the handle element, the basis $(1, \epsilon)$ of D has dual basis $(\epsilon, 1)$, yielding 2ϵ. The matrix units E_{ij} have duals E_{ji}, and $\sum_{ij} E_{ij}E_{ji} = nI_n$. Powers are computed blockwise. □

The algebra B_n supplies noncommutativity, an augmentation, an infinite bar, and an exact cyclic trace in one finite-dimensional model. Its ribbon-graph contractions will serve as an open transverse sector in Chapter 26.

No-go theorem 13: a trace is not an augmentation

A linear functional defines an augmentation precisely when it is unital and multiplicative. The matrix trace satisfies cyclicity and fails multiplicativity, for example $\mathrm{Tr}(E_{12}E_{21}) = 1$ while $\mathrm{Tr}(E_{12})\,\mathrm{Tr}(E_{21}) = 0$. Hence a trace defines cyclic closure and a scalar pairing, whereas an augmentation defines endpoint modules and the outer bar faces.

Chapter 14

Relative quiver bars and boundary conditions

14.1 The natural base

Let Q be a finite quiver, let

$$S = \bigoplus_{i \in Q_0} \Bbbk e_i, \qquad M = \bigoplus_{a:i \to j} e_j M e_i,$$

and let $T_S(M)$ be the path algebra. The map killing all arrows and fixing vertex idempotents is an augmentation to S. Relations generate an S-subbimodule ideal.

The relative bar

$$\mathrm{Bar}_S(A) = S \otimes^{\mathbb{L}}_A S$$

retains object labels and only allows tensors of composable arrows. The relative path algebra is the algebraic form of a category of boundary conditions: vertices are boundaries, arrows are boundary-changing operators, and paths are ordered compositions.

14.2 A complete quadratic example

Consider

$$1 \xrightarrow{a} 2 \xrightarrow{b} 3, \qquad A = \Bbbk Q/(ba).$$

The relative augmentation ideal is spanned by a, b. The only composable length-two tensor is $b \otimes_S a$, and the defining relation makes its boundary zero.

Theorem 14.1 (Relative bar of the relation quiver)**.** *As an S-bimodule,*

$$\mathrm{Tor}_0^A(S,S) = S, \qquad \mathrm{Tor}_1^A(S,S) = \Bbbk a \oplus \Bbbk b, \qquad \mathrm{Tor}_2^A(S,S) = \Bbbk[b|a],$$

and $\mathrm{Tor}_r^A(S,S) = 0$ *for* $r \geq 3$.

Proof. The normalized relative bar has $C_1 = \Bbbk a \oplus \Bbbk b$ and $C_2 = \Bbbk[b|a]$. Every composable arrow sequence has length at most two. The differential sends $[b|a]$ to the relation $ba = 0$, and the source of the differential into C_2 is zero. Adjoining $C_0 = S$ gives the result. □

For the free path algebra, the differential sends $[b|a]$ to the path ba, so $H_2 = 0$. For the quotient $ba = 0$, the same word represents the degree-two relation class. The relative bar therefore detects the imposed relation.

14.3 Simple modules and Ext

Let $S_j^{\mathrm{R}} = e_j S$ be the right augmentation module at vertex j, and let $S_i^{\mathrm{L}} = S e_i$ be the left augmentation module at vertex i. Cutting the relative bar by vertex idempotents computes derived boundary-changing fields:

$$e_j \operatorname{Tor}_r^A(S, S) e_i \cong \operatorname{Tor}_r^A(S_j^{\mathrm{R}}, S_i^{\mathrm{L}}).$$

For the example,

$$\operatorname{Tor}_1^A(S_2^{\mathrm{R}}, S_1^{\mathrm{L}}) = \Bbbk a, \quad \operatorname{Tor}_1^A(S_3^{\mathrm{R}}, S_2^{\mathrm{L}}) = \Bbbk b, \quad \operatorname{Tor}_2^A(S_3^{\mathrm{R}}, S_1^{\mathrm{L}}) = \Bbbk[b|a].$$

The degree-two class is a derived composite whose strict product vanishes. The class $[b|a]$ is the first explicit instance of a relation surviving as a higher boundary field.

14.4 Factorisation families

Let the arrow spaces be factorisable coefficient objects on X, and form the internal tensor algebra over the vertex coefficient S. If the relation morphisms are compatible with disjoint restriction and diagonal extension, the relative path algebra is a transversely ordered chiral algebra. Marked block-Ran sections recover its module categories. A second associative chiral operation on each arrow coefficient produces a doubly associative quiver algebra, with mixed relations imposed by the derived presentation of Chapter 5.

No-go theorem 14: multi-idempotent quivers require a relative base

A scalar augmentation chooses a single linear combination of vertices and forgets the object decomposition. The canonical bar of a quiver category is $S \otimes_A^{\mathbb{L}} S$, not $\Bbbk \otimes_A^{\mathbb{L}} \Bbbk$, unless a separate scalar boundary object has been specified.

Chapter 15

Enveloping algebras and planar enumeration

15.1 Cartan–Eilenberg comparison

Let $\mathfrak{g}$ be a dg Lie algebra object in the coefficient category and $U(\mathfrak{g})$ its augmented enveloping algebra.

Theorem 15.1 (Enveloping bar)**.** *Let $\mathfrak{g}$ be a dg Lie algebra object in a characteristic-zero symmetric monoidal category in which the PBW map*

$$\mathrm{Sym}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{gr}\, U(\mathfrak{g})$$

is an equivalence, tensor product preserves the filtered colimits used by the bar, and the PBW filtration is bounded below and complete in each total weight. Then the antisymmetrisation twisting map induces a natural quasi-isomorphism

$$\mathrm{CE}_*(\mathfrak{g}) = \big(\mathrm{Sym}^c(s\mathfrak{g}), d_{\mathfrak{g}} + d_{[-,-]}\big) \xrightarrow{\sim} \mathrm{Bar}^{\perp}(U(\mathfrak{g})).$$

Proof. The PBW filtrations on both complexes are exhaustive, bounded below in each total weight, and complete by hypothesis. Their associated graded comparison is the classical Koszul quasi-isomorphism

$$\mathrm{Sym}^c(s\mathfrak{g}) \longrightarrow \mathrm{Bar}(\mathrm{Sym}(\mathfrak{g})).$$

The comparison theorem for complete filtered complexes lifts the associated-graded quasi-isomorphism to the stated map. □

For a finite-dimensional semisimple Lie algebra over $\mathbb{C}$,

$$H^*(\mathfrak{g}; \mathbb{C}) \cong \Lambda(\zeta_{2m_1+1}, \dots, \zeta_{2m_r+1}),$$

where m_i are the exponents. By duality the same Poincare polynomial describes homology.

The $\mathfrak{sl}_2$ chain matrix

Take $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$. In bases

$$C_1 = (e, f, h), \qquad C_2 = (e \wedge f, e \wedge h, f \wedge h),$$

the Chevalley differential is

$$d_2 = \begin{pmatrix} 0 & -2 & 0 \\ 0 & 0 & 2 \\ 1 & 0 & 0 \end{pmatrix}, \qquad d_3 = 0.$$

Thus d_2 has rank three and

$$H_0 \cong \mathbb{C}, \quad H_1 = H_2 = 0, \quad H_3 \cong \mathbb{C}, \qquad P(t) = 1 + t^3.$$

The standard exact values are

$$\begin{aligned} U(\mathfrak{sl}_2): &\quad 1 + t^3, \\ U(\mathfrak{sl}_3): &\quad (1+t^3)(1+t^5) = 1 + t^3 + t^5 + t^8, \\ U(\mathfrak{g}_2): &\quad (1+t^3)(1+t^{11}) = 1 + t^3 + t^{11} + t^{14}. \end{aligned}$$

15.2 Three chain models

Three previously compared enumerations belong to three different chain models:

1. the interval bar $\mathrm{Bar}^{\perp}(A)$, whose differential uses transverse multiplication;
2. chiral Chevalley chains, whose differential uses a diagonal-supported chiral Lie bracket;
3. a screen-enhanced bicomplex

$$B^{p,q}_{\mathrm{scr}}(A) = (s\overline{A})^{\otimes_! p} \otimes C^q_{\mathrm{scr}}$$

with total differential $b_A + d_A + d_{\mathrm{scr}} + \delta_{\mathrm{int}}$.

The screen-enhanced bicomplex combines bar words with configuration-space cohomology after an interaction differential has been supplied and its square has been proved to vanish.

Filtering by screen degree gives an E_1-page

$$E_1^{p,q} = H^q(C_{\mathrm{scr}}) \otimes B^{\perp}_p(A),$$

while filtering by bar length gives

$${}'E_1^{p,q} = \mathrm{Tor}^A_p(\mathbf{1}, \mathbf{1}) \otimes C^q_{\mathrm{scr}}.$$

Total homology is determined by the kernels and images of the spectral-sequence differentials; the dimensions of the E_1-page determine it after degeneration has been proved.

15.3 The planar sequences

The Motzkin generating function is

$$M(z) = 1 + zM(z) + z^2 M(z)^2 = \frac{1 - z - \sqrt{1 - 2z - 3z^2}}{2z^2}.$$

For $a_n = M_{n+1} - M_n$,

$$\sum_{n\geq 1} a_n z^n = \frac{4z}{\left(1 - z + \sqrt{1 - 2z - 3z^2}\right)^2},$$

with first values

$$1, 2, 5, 12, 30, 76, 196, 512, 1353, 3610.$$

The Riordan series is

$$R(z) = \frac{1 + z - \sqrt{1 - 2z - 3z^2}}{2z(1 + z)}.$$

Both have dominant singularity $z = 1/3$ and exponential growth rate three. The displayed generating functions are exact combinatorial identities.

Theorem 15.2 (Carrier theorem for the planar series)**.** *The Motzkin-difference and Riordan series are the Euler series of their displayed planar recursions. The transverse bar of $U(\mathfrak{sl}_2)$ is the Chevalley complex and has Poincare polynomial $1 + t^3$. For a fixed Virasoro factorisation coefficient $F_{\mathrm{Vir},c}$, the transverse bars of $B \otimes F_{\mathrm{Vir},c}$ vary with the augmented algebra B. Consequently, a chain-level identification of either planar series with an ordered bar requires a specified transverse multiplication and a quasi-isomorphism from its planar complex to that bar.*

Proof. The equations $M = 1 + zM + z^2M^2$ and $R = (1 + z - \sqrt{1 - 2z - 3z^2})/(2z(1 + z))$ determine the two planar series. The exact Chevalley comparison Theorem 15.1 gives $1 + t^3$ for $U(\mathfrak{sl}_2)$. Taking B successively to be free, square-zero, enveloping, and quantum-plane produces distinct interval bars while preserving the Virasoro coefficient and its central charge. The ordered-bar homology therefore depends on the specified transverse multiplication in addition to the chiral coefficient. $\square$

No-go theorem 15: a scalar sequence is not homology data

Bar homology is the homology of a displayed graded differential. A scalar sequence determines bar dimensions only after a chain model and a proof identify its coefficients with the kernel–image ranks of that differential. Agreement of initial values, discriminants, or growth rates supplies numerical evidence but no chain equivalence.

Chapter 16

Zamolodchikov–Faddeev algebras as doubly noncommutative fields

16.1 Exchange presentation

Let V be finite dimensional and

$$R(u) \in \mathrm{End}(V \otimes V) \otimes \Bbbk(u)$$

be invertible away from a specified pole divisor. Introduce generators $Z_i(u)$ and relations

$$Z_i(u)Z_j(v) = \sum_{k,l} R_{ij}^{kl}(u-v)Z_l(v)Z_k(u).$$

The spectral variable belongs to the chiral/meromorphic direction; the word belongs to the ordered associative direction. The relation reverses order and transports the coefficient.

Fix finitely many distinct spectral labels $u_1, \ldots, u_m$, choose a total order, and orient every relation toward increasing labels.

Theorem 16.1 (Confluence is Yang–Baxter)**.** *Assume*

$$R(u)R_{21}(-u) = 1.$$

The oriented exchange relations are terminating and confluent exactly when

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v).$$

When these equations hold, ordered spectral words form a basis over the localized spectral field.

Proof. The inversion number of a spectral word decreases under each oriented exchange, so reduction terminates. Every irreducible critical overlap has three adjacent labels. The two reduced expressions of the longest permutation in S_3 produce the two sides of the Yang–Baxter equation. A relation followed by its reverse gives the length-two ambiguity, resolved by unitarity. Bergman's diamond lemma gives unique normal forms. $\square$

The theorem constructs a doubly noncommutative algebra by generators and relations. Ordered spectral words give its basis, and the Yang–Baxter defect is the unique three-letter critical ambiguity.

16.2 The rational matrix

Let $P(x \otimes y) = y \otimes x$ and

$$R(u) = 1 - \frac{\hbar P}{u}.$$

The permutation identities

$$P_{12}P_{13} = P_{23}P_{12}, \quad P_{13}P_{23} = P_{12}P_{13}, \quad P_{12}P_{23} = P_{13}P_{12}$$

verify Yang–Baxter after clearing denominators. Since

$$R(u)R_{21}(-u) = 1 - \frac{\hbar^2}{u^2},$$

a formal meromorphic scalar $f(u) = 1 + O(u^{-1})$ satisfying

$$f(u)f(-u) = \left(1 - \frac{\hbar^2}{u^2}\right)^{-1}$$

produces a unitary normalization in the completed Laurent field.

16.3 A scalar braided exterior bar

If V is one-dimensional and the exchange coefficient for labels $a < b$ is a nonzero scalar q_{ab}, the finite-label algebra is multiparameter quantum affine space:

$$z_b z_a = q_{ab} z_a z_b.$$

The multiparameter quantum affine space is a quadratic PBW algebra. Its Koszul dual is the braided exterior algebra generated by ξ_a with

$$\xi_a^2 = 0, \qquad \xi_b \xi_a = -q_{ab}^{-1} \xi_a \xi_b.$$

Hence the bar homology is completely explicit: one basis class for every subset of spectral labels, with the order and braiding recorded in the product. The braided exterior bar gives a direct calculational bridge from ZF exchange to chiral Koszul duality.

No-go theorem 16: an exchange relation without Yang–Baxter has no coherent three-field algebra

Every invertible $R(u)$ defines pairwise rewrite rules, while a three-letter word has two reductions corresponding to the two reduced expressions of the longest element of S_3. Their equality is precisely the spectral Yang–Baxter equation. A nonzero Yang–Baxter defect is therefore an unresolved critical pair and obstructs coherent three-field composition.

Chapter 17

RTT Yangians and the bar filtration

17.1 RTT presentation

Set

$$T(u) = 1 + \sum_{r \geq 1} T^{(r)} u^{-r}, \qquad T^{(r)} = (t_{ij}^{(r)})_{1 \leq i,j \leq n}.$$

The Rees RTT Yangian $Y_\hbar^{\mathrm{RTT}}(\mathfrak{gl}_n)$ is generated over $\Bbbk[\hbar]$ by the coefficients subject to

$$R(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R(u-v), \qquad R(u) = 1 - \frac{\hbar P}{u}.$$

Equivalently, with $t_{ij}^{(0)} = \delta_{ij}$,

$$[t_{ij}^{(r+1)}, t_{kl}^{(s)}] - [t_{ij}^{(r)}, t_{kl}^{(s+1)}] = \hbar\big(t_{kj}^{(r)} t_{il}^{(s)} - t_{kj}^{(s)} t_{il}^{(r)}\big).$$

17.2 Hopf structure

The formulas

$$\Delta T(u) = T_{13}(u)T_{23}(u), \qquad \varepsilon(T(u)) = 1, \qquad S(T(u)) = T(u)^{-1}$$

define a Hopf algebra. The proof moves auxiliary R-matrices through the product using Yang–Baxter. The counit gives a genuine augmentation, so the ordered bar is defined.

For $n = 2$,

$$\operatorname{qdet} T(u) = t_{11}(u)t_{22}(u-\hbar) - t_{12}(u)t_{21}(u-\hbar)$$

is central and group-like. The quotient $\operatorname{qdet} T(u) = 1$ is the $\mathfrak{sl}_2$ RTT Yangian.

17.3 PBW and current shadow

Give $t_{ij}^{(r)}$ loop degree $r-1$. The PBW theorem gives

$$\mathrm{gr}_{\mathrm{loop}}\, Y_\hbar^{\mathrm{RTT}}(\mathfrak{gl}_n) \cong U_\hbar(\mathfrak{gl}_n[t]),$$

and ordered monomials in the $t_{ij}^{(r)}$ form a $\Bbbk[\hbar]$-basis [41]. The $\hbar$-adic completion is topologically free with the same basis.

Theorem 17.1 (Yangian bar spectral sequence)**.** *Filter the normalized bar of the augmented Rees RTT Yangian by total loop degree. Under a complete bounded-below filtration, there is a multiplicative spectral sequence*

$$E^1 \cong H_*\big(C_*^{\mathrm{CE}}(\mathfrak{gl}_n[t])[[\hbar]], \hbar d_{\mathrm{CE}}\big) \Longrightarrow \mathrm{Tor}_*^{Y_\hbar^{\mathrm{RTT}}(\mathfrak{gl}_n)}(\Bbbk[[\hbar]], \Bbbk[[\hbar]]).$$

After adjoining $\hbar^{-1}$, rescaling the Chevalley generators identifies the first page with

$$H_*(\mathfrak{gl}_n[t]; \Bbbk)((\hbar)).$$

For a finite current truncation, the filtration is finite in every total weight.

Proof. The PBW filtration induces a complete filtration on every bar length. The associated graded algebra is the Rees enveloping algebra with relation $xy - yx = \hbar[x, y]$. Its bar is the Chevalley complex with differential $\hbar d_{\mathrm{CE}}$. The complete filtered-bar spectral sequence gives the first assertion. Multiplication of homological degree r by $\hbar^{-r}$ after localization conjugates $\hbar d_{\mathrm{CE}}$ to d_{CE}. □

The higher differentials measure the filtered deformation from the current Rees algebra to the Yangian.

17.4 The exactly calculable abelian case

For $\mathfrak{gl}_1$, RTT is commutative and

$$Y(\mathfrak{gl}_1) \cong \Bbbk[t^{(1)}, t^{(2)}, \ldots].$$

Thus

$$\mathrm{Tor}_*^{Y(\mathfrak{gl}_1)}(\Bbbk, \Bbbk) \cong \Lambda(st^{(1)}, st^{(2)}, \ldots).$$

For the subalgebra on the first N coefficients, the ungraded Poincare polynomial is $(1 + t)^N$. The exterior algebra $\Lambda(st^{(1)}, st^{(2)}, \ldots)$ is the complete bar homology of the abelian Yangian.

17.5 Ordered chiral interpretation

The coefficients $T_{ij}(u)$ are meromorphic fields on the spectral curve. RTT is their ordered multiplication, the coproduct acts on tensor products of line representations, and the rational R-matrix exchanges parallel lines. The Yangian therefore realizes all three noncommutative layers: ordered word, meromorphic field coefficient, and braid transport. A full associative-chiral bar requires the formal-disc/nonlocal structure maps in addition to the Hopf algebra presentation.

Chapter 18

The A_2 Hall algebra in complete detail

18.1 The representation category

Let $Q : 1 \to 2$ over $\mathbb{F}_q$. Its indecomposables are

$$S_1 = (\mathbb{F}_q \to 0), \qquad S_2 = (0 \to \mathbb{F}_q), \qquad M = (\mathbb{F}_q \xrightarrow{1} \mathbb{F}_q),$$

and there is one nonsplit extension

$$0 \to S_2 \to M \to S_1 \to 0.$$

The Euler form is

$$\langle (a,b),(c,d) \rangle = ac + bd - ad.$$

Let $[A] * [B]$ count subobjects isomorphic to B with quotient A, and put $e_i = [S_i]$.

18.2 Binary and triple products

The binary products are

$$e_1 * e_2 = [S_1 \oplus S_2] + [M], \qquad e_2 * e_1 = [S_1 \oplus S_2], \qquad e_1 * e_1 = (q+1)[2S_1].$$

In dimension vector $(2,1)$, let

$$X = 2S_1 \oplus S_2, \qquad Y = S_1 \oplus M.$$

Flag counting gives

$$\begin{aligned} e_1^2 e_2 &= (q+1)([X] + [Y]), \\ e_1 e_2 e_1 &= (q+1)[X] + [Y], \\ e_2 e_1^2 &= (q+1)[X]. \end{aligned}$$

Theorem 18.1 (Untwisted Hall–Serre relation)**.**

$$e_1^2 e_2 - (q+1) e_1 e_2 e_1 + q e_2 e_1^2 = 0.$$

The companion relation is

$$qe_2^2e_1 - (q+1)e_2e_1e_2 + e_1e_2^2 = 0.$$

Proof. For the first relation, the coefficient of $[Y]$ is $(q+1) - (q+1) = 0$, and the coefficient of $[X]$ is

$$(q+1) - (q+1)^2 + q(q+1) = 0.$$

The second follows from the symmetric flag count in dimension vector $(1,2)$. □

At $q = 2$, the coefficient vectors in the basis $([X],[Y])$ are $(3,3)$, $(3,1)$, and $(3,0)$; their Serre combination is $(0,0)$.

18.3 Twist and quantum group

Let $v^2 = q$ and

$$[A] \diamond [B] = v^{\langle \dim A, \dim B \rangle}[A] * [B].$$

The total twisting factors in the first three triple products are $v^{-1}, 1, v$. Therefore

$$e_1 \diamond e_1 \diamond e_2 - (v + v^{-1})e_1 \diamond e_2 \diamond e_1 + e_2 \diamond e_1 \diamond e_1 = 0,$$

with the companion relation obtained by exchanging $1, 2$.

Define

$$e_{12} = e_1 \diamond e_2 - v^{-1}e_2 \diamond e_1 = v^{-1}[M].$$

Every representation is uniquely $S_2^a \oplus M^b \oplus S_1^c$. Ordered divided-power monomials

$$e_2^{(a)}e_{12}^{(b)}e_1^{(c)}$$

are triangular with nonzero diagonal in the isomorphism-class basis ordered by arrow rank. The ordered divided-power monomials form a PBW basis and identify the generated algebra with $U_v^+(\mathfrak{sl}_3)$.

18.4 Ordered chiral Hall coefficients

Let the representation categories vary in a factorisation family over X. Extension correspondences at fixed support define the transverse Hall multiplication; disjoint support gives the chiral coefficient. If pull–push preserves the factorisation system and orientations, the Hall algebra is a transversely ordered chiral algebra. If the sheaf theory itself carries ordered chiral convolution, the derived extension algebra becomes a doubly associative chiral object. The binary and triple finite-field flag counts give the exact local structure constants.

Chapter 19

Necklace Hamiltonians and the mixed trace algebra

19.1 The derived commuting scheme

Let
$$R = \mathbb{C}[[z_1, z_2]], \qquad \{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g,$$
and let $\mathfrak{h} = R/\mathbb{C} \cdot 1$. The shifted cotangent Lie algebra
$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^{\vee}_{\mathrm{cont}}[1]$$
controls formal closed Hamiltonian BF observables.

For N branes, take even matrices $X, Y \in \mathfrak{gl}_N$ and an odd matrix $\psi \in \mathfrak{gl}_N[1]$ with
$$QX = QY = 0, \qquad Q\psi = [X, Y].$$
Adding the conjugation ghost gives functions on the derived quotient of the commuting equation. Gauge-invariant open observables are
$$J_N(f) = \operatorname{Tr} f(X, Y).$$

19.2 Necklace bracket

Let $F = \mathbb{C}\langle x, y\rangle$ and $F_{\mathrm{cyc}} = F/[F, F]$. The cyclic derivative cuts a cyclic word after every occurrence of the indicated letter. Define
$$\delta_f(x) = -\partial_y^{\mathrm{cyc}} f, \qquad \delta_f(y) = \partial_x^{\mathrm{cyc}} f,$$
and
$$\{f, g\}_{\mathrm{neck}} = \Big[\partial_x^{\mathrm{cyc}} f\, \partial_y^{\mathrm{cyc}} g - \partial_y^{\mathrm{cyc}} f\, \partial_x^{\mathrm{cyc}} g\Big]_{\mathrm{cyc}}.$$
The cyclic Euler identity
$$[x, \partial_x^{\mathrm{cyc}} f] + [y, \partial_y^{\mathrm{cyc}} f] = 0$$
implies $\delta_f([x, y]) = 0$, so the action descends to the derived commuting equation.

Theorem 19.1 (Necklace Lie algebra and trace action)**.** *The bracket makes $F_{\mathrm{cyc}}/\mathbb{C}$ a Lie* *algebra and*

$$[\delta_f, \delta_g] = \delta_{\{f,g\}_{\mathrm{neck}}}.$$

Moreover,

$$\delta_f J_N(g) = J_N(\{f,g\}_{\mathrm{neck}}).$$

Proof. Hamiltonian derivations preserve the noncommutative symplectic form $dx\,dy$, so their commutator is Hamiltonian. Evaluating on x, y gives the displayed bracket and Jacobi modulo commutators. Cyclic differentiation of $\mathrm{Tr}\, g(X,Y)$, followed by cyclicity of trace, gives the second identity. □

The theorem gives an exact finite-N Hamiltonian action of cyclic words on trace observables. A chain-level open–closed comparison extends this coordinate action to the full derived centre when its cone is acyclic.

19.3 Stable trace Poisson algebra

On the semisimple commuting locus, write

$$X = \mathrm{diag}(x_1, \ldots, x_N), \qquad Y = \mathrm{diag}(y_1, \ldots, y_N),$$

and define

$$p_{ab} = \sum_{i=1}^{N} x_i^a y_i^b, \qquad a, b \geq 0, \quad a + b > 0.$$

The stable multisymmetric invariant algebra is

$$\Lambda_2 = \mathbb{C}[p_{ab} : a + b > 0].$$

Theorem 19.2 (Stable trace bracket)**.** *For $\{x_i, y_j\} = \delta_{ij}$,*

$$\boxed{\{p_{ab}, p_{cd}\} = (ad - bc)p_{a+c-1,b+d-1}}.$$

The displayed stable trace bracket is the commutative reduction of the necklace bracket.

Proof. Different particles Poisson commute, and for one particle

$$\{x^a y^b, x^c y^d\} = (ad - bc)x^{a+c-1}y^{b+d-1}.$$

Sum over particles. Simultaneous diagonalization identifies cyclic traces with power sums. □

For monomials of total bidegree at most six, direct substitution reduces every Jacobi coefficient to the determinant identity for the lattice vectors $(a,b), (c,d), (e,f)$.

19.4 Two branes and the A_1 cone

For $N = 2$, remove centre-of-mass variables and set $\{x, y\} = 2$. The invariants

$$A = \frac{x^2}{2}, \qquad B = \frac{xy}{2}, \qquad C = \frac{y^2}{2}$$

satisfy

$$B^2 = AC, \qquad \{A, B\} = 2A, \quad \{B, C\} = 2C, \quad \{A, C\} = 4B.$$

After rescaling, this is the Kirillov–Kostant bracket on the nilpotent cone of $\mathfrak{sl}_2$. The first nontrivial finite-rank sector already contains a singular symplectic surface and a nonabelian bracket.

No-go theorem 17: a trace-sector map is not the full open–closed equivalence

The map $f \mapsto J_N(f)$ is a Poisson homomorphism from Hamiltonians to trace observables. Its image is the trace-generated sector and factors through the polynomial identities of $N \times N$ matrices; derived central observables and finite-rank syzygies lie outside this conclusion. A bulk–centre equivalence therefore requires a chain-level open–closed map, a compatible filtration, and acyclicity of its cone.

Chapter 20

Modules, orbifolds, and higher E_n directions

20.1 Pointed bars and interfaces

For a right module M_R and left module M_L,

$$\mathrm{Bar}_A(M_R, M_L) = \left|[r] \mapsto M_R \otimes_! \overline{A}^{\otimes_! r} \otimes_! M_L\right| \simeq M_R \otimes^{\mathbb{L}}_A M_L.$$

A marked chiral insertion gives left and right coactions of $\mathrm{Bar}(A)$ by cutting the interval on either side. Interfaces are therefore bimodules and compose by relative tensor product; after Koszul transform, they compose by cotensor product.

20.2 Crossed products

Let a finite group G act by augmented automorphisms of A. Then

$$\mathrm{Mod}_{A \rtimes G} \simeq (\mathrm{Mod}_A)^G.$$

The crossed-product bar is the diagonal totalization of the group bar and the A-bar. In characteristic zero, averaging contracts positive group-homology rows. The fixed algebra A^G describes invariant operators, while the crossed product $A \rtimes G$ describes equivariant modules and symmetry defects. A Galois or freeness theorem supplies a Morita comparison between them.

20.3 Symmetric products and DMVV

For a graded vector space V with character $f(q)$,

$$\sum_{N \geq 0} p^N \operatorname{ch} \mathrm{Sym}^N(V) = \exp\left(\sum_{r \geq 1} \frac{p^r}{r} f(q^r)\right).$$

Let Y be a compact complex manifold for which the orbifold elliptic genus is defined, and write

$$\mathrm{Ell}(Y;\tau,z)=\sum_{m\geq 0,\ell\in\mathbb{Z}} c(m,\ell)q^m y^\ell.$$

The Dijkgraaf–Moore–Verlinde–Verlinde formula [19] is

$$\sum_{N\geq 0} p^N \,\mathrm{Ell}_{\mathrm{orb}}(\mathrm{Sym}^N Y)=\prod_{n\geq 1,m\geq 0,\ell\in\mathbb{Z}}(1-p^n q^m y^\ell)^{-c(nm,\ell)}.$$

Cycle decomposition in symmetric groups, together with the fixed-cycle contribution to the orbifold trace, proves the product formula. The full orbifold operator algebra is obtained by adjoining twisted sectors, their defect compositions, and their OPE maps.

20.4 Higher transverse and higher holomorphic dimensions

Dunn additivity gives

$$\mathrm{Alg}_{\mathsf{E}_{n+1}}(\mathcal{V})\simeq \mathrm{Alg}_{\mathsf{E}_1}(\mathrm{Alg}_{\mathsf{E}_n}(\mathcal{V})).$$

An E_n-ordered chiral algebra has n locally constant transverse directions and retains its curve-valued chiral coefficient. Its iterated bar is factorisation homology of the pointed n-disk and carries the corresponding higher coalgebra structure under the positive-complete hypotheses. Its derived centre is an E_{n+1}-object.

Higher holomorphic dimension changes the coefficient operad itself. Felder–Gui–Young construct a polysimplicial dg model for configuration functions on $\mathbb{A}^d$, a dg operad of higher chiral operations on a shifted canonical sheaf, and a quasi-isomorphism from L_∞ to that operad [24]. The pair (d,n) therefore records two independent extensions: holomorphic collision dimension d and transverse topological dimension n.

No-go theorem 18: dimension loss does not reconstruct a quantum group

Graded dimensions determine a class in a Grothendieck group and contain no coproduct, antipode, associator, or R-matrix. Tannaka reconstruction requires a tensor category and a fibre functor; Hall reconstruction requires extension correspondences and orientations. A quantum group is therefore reconstructed from structured multiplication and comultiplication data, while a dimension formula is one of its characters.

Part IV

Exchange, reduction, sewing, physics, and arithmetic

Chapter 21

Perturbative BV realization of the ordered chiral object

21.1 Classical data

A perturbative classical BV theory in the Costello formalism is specified locally by an elliptic L_∞-algebra of fields $\mathcal{E}[-1]$, an invariant pairing of degree -1, and an interaction functional S solving the classical master equation

$$dS + \frac{1}{2}\{S, S\} = 0.$$

Classical observables on an open set are Chevalley cochains of compactly supported fields, with differential induced by the L_∞-structure. Disjoint support gives factorisation multiplication.

On $X^{\mathrm{an}} \times \mathbb{R}$, assume the kinetic complex is Dolbeault in X and de Rham or locally constant in the real direction. A boundary condition on $X \times \mathbb{R}_{\geq 0}$ restricts fields to a factorisation algebra on $X \times \mathbb{R}$. The ordered interval direction then supplies a transverse E_1-product.

21.2 Renormalised quantum observables

A gauge fixing produces a propagator P_ϵ^L between ultraviolet and infrared scales. Effective interactions $I[L]$ solve the homotopy renormalisation-group equation and the quantum master equation

$$(Q + \hbar\Delta_L)e^{I[L]/\hbar} = 0.$$

The quantum observable differential is

$$Q_L = Q + \{I[L], -\}_L + \hbar\Delta_L.$$

The QME is exactly $Q_L^2 = 0$.

Feynman weights are differential forms on configuration spaces of vertices. Renormalisation extends them to compactified collision spaces after local counterterms are added. Restriction to a boundary face factorises through contraction of the corresponding subgraph.

Theorem 21.1 (Feynman-weight morphism)**.** *Suppose:*

1. *renormalised graph weights extend to the chosen two-direction compactifications;*
2. *restriction to every face equals the contraction of the corresponding divergent subgraph*

followed by insertion of its renormalisation counterterm;

3. *the renormalised QME holds;*
4. *boundary observables are holomorphically algebraizable on X and locally constant in the ordered direction.*

Then the graph weights define a morphism from the geometric two-coloured forest resolution to the endomorphism operad of boundary observables. Hence the boundary observables form a homotopy-coherent transversely ordered chiral algebra. If ordered chiral collision vertices are also present, they form the corresponding doubly associative object.

Proof. Face factorisation is operadic composition. Stokes' theorem for the extended weights identifies the de Rham boundary with the sum of graph contractions. The QME cancels the full codimension-one boundary, including BV and counterterm faces, so the assignment is a chain map. Algebraisation and local constancy identify the resulting operations with the coefficient and transverse structures. □

No-go theorem 19: a classical local model does not automatically quantize

The obstruction to the next quantum lift is a cohomology class in the quantum deformation complex; a nonzero class prevents the lift. Boundary-compatible gauge fixing and algebraisation to regular holonomic D-modules are independent existence statements. A quantum ordered chiral object is obtained after the obstruction class vanishes, the boundary condition is preserved, and the analytic factorisation algebra admits the stated algebraic realization.

21.3 Higher operations and chiral quantization

Perturbative graph configurations define higher operations on local observables. Gaiotto–Kulp–Wu regularize them in holomorphic–topological theories and derive their quadratic identities from compactified Feynman boundaries [28]. Wang–Williams prove all-orders ultraviolet finiteness and anomaly-vanishing theorems for topological–holomorphic theories on flat space in the ranges stated in their main results [43]. Together these results supply analytic hypotheses for Theorem 21.1.

For a vertex Poisson algebra R, a chiral quantization is an $\hbar$-adic lift of its classical chiral operation. Butson–Nair construct an order-by-order deformation theory in which the obstruction to extending a lift lies in chiral Poisson cohomology $H^2_c(R)$, and the space of extensions is a torsor for $H^1_c(R)$ [11]. For arc algebras of affine symplectic varieties, these groups are identified with de Rham cohomology. The obstruction class is therefore a typed element of the chiral deformation complex.

21.4 Free holomorphic–topological BF theory

Let V be finite dimensional and take fields

$$A \in \Omega^{0,*}(X)\widehat{\otimes}\Omega^*(\mathbb{R}) \otimes V[1], \qquad B \in \Omega^{1,*}(X)\widehat{\otimes}\Omega^*(\mathbb{R}) \otimes V^*[-1]$$

with action

$$S_{\mathrm{BF}} = \int_{X\times\mathbb{R}} \langle B, (\bar\partial + d_t)A\rangle.$$

The theory is Gaussian. The de Rham contraction on an interval proves local constancy in t; the Dolbeault complex gives the holomorphic coefficient. Complementary linear boundary conditions select a polarization. Quantization of paired boundary fields produces chiral Weyl or Clifford systems according to parity. The central Weyl constant becomes the curvature element of the nonhomogeneous Koszul dual described in Chapter 13.

Chapter 22

BRST descent and Drinfeld–Sokolov reduction

22.1 The bar–BRST bicomplex

Let $Q : A \to A[1]$ be a square-zero derivation of the transverse multiplication, augmentation, and every factorisation map. On a bar word define

$$Q_B[a_1|\cdots|a_r] = \sum_{i=1}^{r}(-1)^{\sum_{j<i}(|a_j|+1)}[a_1|\cdots|Qa_i|\cdots|a_r].$$

Proposition 22.1 (BRST compatibility)**.**

$$Q_B^2 = 0, \qquad Q_B d_B + d_B Q_B = 0.$$

Thus $(\mathrm{Bar}(A), d_B, Q_B)$ *is a bicomplex in factorisation coefficients.*

Proof. The first identity is the tensor differential calculation. For the second, compare applying Q before and after an interior face. The two terms agree by the derivation rule and acquire opposite totalisation signs. Outer faces agree because the augmentation is Q-closed. ☐

Theorem 22.2 (Bar–BRST comparison)**.** *Assume tensor products preserve Q-quasi-isomorphisms and the combined filtration is complete and bounded below in every total weight. Then*

$$E_1 \cong \mathrm{Bar}\big(H_Q(A)\big) \Longrightarrow H^* \,\mathrm{Tot}(\mathrm{Bar}(A), d_B + Q_B).$$

If $H_Q(A)$ is concentrated in degree zero, the edge map

$$\mathrm{Bar}(H_Q^0(A)) \xrightarrow{\sim} \mathrm{Tot}(\mathrm{Bar}(A), d_B + Q_B)$$

is a quasi-isomorphism.

Proof. Filter first by BRST degree and then by bar length. The vertical complex is a tensor product of copies of (A, Q). The Kunneth hypothesis identifies its cohomology with tensor powers of $H_Q(A)$, and the induced horizontal differential is the bar differential of the cohomology algebra. Completeness gives convergence. Concentration leaves one row. ☐

22.2 Principal Drinfeld–Sokolov reduction

For an affine vertex algebra $V_k(\mathfrak{g})$, choose a good grading for a nilpotent element f, introduce charged and neutral ghosts, and form the BRST current from the moment map, cubic ghost term, character, and neutral correction. Its residue is a derivation Q with $Q^2 = 0$. The reduced algebra is

$$\mathcal{W}_k(\mathfrak{g}, f) = H^0_Q(V_k(\mathfrak{g}) \otimes \text{Ghosts}).$$

For principal f at generic noncritical level, the standard concentration theorem places cohomology in degree zero. Then Theorem 22.2 identifies the ordered bar of the principal W-algebra with BRST cohomology of the affine-ghost bar.

22.3 The $\mathfrak{sl}_2$ calculation

For affine currents e, h, f at level k, introduce fermionic ghosts b, c and

$$j_{\text{BRST}}(z) = (e(z) - 1)c(z), \qquad Q = \operatorname{Res} j_{\text{BRST}}(z).$$

The e-e and c-c OPEs are regular, hence $Q^2 = 0$. Improving the Sugawara stress tensor and adding the ghost stress tensor gives a Q-closed class with central charge

$$c_{\text{DS}}(k) = 13 - 6(k+2) - \frac{6}{k+2} = 1 - \frac{6(k+1)^2}{k+2}.$$

A filtration by f- and b-modes pairs the constrained currents into contractible quartets. The surviving Heisenberg field has the Feigin–Fuchs background charge, and its invariant Miura subalgebra is Virasoro. The quartet contraction and Miura invariant realize principal $\mathfrak{sl}_2$ reduction in degree zero for generic $k \neq -2$.

Remark 22.3 (Cohomological range). At generic noncritical level, principal Drinfeld–Sokolov reduction is concentrated in degree zero in the standard category. At special levels the derived reduction retains its higher BRST cohomology, and the bar–BRST spectral sequence computes the complete answer.

Chapter 23

From order to exchange: KZ, Tannaka, and four-dimensional Chern–Simons

23.1 Infinitesimal braid relations

Let $\mathfrak{g}$ be a quadratic Lie algebra and $\Omega \in \mathfrak{g} \otimes \mathfrak{g}$ the inverse invariant form. On $V_1 \otimes \cdots \otimes V_n$, write Ω_{ij} for its action in factors i, j. Invariance gives

$$[\Omega_{ij}, \Omega_{kl}] = 0 \quad \text{for disjoint pairs,} \qquad [\Omega_{ij}, \Omega_{ik} + \Omega_{jk}] = 0.$$

The two identities are the infinitesimal braid relations.

The KZ connection on $\mathrm{Conf}_n(\mathbb{C})$ is

$$\nabla_{\mathrm{KZ}} = d - \kappa \sum_{i<j} \Omega_{ij}\, d\log(z_i - z_j).$$

Proposition 23.1 (KZ flatness)**.** *The KZ connection is flat.*

Proof. The logarithmic one-forms are closed, so curvature is the wedge-square of the connection form. Terms on four distinct labels vanish by disjoint commutativity. For three labels, invariance rewrites the three commutators in terms of one common commutator, while the accompanying two-forms sum to

$$\omega_{12} \wedge \omega_{23} + \omega_{23} \wedge \omega_{31} + \omega_{31} \wedge \omega_{12} = 0.$$

The Arnold identity therefore cancels the three-label curvature. □

Monodromy gives a braid-group representation. The Drinfeld–Kohno theorem identifies the formal monodromy category with the braided representation category of the corresponding quantum group after choosing the standard parameter correspondence and a Drinfeld associator. The associator compares parenthesized tensor products in the monodromy category.

23.2 Meromorphic Tannaka reconstruction

Let $\mathcal{C}$ be a rigid meromorphic tensor category of line operators and let $\omega : \mathcal{C} \to \mathrm{Vect}_{\Bbbk((u))}$ be a faithful exact strong monoidal fibre functor. The coend

$$\mathcal{O}(\mathsf{G}_{\mathcal{C}}) = \int^{V \in \mathcal{C}} \omega(V)^* \otimes \omega(V)$$

reconstructs the coordinate bialgebra of tensor automorphisms of ω. Rigidity gives an antipode, and braided exchange gives a coquasitriangular form. A topological completion records infinite spectral expansions. Its continuous dual is the corresponding completed quantum algebra when finite subcoalgebras separate points.

No-go theorem 20: a braided category without a fibre functor does not determine an ordinary Hopf algebra

An ordinary or topological Hopf algebra is reconstructed from a rigid tensor category together with a faithful exact strong monoidal fibre functor to a specified linear base. The braided category alone is the intrinsic invariant; a chosen realization in another base category produces the corresponding internal Hopf object. A scalar extracted from a braid matrix records a character-level value and omits the coend, coproduct, and antipode.

23.3 Four-dimensional Chern–Simons theory

Let Σ be an oriented real surface, C a complex curve with meromorphic one-form ω, and consider the partial connection of four-dimensional Chern–Simons theory on $\Sigma \times C$:

$$S_{4d}(A) = \frac{1}{2\pi\hbar} \int_{\Sigma \times C} \omega \wedge \mathrm{Tr}\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A\right).$$

The theory is topological on Σ and holomorphic on C. A Wilson line in Σ at spectral point $z \in C$ is a line object. Parallel fusion is ordered; exchange in Σ produces an R-matrix. In the rational case, the tree-level exchange is

$$R(z_1 - z_2) = 1 + \hbar \frac{\Omega}{z_1 - z_2} + O(\hbar^2).$$

Gauge invariance at the first nontrivial order imposes the classical Yang–Baxter equation. Quantisation produces the rational Yangian-type exchange algebra under the boundary, anomaly, and representation hypotheses of the model.

The planar exchange geometry produces the ZF/RTT relations and Yang–Baxter compatibility. Tensor generation, PBW, and faithfulness of evaluation modules identify the reconstructed completed algebra with the corresponding Yangian.

Chapter 24

Cyclic traces, modular sewing, and the two-edge master equation

24.1 Closing the transverse interval

Let A be a proper cyclic A_∞-algebra object in the coefficient category. A degree-d Calabi–Yau structure is a negative-cyclic class inducing an equivalence

$$A \xrightarrow{\sim} A^\vee[d].$$

On a finite strict model it is represented by a cyclic trace with perfect pairing $\langle a, b\rangle = \mathrm{Tr}_A(ab)$. The circle value is

$$\int_{S^1} A \simeq \mathrm{HH}_*(A).$$

A ribbon graph is evaluated by cyclic higher multiplications at vertices and the inverse pairing at edges.

Theorem 24.1 (Ribbon-graph action)**.** *A finite perfect cyclic A_∞-algebra defines a dg PROP morphism from the positive-boundary ribbon-graph complex to endomorphisms of its Hochschild chains [13]. Edge contraction is sent to composition of cyclic operations, and gluing boundary circles is sent to insertion of the coevaluation pairing.*

Proof. The graph evaluation is the tensor contraction determined by cyclic order at vertices. The ribbon differential sums internal-edge contractions. The cyclic A_∞-relations are exactly the assertion that this sum intertwines the graph and Hochschild differentials. Disjoint union and gluing agree with tensor product and coevaluation. □

The ribbon-graph PROP is graded by the genus $g_\perp$ of the thickened ribbon surface.

24.2 Chiral sewing over stable curves

A modular chiral coefficient assigns state spaces or complexes to pointed curves and contraction maps to separating and nonseparating clutching morphisms

$$\overline{\mathcal{M}}_{g_1,n_1+1} \times \overline{\mathcal{M}}_{g_2,n_2+1} \to \overline{\mathcal{M}}_{g_1+g_2,n_1+n_2},$$

$$\overline{\mathcal{M}}_{g-1,n+2} \to \overline{\mathcal{M}}_{g,n}.$$

A perfect chiral pairing contracts the two flags of a node. The genus g_X is the genus of the algebraic curve; the transverse ribbon genus $g_\perp$ counts handles in a different geometric carrier.

External theorem 24.2 (Modular functor from strongly rational conformal blocks). *Let V be a strongly rational vertex operator algebra. The vector bundles of conformal blocks of V, together with their factorisation and sewing morphisms, form a modular functor. The corresponding module category is a modular fusion category, and the modular functor extends to a three-dimensional topological field theory [17].*

The theorem supplies a nontrivial chiral-sewing input for the bicoloured construction. Pairing it with a specified proper cyclic algebra supplies an independent transverse ribbon input.

24.3 Bicoloured stable graphs

Give every internal edge a colour X or $\perp$. An X-edge is a chiral clutching node; a $\perp$-edge is a ribbon/trace contraction. Orient a graph by

$$\det E_X \otimes \det E_\perp.$$

Exchanging the two edge colours changes sign. The graph brackets and loop operators are

$$[-,-]_X, \quad [-,-]_\perp, \qquad \Delta_X, \quad \Delta_\perp.$$

The bicoloured Feynman transform is the completed free graph complex generated by common corollas, with differential given by internal differential and contraction of one coloured edge. Its relations are the one-colour sewing identities and the mixed square.

Recognition theorem 24.3 (Two-edge sewing theorem). *Suppose a state object carries:*

1. *cyclic transverse gluing and a perfect transverse pairing;*
2. *chiral clutching and a perfect chiral sewing pairing;*
3. *common genus-zero corollas;*
4. *the one-colour associativity/modular identities;*
5. *a mixed interchange square for one ribbon and one chiral edge, with the product determinant sign.*

Then there is a unique action of the bicoloured stable-graph transform extending the two one-colour actions. Its master element satisfies

$$d\Theta + \frac{1}{2}[\Theta,\Theta]_X + \frac{1}{2}[\Theta,\Theta]_\perp + \hbar_X\Delta_X\Theta + \hbar_\perp\Delta_\perp\Theta = 0.$$

In particular,

$$\Delta_X\Delta_\perp + \Delta_\perp\Delta_X = 0.$$

Proof. Every bicoloured graph is built by gluing common corollas along coloured edges. The one-colour axioms identify all decompositions involving only one colour. The mixed square identifies adjacent contractions of different colours. The one-colour relations and the mixed square connect any two contraction orderings of the same coloured graph, which proves existence and uniqueness of the action. The codimension-two boundary of the bicoloured graph complex is the sum of the two brackets, two loop contractions, and the internal differential. Its square is zero by the determinant orientation, yielding the master equation. □

No-go theorem 21: no universal equality of the two genera

Ribbon genus and chiral genus are the two projections of the bidegree $(g_X, g_\perp)$ in the free bicoloured graph complex. The tensor-product model of Theorem 26.2 contains nonzero amplitudes in independent bidegrees, so the two projections are distinct natural gradings. An equality between selected amplitudes is therefore additional duality data rather than a universal identification of the gradings.

Chapter 25

Prime forms, plumbing, and the analytic chiral genus

25.1 The normalized bidifferential

Let Σ_g be a compact Riemann surface with symplectic homology basis and period matrix Ω. The prime form $E(p, q)$ is a $(-1/2, -1/2)$-form with a simple zero on the diagonal. The normalized fundamental bidifferential has local behaviour

$$B(p, q) = \left(\frac{1}{(z(p) - z(q))^2} + O(1) \right) dz(p)dz(q)$$

and vanishing A-periods. With the prime form normalized by the chosen symplectic basis,

$$B(p, q) = d_p d_q \log E(p, q).$$

For the normalized oscillator block of a rank-one Heisenberg current,

$$\langle J(p)J(q)\rangle^{\mathrm{osc}}_{\Sigma_g} = kB(p, q).$$

Higher oscillator correlators are sums over pairings. Momentum sectors add holomorphic one-form zero modes, and the partition function includes the determinant line and the lattice or Fock zero-mode sum.

25.2 Plumbing a node

For local coordinates z, w at two marked points, plumbing identifies

$$zw = q, \qquad |q| \ll 1.$$

A separating node joins two components; a nonseparating node joins two points on one component. In local coordinates, formal sewing inserts the propagator q^{L_0}:

$$\mathcal{F}_{\mathrm{sew}}(q) = \sum_{\alpha} q^{L_0(\alpha)} \mathcal{F}_1(\cdots, v_\alpha)\mathcal{F}_2(v^\alpha, \cdots).$$

Passing to cylindrical coordinates and trivializing the determinant anomaly line produces the familiar vacuum factor $q^{-c/24}$. Analytic sewing requires convergence, control of the completion, and compatibility of the chosen local coordinates.

No-go theorem 22: formal sewing does not imply analytic convergence

Formal sewing is an identity in a completed power-series object. The series $\sum_{n\geq 0} n!q^n$, for example, is a well-defined formal element and has zero analytic radius of convergence. Analytic conformal blocks therefore require trace-class estimates, energy bounds, or a rationality theorem in addition to the formal modular-operad identities.

25.3 Genus one

For a positive-energy module M, the torus character is

$$\chi_M(\tau) = \mathrm{Tr}_M\, q^{L_0 - c/24}, \qquad q = e^{2\pi i\tau}.$$

For a rank-one Heisenberg algebra,

$$\chi_{\mathrm{Heis}}(\tau) = \eta(\tau)^{-1}.$$

For an even lattice L,

$$\chi_{V_L}(\tau) = \frac{\Theta_L(\tau)}{\eta(\tau)^{\mathrm{rk}\, L}}.$$

The theta numerator records zero modes; the eta denominator is the oscillator determinant. Their modular transformations are sections of the anomaly line rather than ordinary invariant functions until the determinant factor is included.

25.4 Higher genus and Siegel theta series

The lattice zero-mode sum becomes

$$\Theta_L^{(g)}(\Omega) = \sum_{\lambda_1,\ldots,\lambda_g \in L} \exp\left(\pi i \sum_{a,b} (\lambda_a, \lambda_b)\Omega_{ab}\right).$$

The oscillator factor is the corresponding chiral determinant. Separating degeneration factorises the theta series into the products of lower-genus theta series; nonseparating degeneration is a Fourier–Jacobi expansion. Siegel theta series and chiral determinants belong to the chiral genus g_X; ribbon loop count supplies the independent grading $g_\perp$.

Chapter 26

A concrete two-genus model

26.1 The transverse matrix ribbon weight

Let $B_N = M_N(\Bbbk)$ with the trace pairing

$$\langle A, B\rangle = \mathrm{Tr}(AB).$$

Matrix units satisfy $\langle E_{ij}, E_{kl}\rangle = \delta_{il}\delta_{jk}$, so the inverse pairing is the double-line propagator

$$\sum_{i,j} E_{ij} \otimes E_{ji}.$$

Let G be a connected ribbon graph with $b(G)$ distinguished external boundary cycles, and let $F_{\mathrm{int}}(G)$ denote the closed index faces. Matrix-index contraction gives one factor of N for each closed index face; the external boundary cycles remain as normalized trace insertions. With the standard 't Hooft normalization, a vertex contributes N, an internal edge contributes N^{-1}, and a closed index face contributes N. Therefore

$$W_N^{\mathrm{tH}}(G) = N^{V(G)-E(G)+F_{\mathrm{int}}(G)} = N^{2-2g_\perp(G)-b(G)}.$$

The second equality is the Euler characteristic of the thickened ribbon surface with its distinguished boundary cycles left unfilled. The augmented algebra $D\times M_N(\Bbbk)$ of Theorem 13.3 supplies, independently, an interval bar for the same matrix boundary sector.

26.2 A rank-one Deligne–Mumford theory

Let $C_\lambda = \Bbbk\mathbf{1}$ be the one-dimensional commutative Frobenius algebra with

$$\mathbf{1}\cdot\mathbf{1} = \mathbf{1}, \qquad \eta(\mathbf{1},\mathbf{1}) = \lambda \in \Bbbk^\times.$$

The inverse metric is λ^{-1}, and the associated rank-one cohomological field theory assigns

$$\Omega_{g,r} = \lambda^{1-g} \in H^0(\overline{\mathcal{M}}_{g,r}; \Bbbk).$$

Proposition 26.1 (Rank-one clutching)**.** *The classes $\Omega_{g,r}$ satisfy the separating and nonseparating CohFT gluing equations.*

Proof. Separating gluing contracts two classes with the inverse metric:

$$\lambda^{1-g_1}\lambda^{1-g_2}\lambda^{-1} = \lambda^{1-(g_1+g_2)}.$$

Nonseparating gluing gives

$$\lambda^{1-(g-1)}\lambda^{-1} = \lambda^{1-g}.$$

The unit equation follows from $\mathbf{1} \cdot \mathbf{1} = \mathbf{1}$. □

26.3 Product boundary geometry

Let $\mathcal{G}_X$ be the stable-graph complex for the rank-one CohFT and let $\mathcal{G}_\perp$ be the ribbon-graph complex with matrix weights. Their product has codimension-two boundary faces

$$D_X \times D_\perp.$$

Chiral clutching acts on the first tensor factor, and ribbon contraction acts on the second. The two contractions commute on coefficients and anticommute after the product determinant orientation is inserted.

Theorem 26.2 (Realized two-edge recognition)**.** *The external tensor product of the rank-one CohFT and the matrix ribbon theory defines an action of the bicoloured graph transform. For a stable graph Γ_X of genus g_X and a connected ribbon graph G, the amplitude is*

$$\boxed{\mathcal{A}_{\lambda,N}(\Gamma_X, G) = \lambda^{1-g_X}\, N^{2-2g_\perp(G)-b(G)}.}$$

The associated master element satisfies the two-edge equation of Theorem 24.3.

Proof. The rank-one clutching equations give the X-coloured boundary identities. Matrix index contraction gives the ribbon boundary identities and the Euler power of N. On a mixed face, the coefficient maps act on different tensor factors. The determinant orientation changes sign when the two normal directions are exchanged. The product boundary therefore supplies the mixed anticommutator, and the oriented-corner identity gives the master equation. □

The model proves the logical independence of the two genus gradings in a fully calculated example. Replacing the rank-one CohFT by the modular functor of Theorem 24.2 produces a nontrivial chiral factor; replacing the matrix theory by a cyclic A_∞-algebra produces the corresponding open-string ribbon amplitudes.

Chapter 27

Typed anomalies and invertible theories

27.1 Five domains

The word "anomaly" names several obstruction classes.

Anomaly type	Native object
Curved Koszul term	arity-zero element h in a curved dg algebra, with $d^2 = [h, -]$.
BRST anomaly	degree-two square of the quantum BRST operator, represented in the BRST deformation complex.
BV anomaly	degree-one local functional measuring the quantum master equation, modulo exact counterterms.
Conformal/determinant anomaly	curvature or first Chern class of a determinant line over moduli.
Framing anomaly	invertible field theory or central extension measuring dependence on a framing/trivialization.

A specified family index, transgression, boundary, or open–closed morphism transports an anomaly class between two of these complexes. The comparison theorem consists of that morphism and the identity it induces on cohomology.

Theorem 27.1 (Anomaly cancellation by an inverse theory)**.** *Let $\mathcal{L}$ be the invertible anomaly theory of a quantum field theory $\mathcal{T}$. If an inverse $\mathcal{L}^{-1}$ and a trivialization of $\mathcal{L} \otimes \mathcal{L}^{-1}$ are chosen compatibly with gluing, then $\mathcal{T} \otimes \mathcal{L}^{-1}$ has an honest, rather than projective, state assignment on the corresponding bordism/moduli category.*

Proof. The anomalous theory is a relative natural transformation valued in the line assigned by $\mathcal{L}$. Tensoring with the inverse line makes the value canonically scalar after the chosen trivialization. Compatibility with gluing is exactly naturality of the trivialization. □

No-go theorem 23: no universal anomaly scalar

Curved Koszul data, the BV obstruction, determinant curvature, the BRST anomaly, and framing dependence inhabit different complexes and generally have different cohomological degrees. A scalar arises only after one class is paired with a specified cycle, trace, index class, or background. A universal scalar would require natural comparison maps among all five complexes, and the source categories provide no such maps.

No-go theorem 24: central charge does not classify a chiral theory

The Virasoro central charge is one coefficient of the stress-tensor OPE. The Leech lattice vertex operator algebra and the Moonshine vertex operator algebra both have central charge 24, while their weight-one spaces have dimensions 24 and 0, respectively [26]. Their field content and symmetry therefore differ at fixed central charge. Classification of ordered or associative chiral algebras requires the complete chiral operation together with its module, fusion, modular, and transverse data.

Chapter 28

Bar Euler products and Borcherds denominators

28.1 Root-graded Lie superalgebras

Let

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \neq 0} \mathfrak{g}_\alpha, \qquad \mathfrak{n}_+ = \bigoplus_{\alpha \in \Gamma_+} \mathfrak{g}_\alpha,$$

where every root space is finite dimensional and each degree has finitely many decompositions in the chosen completion. PBW and Theorem 15.1 give

$$\mathrm{Bar}^\perp(U(\mathfrak{n}_+)) \simeq \mathrm{CE}_*(\mathfrak{n}_+) = \mathrm{Sym}(\mathfrak{n}_+[1]).$$

Theorem 28.1 (Bar denominator product)**.** *The completed graded Euler supercharacter is*

$$\chi_\Gamma \,\mathrm{Bar}^\perp(U(\mathfrak{n}_+)) = \prod_{\alpha \in \Gamma_+} (1 - e^{-\alpha})^{\mathrm{sdim}\, \mathfrak{g}_\alpha}.$$

If $\mathfrak{g}$ is a Borcherds–Kac–Moody superalgebra with Weyl vector ρ, multiplying by $e^{-\rho}$ gives the product side of its denominator identity.

Proof. An even root vector becomes odd after suspension and contributes $1 - e^{-\alpha}$ to the Euler supercharacter. An odd root vector becomes even and contributes its inverse. Multiplication over a homogeneous basis gives the superdimension exponent. Finite decomposition makes each coefficient of the completed product finite. □

The theorem is an object-to-character map: the Lie bracket controls the Chevalley differential, while the Euler character remembers only signed root multiplicities.

28.2 The Igusa product

Let

$$\phi_{0,1}(\tau, z) = \sum_{n \geq 0, \ell \in \mathbb{Z}} f(n, \ell) q^n y^\ell$$

be the normalized weak Jacobi form of weight zero and index one. In the standard chamber, the monic Borcherds product is

$$D_5(Z) = q^{1/2}y^{1/2}p^{1/2} \prod_{(n,\ell,m)>0} (1 - q^n y^\ell p^m)^{f(nm,\ell)}.$$

For the Gritsenko–Nikulin Borcherds algebra, the signed root multiplicities are $f(nm, \ell)$. After fixing the chamber, Weyl vector, and normalization of the Siegel modular form, the denominator identity reads

$$e^{-\rho}\chi_\Gamma \operatorname{Bar}^\perp(U(\mathfrak{n}_+)) = D_5(Z).$$

The displayed equality uses the monic product normalization and the stated Siegel coordinates. Multiplication of the modular form by a declared scalar or replacement of the Siegel coordinate changes both the normalization and the equation accordingly. In the fixed convention, the equality is the Euler-character identity of the constructed root-graded bar.

28.3 Finite positive windows

Let $h : \Gamma_+ \to \mathbb{N}_{>0}$ be proper and additive. The quotient

$$\mathfrak{n}_+(H) = \mathfrak{n}_+ / \bigoplus_{h(\alpha)>H} \mathfrak{g}_\alpha$$

is finite-dimensional nilpotent. Its level-zero current vertex algebra has PBW character, and the inverse limit over H is separated and complete. Coefficientwise finiteness follows because a fixed root and oscillator degree sees only finitely many positive roots and modes.

28.4 Polarized Gram transform

Let $\Pi : \Gamma_{\mathrm{phys}} \to \Lambda$ be quadratic with polarization

$$B(\gamma, \delta) = \Pi(\gamma + \delta) - \Pi(\gamma) - \Pi(\delta).$$

For a charge-graded algebra $A = \bigoplus_\gamma A_\gamma$, define

$$(a_\gamma u^\lambda) \star_B (b_\delta u^\mu) = a_\gamma b_\delta u^{\lambda+\mu+B(\gamma,\delta)}.$$

Biadditivity gives the cocycle identity and hence associativity. The Gram embedding

$$\iota_\Pi(a_\gamma) = a_\gamma u^{\Pi(\gamma)}$$

is an algebra homomorphism for $\star_B$. The polarized product converts the quadratic Fourier index into an additive exponent while retaining the bilinear correction B in the multiplication law.

Remark 28.2 (Geometric Hall realization). A compact Hall realization consists of a moduli stack, orientation data, an extension correspondence, a PBW comparison, and a homomorphism to the Borcherds algebra. The moduli stack, orientation data, extension correspondence, PBW comparison, and algebra homomorphism lift the denominator character to a geometric multiplication theorem.

Chapter 29

Higher-dimensional sources and factorisation pushforward

29.1 Pushforward along an actual map

Let $f : M \to X$ be a map and $\mathcal{O}$ a factorisation algebra on M. Under properness/support and descent conditions, factorisation pushforward is

$$(f_*^{\mathrm{Fact}}\mathcal{O})(U) = \mathcal{O}(f^{-1}U).$$

Disjoint opens in X have disjoint inverse images, so the factorisation maps descend. Holomorphic or topological reduction to a curve is therefore a functor only after an actual map and support theory are specified.

29.2 Calabi–Yau categories and Hall products

Let $\mathcal{C}$ be a Calabi–Yau category with a chosen brane, proper support for the moduli correspondences used in the Hall construction, and orientation data. The datum defines:

1. an E_1-algebra of endomorphisms of a chosen brane;
2. extension correspondences on moduli of objects, yielding Hall or cohomological Hall multiplication;
3. cyclic pairings and open-string traces;
4. closed-string maps into Hochschild invariants.

Shifted symplectic geometry supplies orientations for the correspondences and controls their quantization. Finiteness, proper support, and compatibility with factorisation turn these operations into an ordered chiral algebra.

29.3 Coulomb branches and shifted Yangians

For a Braverman–Finkelberg–Nakajima three-dimensional $\mathcal{N} = 4$ gauge theory, convolution on the affine-Grassmannian correspondence defines the quantized Coulomb-branch algebra [10]. In the quiver cases covered by shifted-Yangian comparison, line-defect sectors form modules over the corresponding truncated shifted Yangian [36]. A curve-valued chiral realization requires:

1. the curve-valued factorisation coefficient;
2. the ordered line multiplication;
3. the augmentation or relative base used by the bar;
4. the exchange geometry or R-matrix;
5. the comparison to the RTT/Drinfeld presentation.

The five displayed structures make the bar, centre, exchange, and BRST constructions functorial.

29.4 Source-to-chiral comparison

The correct general pattern is a diagram

$$\begin{tikzcd} \text{higher-dimensional geometric category} \arrow[r, "\text{pushforward/restriction}"] \arrow[d, "\text{Hall or convolution}"'] & \text{factorisation category on } X \arrow[d, "\mathrm{RHom}(b,b)"] \\ \text{geometric algebra} \arrow[r, "\text{comparison}"'] & \mathrm{Alg}_{\mathsf{E}_1}(\mathrm{FactD}(X)). \end{tikzcd}$$

Each square is a theorem about correspondences, supports, and descent. The diagram organizes the source geometry, the curve-valued factorisation category, and the ordered endomorphism algebra.

Part V

Geometric and categorical derivations

Chapter 30

The phenomenon and its invariant language

30.1 Two limits of the same experiment

Place local operators on a complex curve X and line defects on an oriented real line. There are two independent short-distance limits. In the first, two points of X approach one another while their transverse positions remain fixed. The product develops a finite principal part along the diagonal. In the second, two transverse intervals become adjacent while their positions on X remain separated. The product is associative and ordered. A theory in which both limits exist therefore carries two operations of different geometric origin:

$$\text{chiral collision on } X, \qquad \text{ordered fusion on } \mathbb{R}. \tag{30.1}$$

The first is encoded by a factorisation D-module. The second is encoded by an E_1-multiplication internal to the category of such modules.

This distinction is forced by topology. A punctured complex disc has a nontrivial fundamental group, so analytic continuation of insertions can produce monodromy. An ordered chamber

$$\mathrm{Conf}_n^{<}(\mathbb{R}) = \{t_1 < \cdots < t_n\}$$

is contractible. The chamber therefore carries order and associativity. Thickening the transverse line to a plane introduces exchange paths and their braid transport. Chiral singularity, associative order, and planar exchange are independent data.

Definition 30.1 (Ordered chiral algebra)**.** Let X be a smooth complex algebraic curve and let $\mathrm{FactD}(X)$ be the symmetric monoidal stable category of factorisable right D-modules defined in Chapter 31. An *ordered chiral algebra* on X is an augmented algebra

$$A \in \mathrm{Alg}_{E_1}^{\mathrm{aug}}(\mathrm{FactD}(X)).$$

Its multiplication is denoted $m : A \otimes A \to A$, and its augmentation is denoted $\varepsilon : A \to \mathbf{1}$.

The word *chiral* belongs to the coefficient category. The word *ordered* belongs to the E_1-operation. Neither adjective is redundant.

30.2 The diagonal and the interval

The diagonal is the carrier of the operator product expansion. Let

$$j : X^2 \setminus \Delta(X) \hookrightarrow X^2, \qquad \Delta : X \hookrightarrow X^2.$$

A chiral singular product is a morphism from an extension off the diagonal to a distribution supported on the diagonal. In the Lie form of Beilinson and Drinfeld it is

$$\mu : j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}. \tag{30.2}$$

A coordinate expansion of (30.2) produces all OPE modes. The morphism itself is coordinate-free.

The interval is the carrier of the ordered product. If $I_1 < I_2$ are disjoint subintervals of an oriented interval I, factorisation gives

$$A(I_1) \otimes A(I_2) \longrightarrow A(I).$$

Shrinking the ordered pair (I_1, I_2) inside I produces the binary multiplication. Three ordered subintervals produce the two composites $(ab)c$ and $a(bc)$; the contractibility of the space of such shrinkings gives their coherent homotopy. After fixing the endpoints and quotienting by affine reparametrisation, the standard real compactification of ordered points is an associahedron.

The two geometries meet on $X \times \mathbb{R}$. A small rectangle $U \times I$ carries a complex of observables. Chiral factorisation varies U in X; ordered factorisation varies I in $\mathbb{R}$. Their interchange is the geometric origin of an E_1-algebra in $\mathrm{FactD}(X)$.

30.3 The unique language

A useful language must satisfy four requirements.

First, it must retain the entire singular product. A logarithmic residue detects a simple normal coefficient. A fourth-order Virasoro pole and a double affine pole require higher normal jets. A factorisation D-module contains all of them at once.

Second, it must retain noncommutative fusion before any symmetrisation. Tensor order, line order, and braid exchange are three different structures. The E_1-multiplication retains the first two; an E_2 enhancement supplies the third.

Third, it must separate algebraic and analytic completion. The bar construction is defined in the stable category. Trace-class sewing and convergence of conformal blocks are additional analytic theorems.

Fourth, it must type every anomaly. Curvature of a determinant line, failure of a BRST differential, failure of a BV quantum master equation, and curvature in a curved Koszul dual are classes in different complexes. They agree only after a comparison morphism has been constructed.

These requirements lead to the following square:

$$\begin{tikzcd}
\mathrm{Alg}_{E_1}(\mathrm{FactD}(X)) \arrow[r, "\mathrm{Bar}_X^{\mathrm{ord}}"] \arrow[d, "\mathrm{an}"'] & \mathrm{Coalg}_{\mathrm{conil}}(\mathrm{FactD}(X)) \arrow[d, "\mathrm{an}"] \\
\mathrm{Fact}^{\mathrm{rect}}_{\mathrm{hol,lc}}(X^{\mathrm{an}} \times \mathbb{R}) \arrow[r, "\int_{([0,1],\partial)}"] & \mathrm{Fact}_{\mathrm{hol}}(X^{\mathrm{an}}).
\end{tikzcd} \tag{30.3}$$

The upper row is the internal associative bar construction. For positive complete objects, cobar reconstructs it. The lower row is interval factorisation of a rectangular holomorphic–topological theory. On regular holonomic objects, Riemann–Hilbert identifies the vertical arrows. The square expresses one phenomenon in algebraic and analytic language.

30.4 What is genuinely noncommutative

A conventional factorisation algebra on a manifold has symmetric multiplication for disjoint opens because disjoint union is unordered. Noncommutativity enters when a codimension-one direction orders the opens. A boundary condition, an interface, or a family of parallel line defects supplies such a direction. The resulting multiplication is not the chiral OPE. It is composition in the category of defects.

Let $\mathcal{C}$ be a factorisation category on X and let b be an object representing a boundary condition. The internal endomorphism object

$$A_b = \mathrm{RHom}_{\mathcal{C}}(b, b) \tag{30.4}$$

is an E_1-algebra in the factorisation coefficient category. Composition of endomorphisms gives the E_1 product. Collision of their insertion points gives the chiral singularity. Formula (30.4) is the native source of noncommutative chiral algebras.

A simple algebraic model is obtained from a factorisation object V and an associative algebra B. If V is commutative as a factorisation object, then

$$A = B \otimes V$$

is an ordered chiral algebra with multiplication $m_B \otimes m_V$. Matrix-valued free fields and Chan–Paton factors have this form locally. The bar construction sees the noncommutative resolution of B while every term remains a chiral coefficient system on X.

30.5 The bar is transverse

The reduced bar construction of an augmented algebra is generated by ordered words

$$[a_1|\cdots|a_r],$$

and its differential multiplies adjacent letters. The word order is the order of intervals on $\mathbb{R}$. Chiral singularities are not extracted by this differential. They are already present in each coefficient a_i and in the factorisation maps of $A^{\otimes r}$.

This observation removes the original ambiguity between collision residues and the associative bar differential. The de Rham differential of a configuration space, the residue differential along a collision divisor, and the simplicial bar differential are three distinct operators. They can coexist in a total complex, but their source and target degrees must be written separately.

Proposition 30.2 (Low-arity bar identity)**.** *Let A be a dg E_1-algebra in a stable symmetric monoidal category. On homogeneous elements of the augmentation ideal,*

$$b_2[a|b|c] = (-1)^{\alpha}[(ab)|c] + (-1)^{\beta}[a|(bc)],$$

with the standard suspension signs, and

$$b_2^2[a|b|c] = \pm[(ab)c - a(bc)].$$

Hence the multiplication part of the bar differential squares to zero exactly by associativity.

Proof. There are two length-two contractions of a word of length three. The suspension orientation gives opposite signs. The two coefficient operations agree by associativity. □

The complete chiral identities are tested elsewhere: by the chiral operad on the formal disc, by factorisation on the Ran space, or by the partition collision complex. Associativity of the transverse product and the Borcherds identity of the chiral coefficient are compatible but not identical.

30.6 From order to exchange

The distinction between order and exchange is the first physical test of the language. Parallel line defects in a topological line can be multiplied, but cannot pass through one another. A second topological direction allows an exchange path. The homotopy class of that path is a braid. Thus

$$E_1 \text{ fusion} \quad \xrightarrow{\text{one more topological direction}} \quad E_2 \text{ exchange}.$$

Dunn additivity identifies the second structure with an E_1-algebra object in E_1-algebras. The universal E_2 object acting on an E_1 algebra is its Hochschild centre.

This yields a precise physical dictionary:

ordered intervals	fusion of parallel defects,
chiral diagonals	singular OPE on the spectral curve,
second transverse direction	braid exchange,
Hochschild centre	universal bulk or exchange algebra,
Tannaka coend	reconstructed quantum symmetry.

Each row denotes a geometric construction and its induced functor.

30.7 The two notions of modularity

A trace on an E_1 algebra closes an interval into a circle. Iterated gluing produces ribbon surfaces. This is the modularity of open topological field theory and Hochschild homology.

A chiral algebra on a varying algebraic curve has a different modularity. Plumbing a node changes the complex curve. The parameter spaces are $\overline{\mathcal{M}}_{g,n}$, and the amplitudes are conformal blocks. The first genus is transverse and ribbon; the second is complex and Deligne–Mumford.

When an ordered chiral algebra carries both sewing structures and their mixed interchange maps, the two contractions supercommute with the product orientation. The resulting master equation has two loop operators. The separation of the two loop parameters is essential in gauge theory: one counts topological colour loops, the other counts handles of the chiral curve.

30.8 Mathematics and physics

The mathematical object is an E_1 algebra in factorisable D-modules. The physical object is a holomorphic–topological factorisation algebra with a boundary or line defect. The comparison is not a metaphor. It is a morphism of factorisation algebras obtained by taking quantum observables and restricting them to the defect.

Let $\mathrm{Obs}_{\mathcal{T}}$ be the quantum observables of a renormalised BV theory $\mathcal{T}$ on $X \times \mathbb{R}_{\geq 0}$. A boundary condition gives a factorisation algebra Obs_{∂} on $X \times \mathbb{R}$. When holomorphic dependence admits a factorisable D-module realization and the boundary theory is locally constant in the ordered direction, Theorem 39.5 gives

$$\mathrm{Obs}_{\partial} \in \mathrm{Alg}_{E_1}(\mathrm{FactD}(X)). \tag{30.5}$$

The renormalized quantum master equation makes the observable differential square to zero and makes the factorisation maps compatible with ultraviolet contractions. A nonzero BV anomaly obstructs this quantum factorisation object.

Perturbative Chern–Simons theory, four-dimensional Chern–Simons theory, holomorphic twists whose translation operators admit topological descent, and open string field theory furnish instances after the corresponding boundary or defect observable maps have been constructed. Their different quantum symmetries arise from different exchange geometries and spectral curves, not from different definitions of the bar construction.

30.9 The theorem spine

The constructions assemble into the following sequence of theorems.

- (i) Ordered block-Ran factorisation objects are equivalent to E_1-algebras in $\mathrm{FactD}(X)$.
- (ii) Their ordered bar is interval factorisation homology and has an associahedral cofibrant model.
- (iii) On connected positive-weight objects and their complete Mittag–Leffler limits, the internal associative bar and cobar reconstruct one another. Separately, Ran cardinality is pro-nilpotent for chiral convolution.
- (iv) The augmentation boundary has endomorphism algebra $A^!$, and the double dual is the augmentation completion.
- (v) An E_1 algebra has no canonical braid; its Hochschild centre is the universal E_2 algebra acting on it.
- (vi) Rigid meromorphic line categories reconstruct quantum symmetries by Tannaka coends.
- (vii) BRST derivations extend to the bar and yield a convergent comparison spectral sequence under boundedness and completeness.
- (viii) Cyclic transverse sewing and chiral curve sewing form a bi-modular master equation with two independent loop operators.
- (ix) Under the algebraisation, locality, gauge-fixing, and anomaly-cancellation hypotheses stated in Part V, factorisation observables realize the preceding constructions in perturbative holomorphic–topological field theory.

The geometry and the exact examples realize every arrow in this sequence. The geometry and the exact examples exhibit these subjects as functorial realizations of one ordered chiral object.

Chapter 31

Factorisable D-modules and the complete chiral coefficient

31.1 The finite-set presentation

Let X be a smooth complex algebraic curve. Write $\mathrm{Fin}^{\mathrm{surj}}$ for the category of nonempty finite sets and surjections. A surjection $\alpha : I \twoheadrightarrow J$ determines a diagonal

$$\Delta_\alpha : X^J \longrightarrow X^I, \qquad (x_j)_{j\in J} \longmapsto (x_{\alpha(i)})_{i\in I}.$$

The Ran prestack is the colimit

$$\mathrm{Ran}(X) = \operatorname*{colim}_{I\in(\mathrm{Fin}^{\mathrm{surj}})^{\mathrm{op}}} X^I.$$

A right D-module on $\mathrm{Ran}(X)$ is represented by a compatible family $\mathcal{F}_I \in D\text{-mod}(X^I)$ with transition morphisms along the diagonals. All functors below are derived.

For a decomposition $I = I_1 \sqcup \cdots \sqcup I_r$, let

$$U(I_1,\ldots,I_r) \subset X^I$$

be the open locus on which points belonging to different blocks are disjoint; points in one block may collide. Denote the inclusion by $j_{I_\bullet}$.

Definition 31.1 (Factorisable D-module)**.** A factorisable right D-module $\mathcal{F}$ on X is a compatible Ran family $\{\mathcal{F}_I\}$ together with equivalences

$$j^!_{I_\bullet}\mathcal{F}_I \simeq j^!_{I_\bullet}(\mathcal{F}_{I_1} \boxtimes \cdots \boxtimes \mathcal{F}_{I_r}) \tag{31.1}$$

for every decomposition of I, compatible with refinement, permutation, and diagonal transition maps.

The equivalence (31.1) expresses independence on separated supports. The extension of $\mathcal{F}_I$ across partial diagonals carries the singular terms of the operator product. The definition uses neither a coordinate nor a propagator.

Let $\mathrm{FactD}(X)$ denote a stable presentable enhancement of this category. Its *coefficient tensor product* is the componentwise !-tensor product

$$(\mathcal{F} \otimes^{\mathrm{pt}} \mathcal{G})_I := \mathcal{F}_I \otimes^! \mathcal{G}_I. \tag{31.2}$$

The unit is the vacuum factorisation module $\mathbf{1}$.

Proposition 31.2. *Assume that the chosen Ran enhancement is closed under componentwise !-tensor product and componentwise colimits. Then* (31.2) *makes* $\mathrm{FactD}(X)$ *a presentable stable symmetric monoidal category, and tensor product preserves colimits separately.*

Proof. The assertion holds on each X^I. The diagonal compatibilities and the restrictions in (31.1) are preserved by !-tensor product. The stated closure hypotheses therefore descend the componentwise structure to the factorisation subcategory. ☐

The category $\mathrm{FactD}(X)$ is the symmetric monoidal category in which the transverse E_1 multiplication is formed. It must be distinguished from the chiral tensor product on $D(\operatorname{Ran} X)$, introduced below.

31.2 The complete singular coefficient

Consider the binary component $\mathcal{F}_{\{1,2\}}$ on X^2. Off the diagonal, factorisation identifies it with $\mathcal{F}_{\{1\}} \boxtimes \mathcal{F}_{\{2\}}$. The extension across the diagonal has local cohomology supported on $\Delta(X)$. In a formal coordinate it is expressed by principal parts

$$\sum_{m\geq 0} \frac{c_m(w)}{(z-w)^{m+1}}.$$

Coordinate changes act triangularly on the individual coefficients and preserve the filtered distribution.

For $D = \Delta(X) \subset X^2$, the pole filtration is

$$F_m \mathcal{O}_{X^2}(*D) = \mathcal{O}_{X^2}(mD),$$

and

$$\operatorname{gr}^F_m \mathcal{O}_{X^2}(*D) \simeq N_{D/X^2}^{\otimes m}. \tag{31.3}$$

Thus the associated graded of a pole is a tensorial normal symbol; the filtered object retains the coordinate-dependent mixing of modes.

Principle 31.3 (Complete coefficient principle). The chiral coefficient of an ordered chiral algebra is the factorisable D-module itself. Residues, normal jets, formal modes, and distributional kernels are functorial realizations of this coefficient.

For an ordered chiral algebra, the transverse multiplication

$$m : A \otimes^{\mathrm{pt}} A \longrightarrow A$$

is a morphism of factorisable D-modules. It therefore respects the whole diagonal filtration and every disjoint-factorisation map. In coordinates it acts on the complete singular expansion, not merely on the simple-pole coefficient.

31.3 Two Lie structures and their comparison

Restriction along the operad map $\mathsf{Lie} \to E_1$ gives the graded commutator

$$[a,b]_\perp = m(a,b) - (-1)^{|a||b|} m(b,a). \tag{31.4}$$

Proposition 31.4 (Internal commutator)**.** *Restriction of operads defines*

$$\mathrm{Alg}_{E_1}\big(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}\big) \longrightarrow \mathrm{Alg}_{\mathsf{Lie}}\big(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}\big).$$

The bracket (31.4) *satisfies the graded Jacobi identity.*

Proof. The displayed bracket is the commutator in an associative algebra object of a symmetric monoidal category. The usual cancellation proof of Jacobi is operadic and therefore applies internally. □

The bracket $[-,-]_\perp$ is a Lie operation on the *transverse multiplication.* A Beilinson–Drinfeld chiral Lie bracket is instead a morphism

$$j_*j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_!\mathfrak{g}$$

in the chiral pseudo-tensor category. The factorisation coefficient of A may carry such a chiral operation, but it is not created by taking (31.4). A comparison between the two Lie structures is an additional morphism of operadic deformation problems. This distinction is essential in boundary field theory: the chiral bracket measures collision on X, while $[-,-]_\perp$ measures the discontinuity between two transverse orders.

31.4 Augmentations and boundary conditions

An augmentation $\varepsilon : A \to \mathbf{1}$ is a morphism of E_1 algebras in $\mathrm{FactD}(X)$. It makes $\mathbf{1}$ a left and right A-module, and the augmentation ideal is

$$\bar{A} = \mathrm{Fib}(A \xrightarrow{\varepsilon} \mathbf{1}).$$

For a right module R and a left module L, the two-sided bar $\mathrm{Bar}(R, A, L)$ computes interval factorisation homology with endpoint labels R and L. The reduced bar is the case $R = L = \mathbf{1}$.

The module formulation is physically primary. A boundary condition is an object, not an augmentation map; its local endomorphisms form an E_1 algebra. The augmentation is the special boundary state represented by the unit.

31.5 Two monoidal products on the Ran space

There are two different tensor products in the subject.

The coefficient tensor $\otimes^{\mathrm{pt}}$ of (31.2) is componentwise. Its support satisfies

$$\mathrm{supp}(\mathcal{F} \otimes^{\mathrm{pt}} \mathcal{G}) \subset \mathrm{supp}(\mathcal{F}) \cap \mathrm{supp}(\mathcal{G}).$$

The pointwise tensor product preserves support cardinality. Its pro-nilpotence, when present, therefore comes from an augmentation-weight filtration rather than support cardinality.

The *chiral tensor product* on $D(\mathrm{Ran}\,X)$ is defined by the disjoint-union correspondence

$$\mathrm{Ran}(X) \times \mathrm{Ran}(X) \supset (\mathrm{Ran} \times \mathrm{Ran})_{\mathrm{disj}} \longrightarrow \mathrm{Ran}(X).$$

For reduced objects supported on subsets of cardinality at least r and s, respectively, chiral convolution is supported in cardinality at least $r+s$:

$$D(\operatorname{Ran} X)^{\geq r} \otimes^{\mathrm{ch}} D(\operatorname{Ran} X)^{\geq s} \subset D(\operatorname{Ran} X)^{\geq r+s}. \tag{31.5}$$

Theorem 31.5 (Ran chiral pro-nilpotence)**.** *For the chiral tensor product $\otimes^{\mathrm{ch}}$, the quotient by objects supported in cardinality $> N$ is nilpotent on reduced objects, and the cardinality completion is pro-nilpotent. This statement concerns $D(\operatorname{Ran} X)$ with chiral convolution; it does not concern the pointwise tensor product of* $\mathrm{FactD}(X)$*.*

Proof. Equation (31.5) implies that an $(N+1)$-fold chiral product of reduced objects is supported in cardinality at least $N+1$ and hence vanishes in the Nth quotient. Taking the inverse limit gives the pro-nilpotent category. This is the support argument used in chiral Koszul duality on the Ran space [82]. ☐

The two products answer different questions. The pointwise tensor permits an E_1 algebra of defects with a fixed chiral coefficient. The chiral tensor assembles disjoint finite subsets and is the ambient product for the Francis–Gaitsgory Lie–commutative-coalgebra equivalence. Any comparison between their bar constructions must name the monoidal functor that relates them.

31.6 Regular holonomic realization

Assume that every $\mathcal{F}_I$ is regular holonomic and that the chosen subcategory is closed under the tensor operations used above. After the standard de Rham shifts, Riemann–Hilbert identifies $\mathcal{F}_I$ with a constructible complex on $X^{I,\mathrm{an}}$. Factorisation equivalences become constructible factorisation equivalences on the disjoint loci.

Proposition 31.6 (Analytic realization)**.** *Under these regular-holonomic closure hypotheses, Riemann–Hilbert gives a symmetric monoidal realization from* $(\mathrm{FactD}(X), \otimes^{\mathrm{pt}})$ *to constructible factorisation sheaves on X^{an}. It carries internal E_1 algebras, modules, bar constructions, and Hochschild cochains to their constructible realizations whenever it commutes with the required geometric realizations and internal Homs.*

Proof. Apply the regular Riemann–Hilbert equivalence on every power X^I, with the conventional de Rham shift. External tensor products, extraordinary pullbacks along diagonals, and restriction to the disjoint loci are preserved on the stated regular-holonomic subcategory. The factorisation isomorphisms therefore descend arity by arity. A symmetric monoidal functor carries algebra and module objects to algebra and module objects. Its commutation with the indicated colimits and internal Homs gives the asserted comparisons for bar constructions and Hochschild cochains. ☐

The analytic realization reveals monodromy already present in a local system on $\mathrm{Conf}_n(X)$. It does not manufacture a braid from the transverse E_1 order.

31.7 Factorisation categories and Morita origin

Let $\mathcal{C}$ be a D-linear factorisation category on X, and let b be a compatible object. Its internal endomorphisms are

$$A_b(U) = \mathrm{RHom}_{\mathcal{C}(U)}(b|_U, b|_U).$$

Composition is ordered, while factorisation in U supplies the chiral coefficient.

Theorem 31.7 (Morita source)**.** *Let $\mathcal{C}$ be a presentable D-linear factorisation category and let b be a dualisable factorisation object. Then $A_b = \mathrm{RHom}_{\mathcal{C}}(b, b)$ is an E_1 algebra in* $\mathrm{FactD}(X)$*. If b is compact and generates each local category, and these local Morita equivalences commute with factorisation descent, then*

$$\mathrm{RHom}_{\mathcal{C}}(b, -) : \mathcal{C} \longrightarrow \mathrm{Mod}_{A_b}(\mathrm{FactD}(X))$$

is a factorisation Morita equivalence.

Proof. Composition gives the E_1 multiplication. Factorisation of b on disjoint opens gives the coefficient factorisation maps. The compact-generator Morita theorem applies locally; the final hypothesis identifies the local equivalences on overlaps and descends them. □

Gauge theory supplies precisely this source: a boundary condition or line defect is b; local endomorphisms form A_b; transverse composition is noncommutative; singular dependence on the holomorphic insertion point is carried by the factorisable D-module.

31.8 Historical position

Dirac's distributional formalism made the singular support of products of quantum fields unavoidable. Wilson organised short-distance singularities by the operator product expansion. Borcherds and Kac encoded their algebraic identities. Beilinson and Drinfeld replaced coordinate series by D-module morphisms on diagonals. Factorisation algebras expressed the same local-to-global law for quantum observables. The present object adds an oriented topological direction and retains its associative order before any exchange or symmetrisation is imposed.

Chapter 32

The duoidal law: locality as a bialgebra identity

The primitive object admits a second, equivalent description which makes its geometry visible at once. The Ran category carries a pointwise tensor product and a convolution product along disjoint union. One product combines coefficients supported at the same finite subset; the other combines observables supported at disjoint finite subsets. Their compatibility is duoidal. An ordered chiral algebra is a monoid for the first product and a factorisation coalgebra for the second. The assertion that multiplication is a morphism of factorisation objects is therefore a bialgebra identity, but the multiplication and comultiplication belong to different monoidal structures.

32.1 The two products and their interchange

Let

$$\mathcal{D}_X = D(\operatorname{Ran} X).$$

Write $\otimes^{\mathrm{pt}}$ for the pointwise !-tensor product. Write $\otimes^{\mathrm{ch}}$ for chiral convolution along the correspondence

$$\operatorname{Ran} X \times \operatorname{Ran} X \xleftarrow{j} (\operatorname{Ran} X \times \operatorname{Ran} X)_{\mathrm{disj}} \xrightarrow{u} \operatorname{Ran} X, \qquad F \otimes^{\mathrm{ch}} G := u_! j^! (F \boxtimes G).$$

The diagonal of the Ran space is compatible with disjoint union. The projection formula and Beck–Chevalley base change therefore give a natural interchange morphism

$$(F_1 \otimes^{\mathrm{ch}} F_2) \otimes^{\mathrm{pt}} (G_1 \otimes^{\mathrm{ch}} G_2) \longrightarrow (F_1 \otimes^{\mathrm{pt}} G_1) \otimes^{\mathrm{ch}} (F_2 \otimes^{\mathrm{pt}} G_2), \tag{32.1}$$

with the middle factors permuted by the symmetry of $\otimes^{\mathrm{pt}}$. In a six-functor realization for which base change and the projection formula are equivalences for the disjoint-union and diagonal squares defining the two products, (32.1) is an equivalence on the disjoint locus. Without those hypotheses it is the corresponding lax interchange map.

Theorem 32.1 (Duoidal Ran theorem)**.** *Fix a sheaf theory on* $\operatorname{Ran} X$ *for which pointwise !-tensor, chiral convolution, the projection formula, and disjoint-union base change are defined. Then*

$$(\mathcal{D}_X, \otimes^{\mathrm{pt}}, \otimes^{\mathrm{ch}})$$

is a lax duoidal stable category. On the full subcategory on which the base-change morphisms *in* (32.1) *are equivalences, it is a strong duoidal category.*

Proof. Associativity and symmetry of $\otimes^{\mathrm{pt}}$ are inherited componentwise from $D(X^I)$. Associativity of $\otimes^{\mathrm{ch}}$ is the associativity of union of pairwise disjoint finite subsets. Apply the two chiral coproduct correspondences, tensor their outputs pointwise, and then use the diagonal base-change square to regroup the first and third, and the second and fourth, factors. The resulting natural transformation is (32.1). The coherence diagrams reduce to the associativity, symmetry, and base-change diagrams of the six-functor formalism. This is precisely the definition of a lax duoidal category; invertibility of the interchange maps gives the strong form. □

The theorem is an instance of the general theory of duoidal ∞-categories [122]. Its particular content here is geometric: the two products are the diagonal product and disjoint-union convolution on the Ran space.

32.2 Factorisation coalgebras

A chiral commutative coalgebra is a commutative coalgebra object for $\otimes^{\mathrm{ch}}$. Its coproduct restricts, for disjoint finite subsets S and T, to a map

$$\Delta_{S,T} : F_{S\cup T} \longrightarrow F_S \boxtimes F_T.$$

The Ran object F is a factorisation coalgebra when this map is an equivalence on every disjointness locus. Francis and Gaitsgory identify factorisation algebras with the corresponding full subcategory of chiral commutative coalgebras [82].

Let

$$\iota : \mathrm{FactD}(X) \hookrightarrow \mathrm{CoCAlg}^{\mathrm{ch}}(\mathcal{D}_X)$$

denote this realization. The pointwise tensor of two factorisation coalgebras is again a factorisation coalgebra: its coproduct is obtained by applying (32.1) to their two coproducts.

Theorem 32.2 (Ordered chirality as a duoidal bimonoid)**.** *Assume the strong duoidal hypotheses of Theorem 32.1 on the chosen factorisation subcategory. For an object $A \in \mathrm{FactD}(X)$, the following data are equivalent:*

- (i) *an E_1 multiplication on A internal to $\mathrm{FactD}(X)$;*
- (ii) *an E_1 monoid structure for $\otimes^{\mathrm{pt}}$ on ιA such that its multiplication and unit are morphisms of chiral commutative coalgebras;*
- (iii) *a bimonoid in the duoidal category $(\mathcal{D}_X, \otimes^{\mathrm{pt}}, \otimes^{\mathrm{ch}})$ whose chiral coproduct is factorising on disjoint loci.*

The compatibility is the duoidal bialgebra identity

$$\Delta_{\mathrm{ch}} \circ m = (m \otimes^{\mathrm{ch}} m) \circ \chi \circ (\Delta_{\mathrm{ch}} \otimes^{\mathrm{pt}} \Delta_{\mathrm{ch}}), \tag{32.2}$$

where χ is the interchange morphism.

Proof. An E_1 algebra in the full subcategory of factorisation coalgebras is, by definition, an E_1 multiplication whose structure maps remain in that subcategory. Saying that multiplication is a morphism of coalgebras is exactly (32.2); saying that the unit is a coalgebra map is the corresponding unit identity. These are the axioms of a bimonoid in a duoidal category. The factorisation condition is the requirement that the chiral coproduct become an equivalence on disjoint loci. □

Equation (32.2) is the exact algebraic form of locality for the noncommutative theory. Separate two collections of insertions in the chiral curve and fuse them in the ordered direction. One obtains the same operation by fusing inside each separated region and then applying factorisation. The equation is not cocommutativity of m, and it is not an R-matrix relation.

32.3 The bar retains factorisation

The simplicial ordered bar is levelwise a chiral commutative coalgebra:

$$B_r(A) = \mathbf{1} \otimes^{\mathrm{pt}} \bar{A}^{\otimes^{\mathrm{pt}} r} \otimes^{\mathrm{pt}} \mathbf{1}.$$

Every face is built from m or the augmentation. By Theorem 32.2, these are coalgebra maps. Every degeneracy is built from the unit and is likewise a coalgebra map.

Theorem 32.3 (Double-coalgebra bar theorem)**.** *Suppose geometric realization in $\mathcal{D}_X$ preserves chiral commutative coalgebras and commutes with the pointwise tensor products appearing in the simplicial bar. Then*

$$C = \mathrm{Bar}_X^{\mathrm{ord}}(A)$$

has two compatible coalgebra structures:

(i) *the coassociative deconcatenation coproduct of the E_1 bar;*

(ii) *the commutative chiral factorisation coproduct inherited levelwise from A.*

Deconcatenation is a morphism of chiral factorisation coalgebras. Equivalently, C is a coassociative coalgebra object in chiral commutative coalgebras.

Proof. The simplicial bar is a simplicial object in chiral commutative coalgebras because every simplicial operator is a coalgebra morphism. Geometric realization gives the second coproduct. Deconcatenation is defined by cutting an ordered word. The chiral coproduct acts independently on each letter, so cutting and factorising commute by the interchange law. The resulting square is (32.2) applied to the iterated pointwise tensor powers of A. □

This theorem explains why the ordered bar is simultaneously an interval object and a Ran-space coefficient. The coassociative coproduct remembers where the interval is cut. The chiral coproduct remembers where the curve is separated. Locality of the original ordered chiral algebra induces the compatibility of the two bar coproducts.

32.4 A Fubini theorem

Let $p : X^{\mathrm{an}} \times \mathbb{R} \to X^{\mathrm{an}}$ and $q : X^{\mathrm{an}} \times \mathbb{R} \to \mathbb{R}$ be the projections. Factorisation pushforward along p integrates the locally constant real direction; pushforward along q integrates the chiral direction when the corresponding chiral homology is defined. Product excision gives a Fubini comparison.

Theorem 32.4 (Ordered chiral Fubini)**.** *Let A be an ordered chiral algebra whose analytic realization is rectangularly generated. Assume that factorisation homology over the interval* *and the curve exists, and that pushforward along both projections preserves the geometric realizations defining the bar. Then there are natural equivalences*

$$\int_X \mathrm{Bar}_X^{\mathrm{ord}}(A) \simeq \mathrm{Bar}^{\mathrm{ord}}\left(\int_X A\right), \tag{32.3}$$

$$\int_X \mathrm{HH}_*(A) \simeq \mathrm{HH}_*\left(\int_X A\right). \tag{32.4}$$

More generally, factorisation homology over a product rectangle may be computed in either order.

Proof. The bar is a geometric realization of the simplicial object $[r] \mapsto \bar{A}^{\otimes^{\mathrm{pt}} r}$. Chiral factorisation homology is a colimit over chiral disks. Under the stated preservation hypotheses, the two colimits commute, giving (32.3). Hochschild chains are factorisation homology over the transverse circle, or equivalently the cyclic realization of the two-sided bar. Commuting that realization with chiral factorisation homology gives (32.4). On the analytic product, both assertions are the Fubini theorem for factorisation homology and follow from excision on product-disk covers. □

The first equivalence says that one may integrate the chiral curve before or after resolving ordered fusion. The second says the same after the interval has been closed into a circle. In conformal field theory this is the equality between first forming the open-string trace and then taking chiral coinvariants, or first taking chiral coinvariants and then closing the ordered state space. Analytic convergence of either trace remains a separate theorem.

32.5 Primitives and quantum symmetry

The chiral factorisation coproduct in Theorem 32.3 is a separation map. It is not the coproduct of a quantum group. A quantum-group coproduct acts on tensor products of line representations and requires a tensor-compatible exchange theory. The two coproducts may be compared in a gauge-theoretic example, but neither determines the other.

The primitive part of the chiral coalgebra records connected support. The primitive part of the coassociative bar coalgebra records indecomposable interval states. Their intersection consists of states that are connected in the curve and indecomposable along the interval. This bidegree is the natural home of a binary collision kernel or a first line-defect interaction. An R-matrix arises only after an exchange functor transports these primitives around a two-dimensional collision.

32.6 Physical meaning

The two products of the duoidal category are two experimental operations. Pointwise tensor places two coefficient systems on the same chiral support and then fuses them in transverse order. Chiral convolution places them on separated supports and then forgets the separation by union. Equation (32.2) says that a local experiment performed in two separated laboratories is independent of whether fusion is described before or after the laboratories are separated.

The displayed bimonoid law is the geometric content of locality in the ordered theory. It is stronger than associativity, because it constrains the dependence on support. It is weaker than braiding, because it contains no exchange path. It is the point at which Beilinson's factorisation language and the noncommutative language of defects become one sentence.

Chapter 33

The block-ordered chiral configuration category

33.1 What is ordered

Let Ch_X denote a fixed operadic enhancement of the Beilinson–Drinfeld chiral pseudo-tensor structure whose algebras in the chosen derived system of right D-modules are the factorisable D-modules of Theorem 31.1. The noncommutative chiral operad is

$$\mathsf{OCh}_X := E_1 \otimes \mathsf{Ch}_X, \tag{33.1}$$

where $\otimes$ is the Boardman–Vogt tensor product of ∞-operads. An algebra over (33.1) has a chiral factorisation structure and an associative transverse multiplication, with the interchange relation built into the operad.

The word *ordered* refers to the E_1 inputs. It does not impose a linear order on the points of one chiral configuration. A typical operation is represented by an ordered sequence

$$(I_1, \dots, I_r), \tag{33.2}$$

where the finite set I_a labels the chiral insertions carried by the ath transverse interval. Points belonging to different blocks may have the same coordinate on X because their transverse coordinates are distinct. Collisions inside the chiral direction are carried by the D-module extension across diagonals; composition in the transverse direction is carried by E_1.

33.2 The operadic Grothendieck construction

The block-ordered chiral configuration category $\mathbf{BRan}_X$ is the operadic Grothendieck construction of the associative operad acting on the chiral configuration category. Its objects are pairs

$$([r]; I_1, \dots, I_r),$$

with $[r]$ a finite ordinal and each I_a a finite chiral label set. Its generating morphisms are of three kinds:

(i) chiral refinements, collisions, and permutations inside one I_a;
(ii) adjacent composition of two E_1 slots;
(iii) insertion or removal of the unit slot.

The relations are the chiral relations, the simplicial identities of the associative nerve, and the interchange squares between the two kinds of operation.

The object space of a fixed block profile (33.2) is

$$X^{I_1} \times \cdots \times X^{I_r}. \tag{33.3}$$

The block-union product carries the operation profiles; the pointwise coefficient tensor remains the monoidal product of $\mathrm{FactD}(X)$. The coefficient morphism attached to an adjacent block composition is obtained from the E_1 multiplication after the chiral inputs have been placed in the common output profile. In a six-functor model, the required pull–tensor–push correspondence is part of the chosen operadic enhancement. There is no canonical theorem obtained by putting arbitrary D-modules on $\mathrm{Ran}(X)^r$ and declaring the union map to be multiplication.

Definition 33.1 (Block-ordered factorisation D-module)**.** A block-ordered factorisation D-module is an algebra over $\mathsf{OCh}_X = E_1 \otimes \mathsf{Ch}_X$ in the chosen derived D-module system. Equivalently, it is a coCartesian operadic section over $\mathbf{BRan}_X$ whose restriction to each block is chiral factorisable, whose restriction to the ordinal direction is Segal, and whose mixed squares satisfy interchange.

Write $\mathrm{OrdFactD}(X)$ for this category. The definition includes the coefficient correspondences needed to compare the different spaces (33.3); it is therefore stronger than the datum of a simplicial family of sheaves on their underlying products.

Theorem 33.2 (Block-ordered recognition)**.** *There are natural equivalences*

$$\mathrm{Alg}_{E_1}\big(\mathrm{FactD}(X)\big) \simeq \mathrm{Alg}_{E_1 \otimes \mathsf{Ch}_X}(\mathcal{V}_X) \simeq \mathrm{OrdFactD}(X), \tag{33.4}$$

where $\mathcal{V}_X$ is the chosen finite-set D-module system presenting $\mathrm{FactD}(X)$.

Proof. By definition, $\mathrm{FactD}(X) \simeq \mathrm{Alg}_{\mathsf{Ch}_X}(\mathcal{V}_X)$. The exponential law for ∞-operads gives

$$\mathrm{Alg}_{E_1}\big(\mathrm{Alg}_{\mathsf{Ch}_X}(\mathcal{V}_X)\big) \simeq \mathrm{Alg}_{E_1 \otimes \mathsf{Ch}_X}(\mathcal{V}_X).$$

Unstraightening an algebra over the tensor-product operad gives a coCartesian section over its operadic Grothendieck construction. The restrictions to the two operadic factors give the chiral and Segal conditions; the tensor-product relation gives interchange. Straightening reverses the construction. □

The theorem is the invariant meaning of factorisation over an ordered chiral configuration space. The block picture is a geometric presentation of the tensor-product operad, not a replacement for it.

33.3 The strict and homotopy-coherent forms

In a strict dg model, an ordered chiral algebra is a factorisable D-module A with maps

$$m : A \otimes^{\mathrm{pt}} A \longrightarrow A, \qquad \eta : \mathbf{1} \longrightarrow A,$$

satisfying strict associativity and unitality. In the invariant ∞-category it is an E_1 algebra. A cofibrant A_∞ model replaces the single multiplication by operations indexed by planar trees. The chiral coefficient is unchanged.

Proposition 33.3 (Interchange)**.** *Let γ be a chiral operation and let m be the transverse multiplication. Whenever the two composites are defined by one product-disk embedding, their images satisfy*

$$\gamma \circ (m \otimes m) = m \circ (\gamma \otimes \gamma), \tag{33.5}$$

with the symmetry isomorphism that places equal-coloured inputs together.

Proof. Equation (33.5) is the defining interchange relation of $E_1 \otimes \mathsf{Ch}_X$. In the analytic product-disk model, both sides are the factorisation map associated with the same embedding, evaluated by composing first in opposite coordinate directions. □

The proposition is stronger than the coexistence of two operations and weaker than braiding. It says that holomorphic collision and transverse composition are compatible. It does not provide a path that exchanges two transverse factors.

33.4 Analytic chamber model

For $r \geq 1$, let

$$C_r^< = \{(t_1, \dots, t_r) \in \mathbb{R}^r \mid t_1 < \cdots < t_r\}.$$

The ordered chamber $\mathrm{Conf}_r^<(\mathbb{R})$ is contractible. A profile (33.2) has analytic carrier

$$X^{I_1,\mathrm{an}} \times \cdots \times X^{I_r,\mathrm{an}} \times C_r^<. \tag{33.6}$$

Local constancy in $C_r^<$ erases the distances while retaining the order. The X-coordinates remain holomorphic and may approach any partial diagonal allowed by the coefficient D-module.

Compactifying the configurations of subintervals in one interval modulo translation and positive dilation gives associahedra. Their strata are indexed by planar rooted trees. This compactification resolves the homotopy coherence of the E_1 multiplication.

Proposition 33.4 (Associahedral resolution)**.** *For a locally constant factorisation algebra on an oriented line, cellular chains on compactified ordered interval configurations act by the A_∞ bar operations. Contracting every associahedral cell to its corolla gives the ordinary simplicial bar construction through a quasi-isomorphism of operadic resolutions.*

Proof. Factorisation assigns an operation to every ordered embedding of disjoint intervals. The corresponding embedding spaces are contractible, and their compactified boundary strata are planar nested collisions. Stokes' formula is the sum of edge contractions, hence the Stasheff identity. The cellular chains form a cofibrant resolution of the associative operad, while the simplicial bar is the standard resolution of an algebra over that operad. The comparison is the augmentation of the cofibrant resolution. □

The simplex and the associahedron therefore play different roles. The simplex is the indexing object of the simplicial bar. The associahedron is a geometric cofibrant resolution of associative coherence.

33.5 Mixed collision geometry

A chain model for OCh_X may be built from two-coloured forests. Chiral edges describe nested partial diagonals in X; transverse edges describe nested intervals in the ordered line.

A face contracts one edge of one colour. On a total complex, the two boundary operators anticommute by the product orientation.

Lemma 33.5 (Mixed square-zero identity). *Let d_X be the differential supplied by a chain model of the chiral coefficient and let $b_\perp$ be the E_1 bar differential. If both are morphisms for the interchange structure, then*

$$d_X b_\perp + b_\perp d_X = 0$$

after the standard totalization sign. Hence

$$(d_X + b_\perp)^2 = d_X^2 + b_\perp^2.$$

Proof. The two operators are boundaries in independent factors of the mixed operadic resolution. Exchanging their conormal orientation lines contributes the totalization sign. Compatibility with interchange identifies the two coefficient composites. □

Associativity proves $b_\perp^2 = 0$. The chiral identities prove $d_X^2 = 0$ in the chosen chiral model. Neither identity repairs the other.

33.6 Modules, interfaces, and marked blocks

A right module is an algebra over the two-coloured operad with ordered algebra inputs followed by one module input. A left module uses the opposite order, and a bimodule has both boundary colours. For a right module R and a left module L, the simplicial two-sided bar is

$$B_r(R, A, L) = R \otimes A^{\otimes r} \otimes L. \tag{33.7}$$

Its realization is interval factorisation homology with endpoint labels R and L.

Proposition 33.6 (Marked-block recognition). *The category of right, left, and bi-modules over A is equivalent to the corresponding category of marked coCartesian sections over the coloured block-ordered configuration category. Relative tensor product of bimodules is gluing of marked intervals.*

Proof. Apply the straightening argument of Theorem 33.2 to the standard coloured associative operads for modules and bimodules. Excision for the glued interval computes the relative tensor product. □

Marked ordered blocks are the native geometry of boundary-changing fields and defect junctions.

33.7 Opposite order

Reflection of the transverse line reverses the ordinal factor while leaving X unchanged. It induces

$$\mathrm{rev} : \mathsf{OCh}_X \longrightarrow \mathsf{OCh}_X$$

and sends multiplication to its opposite.

Proposition 33.7 (Geometric opposite). *Under Theorem 33.2, transverse reflection sends A to A^{op}. Its chiral coefficient is unchanged.*

Proof. An ordered embedding of two intervals is carried by reflection to the embedding with the two components interchanged. Straightening therefore sends the binary operation $m(a,b)$ to $m(b,a)$ with the Koszul symmetry of the coefficient category. Reflection acts only in the transverse real coordinate, so the configuration and diagonal data on X remain fixed. □

Reflection of the oriented interval is the canonical inversion available at the E_1 level. Inverse braiding requires an E_2 structure and is not implied by reflection alone.

33.8 Three distinct forgetful operations

There are three operations that must not be called symmetrisation indiscriminately.

First, forgetting the E_1 multiplication leaves the underlying factorisable D-module. This functor loses all transverse products.

Second, derived commutativisation is the left adjoint

$$\mathrm{Alg}_{E_1}(\mathrm{FactD}(X)) \longrightarrow \mathrm{Alg}_{E_\infty}(\mathrm{FactD}(X))$$

when it exists in the selected presentable category. It imposes coherent commutativity on the transverse product. It is not termwise Reynolds averaging of tensor words.

Third, descent from labelled to unlabelled configurations concerns monodromy of a coefficient local system on $\mathrm{Conf}_n(X)$. It asks for extension from the pure braid group to the full braid group; symmetric exchange further asks that the action factor through Σ_n. This is independent of the existence of a transverse E_1 multiplication.

The separation of these three operations is the first safeguard of noncommutative chiral Koszul duality.

Chapter 34

The mixed operad of chiral collision and ordered fusion

34.1 The invariant operad

Let Ch_X be an ∞-operad whose algebras in the chosen coefficient category are factorisable D-modules on X. The ordered chiral operad is defined by the Boardman–Vogt tensor product

$$\mathsf{OrdCh}_X := E_1 \otimes \mathsf{Ch}_X. \tag{34.1}$$

The tensor product is characterized by interchange: an operation in the E_1 direction commutes with an operation in the chiral direction after the canonical permutation and base-change maps.

Theorem 34.1 (Operadic additivity)**.** *Let $\mathcal{V}$ be a presentable symmetric monoidal ∞-category for which the indicated algebra categories exist. Then*

$$\mathrm{Alg}_{\mathsf{OrdCh}_X}(\mathcal{V}) \simeq \mathrm{Alg}_{E_1}\big(\mathrm{Alg}_{\mathsf{Ch}_X}(\mathcal{V})\big). \tag{34.2}$$

For the D-module realization of Ch_X, the right-hand side is $\mathrm{Alg}_{E_1}(\mathrm{FactD}(X))$.

Proof. Equation (34.2) is the universal property of the tensor product of ∞-operads. It identifies an algebra carrying two compatible operadic actions with an algebra over their tensor product. □

For locally constant operations, Dunn additivity identifies $E_1 \otimes E_n$ with E_{n+1} [40, 72]. The chiral operation on X is not locally constant across collision diagonals, so (34.1) retains a holomorphic singular direction and an ordered topological direction.

34.2 Three geometric models

Three spaces occur for three different reasons.

The simplex indexes the monadic simplicial bar. The associahedron resolves the coherent multiplication of an E_1 algebra. The Fulton–MacPherson compactification resolves relative chiral configurations near collision diagonals. None can replace the other.

Fix an ordered partition $I = I_1 \sqcup \cdots \sqcup I_r$. On that stratum, a local product chart has the form

$$K_r \times \prod_{a=1}^{r} \mathrm{FM}_{I_a}(X), \tag{34.3}$$

together with the base configuration and the normal-bundle factors determined by the contracted edge. When a transverse face merges two blocks, the partition changes and the corresponding Fulton–MacPherson factors are replaced by the factor for their union. Thus the global resolution is assembled from the charts (34.3) over the category of ordered partitions; it is not a single product $\mathrm{FM}_I(X) \times K_r$.

34.3 Two-coloured levelled forests

Choose a cofibrant dg-operadic presentation

$$\mathsf{Q}_X \longrightarrow \mathsf{Ch}_X$$

whose chiral cells are represented locally by Fulton–MacPherson incidence cells with their screen-chain coefficients. Apply a functorial Boardman–Vogt construction to the tensor-product operad:

$$W\big(E_1 \otimes \mathsf{Q}_X\big) \longrightarrow E_1 \otimes \mathsf{Ch}_X. \tag{34.4}$$

A cellular basis element of this resolution is represented by a rooted levelled forest with two kinds of internal edge:

- a transverse edge, recording composition of consecutive E_1 blocks;
- a chiral edge, recording a nested collision inside a chiral coefficient.

A vertex carries the screen-chain coefficient prescribed by Q_X. The determinant line is

$$\mathrm{or}(F) = \det(E_\perp(F)) \otimes \det(E_{\mathrm{ch}}(F)). \tag{34.5}$$

Definition 34.2 (Mixed forest complex)**.** For a finite label set I, let

$$W\mathsf{OrdCh}_X(I) = \bigoplus_F C_*(\mathrm{Screen}(F)) \otimes \mathrm{or}(F), \tag{34.6}$$

where F ranges over the two-coloured levelled forests in the chosen cellular presentation. The differential is the sum of the internal screen differential and the signed contractions of one transverse or one chiral edge.

This definition is a chain model of the operadic resolution. It does not assert the existence of a single smooth compactification whose strata are literally all the summands of (34.6). Such a compactification, when available, gives a geometric realization of the same incidence data.

34.4 The boundary identity

Write

$$d = d_{\mathrm{scr}} + d_\perp + d_{\mathrm{ch}}.$$

Theorem 34.3 (Mixed boundary theorem). *With the orientation convention* (34.5),

$$d_\perp^2 = 0, \qquad d_{\mathrm{ch}}^2 = 0, \qquad d_\perp d_{\mathrm{ch}} + d_{\mathrm{ch}} d_\perp = 0,$$

and the screen differential supercommutes with both edge-contraction differentials. Hence $d^2 = 0$.

Proof. A codimension-two forest is obtained either by contracting two transverse edges, two chiral edges, or one edge of each colour. In the first two cases, the two orders of contraction induce opposite boundary orientations in the corresponding one-colour Boardman–Vogt complex. In the mixed case, the final forest is the same in either order, while exchanging the two one-dimensional determinant factors in (34.5) changes the sign. The screen differential is compatible with every operadic face map and contributes the standard total-complex signs. □

When endomorphism operations are attached to vertices, the transverse part of this identity becomes the Stasheff relations, the chiral part becomes the identities of the chosen chiral coefficient model, and the mixed part says that transverse multiplication is a morphism of chiral objects.

34.5 Cofibrant comparison

Theorem 34.4 (Mixed Boardman–Vogt resolution). *Assume that chain operads are taken in an admissible monoidal model category, that* Q_X *is cofibrant, and that the tensor-product operad and functorial* W*-construction preserve weak equivalences between cofibrant algebras over the displayed operads. Then the augmentation*

$$W\big(E_1 \otimes \mathsf{Q}_X\big) \longrightarrow E_1 \otimes \mathsf{Ch}_X \tag{34.7}$$

is a cofibrant resolution. Its algebras are homotopy-coherent ordered chiral algebras. When rectification holds in the coefficient model category, their localized category is equivalent to $\mathrm{Alg}_{E_1}(\mathrm{FactD}(X))$.

Proof. The W-construction replaces each internal edge by a length parameter and has contractible length fibres. Under the stated model-categorical hypotheses it is cofibrant and its edge-contraction augmentation is an aritywise weak equivalence. The tensor-product universal property identifies its strict algebras with compatible E_1 and chiral structures. Rectification is the standard change-of-operad equivalence along a weak equivalence between admissible cofibrant operads. □

The hypotheses in this theorem are part of the mathematical statement. In particular, one does not infer cofibrancy of a tensor product merely from cofibrancy of two unrelated resolutions.

34.6 Incidence and screen geometry

A collision forest has an incidence type and a screen space. The incidence category records which clusters collide and in what order. Its fibre over a fixed root partition has a terminal corolla and hence a contractible nerve. The screen space records relative positions inside each cluster and generally has nontrivial cohomology.

For three points on an affine complex line modulo translation and dilation, the open screen is $\mathbb{P}^1 \setminus \{0, 1, \infty\}$. Its first cohomology has rank two. Therefore a proper pushforward of the geometric stratum retains more information than the pushforward of its incidence nerve. Arnold classes, configuration-space periods, and propagator forms belong to the screen coefficient in (34.6); they are not produced by the incidence differential.

34.7 Homotopy transfer and Feynman trees

Let a deformation retract of ordered chiral complexes be given. Homological perturbation transfers the structure by summing over the two-coloured forests: vertices carry original operations, internal edges carry the contracting homotopy, and screen chains carry the chiral coefficient. The resulting operations satisfy the complete forest boundary identity because Theorem 34.3 is the differential of the cofibrant resolution.

This is the precise coincidence between Feynman trees and homotopy transfer. The equality is combinatorial once the propagator and vertex maps have been supplied; it is not an identification of every configuration-space form with an OPE coefficient.

34.8 Renormalised Feynman weights

In perturbative field theory, an edge of a graph carries a propagator and a vertex carries an interaction. Renormalisation extends the resulting distribution to a compactified configuration space. Boundary restriction factors through contracted subgraphs.

Proposition 34.5 (Feynman weight morphism)**.** *Let a holomorphic–topological BV theory on $X \times \mathbb{R}$ have renormalised graph weights extending to a geometric realization of the mixed forest model. Assume that restriction to every boundary stratum factorizes by graph contraction and that the renormalised quantum master equation holds. Then the graph weights define a morphism from the mixed cofibrant operad to the endomorphism operad of the boundary observables.*

Proof. Factorisation on a boundary stratum identifies the weight with the composite of the weights of the contracted graph and the quotient graph. Stokes' formula, including the BV boundary and counterterm contributions, is exactly compatibility with the differential of the mixed forest complex. These two identities are the operadic composition and chain-map axioms. □

34.9 Three reversals

Reflection of the transverse line sends an E_1 algebra to its opposite. Reversal of an exchange path, when an E_2 enhancement exists, inverts the braid. Verdier duality reverses !- and *-extensions of the chiral coefficient. These are three different operations. The two-colour grading keeps their domains visible and prevents one from being inferred from another.

Chapter 35

The ordered chiral bar construction

35.1 The simplicial object

Let A be an augmented ordered chiral algebra and let $\bar{A} = \mathrm{Fib}(A \to \mathbf{1})$. The reduced simplicial bar object in $\mathrm{FactD}(X)$ is

$$B_r^{\mathrm{ord}}(A) = \bar{A}^{\otimes r}, \qquad r \geq 0. \tag{35.1}$$

The inner face d_i multiplies the ith and $(i+1)$st factors. The outer faces apply the augmentation. Degeneracies insert the unit in the unreduced model. The ordered chiral bar is the geometric realization

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) := \big|B_\bullet^{\mathrm{ord}}(A)\big|. \tag{35.2}$$

Every term of (35.1) is a factorisable D-module. The differential is a morphism in $\mathrm{FactD}(X)$.

In cochain notation, after desuspension,

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) \simeq \left(\bigoplus_{r\geq 0}(s^{-1}\bar{A})^{\otimes r},\ d_A + b_m\right), \tag{35.3}$$

where b_m is the coderivation induced by the E_1 multiplication. Completion replaces the direct sum by a product.

35.2 The chiral coefficient throughout the interval bar

Formula (35.3) contains no residue map. If a and b have a fourth-order OPE pole, that pole belongs to the factorisation D-module $A \otimes A$ and is carried through the multiplication morphism m. The bar face contracts the ordered factors; it does not extract a particular Laurent coefficient.

A residue complex can be paired with the bar as a separate geometric realization. Such a pairing has the form

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) \longrightarrow \Gamma\big(\mathrm{FM}_I(X), \mathcal{K}_I \otimes A_I\big),$$

where $\mathcal{K}_I$ is a chosen complex of forms, currents, or local cohomology classes. The map must specify how each principal part is paired with a normal jet. Its square-zero identity uses both the bar differential and the differential of $\mathcal{K}_I$. The intrinsic bar exists before this choice.

No-go theorem 35.1 (Residue substitution). There is no natural identification of the internal bar differential with Poincaré residue along the chiral collision divisor. Poincaré residue lowers logarithmic form degree and detects a first normal coefficient. The bar differential lowers transverse word length and uses the entire multiplication morphism.

The source and target of the two maps prove the preceding boundary theorem.

35.3 Coalgebra structure

Deconcatenation gives

$$\Delta[a_1|\cdots|a_r] = \sum_{p=0}^{r}[a_1|\cdots|a_p] \otimes [a_{p+1}|\cdots|a_r]. \tag{35.4}$$

The deconcatenation coproduct is coassociative and conilpotent. The bar differential is a coderivation.

Proposition 35.2 (Internal bar coalgebra)**.** *The ordered chiral bar is a conilpotent coassociative coalgebra object in* $\mathrm{FactD}(X)$*. The coaugmentation is the empty word, and the coradical filtration is the transverse word-length filtration.*

Proof. Deconcatenation is defined componentwise in every symmetric monoidal category. Its coassociativity is the equality of the two ways of cutting a word twice. The multiplication coderivation is local with respect to cuts, so it satisfies the coderivation identity. Iterating the reduced coproduct of a word of length r more than r times gives zero. ☐

The coalgebra structure records interval splitting. It is unrelated to a chiral coproduct of fields. A bialgebra structure requires an additional multiplication compatible with (35.4).

35.4 Interval factorisation homology

Let $([0,1],\partial)$ denote an oriented interval whose endpoints are labelled by the augmentation module $\mathbf{1}$. Excision for factorisation homology gives

$$\int_{([0,1],\partial)} A \simeq \mathbf{1} \otimes^L_A \mathbf{1}. \tag{35.5}$$

The standard two-sided simplicial resolution of the derived tensor product is precisely (35.1).

Theorem 35.3 (Bar as interval factorisation homology)**.** *There is a canonical equivalence in* $\mathrm{FactD}(X)$

$$\mathrm{Bar}^{\mathrm{ord}}_X(A) \simeq \int_{([0,1],\partial)} A. \tag{35.6}$$

For a right module R and a left module L,

$$\mathrm{Bar}(R, A, L) \simeq \int_{([0,1],\{0,1\})} (A; R, L).$$

Proof. Choose a Weiss cover of the interval by an ordered chain of small intervals together with collars at the endpoints. Factorisation excision identifies the colimit of the Čech diagram with the derived relative tensor product. Refining the cover gives the simplicial two-sided bar.

The argument takes place in the coefficient category $\mathrm{FactD}(X)$ and is therefore compatible with chiral factorisation. $\square$

The theorem is the physical meaning of the bar construction. It is the amplitude of an interval whose interior carries the boundary operator algebra and whose endpoints carry the vacuum boundary state.

35.5 The associahedral model

The simplicial bar is the derived tensor product associated with a strict E_1 multiplication. A homotopy-coherent multiplication is an algebra over a cofibrant resolution $W\mathsf{Ass} \to \mathsf{Ass}$. Its bar coderivation contains the transferred operations m_k,

$$b = \sum_{k \geq 1} b_{m_k}, \tag{35.7}$$

and $b^2 = 0$ is the family of Stasheff identities.

Let K_r denote the associahedron in the convention in which its cells are planar trees with r leaves. The cellular chains $C_*(K_r)$ form one presentation of $W\mathsf{Ass}(r)$. They resolve the operation, not the simplicial degree.

Theorem 35.4 (Associahedral comparison)**.** *Assume an admissible monoidal dg model of* $\mathrm{FactD}(X)$*, let A be cofibrant as a* $W\mathsf{Ass}$*-algebra, and let A^{str} be a strictification along* $W\mathsf{Ass} \to \mathsf{Ass}$*. The operadic bar construction of A has a cellular filtration with terms*

$$\bigoplus_{r \geq 0} C_*(K_r) \otimes (s^{-1}\bar{A})^{\otimes r}, \tag{35.8}$$

whose differential contains the cellular boundary and the operations m_k. Change of operad and the simplicial two-sided bar give a natural zigzag of weak equivalences

$$\mathrm{Bar}_{W\mathsf{Ass}}(A) \simeq \mathrm{Bar}_{\mathsf{Ass}}(A^{\mathrm{str}}) \simeq \mathrm{Bar}_X^{\mathrm{ord}}(A^{\mathrm{str}}).$$

Proof. Cellular chains of associahedra present the Boardman–Vogt resolution of the associative operad. The change-of-operad adjunction is a Quillen equivalence under the stated admissibility and cofibrancy hypotheses. Bar constructions are homotopy invariant on cofibrant algebras and compute the same derived relative tensor product $\mathbf{1} \otimes_A^L \mathbf{1}$. This yields the zigzag. $\square$

The theorem identifies the simplicial and associahedral models through their common interval amplitude.

35.6 Low arity

For a strict dg algebra, the normalized reduced bar begins

$$\bar{A} \xleftarrow{b} (s^{-1}\bar{A})^{\otimes 2} \xleftarrow{b} (s^{-1}\bar{A})^{\otimes 3} \xleftarrow{b} \cdots.$$

The binary differential is

$$b[a|b] = (-1)^{|a|}[ab].$$

The length-three component of b^2 is the signed associator. In higher length, every term is indexed by two adjacent multiplications; the two orders of contracting the corresponding planar edges occur with opposite bar signs.

For an A_∞ algebra, m_3 fills the homotopy between the two binary composites. At the first two-dimensional associahedral cell, m_4 and composites of m_2, m_3 fill the pentagon. The geometry is exactly the cellular boundary of associahedra.

The chiral coefficient adds an independent filtration. A word $[a|b]$ can carry an arbitrary OPE singularity in the X variables. The bar differential applies the transverse product ab as a morphism of those singular coefficient systems.

35.7 Functoriality

An augmented morphism $f : A \to B$ of ordered chiral algebras induces

$$\mathrm{Bar}(f) : \mathrm{Bar}_X^{\mathrm{ord}}(A) \longrightarrow \mathrm{Bar}_X^{\mathrm{ord}}(B)$$

by applying f to every bar factor. A homotopy-coherent morphism induces the corresponding morphism of cellular bars.

A square-zero derivation Q of A induces a coderivation Q_{Bar} acting on each factor. Since Q is a derivation of multiplication,

$$[b_m, Q_{\mathrm{Bar}}] = 0.$$

This identity is the basis of BRST descent.

A monoidal functor $F : \mathrm{FactD}(X) \to \mathcal{V}$ preserving geometric realizations satisfies

$$F\big(\mathrm{Bar}_X^{\mathrm{ord}}(A)\big) \simeq \mathrm{Bar}_{\mathcal{V}}\big(F(A)\big). \tag{35.9}$$

Thus restriction to a formal disc, analytic realization, taking a fibre, and factorisation homology commute with the ordered bar when their continuity hypotheses hold.

35.8 Fubini in the two directions

Let $\mathcal{A}$ be the analytic factorisation algebra on $X^{\mathrm{an}} \times \mathbb{R}$ corresponding to A. For a compact one-manifold M with boundary labels, Fubini for factorisation homology gives

$$\int_{U\times M} \mathcal{A} \simeq \int_U \left(\int_M \mathcal{A}\right). \tag{35.10}$$

Taking $M = [0,1]$ with vacuum endpoints yields the analytic realization of (35.6). Taking $M = S^1$ yields internal Hochschild chains whenever the circle factorisation homology exists. A cyclic trace or trace-class analytic completion is additional data needed only to turn those chains into scalar amplitudes.

Formula (35.10) is the first exact relation between chiral homology and topological factorisation. It does not exchange the two directions; it integrates one while retaining the other as a coefficient system.

35.9 Completion

The bar length filtration has finite quotients

$$\mathrm{Bar}_{\leq N}(A) = \bigoplus_{r=0}^{N} (s^{-1}\bar{A})^{\otimes r}.$$

The completed bar is

$$\widehat{\mathrm{Bar}}_X^{\mathrm{ord}}(A) = \prod_{r\geq 0} (s^{-1}\bar{A})^{\otimes r}. \tag{35.11}$$

The coderivation is continuous when every component has finite output in a fixed total filtration degree. This holds automatically for strict E_1 multiplication and for filtered A_∞ operations that raise a complete positive filtration.

If conformal weight is unbounded, one can first restrict to a finite weight window and then complete. The two inverse limits commute under the Mittag–Leffler condition described in Theorem 36.3.

Proposition 35.5 (Cohomology of completion)**.** *Let C_N be the tower of finite bar-length and finite-weight truncations with surjective transition maps on cohomology. Then*

$$H^*(\varprojlim_N C_N) \simeq \varprojlim_N H^*(C_N).$$

Proof. The Milnor exact sequence has a $\varprojlim^1$ term. Surjectivity gives the Mittag–Leffler condition and annihilates that term. $\square$

Completion is the homotopy limit of the displayed inverse system, with convergence governed by its Mittag–Leffler properties.

35.10 The bar of a boundary endomorphism algebra

Let $A_b = \mathrm{RHom}_{\mathcal{C}}(b, b)$ be the endomorphism algebra of a compact boundary condition. The two-sided bar

$$\mathrm{Bar}(b, A_b, b)$$

resolves the identity kernel of the full subcategory generated by b. Under the Morita equivalence of Theorem 31.7, it is the factorisation analogue of the diagonal bimodule resolution.

Applying internal Hom to this resolution computes Hochschild cochains. Applying a trace computes Hochschild chains. Applying a second boundary label computes defect-changing operators. Thus the ordered bar is the common resolution behind the centre, the trace, and the module category.

35.11 The perturbative reading

In a perturbative boundary field theory, a bar word is a sequence of boundary insertions ordered along a collar. The bar differential fuses adjacent insertions. A Feynman propagator on X controls the chiral singularity of that fusion. Counterterms extend the coefficient distribution to the collision diagonal. The two operations are separate but compatible.

For a free Gaussian theory, Wick contraction gives an explicit morphism from the bar to the configuration-space de Rham complex. For an interacting renormalisable theory, Costello's renormalisation constructs the corresponding factorisation map order by order in $\hbar$. The quantum master equation is precisely the condition that the resulting differential squares to zero.

This is the exact sense in which a bar word is an interval amplitude. The bar differential is composition of boundary operations; the propagator supplies the coefficient of that composition.

Chapter 36

Ordered chiral bar–cobar reconstruction

36.1 The internal adjunction

Put

$$\mathcal{C}_X := \bigl(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}, \mathbf{1}\bigr).$$

Fix a stable presentable monoidal category $\mathcal{C}_X$ in which the displayed geometric realizations and totalizations exist and tensor product preserves them separately. These hypotheses hold in the standard presentable derived enhancements after restricting to the complete subcategories on which the indicated limits exist.

For a coaugmented conilpotent coassociative coalgebra C with coaugmentation coideal $\bar{C}$, define

$$\mathrm{Cobar}_X^{\mathrm{ord}}(C) = \left(\bigoplus_{r\geq 0}(s\bar{C})^{\otimes r},\ d_C + d_\Delta\right), \tag{36.1}$$

where d_Δ is the derivation induced by the reduced coproduct. A complete cobar uses the product over tensor lengths and the continuous extension of the same derivation.

A degree-one morphism $\tau : C \to A$ is a twisting morphism when

$$d\tau + m(\tau\otimes\tau)\bar{\Delta} = 0. \tag{36.2}$$

The universal twisting morphisms give natural equivalences of mapping spaces

$$\mathrm{Map}_{\mathrm{Alg}_{E_1}}\bigl(\mathrm{Cobar}_X^{\mathrm{ord}}\, C, A\bigr) \simeq \mathrm{Tw}(C, A) \simeq \mathrm{Map}_{\mathrm{Coalg}_{\mathrm{coAss}}}\bigl(C, \mathrm{Bar}_X^{\mathrm{ord}}\, A\bigr).$$

Hence

$$\mathrm{Cobar}_X^{\mathrm{ord}} : \mathrm{Coalg}_{\mathrm{coAss}}^{\mathrm{conil}}(\mathcal{C}_X) \rightleftarrows \mathrm{Alg}_{E_1}^{\mathrm{aug}}(\mathcal{C}_X) : \mathrm{Bar}_X^{\mathrm{ord}} \tag{36.3}$$

is an adjunction. Existence of the adjunction does not assert that its unit and counit are equivalences.

36.2 The connected positive-weight theorem

Definition 36.1 (Admissible positive object)**.** An augmented algebra A in $\mathcal{C}_X$ is admissibly positive if it has a complete weight decomposition

$$A \simeq \mathbf{1} \oplus \prod_{w \geq 1} A_w, \qquad A_p \otimes A_q \longrightarrow A_{p+q},$$

with each finite weight truncation dualisable and bounded below after every component evaluation $\mathrm{ev}_I : \mathrm{FactD}(X) \to D\text{-mod}(X^I)$. The evaluations are required to be conservative and to commute with the bar and cobar constructions. A coalgebra is admissibly positive when its coaugmentation coideal has the analogous positive, complete, conilpotent weight grading.

The positivity condition makes every fixed weight of a bar or cobar contain only finitely many tensor lengths. It is the internal analogue of a connected weight-graded augmented dg algebra.

Theorem 36.2 (Positive ordered bar–cobar reconstruction)**.** *For an admissibly positive augmented ordered chiral algebra A, the counit*

$$\mathrm{Cobar}_X^{\mathrm{ord}} \mathrm{Bar}_X^{\mathrm{ord}}(A) \longrightarrow A \tag{36.4}$$

is an equivalence. For an admissibly positive conilpotent coalgebra C, the unit

$$C \longrightarrow \mathrm{Bar}_X^{\mathrm{ord}} \mathrm{Cobar}_X^{\mathrm{ord}}(C) \tag{36.5}$$

is an equivalence.

Proof. Filter the two composites by tensor length and then take a fixed total weight. Positivity makes the resulting filtration finite. After applying ev_I, the associated graded is the ordinary connected associative bar–cobar resolution in complexes; its augmentation has the standard extra-degeneracy contraction. Thus each evaluated unit and counit is a quasi-isomorphism in every finite weight. Conservativity of the evaluations gives the equivalence in $\mathcal{C}_X$, and completeness reconstructs the full object from its finite weight truncations. $\square$

Corollary 36.3 (Completed reconstruction)**.** *Let $A \simeq \varprojlim_N A_{\leq N}$ be an inverse limit of admissibly positive finite truncations, and suppose the transition maps are Mittag–Leffler on cohomology and tensor product commutes with the inverse limit. Then the continuous counit*

$$\widehat{\mathrm{Cobar}}_X^{\mathrm{ord}} \widehat{\mathrm{Bar}}_X^{\mathrm{ord}}(A) \longrightarrow A$$

is an equivalence. The analogous assertion holds for a complete conilpotent coalgebra.

Proof. The theorem applies at every finite stage. The Milnor exact sequence identifies cohomology of the inverse limit with the inverse limit of cohomology because the $\varprojlim^1$ term vanishes under the Mittag–Leffler hypothesis. $\square$

The preceding equivalence is the associative bar–cobar theorem internal to the coefficient category. Its convergence is controlled by weight or augmentation completion, not by the cardinality filtration of the pointwise tensor product.

36.3 The separate Ran theorem

The Ran space carries another monoidal structure, the chiral tensor product $\otimes^{\mathrm{ch}}$. By Theorem 31.5, its reduced cardinality quotients are nilpotent. Francis and Gaitsgory prove the enhanced bar–cobar equivalence there for operads satisfying their reduced or pro-nilpotent hypotheses. The reduced associative operad gives the outer E_1 case, while their chiral Lie construction gives the Lie–commutative-coalgebra equivalence [82].

External theorem 36.4 (Enhanced Ran bar–cobar theorem)**.** *In the pro-nilpotent chiral Ran category, the enhanced Koszul adjunction for a reduced derived-Koszul operad restricts to an equivalence between nilcomplete augmented algebras and conilcomplete coalgebras. For the Lie–commutative pair, the chiral Chevalley and primitive functors give the corresponding equivalence. A restriction to a factorisation subcategory is valid when that subcategory is closed under the enhanced constructions and the unit and counit remain factorisation equivalences.* [32, 82]

Theorem 36.4 uses chiral convolution on the Ran space, whereas Theorem 36.2 uses the pointwise coefficient tensor and resolves transverse multiplication. The first uses chiral convolution on the Ran space. The second uses the pointwise coefficient tensor and resolves the transverse E_1 multiplication. A comparison between them is a monoidal comparison functor, not a change of notation.

36.4 The augmentation boundary

The unit is a left A-module through the augmentation. Define

$$A^! := \mathrm{RHom}_A(\mathbf{1}, \mathbf{1}). \tag{36.6}$$

Composition gives $A^!$ an E_1 multiplication. When the bar terms are dualisable, or when continuous internal Hom is used, the bar resolution gives

$$A^! \simeq \mathrm{RHom}_{\mathcal{C}_X}\big(\mathrm{Bar}_X^{\mathrm{ord}}(A), \mathbf{1}\big). \tag{36.7}$$

Theorem 36.5 (Boundary Koszul duality)**.** *Let A be an admissibly positive augmented ordered chiral algebra and let $A^!$ be defined by* (36.6)*. The adjunction*

$$- \otimes^L_{A^!} \mathbf{1} \dashv \mathrm{RHom}_A(\mathbf{1}, -)$$

restricts to an equivalence between the thick subcategory of A-modules generated by $\mathbf{1}$ and the thick subcategory of right $A^!$-modules generated by $A^!$. If $\mathbf{1}$ is a compact generator of the selected A-module category, the equivalence extends to its presentable completion.

Proof. The unit of the adjunction is an equivalence on the free rank-one $A^!$-module, and the counit is an equivalence on $\mathbf{1}$. Exactness and preservation of retracts extend these equivalences to the two thick closures. Compact generation extends the result by Ind-completion. □

Thus $A^!$ is the algebra of local operators on the augmentation boundary condition. It is not the quotient obtained by forgetting transverse order.

36.5 The double dual and completion

There is a canonical map

$$A \longrightarrow A^{!!} := \mathrm{RHom}_{A^!}(\mathbf{1}, \mathbf{1}). \tag{36.8}$$

Theorem 36.6 (Double dual as augmentation completion)**.** *Let A be admissibly positive, with dualisable finite weight pieces. Then (36.8) identifies $A^{!!}$ with the derived completion of A for the augmentation ideal. In particular, if A is complete for its positive weight filtration, then $A \simeq A^{!!}$.*

Proof. At every finite weight, the bar resolution is finite and dualisable. Dualising it once computes $A^!$; dualising the dual bar computes the finite augmentation quotient of A. The maps are compatible with the weight tower. Taking the inverse limit gives the derived augmentation completion, and completeness gives the final equivalence. □

The completion is part of the statement. A direct sum and its weight completion need not be isomorphic even when their finite truncations agree.

36.6 Quadratic comparison

Suppose that A has a connected quadratic presentation

$$A = T_{\mathcal{C}_X}(V)/(R)$$

with dualisable positive weight pieces. The quadratic coalgebra

$$A^{\text{!`}} = C(s^{-1}V, s^{-2}R) \subset T^c(s^{-1}V)$$

has a canonical twisting morphism and a comparison

$$q_A : A^{\text{!`}} \longrightarrow \mathrm{Bar}_X^{\mathrm{ord}}(A). \tag{36.9}$$

Theorem 36.7 (Quadratic recognition)**.** *Assume that component evaluation is conservative and preserves the quadratic twisting constructions. The following conditions are equivalent:*

(i) q_A *is an equivalence;*

(ii) $\mathrm{Cobar}(A^{i}) \to A$ *is an equivalence;*

(iii) *the left quadratic twisted tensor product is acyclic;*

(iv) *the right quadratic twisted tensor product is acyclic;*

(v) *the evaluated bar homology is concentrated on the quadratic diagonal in every finite weight.*

Proof. After component evaluation this is the fundamental theorem for Koszul twisting morphisms. Conservativity lifts the equivalences to $\mathcal{C}_X$. Finite positive weights justify the passage between the quadratic coalgebra and the full bar. □

Quadratic Koszulness concerns the comparison (36.9). It is stronger than the universal reconstruction of Theorem 36.2.

36.7 PBW and nonhomogeneous relations

Let $F_\bullet A$ be a complete exhaustive multiplicative filtration and suppose $\mathrm{gr}_F A \cong A_0$. Filter the bar by total PBW degree.

Theorem 36.8 (Filtered PBW comparison)**.** *Assume that the filtered comparison is complete and separated, its spectral sequence converges strongly in every finite chiral-cardinality and conformal-weight window, and A_0 is quadratic Koszul. Then the nonhomogeneous Koszul coalgebra determined by the quadratic, linear, and constant parts of the relations maps equivalently to* $\mathrm{Bar}^{\mathrm{ord}}_X(A)$.

Proof. The associated graded of the comparison is the quadratic equivalence for A_0. Strong convergence lifts it through the filtration. Linear terms become part of the coalgebra differential, and constant terms become its curvature functional. □

The bar of an inhomogeneous algebra is computed by its nonhomogeneous or curved Koszul coalgebra together with the PBW comparison.

36.8 Curved coalgebras

A curved coalgebra has a coderivation d_C and a curvature functional $h : C \to \mathbf{1}[2]$ satisfying

$$d_C^2 = (h \otimes \mathrm{id} - \mathrm{id} \otimes h)\bar{\Delta}, \qquad hd_C = 0.$$

Its curved cobar has a square-zero total differential because the curvature functional supplies the constant part of the cobar derivation. This curvature is an algebraic feature of an inhomogeneous presentation. It is not a two-form on a moduli space.

When ordinary acyclic complexes are not the correct null system, curved modules belong to the coderived or contraderived category [103]. The choice is determined by the module theory and completion, not by the numerical value of a central charge.

36.9 Verdier duality

Let $\mathbb{D}_{\mathrm{Ran}}$ denote Verdier duality. If the finite bar stages are dualisable and duality commutes with the completion, then

$$A^\vee_{\mathrm{Verdier}} := \mathbb{D}_{\mathrm{Ran}}\, \mathrm{Bar}^{\mathrm{ord}}_X(A) \tag{36.10}$$

acquires an algebra structure by dualising deconcatenation.

The Verdier algebra (36.10) is not automatically the boundary endomorphism algebra (36.6). Their comparison requires a continuous perfect pairing

$$\mathrm{Bar}^{\mathrm{ord}}_X(A) \otimes A^! \longrightarrow \mathbf{1}.$$

Proposition 36.9 (Three dual constructions)**.** *For an augmented ordered chiral algebra, the following are distinct until an explicit equivalence is proved:*

(i) *the covariant reconstruction* $\mathrm{Cobar}\,\mathrm{Bar}(A) \to A$*;*

(ii) *the boundary algebra* $A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$*;*

(iii) *the Verdier dual algebra* $\mathbb{D}_{\mathrm{Ran}}\,\mathrm{Bar}(A)$*, when defined.*

Proof. The first object is the value of a covariant counit in the algebra category. The second is an endomorphism algebra in the module category and is contravariant in the augmentation boundary. The third is obtained by Verdier duality from a coalgebra on the Ran space and requires dualisability. Their targets and variances differ, so an identification can arise only from an additional comparison morphism satisfying aritywise dualisability, continuity, and compatibility with the stated tensor products. □

The four constructions have distinct sources and targets: opposite order, inverse braiding, level reflection, and Verdier duality.

36.10 The complete deformation complex

The deformation complex of the transverse E_1 multiplication is the shifted internal Hochschild complex

$$\mathrm{Def}_{E_1}(A) = \mathrm{HH}^{\bullet}_{\mathrm{FactD}(X)}(A, A)[1]. \tag{36.11}$$

The chiral coefficient has its own deformation complex. On a formal disc it is the chiral operadic complex governing vertex or nonlocal-vertex operations. The deformation theory of the full ordered chiral object is controlled by a cofibrant resolution of the tensor-product operad $E_1 \otimes \mathsf{Fact}_X$.

Proposition 36.10 (Bigraded deformation complex)**.** *Choose an admissible cofibrant resolution of $E_1 \otimes \mathsf{Fact}_X$. The convolution deformation complex of an ordered chiral algebra has a complete filtration by the number of transverse and chiral vertices. Its associated graded is spanned by two-coloured trees; the horizontal axis restricts to the Hochschild deformation complex of the E_1 multiplication, the vertical axis restricts to the deformation complex of the chiral coefficient, and the mixed components impose the interchange law.*

Proof. The cellular Boardman–Vogt resolution is filtered by the two edge colours. Taking convolution with the endomorphism operad sends its cellular boundary to the deformation differential. Forests containing only transverse edges give Hochschild cochains; forests containing only chiral edges give coefficient deformations; forests of both colours give the mixed compatibility terms. □

A Morita equivalence identifies the Hochschild axis. It identifies the full ordered chiral deformation problem only when it also identifies the chiral coefficient and the mixed interchange maps.

Chapter 37

The Koszul–Morita square and the universal boundary state

The bar coalgebra is not only a resolution of the augmentation. It is the coalgebraic form of the universal boundary state. Once this point is made explicit, the interval, the diagonal bimodule, the annulus, Hochschild theory, and the derived centre become different shadows of one two-sided construction.

Let

$$\mathcal{C}_X = (\mathrm{FactD}(X), \otimes^{\mathrm{pt}}, \mathbf{1}),$$

let A be an admissibly positive augmented E_1 algebra in $\mathcal{C}_X$, and put

$$C = \mathrm{Bar}_X^{\mathrm{ord}}(A).$$

Let $\tau : C \to A$ be the universal twisting morphism. All tensor and cotensor products below are derived in the complete subcategories selected by the positive weight filtration.

37.1 One-sided Koszul transforms

For a left A-module M, define the twisted left C-comodule

$$\mathbb{K}_A(M) = \big(C \otimes M, d_C + d_M + d_\tau\big), \tag{37.1}$$

where d_τ applies the reduced coproduct to C, evaluates one factor by τ, and lets the result act on M. For a conilpotent left C-comodule N, define

$$\mathbb{L}_A(N) = \big(A \otimes N, d_A + d_N + d_\tau\big). \tag{37.2}$$

The Maurer–Cartan equation for τ is exactly the statement that both displayed differentials square to zero.

Theorem 37.1 (Module–comodule Koszul equivalence). *The functors $\mathbb{K}_A$ and $\mathbb{L}_A$ are inverse equivalences between the derived category of augmentation-complete left A-modules and the coderived category of conilpotent left C-comodules. The same statement holds for right modules and right comodules.*

Proof. Filter both composites by the coradical degree of C. In each fixed positive weight, the filtration is finite. The associated graded twisting differential vanishes, and the unit and counit reduce to the ordinary bar–cobar contractions. The extra degeneracy of the simplicial bar gives the contraction at each finite weight. Conservativity of component evaluation and completeness lift the equivalence to $\mathcal{C}_X$. When the coalgebra differential is curved or infinite products are essential, the same filtered contraction has coacyclic cone and therefore gives the asserted coderived equivalence. □

The theorem is the module form of ordered chiral Koszul duality. It is stronger than an equality of Ext groups and weaker than a Morita equivalence between all A-modules and all modules over $A^!$. It identifies precisely the sector generated from the augmentation boundary.

37.2 The two-sided transform

For an A-bimodule M, set

$$\mathbb{K}_A^{\mathrm{bi}}(M) = C \otimes_\tau M \otimes_\tau C. \tag{37.3}$$

The two outer factors carry the left and right C-coactions. The differential has four terms: the differentials of C and M, the left twisting action, and the right twisting action. The two twisting terms anticommute because the left and right A-actions commute.

Conversely, for a conilpotent C-bicomodule N, set

$$\mathbb{L}_A^{\mathrm{bi}}(N) = A \otimes_\tau N \otimes_\tau A. \tag{37.4}$$

Theorem 37.2 (Bimodule–bicomodule equivalence)**.** *The functors $\mathbb{K}_A^{\mathrm{bi}}$ and $\mathbb{L}_A^{\mathrm{bi}}$ are inverse equivalences between augmentation-complete A-bimodules and conilpotent C-bicomodules in the corresponding derived or coderived categories.*

Proof. Apply Theorem 37.1 first to the left action and then to the right action. The two transforms commute because their twisting coderivations act on different outer coalgebra factors and the bimodule identity makes their mixed commutator zero. Equivalently, filter (37.3) by the sum of the two coradical degrees and use the two-sided extra-degeneracy contraction on the associated graded. □

37.3 The diagonal bicomodule

The diagonal A-bimodule is A itself. The diagonal C-bicomodule is the coalgebra C equipped with the two coactions

$$C \xrightarrow{\Delta} C \otimes C, \qquad C \xrightarrow{\Delta} C \otimes C,$$

one read as a left coaction and the other as a right coaction. Denote it by C_Δ.

Theorem 37.3 (Diagonal correspondence)**.** *There is a canonical equivalence of C-bicomodules*

$$\mathbb{K}_A^{\mathrm{bi}}(A) \simeq C_\Delta. \tag{37.5}$$

Proof. The two-sided transform of A is

$$C \otimes_\tau A \otimes_\tau C.$$

The bar–cobar counit contracts the middle A successively against the left and right twisting cochains. Filter by the total coradical degree of the two outer factors. On the associated graded, the twisting terms vanish and the contraction identifies the complex with one copy of C. Under this identification the two surviving coactions are the two legs of Δ_C. Finite positive weights and completeness lift the associated-graded equivalence. □

The theorem gives a precise mathematical form to the phrase "universal boundary state." The diagonal bimodule is the identity kernel for tensoring over A; its Koszul transform is the identity kernel for cotensoring over C.

Corollary 37.4 (Composition unit)**.** *For every conilpotent C-bicomodule N,*

$$C_\Delta \,\square^L_C\, N \simeq N, \qquad N \,\square^L_C\, C_\Delta \simeq N.$$

Proof. The diagonal bicomodule is the unit for the cotensor monoidal structure on complete C-bicomodules. In the cobar model for derived cotensor, the coaction of C_Δ is the coaugmentation followed by the coproduct. The extra coaugmentation is contracted by the normalized degeneracy, leaving the cobar complex for N. The same contraction applies on the other side. □

37.4 Tensor becomes cotensor

Let M be an A–B bimodule and N a B–D bimodule, with all algebras admissibly positive and all modules complete. Relative tensor product composes defects. The coalgebraic operation dual to it is derived cotensor product.

Theorem 37.5 (Tensor–cotensor gluing)**.** *Let C_A, C_B, C_D be the ordered bar coalgebras. Then there is a canonical equivalence*

$$\mathbb{K}^{\mathrm{bi}}(M \otimes^L_B N) \simeq \mathbb{K}^{\mathrm{bi}}(M) \,\square^L_{C_B}\, \mathbb{K}^{\mathrm{bi}}(N). \tag{37.6}$$

Proof. Choose two-sided bar resolutions for the relative tensor product and cobar resolutions for the cotensor product. After inserting the universal twisting morphisms, both total complexes are indexed by the same strings of B-operations between M and N. The Eilenberg–Zilber comparison identifies the two bisimplicial totalizations. In fixed positive weight the comparison is finite and is a quasi-isomorphism by the standard two-sided Koszul contraction. Completion gives (37.6). □

This theorem is the algebraic form of cutting and regluing an interval. Tensoring sums over intermediate open states. Cotensoring imposes equality of the two coalgebraic boundary states.

37.5 Hochschild and coHochschild theories

Let

$$C^e = C \otimes C^{\mathrm{cop}}.$$

For a C-bicomodule N, define

$$\mathrm{coHH}_*(C, N) := C_\Delta \,\square^L_{C^e}\, N,$$

and

$$\mathrm{coHH}^*(C, N) := \mathrm{RHom}_{C^e}(C_\Delta, N)$$

in the selected coderived category.

Theorem 37.6 (Chiral Hochschild–coHochschild duality)**.** *For every complete A-bimodule M there are natural equivalences*

$$\mathrm{HH}_*(A, M) \simeq \mathrm{coHH}_*\big(C, \mathbb{K}_A^{\mathrm{bi}}(M)\big), \tag{37.7}$$
$$\mathrm{HH}^*(A, M) \simeq \mathrm{coHH}^*\big(C, \mathbb{K}_A^{\mathrm{bi}}(M)\big). \tag{37.8}$$

In particular,

$$\mathrm{HH}_*(A) \simeq C_\Delta \,\square^L_{C^e}\, C_\Delta,$$
$$\mathrm{HH}^*(A) \simeq \mathrm{RHom}_{C^e}(C_\Delta, C_\Delta).$$

Proof. Hochschild homology is the derived tensor product of the diagonal bimodule with M over A^e. Transport this expression through Theorem 37.2. By Theorem 37.3, the diagonal bimodule maps to C_Δ, and by Theorem 37.5, derived tensor becomes derived cotensor. This proves (37.7). Derived Hom is preserved by an equivalence of derived categories, which proves (37.8). ☐

The annulus is therefore the self-cotensor of the universal boundary state. The derived centre is its coalgebraic self-extension. The two descriptions are not analogous; they are related by an equivalence of categories.

37.6 The Koszul–Morita square

The preceding theorems assemble into a commutative square of functors:

$$\begin{tikzcd} D(A^e\text{-mod})_{\mathrm{comp}} \arrow[r, "\mathbb{K}_A^{\mathrm{bi}}", "\sim"'] \arrow[d, "A\otimes^L_{A^e}(-)"'] & D^{\mathrm{co}}(C^e\text{-comod})_{\mathrm{conil}} \arrow[d, "C_\Delta\square^L_{C^e}(-)"] \\ D(\mathcal{C}_X) \arrow[r, equal] & D(\mathcal{C}_X). \end{tikzcd} \tag{37.9}$$

The corresponding square for derived Hom computes the centre. This is the Koszul–Morita square. Its horizontal maps change algebraic presentation. Its vertical maps close an interval into a circle. The square says that duality commutes with closure.

Corollary 37.7 (Centre as the closed endomorphism of the boundary state)**.** *The ordered chiral centre has the equivalent descriptions*

$$Z_{\mathrm{ord,ch}}(A) \simeq \mathrm{RHom}_{A^e}(A, A) \simeq \mathrm{RHom}_{C^e}(C_\Delta, C_\Delta).$$

Its E_2 operations are transported by the bimodule–bicomodule equivalence to the brace operations of the diagonal bicomodule.

Proof. Apply the two-sided twisting equivalence to the diagonal bimodule. Full faithfulness identifies its derived endomorphisms with the derived endomorphisms of the diagonal bicomodule. Compatibility of the equivalence with relative tensor and cotensor transports the brace operations, hence the complete E_2 structure. ☐

37.7 Calabi–Yau transport

Suppose the perfect A-module category carries a d-Calabi–Yau trace. Equivalently, the diagonal bimodule has a nondegenerate cyclic pairing of degree d compatible with composition. Transporting the pairing through Theorem 37.2 gives a cyclic copairing on C_Δ.

Theorem 37.8 (Transport of the open trace)**.** *Assume that the bimodule–bicomodule equivalence lifts to cyclic objects and commutes with geometric realization of the cyclic bar constructions. Then a perfect d-Calabi–Yau structure on the augmentation-generated A-module category induces a nondegenerate degree-d cyclic structure on the diagonal bicomodule C_Δ. Under the resulting S^1-equivariant Hochschild–coHochschild equivalence, the Connes operator and the pair-of-pants product correspond; when the framed circle action is defined, so do the induced Batalin–Vilkovisky operators.*

Proof. A Calabi–Yau trace is a cyclic natural transformation on the Morita category of perfect modules. Theorem 37.5 transports composition. By the assumed lift to cyclic objects, the cyclic bar construction, its rotation, and its gluing maps are carried to their coHochschild counterparts. The Connes and pair-of-pants operations therefore agree under the equivalence. The same argument applies to the BV operator when the framed circle action has been supplied. ☐

The cyclic two-sided transform is the exact bridge from ordered Koszul duality to ribbon geometry. It generates the open topological surfaces of the transverse trace; Deligne–Mumford sewing enters through an independent chiral modular structure.

37.8 Physical interpretation

Let a factorization-homology realization identify a boundary condition b with the endomorphism algebra $A = \mathrm{RHom}(b, b)$. Under interval excision, the bar coalgebra C is the state object of an interval ending on the augmentation boundary. An A–B bimodule is an interface. Relative tensor product composes interfaces. The Koszul transform turns the same interface into a bicomodule of boundary states, and cotensor product performs the same gluing in coalgebraic variables.

Closing an interface into an annulus gives Hochschild homology. Inserting a local bulk operation gives Hochschild cohomology. Theorem 37.6 says that both amplitudes can be computed before or after ordered chiral Koszul duality. The operation is the same; only the variables have changed.

The square (37.9) is therefore the precise mathematical form of open–closed complementarity on the augmentation-generated sector. It is categorical, not numerical. A scalar equality follows only after a trace is applied to this square.

Chapter 38

Exchange begins beyond the ordered line

38.1 Ordered fusion and planar exchange

For every n, the chamber

$$\mathrm{Conf}_n^<(\mathbb{R}) = \{t_1 < \cdots < t_n\}$$

is convex. It remembers the order of factors but has trivial fundamental groupoid.

Theorem 38.1 (No braid from order)**.** *The topology of ordered configurations on an oriented line supplies no nontrivial braid operator functorially from an E_1 algebra. A braid action requires an E_2 enhancement, a specified flat connection on a two-dimensional configuration space, or equivalent exchange data.*

Proof. Parallel transport inside a contractible chamber is canonically trivial. A braid generator is represented by a path interchanging points in a plane or another two-dimensional configuration space; no such path exists on the line. □

Noncommutativity and braiding are therefore different structures. An associative algebra need not possess an R-matrix, and a braid representation need not arise from multiplication.

38.2 The E_2 enhancement

Replacing the transverse line by an oriented plane gives a forgetful functor

$$\mathrm{Alg}_{E_2}(\mathrm{FactD}(X)) \longrightarrow \mathrm{Alg}_{E_1}(\mathrm{FactD}(X)).$$

An E_2 lift supplies exchange paths. Three-disk isotopy gives the braid relation. Passing to module categories produces the pentagon and hexagon coherences. Analytically, the lift is a factorization algebra holomorphic on X and locally constant in two real transverse directions.

38.3 Internal Hochschild cochains

For an E_1 algebra A in $\mathrm{FactD}(X)$, define

$$Z_{\mathrm{ord,ch}}(A) := \mathrm{RHom}_{A\otimes A^{\mathrm{op}}}(A, A). \tag{38.1}$$

The diagonal bar resolution gives the cup product, braces, and Gerstenhaber operations internally in the coefficient category.

Theorem 38.2 (Universal E_2 centre)**.** *Assume that the diagonal bimodule admits a homotopically correct bar resolution in* $\mathrm{FactD}(X)$*. Then $Z_{\mathrm{ord,ch}}(A)$ is an E_2 algebra. For every E_2 algebra B acting on A through the Swiss-cheese operad, there is a contractibly unique E_2 morphism*

$$B \longrightarrow Z_{\mathrm{ord,ch}}(A)$$

that induces the given action.

Proof. The higher Deligne theorem equips endomorphisms of the unit object A in the monoidal category of A–A bimodules with an E_2 structure. The Swiss-cheese universal property identifies a central action with a morphism to these endomorphisms. The mapping space of such factorizations is contractible by the universal property of the endomorphism object. ☐

The theorem identifies the centre as the universal recipient of every central bulk action. A physical bulk is identified with that centre by an additional open–closed quasi-isomorphism.

38.4 Bimodules and categorical centres

The category $\mathrm{Bimod}_A(\mathrm{FactD}(X))$ is monoidal under relative tensor product, with unit the diagonal bimodule A.

Proposition 38.3 (Hochschild cochains as unit endomorphisms)**.** *There is a natural equivalence*

$$\mathrm{End}_{\mathrm{Bimod}_A}(A) \simeq Z_{\mathrm{ord,ch}}(A),$$

and the E_2 structure on the right is the endomorphism E_2 structure of the unit in the monoidal bimodule category. The Drinfeld centre $Z(\mathrm{Bimod}_A)$ is braided and acts on this unit endomorphism algebra. An equivalence

$$Z(\mathrm{Bimod}_A) \simeq \mathrm{Mod}_{Z_{\mathrm{ord,ch}}(A)}$$

requires a separate monoidal-affineness theorem and is not asserted in general.

Proof. The unit of the monoidal category of A-bimodules is the diagonal bimodule A. Its derived endomorphism object is therefore $\mathrm{RHom}_{A^e}(A, A)$. The little-disks action on endomorphisms of a monoidal unit is the Hochschild E_2 action. The categorical Drinfeld centre acts on every bimodule, hence on the unit and on its endomorphisms. The final equivalence requires precisely the additional monoidal-affineness hypothesis stated in the proposition. ☐

The categorical centre is attached to the monoidal bimodule category, or to another explicitly monoidal line-defect category.

38.5 Meromorphic exchange

Suppose line defects are parametrized by points of a spectral curve. A meromorphic exchange is a family

$$R_{L,M}(z, w) : L_z \otimes M_w \longrightarrow M_w \otimes L_z \tag{38.2}$$

defined away from the collision divisor. Isotopy of three lines gives

$$R_{12}(z_1, z_2)R_{13}(z_1, z_3)R_{23}(z_2, z_3) = R_{23}(z_2, z_3)R_{13}(z_1, z_3)R_{12}(z_1, z_2). \tag{38.3}$$

The poles are chiral singularities on the spectral curve; the exchange path is topological in the transverse plane.

38.6 Pure-braid discrepancy

Let M_n be a complex with a B_n action and let $P_n = \ker(B_n \to \Sigma_n)$. Its derived pure-braid invariants and coinvariants are

$$M_n^{hP_n} = \mathrm{RHom}_{\mathbb{C}[P_n]}(\mathbb{C}, M_n), \qquad (M_n)_{hP_n} = \mathbb{C} \otimes^L_{\mathbb{C}[P_n]} M_n,$$

with residual Σ_n actions.

Definition 38.4 (Pure-braid discrepancy)**.** Writing $q : B_n \to \Sigma_n$ and q^* for inflation, define

$$\mathfrak{C}_n^{\mathrm{inv}}(M) := \mathrm{Cofib}\big(q^* M_n^{hP_n} \longrightarrow M_n\big), \tag{38.4}$$

$$\mathfrak{C}_n^{\mathrm{coinv}}(M) := \mathrm{Fib}\big(M_n \longrightarrow q^*(M_n)_{hP_n}\big). \tag{38.5}$$

The two objects are different derived measures of what is lost when pure-braid transport is removed. They are defined only after a braid action has been supplied.

Proposition 38.5 (Relative group model)**.** *The homology of* (38.5) *is computed by the reduced group-bar complex of P_n with coefficients in M_n. The cohomology of* (38.4) *is computed by reduced group cochains. The residual Σ_n action is induced by conjugation in B_n.*

Proof. Derived coinvariants are $M_n \otimes^L_{\Bbbk[P_n]} \Bbbk$, and the standard free resolution of the trivial module is the normalized group-bar complex. The cofiber of the augmentation is its reduced form. Derived invariants are computed by the dual normalized group-cochain complex. Normality of $P_n \triangleleft B_n$ makes conjugation descend to the quotient Σ_n, which gives the residual action. □

A partition complex may filter this group complex only after a map from collision strata to braid cells has been constructed. It is not the group-bar complex by definition.

38.7 Meromorphic Tannaka reconstruction

Let K be the function field of the spectral curve. Let $\mathcal{L}$ be a small rigid K-linear tensor category with finite-dimensional Hom spaces. A fibre functor may carry either strong monoidal or only quasi-monoidal comparison maps.

Assume first that

$$F : \mathcal{L} \longrightarrow \mathrm{Vect}_K$$

is faithful, exact, and strong monoidal, and that the coend

$$\mathcal{H} = \int^{L \in \mathcal{L}} F(L)^\vee \otimes F(L) \tag{38.6}$$

exists. A meromorphic braiding on $\mathcal{L}$ is understood over the complement of the collision divisor.

Theorem 38.6 (Meromorphic Tannaka reconstruction)**.** *The strong monoidal pair $(\mathcal{L}, F)$ reconstructs a bialgebra $\mathcal{H}$ over K; rigidity supplies an antipode whenever the coend lies in the rigid finite or continuous-dual class. A braiding reconstructs a coquasitriangular structure. If the fibre functor is only quasi-monoidal, the same coend carries a coquasi-bialgebra structure whose coassociator is the transported tensor constraint. When the matrix-coefficient coalgebra has finite graded pieces or a separated continuous dual, these statements dualize to quasitriangular Hopf or quasi-Hopf algebras, respectively.*

Proof. The coend represents natural endomorphisms of the fibre functor. The strong tensor constraints define a coassociative coproduct and counit; rigidity defines the antipode. The braiding gives the universal R-form, and the hexagon identities give coquasitriangularity. For a quasi-monoidal fibre functor, the defect of strict coassociativity is exactly the transported associator, which satisfies the pentagon and hence defines a coquasi-bialgebra. □

A presentation as a Yangian, quantum loop algebra, elliptic quantum group, or Hall double is an additional recognition theorem.

38.8 Morita invariance

Derived Morita equivalent E_1 algebras have equivalent bimodule categories and equivalent Hochschild cochains as E_2 algebras.

Theorem 38.7 (Centre transport through a Morita context)**.** *Let A and B be ordered chiral algebras joined by an invertible A–B bimodule in the Morita $(\infty, 2)$-category of complete factorisation algebras. Then*

$$Z_{\mathrm{ord,ch}}(A) \simeq Z_{\mathrm{ord,ch}}(B)$$

as E_2 algebras, and the induced monoidal equivalence of bimodule categories identifies their Drinfeld centres.

For $B = A^!$, the conclusion applies only when the boundary Koszul equivalence of Theorem 36.5 extends to such an invertible bimodule Morita context. Equivalence of the two augmentation-generated thick subcategories alone does not imply the conclusion.

Proof. Tensoring with the invertible A–B bimodule and its inverse gives a monoidal equivalence between the corresponding bimodule categories. The equivalence carries the diagonal A-bimodule to the diagonal B-bimodule. Derived endomorphisms of these monoidal units are therefore equivalent, and monoidality transports the little-disks operations. Drinfeld centres are invariant under monoidal equivalence. The specialization to $B = A^!$ follows when the Koszul boundary bimodule is invertible on the stated Morita locus. □

The theorem is the precise algebraic content of two boundary presentations having the same universal central bulk. It is not a numerical complementarity law, and it is not automatic for a Koszul-dual pair such as a polynomial algebra and its exterior dual.

38.9 Topological enhancement

The centre is E_2 in the two transverse directions and remains holomorphic on the complex curve. Suppose both real translations underlying the holomorphic coordinate become null-homotopic, coherently with all E_2 operations.

Proposition 38.8 (Dimension of the topological centre)**.** *Under this coherent topologization, the analytic centre is locally constant on the four-real-dimensional manifold $X^{\mathrm{an}} \times \mathbb{R}^2$. It therefore determines an E_4 algebra.*

Proof. The null-homotopies make the factorization algebra locally constant in the two real directions of X. It was already locally constant in two transverse directions. The recognition theorem for locally constant factorization algebras gives an E_4 algebra; equivalently, Dunn additivity combines the two E_2 structures. □

Topologizing the whole complex curve supplies two real directions and gives an E_4 structure; topologizing one additional real direction gives the intermediate E_3 structure. Topologizing only one additional real direction gives an E_3 intermediate structure.

38.10 The bulk-to-centre map

Let a bulk factorization algebra $\mathrm{Obs}_{\mathrm{bulk}}$ act on the boundary algebra A. Bringing a bulk observable to the boundary gives a central action and hence a map

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z_{\mathrm{ord,ch}}(A). \tag{38.7}$$

Criterion 38.9 (Algebraically complete boundary)**.** *The boundary condition reconstructs the chosen local bulk observables exactly when* (38.7) *is an equivalence in the category of local factorisation observables selected by the open–closed map. Without this equivalence, the centre is only the universal recipient of the bulk action.*

A decoupled bulk factor has zero open–closed map. Its observables remain in the fibre of the boundary measurement, so reconstruction requires a generation or nondegeneracy theorem.

Chapter 39

Holomorphic–topological recognition

39.1 Product-generated local observables

Let $M = X^{\mathrm{an}} \times \mathbb{R}$. A product disk is an open set $U \times I$, with $U \subset X^{\mathrm{an}}$ a complex disc and $I \subset \mathbb{R}$ an oriented interval. Let $\mathsf{Disk}^{\mathrm{rect}}_{X\times\mathbb{R}}$ be the symmetric monoidal category generated by finite disjoint unions of product disks and their embeddings. An arbitrary holomorphic–topological factorization algebra need not be determined by this subcategory; we therefore isolate the precise local class used in the recognition theorem.

Definition 39.1 (Rectangular holomorphic–topological factorization algebra)**.** A rectangular holomorphic–topological factorization algebra is a factorization algebra $\mathcal{F}$ on $X^{\mathrm{an}} \times \mathbb{R}$ such that:

(i) $\mathcal{F}$ is the factorization left Kan extension of its restriction to $\mathsf{Disk}^{\mathrm{rect}}_{X\times\mathbb{R}}$;
(ii) for fixed I, the dependence on U is the analytic realization of a factorisable right D-module;
(iii) for fixed U, the dependence on I is locally constant and orientation preserving.

Write $\mathrm{Fact}^{\mathrm{rect}}_{\mathrm{hol,lc}}(X \times \mathbb{R})$ for this category.

The first condition is the generation statement needed to pass from product disks to all opens. The second is the chiral regularity. The third gives the little-interval action. None uses a metric.

39.2 The exponential law

The local operation category of product disks is the tensor product of the chiral disk category with the little-interval operad:

$$\mathsf{Disk}^{\mathrm{ch}}_X \otimes E_1.$$

The tensor-product universal property gives an exponential law for algebras.

Theorem 39.2 (Holomorphic–topological recognition)**.** *There is a natural equivalence*

$$\mathrm{Alg}_{E_1}\big(\mathrm{FactD}(X)\big) \simeq \mathrm{Fact}^{\mathrm{rect}}_{\mathrm{hol,lc}}(X \times \mathbb{R}). \tag{39.1}$$

On the analytic side, the statement is restricted to coefficient systems in the image of the chosen Riemann–Hilbert realization. On the algebraic side, it is an equivalence of local factorization structures generated by product disks.

Proof. An object on the left is an algebra over E_1 in algebras over the chiral disk category. By the tensor-product universal property,

$$\mathrm{Alg}_{E_1}\big(\mathrm{Alg}_{\mathsf{Disk}_X^{\mathrm{ch}}}(\mathcal{V})\big) \simeq \mathrm{Alg}_{E_1\otimes\mathsf{Disk}_X^{\mathrm{ch}}}(\mathcal{V}).$$

The right-hand object is exactly a factorization algebra on the rectangular disk category that is chiral in U and locally constant in I. Factorization left Kan extension gives the right-hand side of (39.1). Restriction back to product disks is inverse because rectangular generation is part of its definition. Riemann–Hilbert identifies the regular-holonomic algebraic coefficient with its analytic constructible realization. □

The theorem identifies local algebraic structures. It neither classifies all field theories on $X \times \mathbb{R}$ nor asserts that every ordered chiral algebra comes from a Lagrangian.

39.3 The two short-distance limits

Let A correspond to $\mathcal{F}$. For disjoint intervals $I_1 < I_2 \subset I$ and discs $U_1, U_2 \subset U$, factorization gives

$$\mathcal{F}(U_1 \times I_1) \otimes \mathcal{F}(U_2 \times I_2) \longrightarrow \mathcal{F}(U \times I). \tag{39.2}$$

Approaching the diagonal in $U_1 \times U_2$ probes the singular chiral coefficient. Bringing I_1 and I_2 together without crossing probes the E_1 multiplication. The two operations satisfy interchange because they are the two factors of one product-disk operation.

Proposition 39.3 (Interchange of short-distance limits)**.** *Let m be the transverse multiplication and let γ be a chiral factorization map. On every product-disjoint locus where both composites are defined,*

$$\gamma \circ (m \otimes m) = m \circ (\gamma \otimes \gamma), \tag{39.3}$$

with the canonical permutation of coefficient factors.

Proof. Both sides are the image of one embedding of four product disks into a larger product disk. They are equal by functoriality of the factorization algebra. □

The interchange law makes the two products compatible; planar motion adds the stronger datum of braiding. Braiding requires motion in a second transverse direction.

39.4 Boundary and line-defect origins

A boundary condition b in a holomorphic–topological theory has a factorisable endomorphism object

$$A_b = \mathrm{RHom}(b, b).$$

Composition in the normal or topological direction is E_1 multiplication; collision of insertion points on X is the chiral coefficient. Boundary-changing operators are bimodules.

Parallel line defects in a theory on $X \times \mathbb{R}^2$ give the same structure after their transverse positions are restricted to an oriented line. Allowing the lines to move in the full plane may promote fusion to an E_2 structure and supply exchange. The E_1 fusion exists before that promotion.

Principle 39.4 (Defect interpretation)*.* The E_1 product is composition of boundary operators or line defects. The factorisable D-module records holomorphic collision. An exchange operator is additional codimension-two data.

39.5 Classical and quantum observables

Let $(\mathcal{E}, Q, \{-,-\})$ be the local fields of a classical BV theory. A local interaction I satisfies

$$QI + \frac{1}{2}\{I, I\} = 0. \tag{39.4}$$

The classical observables have differential $Q + \{I, -\}$. A perturbative quantization is represented by $I_\hbar = I + \hbar I_1 + \cdots$ satisfying the renormalized quantum master equation

$$QI_\hbar + \frac{1}{2}\{I_\hbar, I_\hbar\} + \hbar\Delta I_\hbar = 0. \tag{39.5}$$

The equation makes the renormalized observable differential square to zero and makes graphwise factorization compatible with contraction of ultraviolet subgraphs.

Theorem 39.5 (BV realization of an ordered chiral algebra)**.** *Let a perturbatively renormalized BV theory on $X \times \mathbb{R}$ satisfy the quantum master equation. Assume that its quantum observables are rectangularly generated, locally constant in the real direction, and admit a factorisable D-module realization in the holomorphic direction. Then they determine*

$$\mathrm{Obs}^{\mathrm{q}}_{X\times\mathbb{R}} \in \mathrm{Alg}^{\mathrm{aug}}_{E_1}(\mathrm{FactD}(X)).$$

A weak equivalence of renormalized theories that induces an objectwise quasi-isomorphism of these factorization observables induces an equivalence of ordered chiral algebras.

Proof. The quantum master equation supplies the square-zero differential and compatibility of renormalized factorization maps. The three additional hypotheses place the observable factorization algebra in the right-hand side of Theorem 39.2. Applying that equivalence gives the ordered chiral algebra. Objectwise quasi-isomorphisms become equivalences after localization. □

The converse requires more data: fields, gauge symmetry, a BV pairing, a real form, counterterm choices, state spaces, and integration cycles are not reconstructible from local observables alone.

39.6 Interval evolution

With augmentation boundary conditions at both ends of an interval, excision gives

$$\int_{([0,1],\partial)} A \simeq \mathbf{1} \otimes^L_A \mathbf{1} \simeq \mathrm{Bar}^{\mathrm{ord}}_X(A). \tag{39.6}$$

The ordered bar is therefore the complex of interval amplitudes between the two chosen boundaries. Cutting the interval gives deconcatenation.

Corollary 39.6 (Orientation reversal)**.** *Reflection of the interval identifies the underlying bar objects of A and A^{op} by reversing every word and inserting the Koszul sign of the reversal permutation.*

Proof. Factorisation homology is functorial for framed diffeomorphisms of the interval. Reflection reverses the order of every configuration and replaces multiplication by the opposite multiplication. On normalized bar chains the induced permutation reverses the suspended letters, producing exactly the Koszul sign of that permutation. □

The corollary produces opposite multiplication. It does not invert a braid, because a braid is absent in one real dimension.

Part VI

The chiral coefficient and its resolutions

Chapter 40

The diagonal as the primitive chiral coefficient

40.1 From singular products to geometry

A quantum field is not multiplied at a point. Two fields are first placed at distinct points, where their product is defined, and then the product is extended toward coincidence. The singular part of this extension is the operator product expansion. The regular part is its renormalized continuation. Both are carried by the diagonal.

Let X be a smooth complex curve and let

$$j_2 : X^2 \setminus \Delta(X) \hookrightarrow X^2, \qquad \Delta : X \hookrightarrow X^2$$

be the open complement and the diagonal embedding. A chiral operation on a right D_X-module $\mathfrak{g}$ is a morphism

$$\mu : j_{2,*}j_2^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_!\mathfrak{g}. \tag{40.1}$$

The source allows arbitrary finite poles along the diagonal. The target records that the singular product is supported where the two points meet. Antisymmetry and Jacobi make $\mathfrak{g}$ a chiral Lie algebra in the sense of Beilinson and Drinfeld [49].

Equation (40.1) is the common language of several classical constructions. In a formal coordinate it is the singular part of a vertex operator. In distribution theory it is a distribution supported on the diagonal. In perturbative quantum field theory it is a local counterterm or contact operation. In factorization theory it is the first nontrivial structure map. In representation theory its horizontal sections produce differential equations for conformal blocks. The categories differ, but the support condition does not.

Principle 40.1 (Diagonal principle). The invariant datum of a chiral product is the morphism supported on the diagonal. Laurent coefficients, residues, modes, propagators, and contact terms are presentations of that morphism after auxiliary choices.

The principle has a sharp consequence. No single differential form can encode every operator product coefficient. The logarithmic form $d\log(z-w)$ detects a first normal coefficient. A double pole, a fourth-order pole, and a regular normally ordered term belong to different pieces of the normal filtration. The complete distribution is primary; its coefficients are derived from it.

40.2 The historical line

Dirac's delta distribution made support a mathematical operation. Wilson's operator product expansion made short distance a hierarchy of local fields. Borcherds compressed the consistency of that hierarchy into the vertex identity. Beilinson and Drinfeld replaced formal coordinates by D-modules and diagonals. Fulton and MacPherson resolved simultaneous collisions. Segal, Tsuchiya–Ueno–Yamada, and the theory of conformal blocks organized sewing. Getzler and Kapranov encoded stable sewing by modular operads. Drinfeld and Kohno identified braid monodromy with quantum groups. Costello and Gwilliam expressed perturbative quantum field theory as a factorization algebra of observables. The chiral coefficient is the common diagonal-supported structure in these constructions; the transverse E_1 multiplication is an independent ordered operation on the same factorisable coefficient.

The overlap becomes visible only when the order of the objects is respected:

$$\text{diagonal operation} \xrightarrow{\mathrm{Chev}^{\mathrm{ch}}} \text{factorization coalgebra on } \mathrm{Ran}(X) \xrightarrow{\mathrm{Prim}} \text{chiral Lie algebra} \tag{40.2}$$

The same local operation has three independent enlargements:

$$\begin{tikzcd}
 & \text{labelled configurations} \arrow[d, "\text{monodromy}"] & \\
\text{formal disc} \arrow[r, "\text{coordinate expression}"] & \text{chiral operation} \arrow[l, bend left, "\text{localization}"] \arrow[r, "+\,\text{cyclic sewing data}"] \arrow[d, "\text{Ran globalization}"'] & \text{stable curves} \arrow[d, "\text{master equation}"] \\
 & \text{factorization coalgebra} & \text{modular graph operations}
\end{tikzcd} \tag{40.3}$$

Every arrow in these diagrams is a specified functor, comparison map, or monodromy construction.

40.3 The central theorem

The first theorem identifies the geometry that lies between the diagonal operation and the Ran coalgebra.

Theorem 40.2 (Geometric collision theorem)**.** *Let X be a smooth algebraic curve over a field of characteristic zero and let $\mathfrak{g}$ be a chiral Lie algebra on X. For every nonempty finite set I there are three canonically related complexes.*

(i) *The partition collision complex*

$$C_I^{\mathrm{coll}}(\mathfrak{g}) = \bigoplus_{\pi \in \mathrm{Part}(I)} \Delta_{\pi,!}\left(\bigotimes_{B\in\pi}^{\mathrm{unord}} \mathfrak{g}[1]\right),$$

with differential given by merging two blocks and applying the suspended chiral bracket.

(ii) *The corolla-normalized forest complex $W_I^{\mathrm{FM}}(\mathfrak{g})$, obtained by resolving each block of a partition by the normalized chains of the rooted-tree category and coupling this vertical resolution to the collision differential.*

(iii) *The labelled component $\mathrm{coChev}_I^{\mathrm{ch}}(\mathfrak{g})$ of the chiral Chevalley coalgebra.*

The square of the differential on $C_I^{\mathrm{coll}}(\mathfrak{g})$ is the suspended chiral Jacobiator. There are *canonical quasi-isomorphisms*

$$W_I^{\mathrm{FM}}(\mathfrak{g}) \xrightarrow{\sim} C_I^{\mathrm{coll}}(\mathfrak{g}) \xrightarrow{\cong} \mathrm{coChev}_I^{\mathrm{ch}}(\mathfrak{g}). \tag{40.4}$$

The first map applies the augmentation from each rooted-tree nerve to its terminal corolla. The second identifies the partition decomposition with the cofree commutative coalgebra in the chiral monoidal category.

Proof. For the partition complex, the square of the cover differential is the sum over length-two intervals of the partition lattice. Disjoint mergers cancel with opposite orientation, and triple mergers give the suspended chiral Jacobiator; the chiral Lie hypothesis therefore makes the square zero. For each block, the normalized nerve of its rooted-tree category contracts to the terminal corolla. Tensoring these contractions over the blocks gives the first quasi-isomorphism. The cofree commutative coalgebra in the chiral monoidal category decomposes over surjections $I \twoheadrightarrow \pi$, equivalently over partitions of I, and its Chevalley coderivation merges two blocks by the chiral bracket. This gives the second identification and proves naturality under relabelling and disjoint union. □

The theorem is proved in Chapters 42, 43 and 46. Its force is conceptual. The partition lattice carries the algebraic differential. The Fulton–MacPherson boundary supplies a geometric resolution of the incidence pattern. The Ran space globalizes the result into a factorization coalgebra. Screen configuration spaces and differential forms are genuine additional coefficients; they are not silently collapsed to a point.

40.4 The formal disc and the full operator product

Choose a formal coordinate z on a disc. A weakly translation-equivariant chiral algebra becomes a vertex algebra, and the morphism (40.1) becomes the full operator product

$$Y(a,z)Y(b,w) \sim \sum_{n\geq 0} \frac{Y(a_{(n)}b,w)}{(z-w)^{n+1}}. \tag{40.5}$$

A general coordinate change mixes the modes $a_{(n)}b$ triangularly and preserves the filtered singular distribution. The quotient of pole order $n+1$ by pole order n is a tensor power of the normal line to the diagonal. This is the geometric reason that modes transform triangularly.

The algebraic chiral operad $\mathcal{P}^{\mathrm{ch}}$ makes the statement exact. If V is a vector superspace with translation operator, an odd element $X \in W_1^{\mathrm{ch}}(\Pi V)$ determines a nonunital vertex algebra exactly when

$$[X,X] = 0. \tag{40.6}$$

The associated coderivation obeys

$$d_X^2 = \frac{1}{2} d_{[X,X]}. \tag{40.7}$$

Thus the full vertex identity is the coordinate expression of the chiral Jacobi identity. Arnold's relation contributes to the geometry of configuration-space forms, but it does not replace the Borcherds identity and does not manufacture higher poles from a logarithmic residue.

40.5 Five exact separations

A single language does not mean a single object. The following separations are structural.

First, the chiral Chevalley coalgebra is not a quadratic dual algebra. It exists without finite-dimensionality. A dual algebra appears only after a duality functor has been proved continuous on the finite labelled components and their completed inverse system.

Second, braid monodromy is not a quantum group presentation. Monodromy gives a braided category or a local-system representation. Tannaka reconstruction with a strong or quasi-monoidal fibre functor produces a bialgebra/Hopf or coquasi/quasi-Hopf object; a further recognition theorem produces a named quantum group.

Third, projective curvature on moduli is not the square of a fiberwise differential. It is the curvature of a connection or the class of an Atiyah algebra. Tensoring by an inverse anomaly line cancels it when that line exists.

Fourth, the derived center is not automatically the physical bulk. It is the universal recipient of compatible algebraic actions. A bulk theory determines a map to the center; equality is a separate theorem.

Fifth, numerical invariants are traces of structures. Central charge, level, rank, Hodge classes, theta coefficients, and entropy coefficients are not interchangeable because they inhabit different functors.

These separations remove false universal statements while preserving the full mathematical and physical content.

40.6 Mathematical conventions

All derived categories are stable and $\mathbb{C}$-linear. Right D-modules are used on algebraic curves. For a finite set I, X^I is the labelled Cartesian power, $\mathrm{Part}(I)$ is its partition lattice, and $\Delta_\pi : X^\pi \hookrightarrow X^I$ is the partial diagonal associated with π. Tensor products indexed by unordered finite sets are taken in the symmetric monoidal derived category; the suspension [1] carries the Koszul signs. The symbol $\mathrm{FM}_I(X)$ denotes the algebraic Fulton–MacPherson compactification. Real oriented compactifications are used only in the analytic and field-theoretic sections, where Stokes' theorem is explicit.

For a vertex algebra, modes follow the convention

$$Y(a, z)b = \sum_{n\in\mathbb{Z}} (a_{(n)}b)z^{-n-1}.$$

For a simple Lie algebra $\mathfrak{g}$, the invariant form and the dual Coxeter number are normalized so that the KZ parameter is $\hbar = (k + h^\vee)^{-1}$; other conventions change only the displayed exponent defining q.

Chapter 41

Chiral operations on a curve

41.1 The chiral tensor product

Let X be a smooth algebraic curve. The ordinary tensor product of right D_X-modules describes fields that occupy the same point. A chiral product must first separate the points. For two right D_X-modules M and N, write

$$j : X^2 \setminus \Delta(X) \hookrightarrow X^2.$$

The binary chiral tensor product is represented by $j_* j^*(M \boxtimes N)$, followed by a pushforward to the diagonal when an operation is evaluated. The distinction between j_* and $\Delta_!$ is the distinction between a meromorphic product away from collision and a distribution supported at collision.

For a finite set I, the open configuration locus is

$$U_I = X^I \setminus \bigcup_{i \neq j} \Delta_{ij}, \qquad j_I : U_I \hookrightarrow X^I.$$

An I-ary chiral operation with values in a right D_X-module M is a morphism

$$\mu_I : j_{I,*} j_I^* (\boxtimes_{i \in I} M_i) \longrightarrow \Delta_{I,!} M, \tag{41.1}$$

where $\Delta_I : X \hookrightarrow X^I$ is the small diagonal. Partial compositions are defined by allowing a subset of points to collide first, applying one operation, and then colliding the resulting point with the remaining points. The associativity of these partial compositions is a base-change theorem for the corresponding diagrams of open embeddings and diagonals.

Definition 41.1 (Chiral Lie algebra)**.** A chiral Lie algebra on X is a right D_X-module $\mathfrak{g}$ with an antisymmetric operation

$$\mu : j_* j^* (\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}$$

whose three partial composites satisfy the chiral Jacobi identity.

The word "Lie" refers to the symmetry and Jacobi law of the operation. It does not mean that the coefficients of the operation form an ordinary finite-dimensional Lie algebra. The source contains arbitrary poles and derivatives. On a formal disc, the chiral Jacobi identity becomes the vertex identity.

41.2 The Jacobi identity as a relation among partial diagonals

Let $I = \{1, 2, 3\}$. There are three pair diagonals and one small diagonal. The composite that first collides 1 with 2 and then collides the result with 3 is a morphism supported on Δ_{123}. The two other parenthesizations have the same target. After the symmetry signs are inserted, their sum is the Jacobiator

$$J_\mu = \mu \circ_{12} \mu + \mu \circ_{23} \mu + \mu \circ_{31} \mu. \tag{41.2}$$

The chiral Jacobi identity is $J_\mu = 0$.

The support geometry already predicts the form of the relation. Two disjoint collisions commute. Three points colliding into one admit three intermediate pair collisions. Every square of a collision differential therefore splits into disjoint-pair terms and triple-collision terms. The former cancel by the Koszul sign; the latter are the Jacobiator. This observation is made precise in Chapter 42.

41.3 Units and the vacuum

A unital chiral algebra contains a morphism from the dualizing module ω_X that plays the role of the vacuum. The unit axiom says that a field inserted next to the vacuum extends regularly across the diagonal and returns the field. On the formal disc this becomes

$$Y(\mathbf{1}, z) = \mathrm{id}, \qquad Y(a, z)\mathbf{1} \in a + zV[[z]].$$

The nonunital operation and its Jacobi identity contain the singular and regular binary product. The vacuum is additional nullary data. This separation is useful because chiral Koszul duality is naturally formulated for nonunital or augmented objects, while vertex operator algebras are ordinarily presented with a vacuum.

41.4 Chiral algebras and factorization algebras

A factorization algebra on X assigns an object to each finite configuration and identifies the object on disjoint unions with the tensor product of the objects on the pieces. A chiral algebra supplies the infinitesimal collision law from which this factorization structure is generated. Beilinson and Drinfeld construct the chiral enveloping algebra and the corresponding factorization algebra; Francis and Gaitsgory place the relation in a derived categorical setting [49, 82].

The relation may be read in two directions. Starting from a chiral Lie algebra, the chiral Chevalley construction forms a commutative factorization coalgebra. Starting from a conilpotent factorization coalgebra, primitives recover the chiral Lie algebra. The equivalence is not a linear duality and does not require the underlying D-module to be finite-dimensional. It is a Koszul duality internal to the chiral monoidal category.

Historical note 41.2. Beilinson and Drinfeld's use of right D-modules incorporates the transformation law of fields into the definition. A coordinate is used to calculate, not to define. This is the decisive step that separates a chiral algebra from a list of formal Laurent series.

41.5 Distributions and contact terms

The map (40.1) has an analytic analogue. A distribution on X^2 supported on the diagonal is a finite sum of normal derivatives of the delta distribution with coefficients along the diagonal. In a coordinate z,

$$T(z,w) = \sum_{m=0}^{N} A_m(w)\,\partial_z^m \delta(z-w). \tag{41.3}$$

Cauchy's formula identifies the same coefficients with pole terms in a meromorphic kernel. The formal-distribution identity

$$\partial_w^m \delta(z-w) = \iota_{z,w}\frac{(-1)^m m!}{(z-w)^{m+1}} - \iota_{w,z}\frac{(-1)^m m!}{(z-w)^{m+1}}$$

relates the two expansions. Thus a chiral operation may be read either as a principal part or as a contact distribution. The equality is between two presentations of one diagonal-supported object.

In perturbative field theory, local counterterms are also distributions supported on diagonals. Renormalization extends distributions from configuration space to the compactification. The extension is not unique, but the ambiguity is local and supported on collision strata. Chiral algebra encodes the allowed local extensions and their consistency relations.

41.6 The principal-part hierarchy

Let $D \subset Y$ be a smooth divisor. Poincare residue is a map

$$\mathrm{Res}_D : \Omega_Y^p(\log D) \longrightarrow \Omega_D^{p-1}.$$

Poincaré residue detects the coefficient of dt/t in a logarithmic form. It does not detect the coefficient of t^{-m} for $m > 1$ in a meromorphic function without further differential or jet data. The normal-order filtration gives the following statement.

No-go theorem 41.3 (No universal logarithmic extractor). There is no coordinate-free linear map from logarithmic forms along a smooth divisor to all finite principal parts that agrees, in a coordinate t, with extraction of the coefficient of t^{-m} for every $m \geq 1$. Logarithmic forms contain one normal generator dt/t, whereas the successive principal-part quotients are the mutually distinct line bundles $N_{D/Y}^{\otimes m}$.

Proof. The exact sequence $0 \to \mathcal{O}_Y((m-1)D) \to \mathcal{O}_Y(mD) \to N_{D/Y}^{\otimes m} \to 0$ identifies the order-m symbol with a section of $N_{D/Y}^{\otimes m}$. Under $t' = ut + O(t^2)$ its coordinate coefficient changes by the mth power of the transition of the normal line. By contrast, Poincaré residue is the quotient map $\Omega_Y^p(\log D) \to \Omega_D^{p-1}$ arising from the single logarithmic generator dt/t. The logarithmic complex contains no canonical copies of all $N_{D/Y}^{\otimes m}$, so it cannot perform all coefficient extractions. □

The correct replacement is the full pole filtration, constructed in Chapter 44.

41.7 The affine-line presentation

On $X = \mathbb{A}^1$ with coordinate z, translation-invariant sections of $j_*\mathcal{O}_{X^2\setminus\Delta}$ are Laurent polynomials in $z_1 - z_2$. A chiral operation can therefore be written as a finite sum

$$\{a_\lambda b\} = \sum_{n\geq 0} \frac{\lambda^n}{n!} a_{(n)} b, \tag{41.4}$$

combined with the regular product. The relation

$$\{\partial a_\lambda b\} = -\lambda\{a_\lambda b\}, \qquad \{a_\lambda \partial b\} = (\partial + \lambda)\{a_\lambda b\}$$

is the coordinate form of D-module covariance. The Jacobi identity becomes

$$\{a_\lambda\{b_\mu c\}\} - (-1)^{p(a)p(b)}\{b_\mu\{a_\lambda c\}\} = \{\{a_\lambda b\}_{\lambda+\mu} c\}. \tag{41.5}$$

The binomial coefficients of the Borcherds identity arise by expanding the translated variable $\lambda+\mu$. They are not produced by the Arnold relation. The Arnold relation controls differential forms on configuration space; equation (41.5) controls the coefficient operation.

41.8 Factorization as locality

Suppose $U_1, \dots, U_n \subset X$ are pairwise disjoint discs contained in a larger disc V. A factorization algebra supplies a product

$$\mathcal{F}(U_1) \otimes \cdots \otimes \mathcal{F}(U_n) \longrightarrow \mathcal{F}(V).$$

On the disjointness locus, isotopic motion of the discs transports the products as a local system. When discs collide, the chiral operation describes the singular extension. This is the precise relation between OPE and locality: the OPE is the boundary value of factorization.

The factorization axiom is stronger than associativity at a point. It compares all decompositions by disjoint subdiscs and is compatible with isotopy. On a complex curve the holomorphic dependence of the products adds the D-module structure. The result is a geometric object that can be transported, integrated, and sewn.

41.9 The physical reading

In Euclidean quantum field theory, a local observable supported in a small disc is an element of a cochain complex. Disjoint observables multiply. The product depends smoothly or holomorphically on their positions. A quantization satisfying locality therefore determines a factorization algebra [58, 59]. If one direction is holomorphic and the other directions are topological, restriction to a holomorphic boundary produces a chiral algebra. The local operation (40.1) is then the renormalized collision of boundary observables.

The mathematical and physical statements agree once a comparison map has been supplied between the algebraic D-module and the analytic factorization algebra. Without that map they are parallel realizations. The distinction is essential because a physical theory contains metric, reality, positivity, and global state-space data that a chiral algebra does not remember.

Chapter 42

The partition collision complex

42.1 Partitions and partial diagonals

Let I be a finite nonempty set. A partition π of I is a collection of disjoint nonempty blocks whose union is I. The refinement order is written $\pi \leq \rho$ when every block of π is contained in a block of ρ. The minimal partition has singleton blocks; the maximal partition has one block.

To π associate the partial diagonal

$$\Delta_\pi : X^\pi \hookrightarrow X^I,$$

where the coordinates indexed by a block are required to coincide. If $\pi \lessdot \rho$ is a cover relation, then ρ is obtained from π by merging exactly two blocks. Every saturated chain in $\mathrm{Part}(I)$ is an order of pairwise collisions.

Let $s\mathfrak{g} = \mathfrak{g}[1]$. For a partition π, set

$$C_\pi(\mathfrak{g}) = \Delta_{\pi,!}\left(\bigotimes_{B\in\pi}^{\mathrm{unord}} s\mathfrak{g}\right). \tag{42.1}$$

The unordered tensor product is the coinvariant form of the tensor product in the symmetric monoidal derived category. The suspension is not cosmetic: it turns the antisymmetric chiral bracket into a symmetric degree-one contraction, exactly as in the Chevalley complex of an ordinary Lie algebra.

Definition 42.1 (Partition collision complex)**.** The I-labelled collision object is

$$C_I^{\mathrm{coll}}(\mathfrak{g}) = \bigoplus_{\pi\in\mathrm{Part}(I)} C_\pi(\mathfrak{g}). \tag{42.2}$$

For a cover $\pi \lessdot \rho$ that merges blocks B_1 and B_2, the component

$$d_{\pi,\rho} : C_\pi(\mathfrak{g}) \longrightarrow C_\rho(\mathfrak{g})$$

applies the suspended chiral bracket to the factors indexed by B_1, B_2 and acts identically on the remaining factors. The total differential is the internal differential plus the signed sum over covers.

The sign is determined by moving the two suspended factors next to one another, applying the degree-one bracket, and returning the merged factor to the position determined by the orientation line of the cover. Because the tensor product is unordered, the definition is independent of a chosen linear order on the blocks.

42.2 The square of the collision differential

Every length-two interval $[\pi, \sigma]$ in the partition lattice has one of two forms.

If σ is obtained by merging two disjoint pairs of blocks, there are exactly two intermediate partitions. The coefficient operations act on disjoint tensor factors and commute before suspension. Interchanging the two degree-one contractions changes the sign, so the two paths cancel.

If σ is obtained by merging three blocks into one, there are exactly three intermediate partitions. The three composites are the three terms of the suspended Jacobiator.

Theorem 42.2 (Square of the collision differential). *Let μ be an antisymmetric chiral binary operation on a right D_X-module $\mathfrak{g}$. The collision operator defined from μ satisfies*

$$d_{\mathrm{coll}}^2 = \frac{1}{2} d_{[\mu,\mu]}. \tag{42.3}$$

Consequently $C_I^{\mathrm{coll}}(\mathfrak{g})$ is a complex for every finite I if and only if μ satisfies the chiral Jacobi identity.

Proof. The internal differential squares to zero and its mixed commutator with d_{coll} vanishes precisely when μ is a chain map. The remaining square is a sum over length-two intervals in $\mathrm{Part}(I)$. Disjoint mergers cancel in pairs by the Koszul sign. Triple mergers sum to the three suspended partial composites of μ. This sum is $\frac{1}{2}[\mu, \mu]$ in the convolution Lie algebra of chiral operations. No other length-two interval occurs. □

Corollary 42.3 (Selection by collisions). *An antisymmetric collection of diagonal-supported binary operations defines a chiral Lie algebra exactly when the corresponding partition collision operators form differentials in all arities.*

Proof. By Theorem 42.2, the square in every arity is the coderivation induced by the chiral Jacobiator. Vanishing of the Jacobiator makes every collision operator a differential. Conversely, arity three detects each component of the Jacobiator on the small diagonal, so square-zero in all arities implies the chiral Jacobi identity. □

The partition differential gives the coordinate-free geometric selection statement. It uses the complete chiral operation. It does not identify the Jacobi identity with a relation among differential forms and does not require a central term to repair an ordinary Lie bracket.

42.3 Arity two and three

For $I = \{1, 2\}$ there are two partitions. The complex is

$$\Delta_{1|2,!}(s\mathfrak{g} \boxtimes s\mathfrak{g}) \xrightarrow{s\mu} \Delta_{12,!}(s\mathfrak{g}).$$

The map is the collision operation.

For $I = \{1, 2, 3\}$, the Hasse diagram is

$$\begin{tikzcd}
& 1|2|3 \arrow[dl] \arrow[d] \arrow[dr] & \\
12|3 \arrow[dr] & 13|2 \arrow[d] & 23|1 \arrow[dl] \\
& 123 &
\end{tikzcd} \tag{42.4}$$

The square from the top to the bottom is the sum of three composites. In a coordinate presentation these are the three expansions of the vertex Jacobi identity. In the D-module presentation they are three maps with the same source and target, supported on the small diagonal.

42.4 The cofree commutative coalgebra

Let $\mathcal{C}$ be a symmetric monoidal stable category. The cofree conilpotent commutative coalgebra on an object V has underlying object

$$\mathrm{coSym}(V) = \bigoplus_{n \geq 1} (V^{\otimes n})_{\Sigma_n}.$$

In the chiral monoidal category on the Ran space, the labelled component of $\mathrm{coSym}(s\mathfrak{g})$ decomposes by the ways in which labels are identified. Those ways are the partitions of I. The summand associated with π is exactly (42.1).

Proposition 42.4 (Chevalley identification)**.** *For a chiral Lie algebra $\mathfrak{g}$, the labelled collision complex is canonically the I-component of its chiral Chevalley coalgebra:*

$$C_I^{\mathrm{coll}}(\mathfrak{g}) \cong \mathrm{coChev}_I^{\mathrm{ch}}(\mathfrak{g}). \tag{42.5}$$

Under this identification the collision differential is the Chevalley coderivation induced by the chiral bracket.

Proof. The cofree commutative coalgebra is the sum over finite collections of primitive factors. Pulling its Ran component back to X^I records a surjection from I to the set of primitive factors. A surjection modulo relabeling of its target is a partition of I. The Chevalley coderivation merges two primitive factors by the bracket, hence agrees with the cover differential of the partition lattice. $\square$

42.5 The factorization coproduct

Suppose $I = I_1 \sqcup I_2$. Restriction to the open locus where the points of I_1 remain disjoint from the points of I_2 gives a coproduct

$$C_I^{\mathrm{coll}}(\mathfrak{g}) \longrightarrow C_{I_1}^{\mathrm{coll}}(\mathfrak{g}) \boxtimes C_{I_2}^{\mathrm{coll}}(\mathfrak{g}). \tag{42.6}$$

A partition contributes to this restriction exactly when every block lies entirely in I_1 or entirely in I_2. The coproduct is therefore deconcatenation at the level of primitive blocks.

Coassociativity follows from the associativity of disjoint union.

Proposition 42.5. *The family* $\{C_I^{\mathrm{coll}}(\mathfrak{g})\}_I$*, with the coproducts* (42.6)*, is a conilpotent commutative factorization coalgebra. Its primitive part is* $\mathfrak{g}[1]$*.*

Proof. On the disjointness locus for $I = I_1 \sqcup I_2$, exactly the partitions whose blocks remain inside one side survive. Restriction therefore factors uniquely as the tensor product of the two collision complexes and gives the displayed coproduct. Refining a three-fold disjoint union proves coassociativity. The block-number filtration is the coradical filtration: every reduced coproduct strictly increases the number of tensor factors, so iteration terminates after at most $|I| - 1$ steps. The one-block summand is the primitive copy of $\mathfrak{g}[1]$. □

Conilpotence is measured by the number of blocks. Repeated reduced coproducts eventually reach singleton partitions. This filtration is intrinsic and requires no finite-dimensional dual.

42.6 Incidence geometry and Möbius inversion

The Euler characteristic of the collision complex is governed by the Möbius function of the partition lattice. For the interval from the discrete partition to the one-block partition,

$$\mu_{\mathrm{Part}(I)}(\hat{0}, \hat{1}) = (-1)^{|I|-1}(|I| - 1)!.$$

The same integer is the top reduced Euler characteristic of the proper partition complex. It explains the factorial multiplicities that appear in Lie operad components and in the top cohomology of configuration spaces. The equality is combinatorial; the representations carried by the two sides require their respective coefficient systems.

The proper partition complex is homotopy equivalent to a wedge of $(|I| - 1)!$ spheres of dimension $|I| - 3$. Its top homology is the multilinear part of the Lie operad, up to the sign representation. This is the classical topological shadow of the Lie–commutative Koszul duality that becomes chiral on the Ran space.

42.7 Modules and marked points

Let M be a chiral module over $\mathfrak{g}$. Add a distinguished label $*$ that is never merged with another module label. A partition of $I \sqcup \{*\}$ with a distinguished block containing $*$ indexes a summand in the module collision complex. Merging an ordinary block with the distinguished block applies the module action; merging two ordinary blocks applies the bracket. The same length-two interval analysis proves that the square is the sum of the Lie Jacobiator and the module identity.

This pointed complex is the geometric origin of derived coinvariants and conformal blocks with insertions. It is also the correct place to compare BRST reduction, because the BRST differential acts on both the algebra and the marked module.

42.8 Screen coefficients beyond partition incidence

The partition complex records which groups of points collide and in what incidence pattern. It does not record the relative shape of a collision. For three points on a complex line, the ratio of their relative positions survives after translation and dilation. That ratio belongs to the screen configuration space of the Fulton–MacPherson boundary. Differential forms on screens produce Arnold and Orlik–Solomon classes. Fulton–MacPherson screen coefficients adjoin this relative geometry to the partition incidence complex.

Chapter 43

The Fulton–MacPherson forest resolution

43.1 Compactifying collisions

The configuration space $\mathrm{Conf}_I(X)$ is the complement of the diagonals in X^I. Its Fulton–MacPherson compactification $\mathrm{FM}_I(X)$ is obtained by successively blowing up partial diagonals in an order compatible with inclusion [84]. The result is smooth, and the boundary is a normal-crossing divisor. A boundary point records not only which labels collide but also the limiting relative configuration at every scale.

Boundary strata are indexed by nested collections of subsets, equivalently by rooted forests whose leaves are the labels in I. An internal vertex represents a cluster. An edge represents one scale of collapse. Roots represent the final clusters that remain distinct in X.

Let F be such a forest. The root components determine a partition $q(F)$ of I. The corresponding stratum maps to the partial diagonal $\Delta_{q(F)}(X^{q(F)})$. Its fiber is a product of screen configuration spaces. On a curve, a screen for a cluster S is the compactification of configurations of $|S|$ points on a one-dimensional tangent space modulo translation and dilation.

43.2 Tree categories over collision blocks

For a nonempty finite set B, let $\mathrm{Tree}(B)$ be the category of rooted trees with leaves labelled by B and with no unary internal vertices. Morphisms contract internal edges. The corolla with leaf set B is terminal. Write

$$K(B) = N_*(N\,\mathrm{Tree}(B)), \tag{43.1}$$

for the normalized chain complex of its nerve, placed in nonpositive cohomological degrees. The augmentation and the terminal-corolla inclusion are chain maps

$$K(B) \xrightarrow{\epsilon_B} \mathbb{C}, \qquad \mathbb{C} \xrightarrow{\iota_B} K(B), \qquad \epsilon_B \iota_B = \mathrm{id}\,. \tag{43.2}$$

Because $\mathrm{Tree}(B)$ has a terminal object, ϵ_B is a quasi-isomorphism.

For a partition π of I, define

$$K(\pi) = \bigotimes_{B\in\pi}^{\text{unord}} K(B). \tag{43.3}$$

An element of $K(\pi)$ is a chain of contractions in a forest whose root components are the blocks of π. This is the incidence information carried by the face category of the Fulton–MacPherson boundary; it contains no cohomology of the screen spaces.

43.3 The corolla-normalized forest resolution

Set

$$W_I^{\mathrm{FM}}(\mathfrak{g}) = \bigoplus_{\pi\in\mathrm{Part}(I)} C_\pi(\mathfrak{g}) \otimes K(\pi). \tag{43.4}$$

Its differential has three terms. The internal term is induced by the differential of $\mathfrak{g}$. The vertical term is the normalized nerve differential on the factors $K(B)$. For a cover $\pi \lessdot \rho$ merging blocks B, C, the horizontal term applies the suspended chiral bracket to the B, C coefficient factors and applies the chain map

$$K(B) \otimes K(C) \xrightarrow{\epsilon_B\otimes\epsilon_C} \mathbb{C} \xrightarrow{\iota_{B\cup C}} K(B\cup C) \tag{43.5}$$

while leaving the other tree factors unchanged. Thus a collision of two root components is represented by the terminal corolla of the merged block; all nonterminal tree data remains in the vertical resolution.

The maps (43.5) are symmetric and strictly associative because they factor through the augmentations. They commute with the vertical differential. Consequently the square of the horizontal differential is exactly the square of the partition collision differential, with a common terminal-corolla factor.

There is an augmentation

$$\epsilon_I : W_I^{\mathrm{FM}}(\mathfrak{g}) \longrightarrow C_I^{\mathrm{coll}}(\mathfrak{g}) \tag{43.6}$$

obtained by applying ϵ_B to every tree factor.

Theorem 43.1 (Forest resolution)**.** *For every chiral Lie algebra $\mathfrak{g}$ and finite set I, the differential on* (43.4) *squares to zero and the augmentation* (43.6) *is a quasi-isomorphism. The construction is natural under relabelling and compatible with disjoint union.*

Proof. The vertical differential squares to zero. Its mixed commutator with the horizontal differential vanishes because the augmentations and corolla inclusions in (43.5) are chain maps. Strict associativity of the corolla-normalized merge identifies the horizontal square with the Jacobiator computed in Theorem 42.2; it therefore vanishes for a chiral Lie algebra.

Filter by the total nerve degree of the tree factors. The filtration is finite for fixed I. Its first spectral sequence computes the homology of every $K(B)$. Since each ϵ_B is a quasi-isomorphism, the first page is precisely $C_I^{\mathrm{coll}}(\mathfrak{g})$, and the induced differential is its collision differential. The spectral-sequence comparison proves that ϵ_I is a quasi-isomorphism. Relabelling preserves terminal corollas, and disjoint union tensors the tree complexes, giving naturality and factorization compatibility. ☐

Corollary 43.2. *The corolla-normalized forest complex resolves the chiral Chevalley coalgebra:*

$$W_I^{\mathrm{FM}}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{coChev}_I^{\mathrm{ch}}(\mathfrak{g}).$$

Proof. Compose the quasi-isomorphism of the forest resolution with the canonical identification of the partition collision complex and the labelled chiral Chevalley coalgebra. Both maps respect relabelling and factorisation coproducts, so the composite is a quasi-isomorphism of the labelled coalgebra systems. □

The word "geometric" refers to the tree categories labelling the faces of the compactification. It does not assert that the actual screen fibers are contractible or that their de Rham complexes are removed by the augmentation.

43.4 Screen spaces are genuine coefficients

Let $p_I : \mathrm{FM}_I(X) \to X^I$ be the proper blowdown. The fiber over a point of a partial diagonal contains compactified screen configurations. For $|I| = 3$, the screen modulo translation and dilation has a nontrivial moduli coordinate. Its compactification is not a point. Therefore the derived pushforward of the constant sheaf on $\mathrm{FM}_I(X)$ is not, in general, the partition incidence complex.

No-go theorem 43.3 (No contraction of screen cohomology). The existence of a terminal corolla in the forest category does not imply that the proper geometric fibers of p_I are contractible. Consequently a quasi-isomorphism from a de Rham or constructible complex on $\mathrm{FM}_I(X)$ to the collision complex requires a comparison map that accounts for screen cohomology.

The distinction separates two independent structures:

incidence of collision strata	geometry inside a screen
partition lattice	configuration space
edge contraction	de Rham differential
chiral Jacobi	Arnold–Orlik–Solomon relations
Chevalley coderivation	propagator and period classes

43.5 The stratification spectral sequence

Filter $\mathrm{FM}_I(X)$ by the codimension of its boundary strata. For a constructible complex $\mathcal{F}$ that is locally constant along strata, the standard spectral sequence of a filtered space has first page

$$E_1^{p,q} = \bigoplus_{\operatorname{codim} S = p} H_c^{p+q}(S; \mathcal{F}|_S \otimes \mathrm{or}_S) \Longrightarrow H_c^{p+q}(\mathrm{FM}_I(X); \mathcal{F}). \tag{43.7}$$

The differential is the signed sum of incidence maps between adjacent strata.

For coefficients pulled back from the root diagonal, a stratum splits locally into a base and a product of screens. The Kunneth theorem expresses its cohomology as

$$H_c^*(S; \mathcal{F}|_S) \cong H_c^*(\Delta_{q(F)}; \mathcal{F}_{q(F)}) \otimes \bigotimes_{v \in V_{\mathrm{int}}(F)} H_c^*(\mathrm{Screen}(v)). \tag{43.8}$$

After barycentric replacement of the face incidence category and corolla normalization, the degree-zero screen classes map to the forest incidence resolution. Positive-degree screen classes record the topology of relative configurations.

Theorem 43.4 (Incidence–screen separation)**.** *For a coefficient system constant along Fulton–MacPherson strata, the boundary-codimension spectral sequence separates the collision incidence maps from the cohomology of screen configuration spaces. After replacing the face poset by its normalized nerve and applying the terminal-corolla contractions blockwise, the row obtained from degree-zero screen classes maps quasi-isomorphically to the forest resolution* W_I^{FM}*.*

Proof. Filter the stratified coefficient complex by boundary codimension. The associated graded is the direct sum over strata of the tensor product of the orientation line, the incidence chain of the face category, and the cohomology complex of the screen coefficient. The first differential changes only the incidence factor; the internal screen differential is vertical. Replacing each rooted-tree fibre by its normalized nerve and contracting it to the terminal corolla identifies the incidence factor with the partition collision complex. Screen cohomology remains as the second tensor factor. Hence the spectral sequence separates the two kinds of data and contracts only the incidence resolution. □

This theorem is the precise meeting point of the chiral Chevalley complex and configuration-space forms. The two structures interact through the spectral-sequence differential but are not identified term by term.

43.6 Arnold forms on screens

On an affine coordinate chart, set

$$\eta_{ij} = d\log(z_i - z_j).$$

On $\mathrm{Conf}_3(\mathbb{C})$ the Arnold relation is

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0. \tag{43.9}$$

The substitution $x = z_1 - z_2$, $y = z_2 - z_3$ proves the identity. The Orlik–Solomon algebra of the braid arrangement is generated by the classes of the η_{ij} with these three-term relations [48].

The Arnold relation expresses flatness of the universal logarithmic connection once the coefficients satisfy infinitesimal braid relations. It does not express the full chiral Jacobi identity. In the stratification spectral sequence, (43.9) belongs to the screen coefficient system; the chiral Jacobi identity belongs to the coefficient operation on the root diagonal. A square-zero total differential may use both, but each has its own source.

43.7 Real oriented compactification and Stokes' theorem

For perturbative integrals one uses the real oriented blowup of the diagonals. The boundary hypersurfaces carry outward normal orientations. If α is a compactly supported form of top degree minus one, Stokes' theorem gives

$$\int_{\mathrm{FM}_I^{\mathrm{or}}(M)} d\alpha = \sum_{H \subset \partial\, \mathrm{FM}_I^{\mathrm{or}}(M)} \int_H \alpha.$$

Applying Stokes twice yields cancellation of codimension-two corners because each corner occurs with opposite induced orientations from its two parent hypersurfaces. This is the analytic form of the incidence differential squaring to zero.

When forms carry algebraic coefficients, the total square contains three distinct terms: the de Rham square, the incidence square, and the coefficient Jacobiator. A correct proof specifies the bidegrees and the mixed commutators. The vanishing of boundary-of-boundary alone cannot create the coefficient identities.

43.8 Renormalization forests

The same forest combinatorics appears in the BPHZ forest formula. A divergent subgraph determines a collision subconfiguration; nested divergent subgraphs determine a forest; subtraction is performed recursively from smaller clusters to larger clusters. Fulton–MacPherson compactification turns this recursion into geometry. Counterterms are supported on boundary strata, and locality is the compatibility of their restrictions under further collisions.

This parallel is exact at the level of incidence categories. A particular quantum field theory supplies a propagator, a density, a gauge fixing, and a renormalization prescription. Those analytic data form the screen coefficient system. The universal forest resolution supplies the combinatorics on which they live.

Chapter 44

Pole filtrations and normal geometry

44.1 Principal parts along a divisor

Let Y be a smooth variety and $D \subset Y$ a smooth divisor. For $m \geq 0$ there is an exact sequence

$$0 \longrightarrow \mathcal{O}_Y(mD) \longrightarrow \mathcal{O}_Y((m+1)D) \longrightarrow \mathcal{O}_D((m+1)D) \longrightarrow 0. \tag{44.1}$$

Since $\mathcal{O}_D(D) \cong N_{D/Y}$, the associated graded of the pole filtration is

$$\operatorname{gr}^F_{m+1} \mathcal{O}_Y(*D) \cong N_{D/Y}^{\otimes(m+1)}. \tag{44.2}$$

If t is a local defining equation for D, the class of $t^{-(m+1)}$ is a local generator of this line. Under $t' = ut + O(t^2)$ it transforms by $u^{-(m+1)}$ modulo lower pole order. This is precisely the tensorial transformation law of the normal symbol.

For a normal-crossing divisor $D = \bigcup_{a \in A} D_a$, use the multi-filtration

$$F_{\boldsymbol{m}} \mathcal{O}_Y(*D) = \mathcal{O}_Y\left(\sum_{a \in A} m_a D_a\right).$$

On a stratum $D_J = \bigcap_{a \in J} D_a$, the multi-associated graded is

$$\operatorname{gr}^F_{\boldsymbol{m}} \mathcal{O}_Y(*D)|_{D_J} \cong \bigotimes_{a \in J} N_{D_a/Y}^{\otimes m_a} \otimes \mathcal{O}_{D_J}(*D_J^{\mathrm{res}}). \tag{44.3}$$

The residual divisor records further boundary components meeting the stratum.

44.2 The full operator product

Let D_{12} be the pair-collision divisor in $\mathrm{FM}_2(X) = X^2$. In a coordinate $t = z_1 - z_2$, the singular operator product is

$$\sum_{m \geq 1} A_m(z_2) t^{-m}.$$

The coefficient A_m belongs intrinsically to the mth normal symbol. For a vertex algebra, $A_m = a_{(m-1)}b$. Therefore

$$a_{(r)}b \quad \longleftrightarrow \quad \mathrm{gr}^F_{r+1}\,\mathcal{O}(*D_{12}) \cong N^{\otimes(r+1)}_{D_{12}/X^2}. \tag{44.4}$$

The correspondence is coordinate-free at the associated-graded level. A coordinate chooses a trivialization of the normal line and thereby names the coefficient as a mode.

The zeroth regular coefficient belongs to F_0 and is the normally ordered product; its higher regular Taylor coefficients are obtained from it by translation and normal differentiation. Thus the complete filtered chiral operation contains both the singular modes and the regular multiplication. Keeping only the associated graded forgets extensions between pole orders; these extensions encode the triangular mixing under coordinate changes.

44.3 Nested collisions

Let T be a rooted collision tree indexing a boundary stratum $D_T \subset \mathrm{FM}_I(X)$. Each internal edge e corresponds to one normal direction, hence to a line N_e. Locally the normal bundle splits

$$N_{D_T/\,\mathrm{FM}_I(X)} \cong \bigoplus_{e\in E_{\mathrm{int}}(T)} N_e. \tag{44.5}$$

The multi-associated graded of a meromorphic coefficient is therefore

$$\bigotimes_{e\in E_{\mathrm{int}}(T)} N_e^{\otimes m_e}. \tag{44.6}$$

Assigning a positive integer m_e to every internal edge is the geometric version of assigning a pole order to every intermediate OPE channel.

Tree substitution contracts a subtree and replaces it by one incoming branch. Over an arbitrary curve the root point and its tangent line must remain part of the colour: Fulton–MacPherson spaces are not, by themselves, a single uncoloured operad unless additional translation or framing data is supplied. The intrinsic object is therefore a coloured collision coefficient system over X, whose colours are root points together with their tangent lines.

Theorem 44.1 (Normal-cone collision operad)**.** *The multi-normal symbols of nested collision strata form a coloured operad over X. For a collision tree T, its coefficient line is*

$$\bigotimes_{e\in E_{\mathrm{int}}(T)} N_e^{\otimes m_e},$$

with one positive pole order m_e on every internal edge. Substitution is induced by the canonical morphisms of Rees algebras for nested regular embeddings. It is associative and independent of local coordinates.

Proof. A nested collision stratum is an iterated regular embedding. The associated graded of its formal neighborhood is the symmetric algebra of the conormal bundle, and (44.5) identifies the independent conormal directions with internal edges. For a chain of nested regular embeddings, deformation to the normal cone is functorial: composing the corresponding Rees morphisms either before or after grouping a subtree gives the same morphism. Tensoring the resulting normal lines gives the stated substitution law and its associativity. The construction uses only ideals of diagonals and their normal bundles, hence is coordinate-free. ☐

A strict filtered identification of formal neighborhoods requires choices of compatible local coordinates or tubular neighborhoods. Different choices act by filtration-positive automorphisms. The associated-graded coloured operad is canonical; the filtered operation is canonical as a chiral distribution, not as a chosen Laurent series.

44.4 The collision Rees space

The pole filtration can be assembled into a Rees algebra

$$\mathcal{R}_D(\mathcal{O}_Y(*D)) = \bigoplus_{m\geq 0} F_m\mathcal{O}_Y(*D)\, u^m. \tag{44.7}$$

For $u \neq 0$ its localization recovers the meromorphic coefficient algebra. At $u = 0$ it becomes the normal-symbol algebra. Locally, the deformation is the familiar equation

$$tv = u,$$

where v represents the inverse normal coordinate. The generic fiber contains the full pole; the special fiber records its normal cone.

For the full boundary of $\mathrm{FM}_I(X)$ one obtains a multi-Rees space. Collision scales become independent deformation parameters. The special fiber is a union of normal-cone pieces indexed by forests. This geometry explains why classical limits are controlled by trees and why quantum corrections mix different pole orders.

44.5 The chiral and classical operads

The algebraic chiral operad is filtered by the order of poles along diagonals. Bakalov–De Sole–Heluani–Kac construct the corresponding classical operad and a canonical comparison from chiral symbols to classical operations [50, 51]. Geometrically, a chiral product passes to its normal symbols; when a filtration on the state space is compatible with the products, the regular multiplication becomes commutative and the leading singular symbols define the Poisson vertex bracket.

Theorem 44.2 (Chiral-to-classical symbol map)**.** *On a translation-invariant formal disc, the pole filtration of the chiral operad has a canonical morphism of operads*

$$\mathrm{gr}^F \mathcal{P}^{\mathrm{ch}} \longrightarrow \mathcal{P}^{\mathrm{cl}}.$$

Under the normal-symbol identification (44.4)*, this is the symbol morphism of the geometric pole filtration. If a filtered vertex algebra is compatible with its filtration, the image of its chiral Maurer–Cartan operation is the Poisson vertex algebra operation on the associated graded.*

Proof. Taking the leading normal symbol along each diagonal is compatible with external products and with pull-push composition along nested diagonals: normal cones of nested regular embeddings compose by the canonical map of associated graded Rees spaces. The symbol maps therefore assemble into a morphism of operads. On a translation-invariant disc, the symbol of $(z_i - z_j)^{-m-1}$ is the edge decoration carrying the m-th product, which is precisely the graph operation of the classical operad. Applying the operad morphism to a filtered Maurer–Cartan element sends its leading term to a classical Maurer–Cartan element. □

The theorem identifies the associated-graded operation for a compatible filtration. Flat quantization and isomorphism of the symbol map require the corresponding Rees-flatness and completeness hypotheses. The Li filtration supplies the standard compatible filtration for vertex algebras.

44.6 Residues, jets, and differential operators

Higher coefficients may be extracted by pairing principal parts with jets. If t is a coordinate normal to D, then

$$A_m = \frac{1}{2\pi i} \oint t^{m-1} \left(\sum_{r \geq 1} A_r t^{-r} \right) dt. \tag{44.8}$$

The factor t^{m-1} is a choice of jet. Intrinsically, the pairing is between $N_D^{\otimes m}$ and $N_D^{\vee \otimes m}$. The contour integral is the analytic realization of this pairing. Ordinary residue is $m = 1$.

In a D-module, normal derivatives of delta distributions supply the dual description. The identity between principal parts and delta derivatives is a local Fourier transform in the normal direction. This is why the lambda bracket encodes all singular modes polynomially in λ.

44.7 Coordinate changes and the stress tensor

A projective connection is required to make the stress tensor transform tensorially. Under $z \mapsto w(z)$,

$$T(z) = \left(\frac{dw}{dz} \right)^2 T(w) + \frac{c}{12} \{w, z\}, \tag{44.9}$$

where $\{w, z\}$ is the Schwarzian derivative. The inhomogeneous term is not the failure of $d \log(z_1 - z_2)$ to be invariant. It is the central extension of the action of vector fields, detected by the fourth-order pole in the TT operator product.

The pole filtration makes the distinction visible. The central term belongs to the fourth normal symbol. The transformation law mixes it with lower terms through the choice of projective connection. The invariant object is the chiral operation of the Virasoro algebra together with its coordinate-change action.

44.8 Pole order and homotopy depth as independent filtrations

The maximal pole order of a chosen strong generating set is a local analytic invariant. The length of a minimal resolution, the nonvanishing range of transferred higher operations, and the genus at which a trace appears are homological or modular invariants. There is no general equality among them. A first-order system may have only simple poles and still have rich representation theory; a Virasoro field has a fourth-order self-OPE but its deformation complex is not determined by the number four.

This separation replaces scalar "depth classifications" by canonical filtrations: pole order, coradical degree, collision codimension, graph genus, and cohomological degree. Each filtration has a geometric origin and a convergent spectral sequence under its own boundedness hypotheses.

Chapter 45

The formal disc and vertex algebras

45.1 Translation-equivariant chiral operations

Let V be a vector superspace with an even endomorphism ∂. On the affine line, translation-invariant rational functions on configurations are generated by the differences $z_i - z_j$ and their inverses. The algebraic chiral operad has n-ary component

$$\mathcal{P}^{\mathrm{ch}}_V(n) = \mathrm{Hom}_{\mathcal{D}^T_n}\left(V^{\otimes n} \otimes \mathbb{C}[z_i - z_j, (z_i - z_j)^{-1}], V[\lambda_1, \dots, \lambda_n]/\langle \partial + \textstyle\sum_i \lambda_i \rangle\right), \tag{45.1}$$

where $\mathcal{D}^T_n$ is the algebra of translation-invariant differential operators. The variables λ_i record normal derivatives. The relation in the target expresses simultaneous translation covariance.

The operad (45.1) is the formal-coordinate presentation of the Beilinson–Drinfeld chiral endomorphism operad. It contains the regular multiplication and all finite principal parts. Composition expands rational functions in the domains corresponding to nested collisions.

45.2 The universal Lie superalgebra of an operad

For a linear operad $\mathcal{P}$, set

$$W(\mathcal{P}) = \bigoplus_{k \geq -1} W_k(\mathcal{P}), \qquad W_k(\mathcal{P}) = \mathcal{P}(k+1)^{\Sigma_{k+1}}.$$

Operadic insertion defines a pre-Lie product and hence a Lie superbracket. For $\mathcal{P} = \mathcal{P}^{\mathrm{ch}}_{\Pi V}$, write $W^{\mathrm{ch}}(V)$. An odd binary operation is an element $X \in W^{\mathrm{ch}}_1(\Pi V)$.

The following theorem is the algebraic selection principle.

External theorem 45.1 (Bakalov–De Sole–Heluani–Kac)**.** *An odd element $X \in W^{\mathrm{ch}}_1(\Pi V)$ satisfies*

$$[X, X] = 0 \tag{45.2}$$

if and only if X defines a nonunital vertex algebra structure on V with translation operator ∂. The differential $\mathrm{ad}\, X$ *on $W^{\mathrm{ch}}(V)$ is the vertex-algebra deformation differential. [50]*

The theorem includes all coefficients of the integral lambda bracket. Skewsymmetry, the normally ordered product, the noncommutative Wick formula, and the Borcherds identity are different coefficient forms of the single equation (45.2) [50].

45.3 The coderivation identity

Let $\mathcal{C}$ be a conilpotent dg cooperad and let $\alpha : \mathcal{C} \to \mathcal{P}^{\mathrm{ch}}$ be an operadic twisting morphism, for example the canonical twisting morphism from the bar cooperad. An operation X then determines, through α, a coderivation d_X of the cofree $\mathcal{C}$-coalgebra $\mathrm{cofree}_{\mathcal{C}}(sV)$. The commutator of coderivations corresponds to the convolution bracket.

Proposition 45.2 (Square of the vertex coderivation). *In every such bar-coalgebra model, an odd chiral operation X satisfies*

$$d_X^2 = \frac{1}{2}[d_X, d_X] = \frac{1}{2} d_{[X,X]}. \tag{45.3}$$

Consequently d_X is a differential exactly when the image of X is Maurer–Cartan in the chiral convolution Lie algebra; for the binary operation of Theorem 45.1, this is exactly the vertex-algebra condition.

Proof. Coderivations of a cofree conilpotent coalgebra are determined by their corestriction to cogenerators. The corestriction of $[d_X, d_Y]$ is the signed sum of single insertions represented by the convolution pre-Lie product; antisymmetrizing gives $[X, Y]$. Setting $Y = X$ gives the formula. □

Combining Theorems 44.1, 45.1 and 45.2 gives the geometric statement: the full pole-filtered collision operation produces a square-zero coderivation exactly when the vertex identities hold. The geometry represents the coefficients; the operadic bracket supplies the identity.

45.4 The Borcherds identity

For homogeneous $a, b, c \in V$, the coefficient form of the vertex Jacobi identity is

$$\begin{aligned} &\sum_{j\geq 0} (-1)^j \binom{n}{j} a_{(m+n-j)} b_{(k+j)} c \\ &\quad - (-1)^n (-1)^{p(a)p(b)} \sum_{j\geq 0} (-1)^j \binom{n}{j} b_{(n+k-j)} a_{(m+j)} c \\ &= \sum_{j\geq 0} \binom{m}{j} (a_{(n+j)} b)_{(m+k-j)} c. \end{aligned} \tag{45.4}$$

The binomial coefficients arise from translating one formal variable relative to another. Equation (45.4) is therefore invisible to a complex that retains only a simple residue and discards jets. The normal-symbol filtration retains precisely the powers needed to recover the translated expansions.

At the level of distributions, the same identity is the equality of three expansions of a rational distribution on the complement of the three diagonals. The chiral operad records the expansions and their common meromorphic origin.

45.5 Vacuum and conformal structure

A vacuum vector **1** and the creation axiom extend the nonunital structure to a vertex algebra. A conformal vector ω supplies operators $L_n = \omega_{(n+1)}$ with Virasoro commutation relations and $L_{-1} = \partial$. The central charge is the coefficient of the fourth-order pole in

$$Y(\omega, z)Y(\omega, w) \sim \frac{c/2}{(z-w)^4} + \frac{2Y(\omega, w)}{(z-w)^2} + \frac{\partial_w Y(\omega, w)}{z-w}.$$

The vacuum, translation, and conformal vector are distinct structures. Chiral Koszul duality applies to the underlying chiral Lie object; conformal blocks and modular anomalies use the conformal structure.

45.6 The classical limit

Let $F_\bullet V$ be a filtration compatible with all vertex products, for example the Li filtration. The associated graded is commutative with a derivation and a lambda bracket. The regular normally ordered product becomes the commutative product; the first nontrivial singular symbols become the Poisson vertex bracket.

External theorem 45.3 (Classical symbol operad)**.** *There is a canonical morphism from the pole-associated graded of the chiral operad to the classical operad $\mathcal{P}^{\mathrm{cl}}$. An odd Maurer–Cartan element in $W(\mathcal{P}^{\mathrm{cl}}_{\Pi V})$ is a Poisson vertex algebra structure. For a vertex algebra equipped with a compatible filtration, the image of the principal chiral symbol is the Maurer–Cartan element of the associated Poisson vertex algebra. [51]*

The theorem makes quantization a lifting problem in the chiral deformation complex. Obstructions lie in the second cohomology of the classical differential; choices of lifts lie in the first. This is ordinary Maurer–Cartan obstruction theory applied to the filtered operad.

45.7 Deformations

Fix a vertex operation X. The complex

$$\big(W^{\mathrm{ch}}(\Pi V), d_X = [X, -]\big) \tag{45.5}$$

controls formal deformations. If

$$X_\hbar = X + \hbar X_1 + \hbar^2 X_2 + \cdots,$$

then the coefficient of $\hbar$ in $[X_\hbar, X_\hbar] = 0$ says $d_X X_1 = 0$. Gauge-equivalent first-order deformations differ by a coboundary. The second-order obstruction is the class of $\frac{1}{2}[X_1, X_1]$ in H^2.

This cohomology classifies local infinitesimal deformations, their obstructions, and their gauge symmetries in the displayed chiral deformation complex. It is not automatically the Hochschild cohomology of a mode algebra and not the cohomology of the moduli space of curves. Comparison maps may relate these theories in specific settings.

45.8 Homotopy chiral structures

Let $\mathcal{P}_\infty \to \mathcal{P}^{\mathrm{ch}}$ be a cofibrant resolution and let $\mathcal{C}_\infty$ be a conilpotent cooperadic model carrying its canonical twisting morphism to $\mathcal{P}_\infty$. A homotopy chiral algebra is an algebra over $\mathcal{P}_\infty$, equivalently a Maurer–Cartan element in the completed convolution L_∞-algebra

$$\mathrm{Hom}_\Sigma(\overline{\mathcal{C}_\infty}, \mathrm{End}_V).$$

The higher operations are indexed by cells in a resolution of collision composition. Their identities are the components of the Maurer–Cartan equation. This is the correct setting for transferred structures and for BRST models before cohomology concentration.

A strict vertex algebra is a homotopy chiral algebra whose higher generators vanish in a chosen model. Homotopy transfer preserves the quasi-isomorphism type but need not preserve strictness. Therefore the nonvanishing of a transferred higher operation is a property of the contraction as well as of the quasi-isomorphism class; its cohomology class is the invariant obstruction.

45.9 The meaning of “chiral Koszul”

The adjective “chiral” refers to the monoidal category and the diagonal operation. The term “chiral Koszul duality” denotes the Chevalley–primitives equivalence on the Ran space. A vertex algebra is a formal-coordinate manifestation of a chiral algebra. A quadratic presentation of a vertex algebra may admit a smaller quadratic dual, but that is an additional Koszul property, not the definition of the subject.

Chapter 46

The chiral Chevalley coalgebra on the Ran space

46.1 The Ran space

The Ran space of X is the prestack of nonempty finite subsets of X:

$$\mathrm{Ran}(X) = \operatorname*{colim}_{I \in \mathrm{Fin}^{\mathrm{surj,op}}} X^I.$$

A point of X^I maps to the subset of its coordinate values; coincident coordinates are allowed in the colimit. The transition maps are partial diagonals. Thus the Ran space remembers every finite collision simultaneously.

The category $D(\mathrm{Ran}\, X)$ carries two monoidal structures. The star tensor product is induced by union of finite subsets without a disjointness condition. The chiral tensor product is induced by union restricted to disjoint subsets. Chiral Lie algebras are Lie algebra objects for the chiral tensor product. Factorization objects are those whose restriction to disjoint configurations is the tensor product of their one-point pieces.

The Ran space is not a finite-dimensional variety, but its category of D-modules is controlled by the Cartesian powers X^I and their diagonal transition maps. Every formula in the preceding sections is therefore a labelled chart of a Ran object.

46.2 Chevalley chains

Let $\mathfrak{g}$ be a chiral Lie algebra. Its chiral Chevalley chains are the cofree commutative coalgebra on $\mathfrak{g}[1]$ with the coderivation induced by the chiral bracket:

$$\mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g}) = \Big(\mathrm{coSym}^{\mathrm{ch}}(\mathfrak{g}[1]), d_{\mathfrak{g}} + d_{\mu}\Big). \tag{46.1}$$

Pullback to X^I gives the partition decomposition of Theorem 42.4. The factorization coproduct is induced by the decomposition of a finite set into disjoint subsets.

The primitive filtration is the filtration by the number of factors. Its associated graded is the cofree commutative coalgebra with only the internal differential. This filtration is complete in the conilpotent direction: every element has finite primitive length in a labelled finite component.

Proposition 46.1 (Geometric realization of Chevalley chains)**.** *The forest incidence objects* $W_I^{\mathrm{FM}}(\mathfrak{g})$ *assemble, under disjoint union of labels and diagonal pullback, into a simplicial resolution of* $\mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g})$ *on* $\mathrm{Ran}(X)$*.*

Proof. Let $\iota_I : C_I^{\mathrm{coll}}(\mathfrak{g}) \to W_I^{\mathrm{FM}}(\mathfrak{g})$ insert the terminal corolla in every block. Then $\epsilon_I \iota_I = \mathrm{id}$. For a surjection of label sets $\phi : I \twoheadrightarrow J$, let T_ϕ be the canonical Ran transition on the partition collision complexes and define the transition on the resolution by

$$\widetilde{T}_\phi = \iota_J \, T_\phi \, \epsilon_I.$$

Because $\epsilon\iota = \mathrm{id}$, the maps $\widetilde{T}_\phi$ are strictly functorial under composition, and ϵ_I is a natural transformation. Disjoint union tensors the tree complexes and preserves terminal corollas, so these transitions are compatible with the factorization coproduct. They therefore assemble the labelled resolutions over the colimit presentation of $\mathrm{Ran}(X)$. □

46.3 Primitives

Let C be a coaugmented conilpotent commutative coalgebra in the chiral monoidal category. The primitive functor is defined invariantly as the right adjoint to the chiral Chevalley functor. With the Chevalley convention of (46.1), its value is shifted by $[-1]$ to produce the corresponding chiral Lie object.

In a strict conilpotent dg-coalgebra model, let

$$\overline{C} = \mathrm{Cofib}(\mathbf{1} \longrightarrow C)$$

be the coaugmentation coideal and let $\overline{\Delta} : \overline{C} \to \overline{C} \otimes^{\mathrm{ch}} \overline{C}$ be the reduced coproduct. When strict primitives compute the derived right adjoint—in particular for cofree conilpotent models and their fibrant replacements—one may represent the primitive complex by

$$\mathrm{Prim}(C) \simeq \mathrm{Fib}\Big(\overline{C} \xrightarrow{\overline{\Delta}} \overline{C} \otimes^{\mathrm{ch}} \overline{C}\Big). \tag{46.2}$$

For a cofree conilpotent coalgebra this complex is the cogenerator complex.

The Chevalley and primitive constructions form an adjunction

$$\mathrm{coChev}^{\mathrm{ch}} : \mathsf{Lie}_{\mathrm{ch}}(X) \rightleftarrows \mathrm{Coalg}^{\mathrm{ch,conil}}_{\mathsf{Com}}(\mathrm{Ran}\, X) : \mathrm{Prim}[-1]. \tag{46.3}$$

The shift depends on the Chevalley convention. The substantive point is that the bracket and the reduced coproduct are Koszul-dual operations.

46.4 Francis–Gaitsgory duality

The chiral monoidal category on the Ran space is pro-nilpotent with respect to cardinality. This property removes the convergence obstruction that ordinary Lie–commutative Koszul duality encounters in an arbitrary monoidal category.

External theorem 46.2 (Francis–Gaitsgory chiral Koszul duality)**.** *In the factorization and conilpotent subcategories of the pro-nilpotent chiral monoidal category of D-modules on* $\mathrm{Ran}(X)$ *considered by Francis and Gaitsgory, the chiral Chevalley functor and the primitive* *functor are inverse equivalences between chiral Lie algebras and conilpotent factorization coalgebras:*

$$\mathsf{Lie}_{\mathrm{ch}}(X) \simeq \mathrm{CoFact}_{\mathrm{conil}}(\mathrm{Ran}\, X). \tag{46.4}$$

The unit and counit of the adjunction are equivalences. [82]

This theorem is the invariant core of chiral Koszul duality [82]. It is stronger than the statement that a quadratic algebra is Koszul. No presentation is chosen and no linear dual is taken.

46.5 Chiral homology

The chiral homology of $\mathfrak{g}$ on a complete curve X is the de Rham cohomology of its factorization coalgebra:

$$\int_X^{\mathrm{ch}} \mathfrak{g} := R\Gamma_{\mathrm{dR}}(\mathrm{Ran}\, X, \mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g})). \tag{46.5}$$

The cardinality filtration produces a spectral sequence whose first page is a sum of cohomologies of configuration spaces with coefficients in external tensor powers of $\mathfrak{g}$. The collision differential is induced by the chiral bracket.

For an ordinary Lie algebra $\mathfrak{g}_0$ placed constantly on a point, this reduces to the usual Chevalley complex. For a current chiral algebra on a curve, chiral homology is Lie algebra homology globalized over the curve with the diagonal singularities retained by the coefficient D-module. For modules inserted at marked points, the pointed collision complex computes derived coinvariants.

46.6 The factorization envelope

Beilinson and Drinfeld associate to a chiral Lie algebra its chiral enveloping algebra $U^{\mathrm{ch}}(\mathfrak{g})$. Its factorization structure is generated by the same local operation μ, but the output is associative rather than coalgebraic. The Chevalley coalgebra and the enveloping factorization algebra play different roles:

$$\begin{aligned} \mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g}) &\quad \text{resolves primitives and homology,} \\ U^{\mathrm{ch}}(\mathfrak{g}) &\quad \text{acts on modules and defines centers.} \end{aligned}$$

The Poincare–Birkhoff–Witt theorem identifies the associated graded of $U^{\mathrm{ch}}(\mathfrak{g})$ with the symmetric chiral algebra under the standard filtration. This is a theorem about the enveloping construction, not an assertion that every vertex algebra is a universal enveloping algebra.

46.7 Analytic realization

Over $\mathbb{C}$, regular holonomic D-modules correspond under Riemann–Hilbert to constructible sheaves. The chiral operation becomes a morphism of constructible objects with singular support along diagonals. On the open configuration space, horizontal sections form a local system. The difference between the algebraic and analytic descriptions is then the difference between a regular singular connection and its monodromy representation.

This passage explains the ordered theory. The Ran coalgebra retains symmetric factorization. Pulling its analytic realization to labelled configurations and taking horizontal sections retains paths and braid monodromy. The two constructions are related but not identical: de Rham cohomology forgets a chosen basis of horizontal solutions, while monodromy acts on it.

46.8 Completion and cardinality

The Ran category is pro-nilpotent by cardinality, but additional completions may enter from conformal weight, pole order, or infinite-dimensional state spaces. These are independent filtrations. If C_N is an inverse system of finite-window complexes with surjective transition maps on cohomology, then the Milnor sequence

$$0 \longrightarrow \varprojlim{}^1 H^{m-1}(C_N) \longrightarrow H^m(\varprojlim C_N) \longrightarrow \varprojlim H^m(C_N) \longrightarrow 0 \tag{46.6}$$

shows that completion is exact when the derived limit vanishes. This is the precise content of a Mittag–Leffler argument.

No unrestricted completion theorem is needed for Theorem 46.2. Completion enters when one replaces abstract Ran objects by explicit infinite products of modes or conformal weights.

46.9 The Beilinson hierarchy

The objects generated by one chiral operation form a hierarchy:

$$\mathfrak{g} \longmapsto \mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g}) \longmapsto \int_X^{\mathrm{ch}} \mathfrak{g}, \qquad \mathfrak{g} \longmapsto U^{\mathrm{ch}}(\mathfrak{g}) \longmapsto \mathrm{Mod}(U^{\mathrm{ch}}(\mathfrak{g})). \tag{46.7}$$

The first branch is coalgebraic and homological. The second is algebraic and representation-theoretic. Centers, traces, and conformal blocks are built from the second branch; primitives and factorization homology are built from the first. Chiral Koszul duality connects the primitive object to its factorization coalgebra without collapsing the two branches.

Chapter 47

Chiral Koszul duality, bar–cobar, and presentations

47.1 Three meanings of Koszul duality

The phrase "Koszul duality" is used for three related constructions.

The first is the Lie–commutative equivalence of Theorem 46.2. This Chevalley–primitives equivalence is the chiral Koszul duality used here.

The second is bar–cobar reconstruction for augmented associative or operadic algebras. An augmented algebra A has a conilpotent bar coalgebra $B(A)$, and a conilpotent coalgebra C has a cobar algebra $\Omega(C)$. Under the usual conilpotence and completeness conditions, the unit and counit are equivalences.

The third is quadratic Koszul duality. A quadratic presentation $A = T(V)/(R)$ determines a smaller coalgebra $A^{¡}$ generated by sV with corelations s^2R. The algebra is Koszul when the canonical map $A^{¡} \to B(A)$ is a quasi-isomorphism.

The first construction is presentation-free. The second is a universal resolution. The third is a property of a presentation. They coincide only when comparison maps have been constructed.

47.2 Twisting morphisms

Let C be a conilpotent dg coalgebra and A an augmented dg algebra in a complete symmetric monoidal category. A twisting morphism is a degree-one map $\tau : C \to A$ satisfying

$$d\tau + \mu_A(\tau \otimes \tau)\Delta_C = 0. \tag{47.1}$$

Equation (47.1) is the Maurer–Cartan equation in the convolution algebra $\mathrm{Hom}(C, A)$. It determines a coalgebra map $C \to B(A)$ and an algebra map $\Omega(C) \to A$.

External theorem 47.1 (Fundamental theorem of twisting morphisms)**.** *Assume that the bar and cobar constructions are defined and that the coradical filtrations converge. For a twisting morphism $\tau : C \to A$, the following are equivalent:*

(i) *the map $C \to B(A)$ is a quasi-isomorphism;*

(ii) *the map $\Omega(C) \to A$ is a quasi-isomorphism;*

(iii) *the left twisted tensor product $A \otimes_\tau C$ is acyclic;*

(iv) *the right twisted tensor product $C \otimes_\tau A$ is acyclic.*

[98]

The theorem is classical over a field and extends to filtered chiral settings when tensor product preserves the coradical and weight-complete limits and the displayed comparison spectral sequences converge strongly [98]. Every hypothesis in a chiral application concerns the ambient category, not the formal equation (47.1).

47.3 Quadratic recognition

Let $A = T(V)/(R)$ be a connected weight-graded quadratic algebra. Its quadratic coalgebra $A^{¡}$ comes with a canonical twisting morphism. The usual Koszul criterion is diagonal concentration of bar homology.

External theorem 47.2 (Quadratic Koszul criterion)**.** *For a connected weight-graded quadratic algebra A, the following are equivalent:*

(i) *$A^{i} \to B(A)$ is a quasi-isomorphism;*

(ii) *$\Omega(A^{i}) \to A$ is a quasi-isomorphism;*

(iii) *$\mathrm{Tor}^{A}_{p,q}(\mathbb{C}, \mathbb{C}) = 0$ for $p \neq q$;*

(iv) *the quadratic twisted tensor complexes are acyclic.*

[98]

For a chiral algebra presented by generators and relations, the same criterion applies in each finite window once the free chiral algebra, the quadratic relation object, and the filtered comparison have been defined. It does not imply that universal affine, Virasoro, lattice, and W-algebras are all Koszul. Each family requires its own presentation and spectral-sequence proof.

47.4 PBW deformations

A PBW theorem identifies the associated graded of a filtered algebra with a prescribed graded algebra. By itself, this does not identify the bar complex of the filtered algebra with a quadratic dual: lower-order linear and constant terms alter the Koszul differential and may introduce curvature. The invariant comparison statement is the following filtered lemma.

Criterion 47.3 (Filtered Koszul comparison)**.** *Let $q : C \to B(A)$ be a morphism from a filtered conilpotent coalgebra to the bar coalgebra of a filtered augmented algebra. Assume that*

(a) *the filtrations on C and $B(A)$ are exhaustive, separated, complete, and preserved by q;*

(b) *the induced map $\mathrm{gr}\, q : \mathrm{gr}\, C \to B(\mathrm{gr}\, A)$ is a quasi-isomorphism;*

(c) *the two comparison spectral sequences converge strongly.*

Then q is a quasi-isomorphism.

Proof. The cone of q inherits a complete exhaustive filtration. Its associated graded is the cone of $\mathrm{gr}\, q$ and is acyclic by hypothesis. Strong convergence therefore gives zero cohomology for the filtered cone, so q is a quasi-isomorphism. □

For a quadratic-linear-constant PBW deformation of a quadratic Koszul algebra, the coalgebra C is the nonhomogeneous or curved Koszul coalgebra defined by the quadratic, linear, and constant parts. The PBW compatibility identities ensure that its coderivation squares to the prescribed curvature, and the associated-graded comparison is the ordinary quadratic Koszul resolution. Thus PBW, classical Koszulness, and the curved compatibility identities together imply the filtered comparison. PBW alone does not.

In a vertex algebra, an application must additionally control derivatives, infinitely many modes, and the completion in conformal weight or pole order. No family is declared Koszul merely from the existence of a PBW spanning set.

47.5 Verdier duality and a dual algebra

The chiral Chevalley coalgebra is not a linear dual. Nevertheless, if its labelled components are Verdier dualizable, duality exchanges coalgebra maps with algebra maps. Let $\mathbb{D}_{\mathrm{Ran}}$ denote Verdier duality on the full subcategory of constructible or holonomic Ran objects whose labelled components are Verdier dualisable and whose factorisation maps commute with duality.

Proposition 47.4 (Verdier-dual algebra)**.** *Let C be a factorization coalgebra whose finite labelled components are Verdier dualizable, and suppose Verdier duality commutes with the factorization restriction maps. Then*

$$C^\vee := \mathbb{D}_{\mathrm{Ran}} C$$

is a factorization algebra. If biduality holds componentwise and is compatible with factorization, then $C \to \mathbb{D}_{\mathrm{Ran}}\mathbb{D}_{\mathrm{Ran}} C$ is an equivalence of factorization coalgebras.

Proof. Verdier duality reverses arrows and exchanges external tensor products with their dual tensor products on dualisable components. Dualizing the factorisation coproduct of C therefore gives multiplication maps on $\mathbb{D}_{\mathrm{Ran}} C$. Coassociativity and counitality dualize to associativity and unitality. The componentwise biduality maps commute with the factorisation restrictions by hypothesis, giving the final equivalence. □

Applied to $C = \mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g})$, this produces a Verdier-dual factorization algebra. Identifying it with a familiar vertex algebra, a reflected level, or a quantum group is a separate recognition theorem.

47.6 Completion

Let $F^1 A$ be the augmentation ideal. The completed bar and cobar constructions use products over tensor length rather than direct sums. Their differentials converge when they raise the filtration. The completed counit

$$\widehat{\Omega}\widehat{B}(A) \longrightarrow A$$

is an equivalence for complete augmented algebras in the standard pronilpotent setting.

Curved algebras require another distinction. If a curvature element m_0 is present, the unary operation satisfies $m_1^2 = [m_0, -]$ rather than zero. A central scalar has zero commutator and therefore cannot be the same datum as a nonzero inner curvature. Curved bar–cobar duality belongs to the coderived and contraderived categories of Positselski [103]. It does not alter the square-zero chiral differential of an honest vertex algebra.

47.7 Koszul dual modules

A twisting morphism $\tau : C \to A$ defines adjoint functors between A-modules and C-comodules. Under the hypotheses of Theorem 47.1, these functors give an equivalence on the complete module and conilpotent comodule categories specified in the theorem. For a quadratic Koszul algebra, the resolution is linear and yields the familiar exchange of Ext and Tor.

In the chiral setting, modules are factorization modules at marked points. The pointed collision complex is the bar resolution. A module-level Koszul duality therefore requires the algebra comparison and the compatibility of marked-point factorization. Fusion products and conformal blocks are preserved only when a braided tensor equivalence of module categories has been established.

47.8 Maurer–Cartan moduli

Twisting morphisms form a Maurer–Cartan space in a convolution dg Lie or L_∞-algebra. Gauge equivalence is homotopy of twisting data. The tangent complex at τ is the twisted convolution complex

$$(\mathrm{Hom}(C, A), d + [\tau, -]) .$$

Its first cohomology contains infinitesimal deformations and its second contains obstructions.

This is the natural moduli object associated with Koszul presentations. Koszulness itself is the open condition that the comparison cone be acyclic in a finite perfect family; globally it is a derived condition on the vanishing of that cone. There is no canonical finite list of scalar invariants that classifies the bar coalgebra.

47.9 Exact limits of the notion

The following implications do not hold without additional hypotheses:

$$\begin{gathered} \text{PBW presentation} \not\Rightarrow \text{Koszul}, \qquad C_2\text{-cofinite} \not\Rightarrow \text{Koszul}, \\ \text{Verdier dualizable} \not\Rightarrow \text{self-dual}, \qquad \text{braided monodromy} \not\Rightarrow \text{specified Yangian}, \\ \text{equal central charges} \not\Rightarrow \text{equivalent chiral algebras}. \end{gathered} \tag{47.2}$$

Each failed implication confuses a coarse shadow with a structured object. The correct replacements are comparison maps and recognition theorems.

Chapter 48

Duality, Hochschild theory, and the derived center

48.1 The center requires an associative object

A chiral Lie algebra has a bracket, not a bimodule category. The expression $\mathrm{RHom}_{A\text{-}A}(A, A)$ is therefore undefined when A denotes only a chiral Lie object. To form a center, one first passes to an associative chiral or factorization algebra, such as the chiral enveloping algebra $U^{\mathrm{ch}}(\mathfrak{g})$.

Definition 48.1 (Derived chiral center)**.** For a chiral Lie algebra $\mathfrak{g}$ for which the chiral enveloping factorization algebra is defined, set

$$Z^{\mathrm{der}}_{\mathrm{ch}}(\mathfrak{g}) := \mathrm{HH}^*_{\mathrm{Fact}}\left(U^{\mathrm{ch}}(\mathfrak{g}), U^{\mathrm{ch}}(\mathfrak{g})\right). \tag{48.1}$$

For an associative factorization algebra A, write $Z^{\mathrm{der}}_{\mathrm{ch}}(A) = \mathrm{HH}^*_{\mathrm{Fact}}(A, A)$.

Hochschild cochains are derived endomorphisms of the diagonal bimodule. The bar resolution supplies a concrete model. In the chiral setting the resolution is supported on configurations with an incoming and an outgoing boundary component, and insertion of configurations produces brace operations.

48.2 The E_2 structure

The higher Deligne theorem gives Hochschild cochains of an E_1 algebra an E_2 structure. The same statement holds for associative factorization algebras in a stable presentable symmetric monoidal category whose tensor product preserves the bar realizations used to form bimodules. The product is cup product; the shifted Lie bracket is the Gerstenhaber bracket; higher braces encode homotopies.

External theorem 48.2 (Derived center structure)**.** *Let A be an associative factorization algebra whose bimodule category admits the diagonal bar resolution. Then $Z^{\mathrm{der}}_{\mathrm{ch}}(A)$ is an E_2 algebra. Its action on A is universal among homotopy-coherent central actions. [40]*

The universal property is algebraic. An object B acting on A through the Swiss-cheese operad determines a morphism of E_2 algebras

$$B \longrightarrow Z^{\mathrm{der}}_{\mathrm{ch}}(A). \tag{48.2}$$

The map sends a closed operation to the family of open operations obtained by inserting it next to boundary inputs.

48.3 The center and a physical bulk

Let $\mathrm{Obs}_{\mathrm{bulk}}$ be the factorization algebra of local observables of a field theory with boundary algebra A. The bulk-to-boundary operator product defines a map

$$\mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z^{\mathrm{der}}_{\mathrm{ch}}(A). \tag{48.3}$$

If the map is a quasi-isomorphism, the bulk local observables are reconstructed from the boundary algebra. If a decoupled bulk sector has zero boundary action, it lies in the kernel of every boundary measurement and prevents (48.3) from being an equivalence.

No-go theorem 48.3 (Boundary data do not determine a decoupled bulk). A boundary chiral algebra and all of its algebraic centers are unchanged by tensoring the bulk theory with a decoupled theory whose boundary action is trivial. Therefore no theorem can identify an arbitrary physical bulk with the derived center of its boundary without a nondegeneracy or generation hypothesis.

This statement is the exact mathematical limit of holographic reconstruction from local boundary operations.

48.4 The annulus and Hochschild homology

Factorization homology of the circle with coefficients in an E_1 algebra is Hochschild homology:

$$\int_{S^1} A \simeq A \otimes^L_{A\otimes A^{\mathrm{op}}} A. \tag{48.4}$$

For a boundary theory, the annulus is obtained by thickening the circle. Its state space is therefore a trace of the boundary algebra. The pairing between Hochschild cochains and Hochschild chains is the algebraic form of inserting a bulk local operator into an annulus amplitude.

A cyclic trace on A thickens planar trees to ribbon graphs. This is the point at which genus enters. Without a cyclic pairing, the derived center has an E_2 structure but no canonical loop contraction.

48.5 Morita invariance

Hochschild cochains and chains are Morita invariant. If A and B are derived Morita equivalent associative factorization algebras, then

$$\mathrm{HH}^*_{\mathrm{Fact}}(A, A) \simeq \mathrm{HH}^*_{\mathrm{Fact}}(B, B)$$

as E_2 algebras, and their Hochschild homologies agree.

A Koszul duality between module categories may therefore induce an equivalence of centers. The implication runs through Morita equivalence, not through a scalar complementarity rule. A coalgebraic Koszul dual without a module-category equivalence does not by itself identify centers.

48.6 A small convolution model

Assume A is augmented, complete, and Koszul with a perfect bar coalgebra $B(A)$. The diagonal bar resolution identifies Hochschild cochains with a twisted convolution complex

$$Z^{\mathrm{der}}_{\mathrm{ch}}(A) \simeq \mathrm{Hom}^{\tau}(B(A), A)\,, \tag{48.5}$$

where the differential is

$$df = d_A f - (-1)^{|f|} f d_B + [\tau, f].$$

The braces insert one coalgebra input into another before applying f. This model is finite and explicit when the bar coalgebra is locally finite.

The formula (48.5) explains the relation between deformation theory and the center. A first-order associative deformation is a degree-two Hochschild cocycle; its obstruction is the Gerstenhaber square. The chiral deformation complex of Equation (45.5) and the factorization Hochschild complex may be compared, but they govern different structures: one deforms the chiral operation, the other the associative factorization envelope.

48.7 Verdier duality and centers

Verdier duality exchanges !- and *-extensions across diagonals. For a perfect factorization algebra A, duality produces an opposite or dual factorization algebra. The induced map on bimodule categories reverses left and right actions. Consequently the Hochschild complex of the dual is the dual of the Hochschild chain complex of A under a Calabi–Yau pairing.

A degree-d Calabi–Yau structure on A gives a nondegenerate pairing

$$\mathrm{HH}^i(A, A) \otimes \mathrm{HH}_{d-i}(A) \longrightarrow \mathbb{C}.$$

The cyclic structure determines the degree independently of any later moduli-space realization. This pairing is the algebraic origin of BV operators on Hochschild cohomology.

48.8 Topological enhancement

The derived centre is E_2 in the transverse directions. A coherent null-homotopy of one further real translation gives an E_3 enhancement. A coherent topologization of the whole complex curve supplies two additional real topological directions and gives an E_4 algebra, as in Proposition 38.8. A holomorphic centre is not topological merely because it is a centre.

In three-dimensional Chern–Simons theory, local bulk observables are E_3 because the bulk has three topological directions. Comparison with a boundary centre therefore uses the locally constant or topological factorisation object produced by the stated topologization functor. In a four-real-dimensional holomorphic–topological theory whose holomorphic direction is also topologized, the corresponding local structure is E_4. In either dimension, proving that the bulk-to-centre map is an equivalence is a field-theoretic calculation.

48.9 Centers at critical level

For the universal affine vertex algebra at the critical level, the ordinary vertex-algebra center is large: the Feigin–Frenkel theorem identifies it with functions on opers for the Langlands dual group on the formal disc [78]. This center is a commutative chiral algebra inside the affine algebra. It is not the same object as factorization Hochschild cochains, although the latter receives operations from it.

The distinction is visible in degrees. The vertex center consists of fields commuting with all fields under nonnegative modes. The derived center includes higher self-extensions of the diagonal bimodule. At noncritical level the ordinary center may be scalar while the derived center remains nontrivial.

Part VII

Exchange, reduction, and quantum symmetry

Chapter 49

Labelled chiral configurations and monodromy

49.1 Ordered configurations and exchange geometry

The order in an E_1 algebra comes from chambers of configurations on a real line. Monodromy arises from moving labelled chiral insertion points in the complex curve or from an E_2 transverse enhancement. Nothing below is implied by the bare E_1 multiplication of an ordered chiral algebra.

49.2 Labels retain paths

The labelled configuration space of n points is

$$\mathrm{Conf}_n(X) = \{(x_1, \ldots, x_n) \in X^n \mid x_i \neq x_j \text{ for } i \neq j\}.$$

The symmetric group acts freely, and the quotient is the unordered configuration space

$$\mathrm{UConf}_n(X) = \mathrm{Conf}_n(X)/\Sigma_n.$$

For $X = \mathbb{C}$,

$$\pi_1(\mathrm{Conf}_n(\mathbb{C})) = P_n, \qquad \pi_1(\mathrm{UConf}_n(\mathbb{C})) = B_n,$$

the pure braid and braid groups. A local system on the labelled space therefore retains pure braid monodromy; an equivariant local system descends to the unordered space and retains full braid monodromy.

A symmetric factorization coalgebra is not obtained by forgetting labels in the naive set-theoretic sense. It carries coherent descent data under all relabelings and diagonals. Labelled analytic continuation is extra information: it is a representation of the fundamental groupoid of the configuration space.

49.3 Derived descent

Let M be a chain complex with an action of a finite group G. The homotopy coinvariants are

$$M_{hG} = M \otimes^{L}_{\mathbb{C}[G]} \mathbb{C}.$$

The augmentation ideal $I_G = \ker(\mathbb{C}[G] \to \mathbb{C})$ fits into a distinguished triangle

$$M \otimes^L_{\mathbb{C}[G]} I_G \longrightarrow M \longrightarrow M_{hG}. \tag{49.1}$$

Thus the homotopy fiber of descent is canonical. In characteristic zero and for finite G, the group algebra is semisimple, so derived coinvariants agree with ordinary coinvariants. The fiber still records all nontrivial isotypic components.

For a braid-group local system, descent consists of the monodromy representation together with its equivariance under the extension

$$1 \longrightarrow P_n \longrightarrow B_n \longrightarrow \Sigma_n \longrightarrow 1.$$

Discarding the pure braid action loses analytic continuation even when the underlying vector space is unchanged.

49.4 The descent fiber and collision filtrations

For a coefficient system M_n carrying a Σ_n-action, define the descent fiber

$$\mathcal{C}_n(M) = \mathrm{Fib}(M_n \to M_{n,h\Sigma_n}). \tag{49.2}$$

By (49.1), it is canonically $M_n \otimes^L_{\mathbb{C}[\Sigma_n]} I_{\Sigma_n}$. A universal chain model is obtained from the ordinary group bar resolution of the trivial $\mathbb{C}[\Sigma_n]$-module. In characteristic zero the group algebra is semisimple, so this derived fiber is represented by the direct sum of the nontrivial isotypic components.

The proper partition complex is a different universal object. It is the incidence and Koszul complex of collision partitions, and its top homology is the Lie representation. It does not resolve the descent fiber of an arbitrary Σ_n-module. When M_n carries a functorial filtration indexed by collision partitions, an explicit comparison morphism from the partition-incidence complex to the group-bar model may filter $\mathcal{C}_n(M)$ by binary collisions, ternary collision coherence, and higher strata. Such a filtration belongs to the coefficient system and must be constructed; it is not automatic from the group action.

The descent fiber need not contain an r-matrix. It contains whatever nontrivial isotypic and braid data the coefficient system actually carries. An r-matrix appears when the first-order monodromy of a flat connection has the form required by the classical Yang–Baxter equation.

49.5 Dual local systems

Let M be an oriented manifold and $\mathcal{L}$ a finite-rank local system with monodromy $\rho : \pi_1(M) \to GL(V)$. Its Verdier dual is

$$\mathbb{D}\mathcal{L} \cong \mathcal{L}^\vee[\dim M],$$

up to the orientation line. The monodromy of the dual local system is

$$\rho^\vee(\gamma) = \rho(\gamma^{-1})^{\mathsf{t}}. \tag{49.3}$$

The inverse occurs because parallel transport on the dual is contragredient; the transpose is the linear dual.

Corollary 49.1. *Verdier duality sends a braid operator R to the inverse transpose $R^{-\mathrm{t}}$. If the representation is identified with its dual by a nondegenerate invariant pairing, this becomes inverse braiding.*

Proof. Parallel transport on the dual local system is the inverse transpose of parallel transport on the original local system. A braid generator is such a transport operator, so Verdier duality sends R to R^{-t}. An invariant nondegenerate pairing identifies the dual representation with the original and converts inverse transpose to inverse braiding. ☐

The further notation $q \mapsto q^{-1}$ is justified only when a family of braid operators has been identified with a quantum-group family parametrized by q.

49.6 Associators

The monodromy of a flat connection on the configuration space of three points depends on comparing asymptotic zones. A Drinfeld associator is a group-like formal series satisfying the pentagon and hexagon equations. Associators form a torsor under the Grothendieck–Teichmuller group GRT_1 after the usual normalization [74, 120].

An arbitrary linear splitting of a symmetrization map is not an associator. The pentagon equation is a nonlinear condition in arity four; the hexagon relates the associator to the braiding. The torsor arises only inside the solution space of these equations.

In the de Rham realization, the Knizhnik–Zamolodchikov associator is the regularized parallel transport between tangential base points at 0 and 1 on $\mathbb{P}^1 \setminus \{0, 1, \infty\}$. Its coefficients are multiple zeta values. This arithmetic belongs to chiral configuration-space monodromy, not to a scalar obtained by averaging the underlying tensor space.

49.7 Tannaka reconstruction

Let $\mathcal{C}$ be a rigid braided tensor category and let $F : \mathcal{C} \to \mathrm{Vect}$ be a faithful exact fibre functor. The coend

$$H = \int^{M \in \mathcal{C}} F(M)^\vee \otimes F(M) \tag{49.4}$$

reconstructs a bialgebra object when F is strong monoidal; rigidity supplies an antipode when the coend belongs to the finite or continuous-dual class. If F is only quasi-monoidal, the associativity constraint produces a coassociator and H is a coquasi-bialgebra. The braiding produces a coquasitriangular structure. When every fibre is finite dimensional, or when the completed matrix-coefficient coalgebra admits a separated continuous dual, one obtains a quasitriangular Hopf or quasi-Hopf algebra, respectively.

Theorem 49.2 (Ordered Tannaka principle)**.** *A rigid tensor-compatible monodromy category with a faithful strong monoidal fibre functor reconstructs a coquasitriangular bialgebra, and a Hopf algebra whenever the antipode is defined in the chosen finite or continuous-dual class. With a faithful quasi-monoidal fibre functor, it reconstructs a coquasitriangular coquasi-bialgebra. A quasi-Hopf presentation becomes a strict Hopf presentation only after a twist that trivializes the coassociator.*

Proof. Form the coend $H = \int^M F(M)^\vee \otimes F(M)$. Tensor product in the source category induces the coproduct and multiplication of matrix coefficients, while the unit object gives the unit and counit. Braiding defines the universal coquasitriangular form. Rigidity constructs the antipode whenever the required duals remain in the chosen finite or continuous class. If the tensor constraint of F is quasi-monoidal, its pentagon is transported to a coassociator, producing the coquasi-bialgebra and, under rigidity, quasi-Hopf structures. Faithfulness gives the stated reconstruction on the tensor subcategory generated by the image. □

The theorem is universal. Identifying the reconstructed object with $U_q(\mathfrak{g})$, a Yangian, an elliptic quantum group, or a Hall algebra is a recognition theorem using the specific connection and representation category.

49.8 Surface braid groups

For a curve of genus $g > 0$, $\pi_1(\mathrm{Conf}_n(X))$ is a surface pure braid group. Its generators include collisions and transport around the A- and B-cycles of the curve. Flat connections such as the KZB connection represent these generators by operators satisfying elliptic braid relations.

The elliptic theory is not determined by the genus-zero associator. Quasi-periodicity, the choice of a dynamical parameter, and the mapping-class-group action supply additional data. At higher genus, conformal-block monodromy is a representation of the mapping class group of the punctured surface, not merely an iteration of B_n.

49.9 What symmetrization preserves

Symmetrization preserves invariant traces, dimensions of invariant subspaces, and any operation that factors through the quotient. It may preserve nontrivial periods when those periods lie in the invariant part of the local system. Therefore it is false that symmetrization erases all arithmetic or all monodromy. The exact statement is that it kills the non-invariant quotient measured by (49.2).

For conformal blocks, passing from vectors to their dimension preserves the Verlinde number and loses the mapping-class-group representation. Passing from a braided category to its Grothendieck ring preserves fusion coefficients and loses the associator and twists. These are concrete instances of the same descent.

Chapter 50

KZ geometry, quantum groups, Gaudin systems, and lines

50.1 The logarithmic connection

Let $\mathfrak{g}$ be a finite-dimensional complex simple Lie algebra with invariant tensor $\Omega \in \mathfrak{g} \otimes \mathfrak{g}$. On $V_1 \otimes \cdots \otimes V_n$, define

$$\nabla_{\mathrm{KZ}} = d - \hbar \sum_{i<j} \Omega_{ij}\, d\log(z_i - z_j), \qquad \hbar = \frac{1}{k + h^\vee}. \tag{50.1}$$

The parameter depends on the normalization of the invariant form; the shifted level is invariantly determined by the Sugawara construction.

The infinitesimal braid relations are

$$[\Omega_{ij}, \Omega_{kl}] = 0 \quad \text{if } \{i, j\} \cap \{k, l\} = \varnothing, \qquad [\Omega_{ij}, \Omega_{ik} + \Omega_{jk}] = 0 \tag{50.2}$$

for distinct i, j, k.

Theorem 50.1 (Flatness of the KZ connection)**.** *The connection* (50.1) *is flat.*

Proof. The logarithmic forms are closed, so for the convention $\nabla = d - A$ the curvature is $A \wedge A$. Terms with disjoint index pairs vanish by the first relation in (50.2). For every triple i, j, k, the remaining terms factor as the Arnold relation

$$d\log z_{ij} \wedge d\log z_{jk} + d\log z_{jk} \wedge d\log z_{ki} + d\log z_{ki} \wedge d\log z_{ij} = 0$$

times the second infinitesimal braid relation. Hence the curvature vanishes. □

This proof displays the exact division of labor. The differential forms supply the Arnold identity. The coefficient tensor supplies the infinitesimal braid identity. Neither identity alone implies flatness.

50.2 From affine currents to KZ

For the affine vertex algebra $V_k(\mathfrak{g})$, current fields satisfy

$$J^a(z)J^b(w) \sim \frac{k\,(a,b)}{(z-w)^2} + \frac{J^{[a,b]}(w)}{z-w}. \tag{50.3}$$

Ward identities for primary insertions convert the simple-pole action into the differential equation (50.1); the double pole and the Sugawara stress tensor produce the shift by $h^\vee$. The KZ connection is therefore a representation-theoretic shadow of the affine chiral operation, not the bar differential of the entire vertex algebra.

50.3 Drinfeld–Kohno

Let $q = \exp(\pi i \hbar)$ in the standard convention. The Drinfeld–Kohno theorem identifies the braided tensor category defined by KZ monodromy with the category of finite-dimensional modules over the Drinfeld–Jimbo quantum group $U_q(\mathfrak{g})$, with an associator determined by KZ parallel transport [74, 94, 95].

External theorem 50.2 (Drinfeld–Kohno)**.** *For generic $\hbar$, the braid-group representations obtained from the monodromy of the KZ connection are naturally equivalent, as braided tensor data, to those obtained from the universal R-matrix of $U_q(\mathfrak{g})$ with $q = e^{\pi i \hbar}$. [74, 94, 95]*

The theorem is the rigorous statement that a quantum group is present in the ordered affine theory. It includes the associator. It does not identify the kernel of symmetrization with a Hopf algebra.

50.4 The classical r-matrix

The one-form in (50.1) has coefficient

$$r_{ij}(z_i - z_j) = \hbar \frac{\Omega_{ij}}{z_i - z_j}. \tag{50.4}$$

The flatness equation at first order in $\hbar$ is the classical Yang–Baxter equation. The monodromy is a path-ordered exponential of the connection; its semiclassical expansion begins with the classical r-matrix.

The same rational kernel appears in several constructions once their normalizations are matched: the KZ connection, the rational Gaudin model, and the line-operator OPE of four-dimensional Chern–Simons theory. The equality is a theorem only after the coefficient spaces and coupling constants have been identified.

50.5 Gaudin Hamiltonians

Fix distinct points $z_1, \ldots, z_n$. The Gaudin Hamiltonians are

$$H_i = \sum_{j \neq i} \frac{\Omega_{ij}}{z_i - z_j}. \tag{50.5}$$

The infinitesimal braid relations imply

$$[H_i, H_j] = 0.$$

The Gaudin Hamiltonians are the residues of the KZ connection in the z_i variable. Their commutativity is therefore another coefficient of flatness.

For highest-weight modules, the Bethe ansatz constructs simultaneous eigenvectors from critical points of a master function. In the simplest $\mathfrak{sl}_2$ case, Bethe roots t_a satisfy

$$\sum_{i=1}^{n} \frac{\lambda_i}{t_a - z_i} - \sum_{b \neq a} \frac{2}{t_a - t_b} = 0. \tag{50.6}$$

The Schechtman–Varchenko hypergeometric solutions of the KZ equations relate these critical points to horizontal sections. Thus KZ monodromy and the Gaudin spectrum are two readings of the same flat connection.

50.6 Four-dimensional Chern–Simons theory

Costello–Witten–Yamazaki construct a four-dimensional gauge theory that is topological in two directions and holomorphic in one complex direction. Wilson lines at distinct holomorphic positions have an operator product governed by a spectral-parameter R-matrix. In the rational case, the algebra of line operators is a Yangian; trigonometric and elliptic choices produce quantum loop and elliptic structures [63, 64].

The spectral parameter is geometric: it is the difference of holomorphic positions of the lines. The Yang–Baxter equation follows from isotopy of three lines. The one-loop computation fixes the first correction to the R-matrix. This is a physical construction of ordered monodromy, not a consequence of chiral Koszul duality alone.

50.7 Yangians and recognition

A Yangian $Y_\hbar(\mathfrak{g})$ is a filtered deformation of $U(\mathfrak{g}[t])$ with a rational R-matrix on the tensor category generated by finite-dimensional evaluation representations. To recognize a reconstructed quasi-bialgebra as a Yangian one needs, at minimum, a PBW theorem, evaluation representations, the rational difference property, and a proof that no additional relations occur.

Criterion 50.3 (Yangian recognition)**.** *Let H be a filtered quasi-bialgebra reconstructed from an ordered tensor category. If*

(i) $\operatorname{gr} H \cong U(\mathfrak{g}[t])$ *as a bialgebra;*
(ii) *the fundamental evaluation modules are faithful on the completed algebra;*
(iii) *their R-matrices agree with the rational Yangian R-matrices;*
(iv) *the defining RTT or Drinfeld relations generate the full kernel of the representation map;*

then H is isomorphic, up to the associator twist already present in the tensor functor, to the corresponding completed Yangian.

The criterion states exactly what an ordered chiral construction must prove before the word “Yangian” is justified.

50.8 Elliptic transport

On an elliptic curve, the KZB connection replaces $d\log(z_i - z_j)$ by elliptic kernels with prescribed quasi-periodicity. Its flatness uses Fay identities and dynamical correction terms. The monodromy represents the elliptic braid group and the modular group. Depending on the setting, reconstruction leads to elliptic or dynamical quantum groups.

No universal formula sends genus-zero Yangians to genus-one quantum groups. Rational, trigonometric, and elliptic structures are different degenerations of connections with different base geometry. Their relation is controlled by limits of the curve and of the coefficient category.

50.9 Knots and ribbon traces

A ribbon category assigns invariants to framed links by composing braidings, duality maps, and twists. The Reshetikhin–Turaev construction applied to $U_q(\mathfrak{g})$ produces quantum link invariants [106]. Chern–Simons theory supplies the physical path-integral interpretation. The KZ connection supplies a monodromy realization of the same braided tensor category.

The closure of a braid uses a categorical trace. This is the precise point at which ordered monodromy becomes a scalar knot invariant. The scalar does not retain the individual braid representation, just as a conformal-block dimension does not retain mapping-class-group monodromy.

50.10 Duality of monodromy

By (49.3), dualizing the KZ local system sends the monodromy to its inverse transpose. Under the invariant form on representations and the Drinfeld–Kohno equivalence, this agrees with the dual quantum-group representation and inverse braiding. A level reflection such as $k \mapsto -k - 2h^\vee$ has the same effect on the parameter $\hbar = (k + h^\vee)^{-1}$, namely $\hbar \mapsto -\hbar$, hence $q \mapsto q^{-1}$. This is a statement about the KZ family. It is not a universal formula for every chiral Koszul dual.

Chapter 51

Quantum line defects and meromorphic tensor categories

51.1 Fusion before exchange

Let $A \in \mathrm{Alg}_{E_1}(\mathrm{FactD}(X))$. Its modules can be placed in a fixed order along the transverse line. Relative tensor product gives fusion:

$$M \star_A N := M \otimes_A^L N.$$

Associativity follows from excision for three consecutive intervals. There is no canonical map $M \star_A N \to N \star_A M$ because an isotopy in one dimension cannot exchange the intervals without a collision.

No-go theorem 51.1 (Order does not imply an R-matrix). The forgetful functor

$$\mathrm{Alg}_{E_2}(\mathrm{FactD}(X)) \longrightarrow \mathrm{Alg}_{E_1}(\mathrm{FactD}(X))$$

admits no natural section and its essential image excludes noncommutative discrete algebras. Consequently an ordered chiral algebra has no functorially determined braiding, classical r-matrix, or quantum group.

Proof. The multiplication on the homology of an E_2 algebra is graded commutative. A noncommutative algebra concentrated in cohomological degree zero, such as $M_2(\mathbb{C})$ or a free associative algebra on two generators, therefore cannot be the underlying discrete E_1 algebra of an E_2 algebra with the same product. A natural section would provide such a lift and is impossible. Geometrically, the space of two ordered intervals in an interval has two disconnected order components and contains no exchange path; the path exists only after a second transverse direction is supplied. $\square$

The theorem fixes the location of quantum symmetry. Fusion belongs to the ordered chiral algebra. Exchange belongs to an E_2 enhancement, to the Hochschild centre, or to a separately supplied meromorphic local system.

51.2 The meromorphic line category

Let $\mathcal{L}$ be a category of line defects varying over the spectral curve X. For distinct spectral points, fusion is defined and is compatible with factorization. An exchange structure consists of isomorphisms

$$R_{L,M}(x,y) : L_x \star M_y \longrightarrow M_y \star L_x, \qquad x \neq y,$$

which are horizontal for the D-module connection and satisfy the braid relation on $\mathrm{Conf}_3(X)$.

Definition 51.2 (Meromorphic tensor category)**.** A meromorphic tensor category on X consists of a factorization category $\mathcal{L}$ on the ordered configuration spaces of X, an associative fusion product on disjoint ordered blocks, and a braid-group descent datum on the locus of distinct points. The descent datum is meromorphic along the collision divisors and satisfies the pentagon and hexagon identities.

The associator compares the three asymptotic regions of four ordered insertions. The braiding compares two chambers after a transverse exchange. The pole of the exchange operator is a chiral singularity; its homotopy class is topological.

Theorem 51.3 (Line-category recognition)**.** *Let $\mathcal{T}$ be a renormalized quantum field theory on $X \times \mathbb{R}^2$ that is holomorphic on X and topological on $\mathbb{R}^2$. Assume that the fusion and exchange Ward identities are anomaly-free on a chosen class of line defects, that their morphism complexes are perfect over the meromorphic functions on X, that the line OPE extends meromorphically across the collision divisors, and that the class is closed under fusion and duality. Then these line defects form a rigid meromorphic tensor category in the sense of Theorem 51.2.*

Proof. Fusion is factorization in one transverse direction. Associativity is isotopy of three parallel lines. The exchange of two lines is a path in the configuration space of two points in the transverse plane. Isotopy of three exchange paths gives the braid relation. Isotopy of decompositions of four lines gives the pentagon; compatibility of exchange with fusion gives the hexagons. Holomorphic dependence on spectral position and locality of counterterms make the structure meromorphic along diagonals. Dual line orientations give categorical duals under the perfectness hypothesis. □

51.3 The centre and categorical exchange

The internal Hochschild centre $Z_{\mathrm{ord,ch}}(A)$ is an E_2 algebra in $\mathrm{FactD}(X)$. This means that its local multiplication carries the homotopy-coherent exchange of little two-disks. It does not mean that its ordinary module category is braided: modules over an E_2 algebra are naturally E_1-monoidal. A braided line category is instead supplied by the Drinfeld centre of a monoidal line category, or by modules over an E_3 enhancement.

Corollary 51.4 (Exchange through a complete centre)**.** *Suppose the bulk-to-boundary map*

$$\beta_{\mathcal{T}} : \mathrm{Obs}_{\mathrm{bulk}} \longrightarrow Z_{\mathrm{ord,ch}}(A)$$

is an equivalence of local E_2 algebras. Suppose in addition that a monoidal-affineness theorem identifies the selected bulk line category with the Drinfeld centre $Z(\mathrm{Bimod}_A)$, and identifies this categorical centre with modules over an E_3 topological enhancement of $Z_{\mathrm{ord,ch}}(A)$. Then the corresponding perfect line categories are equivalent as braided monoidal categories.

Proof. Compose the bulk-to-centre equivalence with the assumed monoidal-affineness equivalence between bulk lines and the Drinfeld centre of boundary modules. The final identification with modules over the topological enhancement then transports the braided tensor product and its exchange operators. Each arrow is an equivalence by the displayed hypotheses, so their composite is the asserted reconstruction. □

The extra hypotheses are necessary. Equality of local bulk observables does not by itself classify extended line defects, and an E_2 algebra alone does not make its module category braided.

51.4 Tannaka reconstruction and its exact output

Choose a faithful exact meromorphic fibre functor

$$F : \mathcal{L} \longrightarrow \mathrm{Vect}_K, \qquad K = \mathbb{C}(X).$$

The coend

$$H_{\mathcal{L}} = \int^{L \in \mathcal{L}} F(L)^\vee \otimes_K F(L)$$

represents tensor automorphisms of F. The tensor product, associator, and braiding produce respectively a coproduct, coassociator, and universal R-form.

Theorem 51.5 (Quantum symmetry of line defects). *If F is strong monoidal, $H_{\mathcal{L}}$ is a coquasitriangular bialgebra over K, and rigidity supplies a Hopf antipode in the finite or continuous-dual class. If F is only quasi-monoidal, $H_{\mathcal{L}}$ is a coquasitriangular coquasi-bialgebra whose coassociator is the transported tensor constraint. When the matrix-coefficient coalgebra has finite graded pieces or a separated continuous dual, the dual objects are quasitriangular Hopf or quasi-Hopf algebras, respectively. The construction is invariant under braided tensor equivalence compatible with the chosen fibre functor.*

Proof. The coend construction represents natural matrix coefficients of the fibre functor. Strong monoidality supplies associative multiplication and coassociative coproduct; braiding supplies the coquasitriangular form. Duals in the line category give the antipode. For a quasi-monoidal functor, the transported tensor constraint is a coassociator satisfying the pentagon, and the same construction yields the coquasi and quasi-Hopf variants. Passage to continuous duals gives the dual quantum algebra when the stated completeness and separation conditions hold. □

The theorem reconstructs the coordinate quantum symmetry from the tensor category and fibre functor. The names Yangian, quantum loop algebra, elliptic quantum group, or Hall algebra require additional relations and a PBW theorem.

51.5 Rational gauge theory and the Yangian

In four-dimensional Chern–Simons theory on $\Sigma \times X$, where the theory is topological on Σ and holomorphic on X, parallel Wilson lines form a meromorphic tensor category. In the rational theory $X = \mathbb{C}$, perturbation theory produces

$$R_{V,W}(z) = 1 + \hbar \frac{\Omega_{V,W}}{z} + O(\hbar^2). \tag{51.1}$$

The one-loop anomaly of a line is cancelled by deforming the classical current algebra to the Yangian. The resulting line OPE is the RTT or Drinfeld coproduct of $Y_\hbar(\mathfrak{g})$.

External theorem 51.6 (Rational Wilson-line algebra)**.** *For the perturbative Wilson-line categories constructed in anomaly-free rational four-dimensional Chern–Simons theory by Costello–Witten–Yamazaki, the algebra of line operators is the completed Yangian deformation of $U(\mathfrak{g}[z])$ on the stated class of representations; exchange is given by the corresponding rational universal R-matrix. The spectral parameter is the difference of the holomorphic positions. The assertion belongs to the completed formal ħ-adic category in which the line anomaly and its cancellation are defined [63, 64].*

The theorem is a field-theoretic recognition theorem. It uses the gauge-theory Ward identities and anomaly calculation, not merely the existence of an ordered bar construction.

51.6 Trigonometric and elliptic geometries

Changing the holomorphic curve changes the meromorphic tensor category. For $X = \mathbb{C}^\times$, the trigonometric kernel and its periodicity lead to a quantum loop algebra. For an elliptic curve, quasi-periodicity and a dynamical parameter lead to elliptic or dynamical quantum groups. These structures are not obtained by exponentiating a universal scalar. They are reconstructed from different meromorphic connections.

Principle 51.7 (The spectral curve selects the quantum group)*.* For the standard rational, trigonometric, and elliptic meromorphic line categories, Tannaka reconstruction produces the corresponding rational, quantum-loop, and elliptic or dynamical quantum symmetries after their presentation-recognition theorems are applied. A degeneration of the spectral curve induces a degeneration of the exchange kernel only when the line category, fibre functor, and meromorphic connection extend flatly over the degenerating family.

51.7 Shifted Yangians and boundary conditions

A line defect may end on a boundary or meet an 't Hooft defect. Magnetic charge changes the allowed pole orders of the Drinfeld currents. Algebraically this produces shifted Yangians. Geometrically the shift is a modification of the factorization module at a marked transverse boundary condition.

Let μ be a coweight. A shifted object is represented by a pointed module bar

$$\mathrm{Bar}_X^{\mathrm{ord}}(M_-, A, M_+) = M_- \otimes_A^L M_+,$$

where the endpoint modules encode the two boundary charges. The unshifted algebra is recovered when both modules are the augmentation boundary.

Criterion 51.8 (Shifted recognition)**.** *A line algebra is identified with a shifted Yangian only after the following data agree: the coweight filtration of currents, the shifted coproduct, the PBW associated graded, the fundamental line modules, and the rational exchange kernel. Agreement of the unshifted R-matrix alone does not determine the shift.*

51.8 KZ and gauge-theory exchange

For affine conformal blocks, the KZ connection gives braid monodromy with coupling $(k + h^\vee)^{-1}$. For four-dimensional Chern–Simons lines, the gauge coupling fixes the rational R-matrix. These two constructions are related through the Drinfeld–Kohno theorem after the representation categories and normalizations have been identified. The equality of kernels is not formal: the affine theory is a two-dimensional chiral theory, whereas the gauge theory is a four-dimensional holomorphic–topological theory.

Theorem 51.9 (Comparison through the line category)**.** *Let $\mathcal{L}_{\mathrm{KZ}}$ be a rigid braided tensor subcategory of the Drinfeld–KZ category at a nonresonant shifted level, and let $\mathcal{L}_{\mathrm{gauge}}$ be a rigid Wilson-line subcategory of a rational gauge theory. A braided tensor equivalence matching tensor generators and the first-order Casimir kernel identifies their braid monodromies. Under compatible Tannaka fibre functors, both reconstruct the same quasitriangular Hopf or quasi-Hopf object up to the transported associator twist.*

Proof. A tensor equivalence preserves the coend reconstruction. Matching the Casimir coefficient fixes the formal deformation parameter. The braid operator on tensor generators then agrees, and rigidity extends the agreement to the generated category. The associator records the difference between strict and KZ tensor structures. □

The theorem isolates the actual comparison datum. Without the tensor equivalence, similarity of rational kernels is evidence rather than identity.

51.9 Koszul duality of line couplings

Let A be the algebra of local observables near a line and let $\varepsilon : A \to \mathbf{1}$ describe the distinguished trivial line. Formal deformations supported on that line are governed by the derived deformation theory of the augmentation module. Its endomorphism algebra is the Koszul dual

$$A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1}).$$

Theorem 51.10 (Koszul duality for the augmentation-generated line sector)**.** *Let A be a complete augmented E_1 algebra of local observables in the formal neighbourhood of a line defect, and put $A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$. The adjunction of Theorem 36.5 identifies the thick, or presentably completed, sector of A-modules generated by the augmentation with the corresponding sector of $A^!$-modules. The bar complex $\mathbf{1} \otimes^L_A \mathbf{1}$ computes the derived matrix coefficients of the augmentation line.*

Proof. The augmentation module is the distinguished line. Its derived endomorphisms form $A^!$. The endomorphism adjunction is an equivalence on the generator and therefore on its thick closure; compact generation extends it to the stated presentable completion. The two-sided bar computes the derived tensor product of the two augmentation modules. □

This theorem is the precise common content of Koszul duality in defect physics. Identifying $A^!$ with a named Yangian is a further calculation.

51.10 Transfer matrices and scalar observables

Let V be an auxiliary object of a rigid line category and let $M_1, \ldots, M_n$ be quantum spaces. The monodromy operator

$$T_V(z) = R_{V,M_n}(z - z_n) \cdots R_{V,M_1}(z - z_1)$$

obeys the RTT relation. Taking the categorical trace over V gives commuting transfer matrices:

$$[t_V(z), t_W(w)] = 0.$$

The proof is the braid relation followed by cyclicity of the trace.

This passage loses information. The transfer matrix is a scalar or endomorphism-valued trace of ordered exchange. Bethe equations and Baxter relations describe its spectrum in selected representations. They are not universal consequences of an E_1 chiral algebra; they follow from the particular reconstructed quantum group and its representation theory.

51.11 Knots and framing

Closing a braid requires duals and a ribbon twist. The Reshetikhin–Turaev invariant is the categorical trace of the closed braid. In Chern–Simons theory the twist is the framing anomaly. Thus three structures enter in order:

$$\text{fusion} \quad \longrightarrow \quad \text{braiding} \quad \longrightarrow \quad \text{ribbon trace.}$$

Fusion, braiding, and ribbon trace are governed respectively by E_1, E_2, and cyclic or framed data. The distinction is the categorical form of the distinction between an ordered word, an exchange path, and a closed world-line.

Chapter 52

BRST descent and W-algebras

52.1 A differential compatible with chirality

Let (C, Q) be a cochain complex equipped with a chiral operation X. The statement that Q is a derivation of the chiral structure is

$$[Q, X] = 0 \tag{52.1}$$

in the chiral deformation Lie algebra. If $Q^2 = 0$ and $[X, X] = 0$, then

$$[Q + X, Q + X] = 0.$$

Thus the BRST differential and the chiral operation form one Maurer–Cartan element with unary and binary parts.

Proposition 52.1 (BRST descent of chiral operations)**.** *If Q is a square-zero derivation of a chiral algebra C, then $H_Q(C)$ inherits a chiral algebra structure. The induced operation is independent of the choice of cocycle representatives.*

Proof. Equation (52.1) says that the operation sends cocycles to cocycles and that changing one input by a boundary changes the output by a boundary. The chiral Jacobi identity descends because it holds before taking cohomology. □

The proposition applies equally to vertex algebras in a formal coordinate. It is the algebraic core of Hamiltonian reduction.

52.2 The BRST–collision bicomplex

The collision complex of C carries the internal differential induced by Q and the collision differential induced by X. They anticommute by (52.1). Hence

$$\left(C_I^{\mathrm{coll}}(C), Q + d_{\mathrm{coll}}\right) \tag{52.2}$$

is a bicomplex.

Filter by BRST degree. The first spectral sequence has

$$E_1 \cong C_I^{\mathrm{coll}}(H_Q(C)).$$

Filter by collision degree. The other spectral sequence computes BRST cohomology of the collision resolution.

Theorem 52.2 (Bar–BRST comparison)**.** *Assume that the tensor products and sums forming the coefficient objects are Q-acyclic outside degree zero, and that the total bicomplex (52.2) is bounded below in one grading and complete with respect to an exhaustive filtration in the other. Then the canonical map of collision complexes*

$$C_I^{\mathrm{coll}}(H_Q^0(C)) \longrightarrow H_Q^0\big(C_I^{\mathrm{coll}}(C), Q\big) \tag{52.3}$$

is an isomorphism, while $H_Q^j(C_I^{\mathrm{coll}}(C), Q) = 0$ for $j \neq 0$. Consequently the total complex is quasi-isomorphic to $C_I^{\mathrm{coll}}(H_Q^0(C))$ with its induced collision differential.

Proof. Taking vertical Q-cohomology commutes with the coefficient tensor products under the stated acyclicity hypothesis, which proves (52.3) and the higher vanishing. The BRST-first spectral sequence therefore has one nonzero row. Boundedness and completeness identify its abutment with the total cohomology, giving the final quasi-isomorphism. □

The theorem is exact and strong, but its concentration hypothesis is substantive. Without it, the result is a derived BRST object rather than the ordinary bar complex of degree-zero cohomology.

52.3 Quantum Drinfeld–Sokolov reduction

Let $\mathfrak{g}$ be a simple Lie algebra and $f \in \mathfrak{g}$ nilpotent. Quantum Drinfeld–Sokolov reduction forms a BRST complex from the affine vertex algebra $V_k(\mathfrak{g})$, charged ghosts, neutral fields when required, and a differential determined by f. The affine W-algebra is

$$\mathcal{W}^k(\mathfrak{g}, f) = H^0_{\mathrm{BRST}}(C^k_{\mathrm{DS}}(\mathfrak{g}, f)). \tag{52.4}$$

For the universal W-algebra, the BRST cohomology is concentrated in degree zero in the standard construction; the resulting vertex algebra has an associated graded described by functions on an arc space of the Slodowy slice [46, 47].

The chiral operation on the BRST complex descends by Theorem 52.1. The complicated nonlinear OPEs of W-generators are therefore transferred collision operations. Their composite fields are not exceptions to chiral geometry; they are the output of homological reduction.

52.4 Homotopy transfer

Choose a contraction

$$(H, 0) \underset{p}{\overset{i}{\rightleftarrows}} (C, Q), \qquad Qi = 0, \quad pQ = 0, \quad Qh + hQ = \mathrm{id} - ip.$$

Homotopy transfer expresses the induced higher operations on H as sums over rooted trees. Vertices carry the original chiral operation, internal edges carry h, leaves carry i, and the root carries p.

If the transferred structure is strict, the tree sums beyond the binary operation cancel. If it is not strict, H carries a homotopy chiral structure. The cohomology classes of the higher operations measure the obstruction to strictification; their raw coefficients depend on the contraction.

In Drinfeld–Sokolov reduction, the cohomology is known independently to carry a strict vertex-algebra structure. This fact alone does not trivialize the higher operations produced by an arbitrary transfer. If the contraction extends to a $\mathcal{P}_\infty$ quasi-isomorphism from the transferred homotopy chiral structure to the strict BRST-cohomology model, then the two Maurer–Cartan elements are gauge equivalent and the transferred higher operations are removable in that model. Without such a comparison, they remain legitimate transfer data.

52.5 Feigin–Frenkel duality for principal W-algebras

For principal nilpotent f, the Feigin–Frenkel duality relates the principal W-algebras of Langlands dual Lie algebras. With lacing number $r^\vee$, the levels k and ℓ satisfy

$$r^\vee(k+h^\vee)(\ell+{}^Lh^\vee)=1. \tag{52.5}$$

Under this relation there is an isomorphism

$$\mathcal{W}^k(\mathfrak{g})\cong\mathcal{W}^\ell({}^L\mathfrak{g}). \tag{52.6}$$

Equation (52.6) is the standard principal W-algebra duality; its multiplicative shifted-level relation differs from affine level reflection.

The duality may be constructed through screening operators or free-field realizations. Its relation to a bar coalgebra requires a separate comparison between the BRST resolution and the chiral Chevalley construction.

52.6 Screening operators

A free-field realization embeds a W-algebra into an intersection of kernels of screening operators. The screenings are contour integrals of vertex operators and commute with the W-algebra up to total derivatives. Their kernels form a complex whose cohomology realizes the reduced algebra.

Geometrically, a screening contour is a cycle in a configuration space. Its monodromy and intersection pairing depend on the charges of the free fields. Multiple screenings produce hypergeometric local systems and quantum-group actions. This is an ordered configuration-space realization of BRST reduction.

52.7 Critical level

At $k=-h^\vee$, the Sugawara stress tensor is undefined because its denominator vanishes. On the formal disc D, the center of the critical vacuum vertex algebra is

$$Z(V_{-h^\vee}(\mathfrak{g}))\cong\mathcal{O}(\mathrm{Op}_{{}^LG}(D))\,. \tag{52.7}$$

The center of the completed critical enveloping algebra over the punctured disc is described instead by functions on opers on $D^\times$, with completion in the loop-rotation filtration and central reduction at a chosen oper [78, 80]. These are related but distinct statements. The vertex center is a commutative chiral algebra and underlies the local geometric Langlands construction.

The critical level is singular for KZ because $\hbar = (k + h^\vee)^{-1}$ diverges. It is not a limit in which q becomes one. The affine chiral algebra remains well defined, but its conformal structure and monodromy description change character.

52.8 What BRST reduction preserves

BRST reduction preserves any structure represented by a chain map commuting with Q. It preserves the chiral operation, module actions, and pairings that satisfy this condition. It does not automatically preserve:

- a chosen quadratic presentation;
- Koszulness of that presentation;
- a particular braid local system;
- a Verdier-dual identification;
- positive-genus sewing before compatibility with Q is proved.

The precise theorem is Theorem 52.2. Every stronger statement is a comparison theorem for the corresponding additional structure.

52.9 The physical meaning

Classical Hamiltonian reduction imposes constraints and divides by gauge transformations. BRST replaces this quotient by a cochain complex whose differential combines constraints and ghosts. Quantum Drinfeld–Sokolov reduction applies the same BRST mechanism to the affine chiral algebra and its ghost system. The Virasoro or W-algebra is the algebra of gauge-invariant boundary observables after reduction.

From the field-theoretic viewpoint, the ghost central charge is part of the anomaly calculation. BRST nilpotence requires cancellation of the total anomaly in a coupled theory. This is an equality of central extensions in the BRST complex, not a universal formula for a Koszul dual.

Part VIII

Two sewing geometries and their anomalies

Chapter 53

Two genera and the two-edge master equation

53.1 Two loop-forming operations

Let A be an ordered chiral algebra. A cyclic trace on its transverse E_1 multiplication closes an interval into a circle and produces Hochschild chains, annuli, ribbon graphs, and open topological surfaces. A modular sewing structure on the chiral coefficient glues marked points of algebraic curves by plumbing and produces conformal blocks. Both operations contract pairs of flags, but their moduli spaces and anomaly lines are different.

Definition 53.1 (Transverse Calabi–Yau structure)**.** A transverse Calabi–Yau structure of degree $d_\perp$ is a trace

$$\mathrm{tr}_\perp : \mathrm{HH}_0^{\mathrm{FactD}(X)}(A) \longrightarrow \mathbf{1}[-d_\perp]$$

whose pairing $\langle a, b\rangle_\perp = \mathrm{tr}_\perp(ab)$ is perfect and cyclic.

Definition 53.2 (Chiral modular sewing structure)**.** A chiral modular sewing structure of degree d_X consists of perfect pairings on the insertion and module objects decorating the flags and operations assigned to stable pointed complex curves, compatible with separating and nonseparating clutching, the factorisation D-module maps, and the determinant twists.

The transverse cyclic structure and the chiral modular sewing structure are independent. The first is an open two-dimensional topological structure internal to the chiral coefficient; the second varies the complex curve.

53.2 Bicoloured stable graphs

A bicoloured stable graph Γ has two sets of internal edges,

$$E(\Gamma) = E_X(\Gamma) \sqcup E_\perp(\Gamma).$$

An X-edge represents a chiral plumbing contraction. A $\perp$-edge represents a transverse cyclic contraction. Vertices incident only to chiral flags carry the stable-curve operation; vertices incident to transverse flags carry the cyclic E_1 operation, with the transverse flags in their ribbon order. The orientation line is

$$\mathrm{or}(\Gamma) = \det(E_X(\Gamma)) \otimes \det(E_\perp(\Gamma)). \tag{53.1}$$

Contracting a loop of colour X raises the chiral genus g by one; contracting a loop of colour $\perp$ raises the ribbon genus h by one.

Let G^{bi} be the groupoid of these graphs with isomorphisms preserving the edge colour and cyclic orders. The free completed graph object generated by the two sewing systems is

$$\mathbb{F}^{\mathrm{bi}}(A) = \prod_{[\Gamma]\in\mathsf{G}^{\mathrm{bi}}} (\mathrm{or}(\Gamma)\otimes A(\Gamma))_{\mathrm{Aut}(\Gamma)}. \tag{53.2}$$

The bicoloured graph object directly records the two edge types and their common vertex operations.

53.3 The two boundary operators

Let ∂_X be the signed sum of contractions of one X-edge and $\partial_\perp$ the signed sum of contractions of one $\perp$-edge.

Proposition 53.3 (Independent graph boundaries)**.** *With the orientation convention* (53.1)*,*

$$\partial_X^2 = 0, \qquad \partial_\perp^2 = 0, \qquad \partial_X\partial_\perp + \partial_\perp\partial_X = 0.$$

Hence $\partial_X + \partial_\perp$ is square-zero.

Proof. For two edges of the same colour, the two contraction orders induce opposite orientations in the determinant of that edge set. For edges of different colours, the final graph is the same in either order, while exchanging the two odd determinant factors in (53.1) changes the sign. □

The proposition is a graph-theoretic identity. A geometric realization must additionally show that the chosen ribbon and Deligne–Mumford boundary maps represent these edge contractions.

53.4 The bicoloured Feynman transform

Let $\mathcal{P}_X$ be the modular or cyclic object governing chiral sewing, and let $\mathcal{P}_\perp$ govern transverse cyclic sewing. Assume that their endomorphism actions on A obey the interchange identities on common flags. Define the *two-edge Feynman transform* to be the free completed bicoloured graph object with differential

$$d_{\mathbb{F}} = d_{\mathrm{int}} + \partial_X + \partial_\perp.$$

By Proposition 53.3, this is a differential.

The corresponding completed convolution complex has components

$$\mathfrak{g}_A^{\mathrm{bi}} = \prod_{g,h,n} \mathrm{Hom}_{\Sigma_n}\left(\mathcal{P}^{\mathrm{bi}}(g,h,n), \mathrm{End}_A(g,h,n)\right), \tag{53.3}$$

completed by the number of internal edges and vertices. There are brackets $[-,-]_X$ and $[-,-]_\perp$ from joining two vertices with an edge of the indicated colour, and loop operators Δ_X and $\Delta_\perp$ from joining two flags of one vertex.

Theorem 53.4 (Two-edge master equation)**.** *A morphism from the two-edge Feynman transform to the cyclic endomorphism object of A is equivalent to an element*

$$\Theta_A^{\mathrm{bi}} \in \mathfrak{g}_A^{\mathrm{bi}}[[\hbar_X, \hbar_\perp]]$$

satisfying

$$d\Theta_A^{\mathrm{bi}} + \frac{1}{2}[\Theta_A^{\mathrm{bi}}, \Theta_A^{\mathrm{bi}}]_X + \frac{1}{2}[\Theta_A^{\mathrm{bi}}, \Theta_A^{\mathrm{bi}}]_\perp + \hbar_X \Delta_X \Theta_A^{\mathrm{bi}} + \hbar_\perp \Delta_\perp \Theta_A^{\mathrm{bi}} = 0. \tag{53.4}$$

Proof. The free transform is generated by corollas, and its differential is the sum of the internal differential and all one-edge contractions. Evaluating a morphism on the generators gives Θ_A^{bi}. Compatibility with $d_{\mathbb{F}}$ is exactly (53.4). Conversely, the equation extends the assignment on corollas uniquely to a chain morphism on the free completed graph object. □

The hypotheses include both sewing systems. The theorem identifies their common master equation; no sewing operation is inferred from a bare ordered chiral algebra.

53.5 The two loop operators

On the suspended convolution complex, both loop contractions are odd.

Proposition 53.5 (Supercommutation of loop contractions)**.** *With the determinant convention* (53.1)*,*

$$\Delta_X \Delta_\perp + \Delta_\perp \Delta_X = 0.$$

After passing to unsuspended scalar amplitudes, the induced even contractions commute.

Proof. The orientation line of a bicoloured graph is the ordered tensor product of the determinant lines of its X-edges and its transverse edges. Contracting one edge of each colour in the two possible orders exchanges two odd determinant generators and changes the sign. The underlying coefficient contractions act on independent tensor factors and therefore agree. Their signed sum is zero. Desuspension removes the odd determinant degrees, so the induced scalar contractions commute. □

At bidegree $(g, h) = (0, 1)$, one sees the annulus and the Hochschild trace. At $(1, 0)$, one sees genus-one chiral sewing and its projective anomaly. At $(1, 1)$, both loop operations occur; the mixed codimension-two identity is their supercommutation.

53.6 Geometric realization

A geometric realization of the $\perp$-edges uses ribbon surfaces or cyclically ordered boundary configurations. A realization of the X-edges uses stable curves and clutching morphisms. Compatibility requires a Cartesian or correspondential model in which performing one degeneration does not alter the local definition of the other except through the prescribed interchange map.

Criterion 53.6 (Geometric two-edge realization)**.** *A pair of geometric sewing theories realizes the two-edge master equation when:*

(i) *every codimension-one boundary of either type has a coefficient contraction;*

(ii) *same-colour codimension-two boundaries satisfy the usual modular or ribbon boundary identity;*

(iii) *mixed codimension-two boundaries form Cartesian comparison squares and carry the product orientation* (53.1)*;*

(iv) *the coefficient contractions commute over those squares;*

(v) *the graph filtration is complete.*

Under these conditions, the geometric boundary differential maps to $d_{\mathbb{F}}$ and Theorem 53.4 applies.

53.7 Open topological field theory and chiral conformal theory

A smooth proper Calabi–Yau E_1 algebra defines an oriented open topological field theory; its value on the circle is Hochschild homology,

$$\int_{S^1} A \simeq \mathrm{HH}_\bullet(A).$$

Ribbon graphs compute its amplitudes. This is the $g = 0$ row with arbitrary h.

A chiral conformal theory assigns blocks to complex curves and varies with complex structure through a projectively flat connection. This is the $h = 0$ column with arbitrary g.

No-go theorem 53.7 (No universal equality of the genera). There is no natural identity $g = h$. A cyclic associative algebra may have nontrivial ribbon amplitudes without a family of algebraic curves, while a chiral conformal theory may have higher-genus blocks without being determined by one finite-dimensional cyclic E_1 algebra.

A string duality may relate the two expansions, but that relation is a theorem of the specific dynamics.

53.8 The double expansion

For a matrix realization, a connected ribbon graph is weighted by a power of N determined by h and the number of boundaries. Chiral sewing is weighted by a string coupling determined by g. When both exist, the connected potential has the form

$$\mathcal{F}_A(N, g_s) = \sum_{g,h,b} N^{2-2h-b} g_s^{2g-2} \mathcal{F}_{g,h,b}(A). \tag{53.5}$$

The coefficients are traces of the bidegree components of Θ_A^{bi}. Keeping the two parameters separate is the exact algebraic expression of the distinction between matrix-planar and worldsheet-genus expansions.

53.9 Duality of the two-edge theory

Theorem 53.8 (Duality of bicoloured amplitudes)**.** *Suppose:*

(i) A and $A^!$ are joined by an invertible bimodule Morita context on the selected complete perfect sector;

(ii) their transverse Calabi–Yau traces correspond;

(iii) their chiral sewing systems are identified by a tensor equivalence preserving projective connections, pairings, and determinant twists.

Then the equivalence induces an isomorphism of two-edge convolution complexes carrying Θ_A^{bi} to $\Theta_{A^!}^{\mathrm{bi}}$.

Proof. Morita invariance identifies Hochschild chains, ribbon contractions, and $\Delta_\perp$. The sewing equivalence identifies the chiral contractions and Δ_X. Preservation of determinant lines identifies the orientation signs. Hence every term of (53.4) is intertwined. □

A numerical complementarity statement follows only after a trace has been applied to this structured equivalence.

53.10 Physical meaning

The operator $\Delta_\perp$ contracts two open or line-defect states and creates a loop of the transverse topological theory. The operator Δ_X contracts two chiral insertions through a plumbing fixture and creates a handle on the complex curve. Their supercommutation is locality for two independent degenerations. The universal object is therefore not a scalar curvature but a bicoloured graph operation that remembers which edge was contracted and on which moduli space the contraction lives.

Chapter 54

Cyclicity, stable graphs, and the master equation

54.1 Why genus requires a pairing

A genus-zero operation has outputs and inputs. To create a loop, one must identify two flags. Such an identification requires a pairing. Let V be a complex with a nondegenerate graded-symmetric pairing

$$\langle -, - \rangle : V \otimes V \longrightarrow \mathbb{C}[d] \tag{54.1}$$

that is invariant under the genus-zero operations. The pairing converts an output into an input and permits contraction along internal edges.

Without (54.1), there is no canonical nonseparating sewing. A chiral algebra alone therefore does not determine a modular object. A cyclic chiral algebra, a conformal-block pairing, or a field-theoretic propagator supplies the additional datum.

54.2 Modular operads

A modular operad has operations indexed by genus g and a finite set of legs. It allows two kinds of composition: joining a leg of one operation to a leg of another, and joining two legs of the same operation. The first adds genera; the second raises genus by one. Stable graphs encode iterated compositions [87].

Let $\mathcal{P}$ be a dg modular cooperad and let $\mathrm{End}_V^{\mathrm{mod}}$ be the endomorphism modular operad determined by the pairing (54.1). The completed convolution space

$$\mathfrak{g}_{\mathcal{P},V} = \prod_{g,n} \mathrm{Hom}_{\Sigma_n}\Big(\mathcal{P}(g,n), \mathrm{End}_V^{\mathrm{mod}}(g,n)\Big) \tag{54.2}$$

contains a differential, a bracket joining two vertices, and a loop operator Δ joining two legs of one vertex.

54.3 The Feynman transform

The Feynman transform $F\mathcal{P}$ is the free modular operad on the shifted dual of $\mathcal{P}$, with the standard determinant twists, and with differential given by the internal differential and the signed sum over edge contractions. A morphism $F\mathcal{P} \to \mathrm{End}_V^{\mathrm{mod}}$ is determined by its values on generators. Compatibility with the differential is the master equation.

Theorem 54.1 (Feynman transform and master equation)**.** *Giving V the structure of an algebra over the Feynman transform $F\mathcal{P}$ is equivalent to giving an element $\Theta \in \mathfrak{g}_{\mathcal{P},V}$ satisfying*

$$d\Theta + \frac{1}{2}[\Theta, \Theta] + \hbar\Delta\Theta = 0. \tag{54.3}$$

Proof. The free modular operad is generated by corollas. The internal part of its differential gives $d\Theta$. Contracting an edge between two vertices gives the bracket term, with the factor $1/2$ accounting for interchange of the vertices. Contracting a loop at one vertex gives $\Delta\Theta$. Equality with the endomorphism differential is exactly (54.3). □

The theorem is formal and unconditional. Its hypothesis is the existence of a morphism from the Feynman transform, equivalently the operations that Θ is supposed to encode. Boundary-of-boundary proves the equation for constructed sewing maps; it does not construct the maps.

54.4 Stable curves

The boundary strata of $\overline{\mathcal{M}}_{g,n}$ are indexed by stable graphs. Separating nodes correspond to edges joining distinct vertices. Nonseparating nodes correspond to loops. Clutching morphisms realize graph contraction geometrically.

Borel–Moore chains on the clutching spaces, with the usual determinant twists, form a modular operad under proper pushforward along gluing. After aritywise finite duality, or in a specified completed continuous dual, the corresponding cochains form a modular cooperad. If a system of multilinear maps on V is compatible with every clutching morphism, it defines an algebra over the Feynman transform of the chosen modular cooperad with its determinant twist. The master equation then states that the two ways of reaching every codimension-two boundary stratum cancel with orientation signs.

Principle 54.2 (Sewing principle)*.* The topology of $\overline{\mathcal{M}}_{g,n}$ supplies the relations among sewing maps. The coefficient theory must supply the sewing maps themselves.

54.5 Graph filtration and obstruction theory

Filter $\mathfrak{g}_{\mathcal{P},V}$ by the number of internal edges, or more generally by stable-graph complexity. The filtration is complete and compatible with the bracket. Suppose a genus-zero Maurer–Cartan element Θ_0 is fixed. A partial lift

$$\Theta_{\le r} = \Theta_0 + \Theta_1 + \cdots + \Theta_r$$

has curvature in filtration $r+1$. Its leading term is a cocycle for the twisted differential $d_{\Theta_0} = d + [\Theta_0, -]$.

Theorem 54.3 (Filtered obstruction theorem)**.** *The obstruction to extending $\Theta_{\le r}$ to filtration $r+1$ is a canonical class*

$$\mathfrak{o}_{r+1}(\Theta_{\le r}) \in H^2\big(\mathrm{gr}_F^{r+1}\, \mathfrak{g}_{\mathcal{P},V}, d_{\Theta_0}\big)\,. \tag{54.4}$$

The obstruction class vanishes exactly when an extension exists. When it vanishes, the set of extensions modulo gauge transformations trivial in lower filtration is a torsor for the corresponding H^1.

Proof. The complete filtered Maurer–Cartan equation supplies the stated obstruction theory. The Bianchi identity implies that the leading curvature is closed. Changing a lift alters it by a coboundary. Solving the next equation is equivalent to choosing a primitive; two primitives differ by a closed degree-one element. □

This graph obstruction tower is invariant. Restricting to an arbitrary primary line need not preserve the bracket and does not produce an invariant tower unless the restriction is a morphism of deformation complexes.

54.6 Connected graph cumulants

Suppose a cyclic trace assigns a scalar weight $W(\Gamma)$ to every stable graph and the graph sum converges formally in $\hbar$:

$$Z = 1 + \sum_{\Gamma} \frac{\hbar^{g(\Gamma)}}{|\operatorname{Aut}\Gamma|} W(\Gamma). \tag{54.5}$$

The exponential formula for combinatorial species gives

$$\log Z = \sum_{\Gamma \text{ connected}} \frac{\hbar^{g(\Gamma)}}{|\operatorname{Aut}\Gamma|} W(\Gamma). \tag{54.6}$$

Thus connected amplitudes are cumulants of the full partition function. This is a canonical scalar projection once the trace is fixed. It does not classify the underlying chiral algebra.

54.7 Graphwise duality

Let V and $V^\vee$ be paired cyclic objects. Transposing every vertex operation and using the inverse pairing on every internal edge sends a graph weight for V to a graph weight for $V^\vee$. Orientation and determinant lines contribute the standard signs.

Proposition 54.4 (Graphwise transpose)**.** *Assume that the coefficient systems are aritywise perfect, that continuous duality commutes with the stable-graph completion, and that the cyclic pairing identifies their endomorphism modular operads. Then transposition at every vertex and contraction with the inverse pairing on every edge induce a filtered isomorphism between the two modular convolution complexes. It intertwines their master equations and sends connected graph weights to their transposes.*

Proof. On each stable graph, dualize every vertex operation and contract each internal edge with the inverse cyclic pairing. Perfectness identifies the resulting tensor with the coefficient assigned to the same graph for the dual theory. Edge contraction and graph differential are preserved because evaluation and coevaluation satisfy the zig-zag identities. Continuity permits passage from each finite graph filtration quotient to the completed product. The construction is therefore a filtered isomorphism of the modular convolution complexes. □

Any scalar complementarity formula is a trace of this graphwise statement and depends on the chosen duality and trace. There is no universal scalar conductor shared by levels, central charges, and Hodge classes.

54.8 The Batalin–Vilkovisky reading

In the BV formalism, fields form a shifted symplectic space, the bracket is induced by the symplectic pairing, and Δ is the second-order operator corresponding to contraction of a loop. The quantum action S satisfies

$$(Q + \hbar\Delta)e^{S/\hbar} = 0, \tag{54.7}$$

which expands to

$$QS + \frac{1}{2}\{S, S\} + \hbar\Delta S = 0.$$

The BV equation has the same algebraic shape as (54.3). Equality of the two equations for a particular theory requires a map from the renormalized BV complex to the modular endomorphism complex. Compactified configuration spaces supply that map in perturbative models where the Feynman weights extend to the boundary.

54.9 Anomalies

If the quantum master equation fails, the failure is a degree-one local cocycle, the anomaly. Its cohomology class is independent of the choice of counterterm. An anomaly can be cancelled by adding a local counterterm exactly when the class vanishes. A projective connection on conformal blocks is a different but related manifestation: the curvature is central rather than zero. Tensoring by a line with opposite curvature converts projective flatness into flatness.

On the moduli space of curves, the anomaly line is the determinant or Atiyah line whose curvature equals the scalar projective curvature of the conformal-block connection.

Chapter 55

Conformal blocks, sewing, and the anomaly line

55.1 Coinvariants and blocks

Let V be a conformal vertex algebra and let $M_1, \ldots, M_n$ be modules inserted at marked points of a smooth curve C. The global chiral algebra acts on $M_1 \otimes \cdots \otimes M_n$. The coinvariants are the quotient by this action; conformal blocks are their duals. Varying the pointed curve produces sheaves on moduli.

Damiolini–Gibney–Tarasca construct sheaves of coinvariants for vertex algebras on moduli of pointed stable curves and prove factorization and vector-bundle results under finiteness and semisimplicity hypotheses [66, 68]. For strongly rational vertex operator algebras, Damiolini–Woike prove that the conformal blocks define a modular functor [69].

The modular functor is the positive-genus coefficient system required by Theorem 54.2. Its genus-zero restriction is built from intertwining operators and the vertex tensor category. Its sewing maps are convergent analytic operations and satisfy the mapping-class-group relations.

55.2 Projective flatness

The sheaf of conformal blocks carries a projectively flat connection. Infinitesimal changes of complex structure act through the stress tensor. The central term in the Virasoro algebra makes the action projective. Algebraically, the sheaf is a module for an Atiyah algebra rather than for the untwisted tangent sheaf.

Let $\lambda = c_1(\mathbb{E})$ be the Hodge class and ψ_i the cotangent-line classes at the markings. If a_i is the conformal weight of M_i, the interior projective class has the form

$$\mathrm{rk}(\mathbb{V}) \left(\frac{c}{2}\lambda + \sum_i a_i \psi_i \right). \tag{55.1}$$

On the compactification, boundary terms are determined by factorization through intermediate modules. Their coefficients depend on fusion rules and conformal weights [66, 67].

External theorem 55.1 (Chern-class form of the anomaly)**.** *For a conformal-block bundle satisfying the factorization hypotheses, its first Chern class is the sum of the interior class* (55.1) *and explicit boundary divisor contributions obtained by sewing through simple modules. In particular, the Hodge coefficient is* $\mathrm{rk}(\mathbb{V})c/2$*, while the full class is not determined by* c

alone. [66, 67]

This theorem separates the local central charge from the global geometry. Two theories with the same central charge may have different ranks, conformal weights, fusion rules, and boundary classes.

55.3 The anomaly line

A projectively flat bundle E has curvature

$$\nabla_E^2 = \Omega \ \mathrm{id}_E$$

for a scalar two-form Ω. Let L be a line bundle with connection satisfying $\nabla_L^2 = -\Omega$. Then

$$\nabla_{E\otimes L}^2 = 0.$$

If only a rational multiple of the required line exists, one works with a twisted D-module or on the corresponding gerbe.

Proposition 55.2 (Anomaly cancellation by a line)**.** *Tensoring a projectively flat conformal-block system by a line or twisted line of opposite Atiyah class produces a flat system. The construction is compatible with sewing when the anomaly lines satisfy the clutching multiplicativity dictated by the Hodge and cotangent-line bundles.*

Proof. Curvatures add under tensor product of connections. The curvature of the conformal-block system is scalar, say $\Omega\,\mathrm{id}$, while the chosen line has curvature $-\Omega$. Their tensor product is flat. On a clutching divisor, multiplicativity of the anomaly line identifies the tensor product of the two component lines with the restriction of the global line, so the same cancellation commutes with sewing. □

The proposition twists the connection on the family of conformal blocks while preserving the internal vertex-algebra differential.

55.4 Sewing and the master equation

Choose a perfect pairing between a module and its contragredient. Sewing two marked points inserts the inverse pairing and sums over a homogeneous basis. In a local sewing parameter q, the operation contains the propagator

$$q^{L_0-c/24}.$$

Convergence is analytic; factorization identifies the limit at $q = 0$ with the direct sum over intermediate modules.

The modular-functor sewing maps define an algebra over the chains of the surface operad. After the anomaly twist of Theorem 55.2, they determine a solution of the master equation in the modular convolution algebra. This conclusion follows from Theorem 54.1; no additional universal scalar is introduced.

55.5 Genus one

At genus one, Zhu's theorem gives modular transformation laws for trace functions of rational vertex operator algebras [121]. For a holomorphic VOA,

$$Z_V(\tau) = \mathrm{Tr}_V\, q^{L_0 - c/24}$$

is a modular function with the multiplier determined by the central charge and conformal weights. For a rational VOA, the vector of irreducible characters transforms under a finite-dimensional representation of $SL_2(\mathbb{Z})$.

The projective anomaly appears as the factor $q^{-c/24}$ and the multiplier system. The modular S-matrix and fusion coefficients are related by the Verlinde formula when the representation category is modular [91, 116]. The dimension of a conformal-block space is a trace of the modular functor; the $SL_2(\mathbb{Z})$ representation is strictly richer.

55.6 Prime forms and analytic kernels

On a compact Riemann surface, the prime form $E(z, w)$ is a canonical section, after the choice of a theta characteristic, that vanishes simply on the diagonal. The differential $d_z \log E(z, w)$ has the correct local simple pole but is quasi-periodic. Adding a harmonic correction produces a single-valued Green kernel. Fay's identities control products of prime forms and Szego kernels [76].

These kernels are analytic representatives of propagators and connections. Their quasi-periodicity produces monodromy and projective curvature. They do not define a universal fiberwise differential whose square is $c_1(\mathbb{E})$. The base and fiber degrees must remain distinct.

55.7 Free bosons and determinant lines

For a rank-r free boson, the oscillator partition function is

$$\eta(\tau)^{-r}.$$

The factor η^{-r} is the regularized determinant of the chiral Laplacian, while the nonchiral partition function includes $(\mathrm{Im}\,\tau)^{-r/2}|\eta(\tau)|^{-2r}$. On higher-genus surfaces, the determinant line of the $\bar{\partial}$ operator carries the Quillen metric and curvature given by the family index theorem.

This is the precise meeting point of Polyakov's conformal anomaly and the geometry of determinant lines. The coefficient is the central charge of the free fields. Interacting theories add non-Gaussian conformal blocks; they are not determined by the determinant line.

55.8 Lattice theories and Siegel theta series

For an even positive-definite lattice L of rank r, the genus-one character of the lattice VOA is

$$Z_{V_L}(\tau) = \frac{\Theta_L(\tau)}{\eta(\tau)^r}. \tag{55.2}$$

At higher genus the zero modes contribute the Siegel theta series $\Theta_L^{(g)}(\Omega)$, while oscillators contribute determinant factors. Equal genus-one theta series need not imply equal lattices or equal VOAs.

For the two even unimodular lattices of rank sixteen, the degree-g Siegel theta series agree for $g \leq 3$. Their degree-four difference is proportional to the Schottky form and vanishes on the Jacobian locus. Hence genus-two does not separate the corresponding lattice theories. This example shows why every arithmetic claim must specify the genus and the moduli locus.

55.9 Genus two and beyond

Genus two introduces a three-dimensional Siegel upper half-space, separating and nonseparating boundary divisors, and stable graphs with loops. The period matrix and the off-diagonal sewing parameter couple channels that are independent at genus one. For rational VOAs, the modular functor gives finite-dimensional blocks with mapping-class-group monodromy. For lattice VOAs, Siegel theta series provide explicit sections.

No universal genus expansion follows from central charge alone. The stable-graph sum depends on all vertex and propagator data. The central charge controls the anomaly line; fusion and OPE coefficients control the non-scalar graph weights.

Chapter 56

Anomalies, curvature, and invertible field theories

56.1 An anomaly has a domain

The word *anomaly* denotes an obstruction to making a classical symmetry act strictly after quantization. Its mathematical type depends on the symmetry and on the object on which the symmetry acts. The following classes occur in the present subject:

central extension of a local operator algebra,	$[\omega_{\mathrm{loc}}] \in H^2_{\mathrm{Lie}}(\mathfrak{s};\mathbb{C})$,
projective curvature of a family of states,	$[\Omega_{\mathrm{fam}}] \in H^2(B;\mathbb{C})$,
BV quantization obstruction,	$[\mathfrak{A}_{\mathrm{BV}}] \in H^1_{\mathrm{loc}}(\mathrm{Obs}^{\mathrm{cl}})$,
BRST obstruction,	$[Q^2] \in H^2(\mathrm{End}\, C)$,
framing anomaly,	$\alpha_{\mathrm{fr}} \in \pi_0\, \mathrm{InvFT}^{\mathrm{fr}}_{d+1}$,
curvature of a curved Koszul object,	$m_0 \in C^2$.

The symbols lie in different complexes. A theorem can identify their images under transgression or a comparison map. No scalar is their common definition.

56.2 Local central extensions

The affine current algebra has central term

$$J^a(z)J^b(w) \sim \frac{k(a,b)}{(z-w)^2} + \frac{J^{[a,b]}(w)}{z-w}.$$

The Virasoro algebra has central term

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}.$$

The parameters k and c classify different central extensions. The Sugawara construction relates them for a chosen affine theory:

$$c_{\mathrm{Sug}} = \frac{k\, \dim \mathfrak{g}}{k+h^\vee}.$$

The affine level is k, whereas the Sugawara central charge is the displayed rational function with a pole at $k = -h^\vee$.

A central extension is part of the chiral operation. It does not prevent the chiral or ordered bar differential from squaring to zero. Square-zero follows from the complete algebra identities, including invariance of the cocycle.

Proposition 56.1 (Central terms and the ordered bar)**.** *Let A be an augmented dg E_1 algebra in* $\mathrm{FactD}(X)$ *whose chiral coefficient carries a central extension. The ordered bar coderivation satisfies $b^2 = 0$. The central term contributes to b through the multiplication of A or through its coefficient differential; it is not a scalar curvature of the simplicial bar.*

Proof. The simplicial identities of the bar construction use only the associativity and chain-map property of multiplication. A central extension satisfying the algebra identities is already included in that multiplication. Hence the standard proof of $b^2 = 0$ is unchanged. □

56.3 Projective curvature over moduli

Let $\mathbb{V} \to B$ be a locally free sheaf of conformal blocks with its projectively flat connection. A projectively flat connection satisfies

$$\nabla_{\mathbb{V}}^2 = \Omega \ \mathrm{id}_{\mathbb{V}} .$$

The class $[\Omega]$ is the curvature of a line-valued central extension of the tangent action on B. For pointed stable curves, the interior contribution to the first Chern class is

$$\mathrm{rk}(\mathbb{V}) \left(\frac{c}{2}\lambda + \sum_i a_i \psi_i \right),$$

and factorization supplies boundary terms. The central charge controls one coefficient of this class; the conformal weights and fusion channels control the rest.

Theorem 56.2 (Type of the conformal anomaly)**.** *The conformal anomaly of a family of chiral blocks is the central extension encoded by the Atiyah algebra of the block bundle. After a projective connection is chosen, the same class is represented by its scalar curvature. It acts in the moduli direction and is not the square of the internal differential on an individual block complex.*

Proof. The internal differential is defined before the base curve varies and has cohomological degree one in the fibre. The projective curvature is a two-form on the base acting by a scalar endomorphism of the fibre. They belong to the vertical and horizontal parts of the superconnection. The family Ward identity identifies the horizontal curvature with the central extension of vector fields represented by the stress tensor. □

56.4 The superconnection equation

The correct object combining fibre and base is a superconnection

$$\mathbb{D} = d_{\mathrm{fib}} + \nabla + \mathbb{A}_2 + \mathbb{A}_3 + \cdots .$$

Its square has total degree two. Flatness means $\mathbb{D}^2 = 0$; projective flatness means $\mathbb{D}^2 = \Omega\, \mathrm{id}$.

Proposition 56.3 (Cancellation by an anomaly line)**.** *If a line bundle L with connection satisfies $\nabla_L^2 = -\Omega$, then the tensor-product superconnection on $\mathbb{V} \otimes L$ is flat. If the class of Ω is only rationally integral, the same statement holds in the corresponding gerbe of twisted bundles.*

Proof. For the tensor-product superconnection, the two curvature terms act on separate factors and the mixed commutator vanishes. Hence $(\mathbb{D} \otimes 1 + 1 \otimes \nabla_L)^2 = (\Omega - \Omega)\,\mathrm{id} = 0$. A rationally integral class defines the corresponding twisted line on a gerbe, and the identical curvature computation holds there. $\square$

The line L must exist. Writing a formal symbol L^{-1} does not construct a root of a determinant line.

56.5 Polyakov's Weyl anomaly

Let $g = e^{2\sigma}\hat{g}$ be two metrics in the same conformal class on a closed surface. For a two-dimensional conformal field theory of central charge c, the change of the renormalized effective action has the local form

$$W[e^{2\sigma}\hat{g}] - W[\hat{g}] = \frac{c}{24\pi} \int \left(|\nabla_{\hat{g}}\sigma|^2 + R_{\hat{g}}\sigma \right) d\mathrm{vol}_{\hat{g}}, \tag{56.1}$$

up to convention and local topological terms. Differentiating gives the trace anomaly

$$\langle T^{\mu}{}_{\mu} \rangle = -\frac{c}{24\pi} R.$$

The family index theorem identifies the curvature of the corresponding determinant line over moduli. This is the analytic origin of the $c\lambda/2$ term in the chiral block connection.

The Polyakov action concerns the response to a Weyl transformation. The ordered E_1 multiplication concerns fusion in a transverse line. Their relation passes through the stress tensor and the determinant line, not through the associative bar differential.

56.6 BRST anomaly

A BRST operator Q is required to satisfy $Q^2 = 0$. In the bosonic string, the total BRST charge squares to a multiple of the total central-charge defect and the intercept defect. Matter and ghosts cancel the central extension when

$$c_{\text{matter}} + c_{bc} = 0, \qquad c_{bc} = -26.$$

The cancellation equation is additivity of anomaly classes under tensor product of chiral theories.

Theorem 56.4 (BRST cancellation is tensorial)**.** *Let V and G be conformal complexes with commuting BRST data. The BRST operator on $V \otimes G$ is $Q_V \otimes 1 + 1 \otimes Q_G$. Its square is the sum of the two anomaly endomorphisms. Hence nilpotence is cancellation of anomaly classes under tensor product. It is not a general identity between V and a Koszul dual of V.*

Proof. The two summands of the total BRST operator are odd and act on separate tensor factors, so their supercommutator is zero. Squaring gives $Q_V^2 \otimes 1 + 1 \otimes Q_G^2$. If each square is a scalar anomaly endomorphism, the total square is their sum. Nilpotence is therefore exactly cancellation of these two classes. ☐

56.7 BV anomaly

For a classical interaction I, define the renormalized quantum master defect

$$\mathfrak{A}(I_\hbar) = QI_\hbar + \frac{1}{2}\{I_\hbar, I_\hbar\} + \hbar\Delta I_\hbar.$$

At the first unsolved order in $\hbar$, $\mathfrak{A}$ is closed for the classical BV differential. Its cohomology class is independent of counterterm representatives. A quantization exists to the next order precisely when this class vanishes.

Proposition 56.5 (Locality of the BV anomaly)**.** *The BV anomaly is represented by a local functional. In a configuration-space renormalisation scheme, its graph representatives are singular near partial diagonals before extension; after Fulton–MacPherson or Axelrod–Singer compactification, the obstruction is represented by the uncancelled boundary terms of the renormalised graph weights together with local counterterm faces.*

Proof. The renormalized quantum master defect is assembled graph by graph from the same local vertices, propagator, and counterterms as the effective interaction. On pairwise disjoint supports, factorisation separates every graph into the tensor product of its connected components; the connected defect therefore has support on the small diagonals and defines a local functional. Extend each graph weight to the compactified configuration space. Stokes' theorem expresses the master defect as the oriented sum of boundary restrictions. Faces that split an internal edge reproduce the classical bracket and differential terms, while divergent collision faces are exactly the faces corrected by local counterterms. The remaining oriented boundary sum represents the anomaly class. Thus both the support and the graph representative are local. This is the configuration-space form of locality of renormalisation in perturbative BV theory [14, 43]. ☐

Fulton–MacPherson or Axelrod–Singer compactification supplies the boundary geometry of the quantum anomaly calculation. The coefficient system consists of propagators and graph weights, not only the incidence complex of collisions.

56.8 Framing anomaly

Chern–Simons theory is invariant under orientation-preserving diffeomorphisms only after a choice of framing or a compensating gravitational Chern–Simons term. The framing change multiplies the partition function and Wilson-line observables by phases determined by the central charge of the associated modular tensor category. This anomaly lives in three-dimensional geometry.

The ribbon twist of the line category records the same phase on framed knots. Its relation to the two-dimensional WZW central charge follows from the Chern–Simons/WZW correspondence. It is not determined by an arbitrary E_1 chiral algebra.

56.9 Curved Koszul duality

A curved dg algebra or coalgebra has a curvature element m_0 satisfying

$$d^2(x) = [m_0, x]$$

in the associative case, with the corresponding A_∞ identities in the homotopy case. If m_0 is a central scalar, the commutator vanishes. Such a scalar may still change the Maurer–Cartan equation, but it does not make d^2 nonzero.

Curvature arises naturally when an inhomogeneous relation has a constant term. For example, quadratic-linear-constant Koszul duality places the constant term in m_0. This algebraic curvature is not $c_1(\mathbb{E})$ unless a family comparison sends it to that class.

No-go theorem 56.6 (No universal anomaly scalar). There is no functorial scalar $\kappa(A)$ that simultaneously determines the affine level, the Virasoro central charge, the projective Chern class of all conformal-block bundles, the BRST anomaly, the framing anomaly, and the curvature of every Koszul dual.

Proof. The affine level and Virasoro central charge vary independently before a stress tensor is chosen. Conformal-block Chern classes also depend on module weights and boundary fusion coefficients. BRST anomalies depend on the ghost complex; framing anomalies depend on a three-dimensional ribbon structure; curved Koszul terms depend on the chosen inhomogeneous presentation. Since these data can be varied independently while holding the others fixed, no such scalar is functorial. ☐

56.10 Anomalies as invertible theories

For local anomalies admitting the standard anomaly-field-theory description, the anomaly is represented by an invertible field theory one dimension higher. The anomalous theory is relative to its state lines, and cancellation is a trivialisation of the tensor product anomaly.

For chiral conformal blocks, the invertible theory is represented by a determinant or Hodge line. For fermions it is the determinant line of the Dirac operator. For Chern–Simons theory it includes the framing line. This viewpoint explains additivity: tensoring theories tensors anomaly lines and adds their characteristic classes.

Principle 56.7 (Anomaly comparison). Two anomalies are the same only when a morphism of invertible theories identifies their state lines, connections, and gluing laws. Equality of one numerical coefficient is a consequence of such an identification, not its substitute.

56.11 The anomaly square

The complete theory contains four compatible arrows:

$$\begin{tikzcd}
\text{local central extension} \arrow[r, "\text{Ward identity}"] \arrow[d, "\text{BV descent}"'] & \text{Atiyah algebra of blocks} \arrow[d, "c_1"] \\
\text{local BV anomaly} \arrow[r, "\text{family index}"'] & \text{curvature of determinant line.}
\end{tikzcd} \tag{56.2}$$

Each arrow is constructed in a specified field theory. For free chiral fields and WZW models, the standard Ward-identity and family-index calculations construct the arrows of this square. For a general ordered chiral algebra, the upper-left corner exists while the other corners require a stress tensor, a family of states, and a BV realization.

The square is the precise form of the unity sought by the subject. It gives one language without declaring unlike objects equal.

Part IX

Exact models, physics, and arithmetic

Chapter 57

Exact examples

57.1 Heisenberg

Let $\mathcal{H}_k$ be generated by a current J with

$$J(z)J(w) \sim \frac{k}{(z-w)^2}. \tag{57.1}$$

The Heisenberg chiral operation begins with the second normal symbol of the diagonal, carrying the level k. The corresponding ordinary Lie conformal bracket is

$$[J_\lambda J] = k\lambda.$$

Every iterated bracket with the central value vanishes, so the three Jacobi terms are zero.

The chiral Chevalley coalgebra is cofree on the abelian chiral Lie algebra with the central extension built into the operation. Ordered monodromy is scalar. For vacuum and Fock-module insertions on a compact curve, the conformal blocks admit Fock-space descriptions and their partition functions are determinant expressions. The rank-one conformal vector has central charge one, independent of the nonzero normalization k after rescaling the current. This already separates level from central charge.

57.2 Affine Kac–Moody

For $V_k(\mathfrak{g})$ the current OPE is (50.3). The simple pole is the Lie bracket; the double pole is the invariant bilinear form. The chiral Jacobi identity splits into the ordinary Jacobi identity and invariance of the form:

$$([a,b],c) = (a,[b,c]).$$

The Chevalley differential formed from the simple-pole bracket squares to zero by itself. The level term is not required to repair Jacobi. It supplies a central extension and enters conformal and representation-theoretic constructions.

At noncritical level the Sugawara vector has central charge

$$c_{\mathrm{aff}} = \frac{k \dim \mathfrak{g}}{k+h^\vee}. \tag{57.2}$$

The KZ connection and its quantum-group monodromy are described in Chapter 50. At positive integral level, the integrable modules of the simple affine VOA $L_k(\mathfrak{g})$ form a modular tensor category, and their conformal blocks satisfy the Verlinde formula.

At level one for simply laced $\mathfrak{g}$, the Frenkel–Kac lattice realization expresses the affine currents inside a lattice VOA [79]. The nonabelian simple-pole bracket remains present; it is not killed by the realization.

57.3 Virasoro

The Virasoro vertex algebra is generated by T with

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \tag{57.3}$$

The three singular terms lie in the fourth, second, and first normal symbols. Their relation is fixed by conformal covariance. The central charge c is the coefficient of the fourth-order symbol and the central extension of the vector-field algebra.

The vacuum Verma module has a level-four quasi-primary

$$\Lambda =: TT : -\frac{3}{10}\partial^2 T$$

whose norm is proportional to $c(5c+22)$. The zeros at $c=0$ and $c=-22/5$ signal null vectors. This Gram determinant is representation-theoretic data. It does not define a universal quartic obstruction tower.

Conformal blocks carry a projective connection with Hodge coefficient $c/2$. At $c=26$, coupling to the (b,c) ghost system of weights $(2,-1)$ cancels the total central charge. At $c=13$ no universal self-duality theorem for Virasoro algebras follows; 13 is merely the midpoint of the reflection $c \mapsto 26-c$ used in bosonic string bookkeeping.

57.4 Free fermions

A complex free fermion has fields ψ, ψ^* with

$$\psi(z)\psi^*(w) \sim \frac{1}{z-w}.$$

The chiral operation lies in the first normal symbol. Wick's theorem determines all higher correlators as determinants. The representation category, spin structures, and modular transformation law depend on whether one uses a real, complex, or lattice extension.

Boson–fermion correspondence identifies the charged free fermion VOA with a rank-one lattice VOA. This is an isomorphism of vertex algebras after the lattice extension; it is not the statement that the Heisenberg and fermion chiral Lie algebras are linear Koszul duals.

57.5 The $\beta\gamma$ and bc systems

A first-order bosonic system has

$$\beta(z)\gamma(w) \sim \frac{1}{z-w},$$

while a fermionic bc system has the same singular kernel with odd parity. Their oscillator algebras are Weyl and Clifford algebras. For fields of weights $(\lambda, 1-\lambda)$, the central charges are

$$c_{\beta\gamma} = 2(6\lambda^2 - 6\lambda + 1), \qquad c_{bc} = -2(6\lambda^2 - 6\lambda + 1). \tag{57.4}$$

The opposite signs reflect statistics. The determinant lines cancel when the systems are paired with identical weights.

The simple pole does not imply that the full algebra has only one homotopical layer. Composite fields and modules may generate nontrivial higher operations. Pole order and deformation cohomology are independent.

57.6 Lattice vertex algebras

Let L be an even positive-definite lattice. The lattice VOA V_L has central charge $\operatorname{rk} L$. Its fields include Heisenberg currents and exponentials e^α. Their operator products contain powers

$$(z-w)^{(\alpha,\beta)}$$

and a cocycle $\epsilon(\alpha, \beta)$. Locality requires the commutator of the cocycle to match the parity determined by the lattice form.

If L is even unimodular, V_L is holomorphic. The character is (55.2). Inverting the cocycle gives the contragredient braiding; a symmetric cocycle may make the theory self-contragredient. This is a statement about the lattice extension, not about all chiral Koszul duals.

57.7 Leech and Moonshine

The Leech lattice VOA V_Λ and the Moonshine VOA $V^\natural$ both have central charge 24. Their weight-one spaces differ:

$$(V_\Lambda)_1 \cong \mathbb{C}^{24}, \qquad (V^\natural)_1 = 0.$$

Their characters also differ in the constant term after the vacuum normalization. Thus central charge does not classify holomorphic VOAs, their chiral operations, or their factorization coalgebras.

The Monster action on $V^\natural$ produces McKay–Thompson series. These are traces of automorphisms on the state space. They are arithmetic shadows of the VOA, not coefficients of a universal modular Koszul invariant.

57.8 Principal W-algebras

The principal W-algebra $\mathcal{W}^k(\mathfrak{g})$ is obtained by BRST reduction. At generic level its standard strong generators have conformal weights equal to the exponents of $\mathfrak{g}$ plus one; special levels may impose decoupling relations or collapse generators. Their OPEs are nonlinear and contain normally ordered composites. The full chiral operation nevertheless satisfies the same Maurer–Cartan equation (45.2).

For $\mathfrak{sl}_2$, the principal W-algebra is Virasoro. For $\mathfrak{sl}_3$, it is generated by T and a weight-three field W; the WW OPE contains T, its derivatives, and the quasi-primary Λ. These terms are fixed by Jacobi and conformal covariance. No single scalar extracted from the WW OPE determines the algebra.

Feigin–Frenkel duality uses (52.5). The central charges of dual pairs agree because the vertex algebras are isomorphic. This duality is distinct from KZ inverse braiding.

57.9 Critical affine algebras

At $k = -h^\vee$, the affine OPE remains (50.3), but the Sugawara conformal vector ceases to exist. The Feigin–Frenkel center becomes large and is identified with opers. KZ equations in their usual WZW normalization become singular. The critical level is therefore a change of structure, not an uncurved point of a universal bar differential.

57.10 A comparison table

family	highest singular symbol	ordered structure	positive-genus structure
Heisenberg	second normal symbol k	scalar monodromy	determinant line and Fock sewing
affine $V_k(\mathfrak{g})$	bracket plus second symbol	KZ braid monodromy	WZW conformal blocks
Virasoro	fourth symbol $c/2$	conformal-block monodromy	projective anomaly $c\lambda/2$ plus boundary data
free first-order	first symbol	Weyl/Clifford statistics	determinant lines
lattice V_L	lattice powers and cocycle	lattice braiding	Siegel theta series
principal W	nonlinear finite OPEs	screening monodromy	W-conformal blocks when finite
critical affine	bracket and critical central extension	oper geometry replaces KZ	Feigin–Frenkel center

The table records the full structures realized by each family. Each row realizes the diagonal principle in a different representation category.

Chapter 58

Strictly noncommutative chiral algebras

58.1 Free ordered chiral algebras

Let $M \in \mathrm{FactD}(X)$. The free augmented ordered chiral algebra is

$$T_X(M) = \mathbf{1} \oplus M \oplus M^{\otimes 2} \oplus M^{\otimes 3} \oplus \cdots, \tag{58.1}$$

with concatenation. The tensor products are taken in $\mathrm{FactD}(X)$, so each word retains the full chiral coefficient system of its letters.

Proposition 58.1 (Bar homology of the free algebra)**.** *Assume the tensor products in* (58.1) *are exact. The reduced bar of $T_X(M)$ contracts onto $M[1]$. Equivalently,*

$$\mathbf{1} \otimes^L_{T_X(M)} \mathbf{1} \simeq \mathbf{1} \oplus M[1]$$

as a conilpotent coalgebra, with zero reduced coproduct on $M[1]$.

Proof. The word-length decomposition gives a fibre sequence of left $T_X(M)$-modules

$$T_X(M) \otimes M \xrightarrow{\text{mult}} T_X(M) \longrightarrow \mathbf{1}.$$

On the summand of words of positive length, multiplication is the canonical identification obtained by separating the last letter; the cofiber is the empty word. Applying $\mathbf{1} \otimes^L_{T_X(M)} -$ leaves the two-term complex $M \to \mathbf{1}$ with zero reduced differential, hence the class $M[1]$. Equivalently, the standard extra degeneracy of the normalized bar contracts every bar word having an internal cut and leaves the one-letter cogenerator. $\square$

Thus the Koszul dual algebra of a free associative object is the square-zero extension

$$T_X(M)^! \simeq \mathbf{1} \oplus M^\vee[-1]$$

when M is dualizable. Noncommutative freedom on one side becomes a first-order infinitesimal object on the other.

58.2 Square-zero chiral algebras

Conversely, let

$$A = \mathbf{1} \oplus M, \qquad M^2 = 0.$$

The bar multiplication differential vanishes, and

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) = \bigoplus_{r \geq 0} M[1]^{\otimes r}$$

with deconcatenation. Its continuous dual is the completed tensor algebra

$$A^! \simeq \widehat{T}_X(M^\vee[-1]).$$

This pair is the local model of an infinitesimal boundary coupling and its algebra of iterated defect insertions.

58.3 Quiver chiral algebras

Let Q be a finite quiver, let

$$S = \bigoplus_{i \in Q_0} \mathbf{1} e_i,$$

and assign an object $M_a \in \mathrm{FactD}(X)$ to every arrow $a : i \to j$. Put

$$M = \bigoplus_{a \in Q_1} e_{t(a)} M_a e_{s(a)}.$$

The relative tensor algebra

$$A_Q = T_S(M) \tag{58.2}$$

is an E_1 algebra in factorization D-modules. Multiplication concatenates composable arrows and is zero otherwise. The augmentation $A_Q \to S$ forgets nontrivial paths.

The relative bar

$$B_S(A_Q) = S \otimes^L_{A_Q} S$$

retains the ordered path structure. If Q has no relations, it contracts to $S \oplus M[1]$. If relations $R \subset M \otimes_S M$ are imposed, the quadratic coalgebra begins

$$S \oplus M[1] \oplus R[2] \oplus \cdots .$$

Example 58.2 (Two boundary conditions)**.** For the quiver $1 \rightleftarrows^{v}_{u} 2$, the objects M_u and M_v are boundary-changing fields. The composites uv and vu are endomorphisms of different boundaries and need not agree. Their chiral insertion points may collide on X, while the order of composition is fixed by the transverse interval. This is strict noncommutative chirality in its smallest form.

A cyclic potential W determines a cyclic functional on the completed path algebra. After choosing the standard Ginzburg or cyclic A_∞ construction, its cyclic derivatives enter the differential and are the classical equations of motion of the corresponding open-string field theory. The ribbon genus of Chapter 53 is then the genus of the resulting open-string Feynman surface.

58.4 Matrix and Chan–Paton extensions

Let V be a commutative algebra object in $\mathrm{FactD}(X)$ and let E be a finite-dimensional vector space. The matrix extension

$$A = \mathrm{End}(E) \otimes V \tag{58.3}$$

is an E_1 algebra in $\mathrm{FactD}(X)$. Its multiplication is matrix multiplication times the chiral factorization product. The OPE coefficients are $\mathrm{End}(E)$-valued and preserve their order.

The algebra (58.3) is Morita equivalent to V. Hence

$$Z_{\mathrm{ord,ch}}(A) \simeq Z_{\mathrm{ord,ch}}(V)$$

and their perfect module categories have equivalent centres. The open amplitudes differ by Chan–Paton traces, producing powers of $\dim E$ on ribbon graphs.

Proposition 58.3 (Chan–Paton weight)**.** *For a connected ribbon graph Γ contributing to a cyclic amplitude of* (58.3)*, contraction of matrix indices contributes*

$$(\dim E)^{F(\Gamma)},$$

where $F(\Gamma)$ is the number of index faces. For a closed ribbon surface of genus h with b boundary components, this is $(\dim E)^{2-2h-b}$ after the standard normalization of vertices and propagators.

Proof. Write the matrix propagator with two oriented index strands. Every closed index strand contributes the sum of one free index, hence a factor $\dim E$. The closed strands are the faces of the ribbon graph, giving $(\dim E)^{F(\Gamma)}$. With the conventional factor $(\dim E)^{-1}$ at each propagator and $(\dim E)$ at each vertex, the exponent becomes $V - E + F = 2 - 2h - b$ by the Euler formula for the thickened ribbon surface. □

The Chan–Paton index loops are the exact algebraic origin of the 't Hooft expansion. It belongs to the transverse ribbon genus and is independent of the genus of the chiral curve.

58.5 Universal enveloping ordered chiral algebras

Let $\mathfrak{g}$ be a Lie algebra object in $\mathrm{FactD}(X)$. Its universal enveloping algebra

$$U_X(\mathfrak{g}) \in \mathrm{Alg}_{E_1}(\mathrm{FactD}(X))$$

is defined by the usual coequalizer of the tensor algebra by

$$x \otimes y - (-1)^{|x||y|} y \otimes x - [x, y].$$

The Poincare–Birkhoff–Witt filtration has

$$\mathrm{gr}\, U_X(\mathfrak{g}) \simeq \mathrm{Sym}_X(\mathfrak{g})$$

under the standard flatness hypotheses.

Theorem 58.4 (Associative bar and internal Chevalley chains)**.** *For a flat Lie object $\mathfrak{g}$ in*

FactD(X), *the canonical twisting morphism induces a quasi-isomorphism*

$$\mathrm{Chev}_X(\mathfrak{g}) \xrightarrow{\sim} \mathrm{Bar}_X^{\mathrm{ord}}\big(U_X(\mathfrak{g})\big).$$

Proof. Filter both sides by symmetric degree. The associated graded map is the ordinary Koszul resolution of $\mathrm{Sym}_X(\mathfrak{g})$ by the exterior coalgebra on $\mathfrak{g}[1]$. PBW identifies the associated graded of the enveloping algebra. Completeness and exactness of the filtration lift the associated-graded quasi-isomorphism. □

This theorem is the internal Lie–associative Koszul comparison in the pointwise coefficient category. It is distinct from the Beilinson–Drinfeld chiral Chevalley construction on the Ran space. The internal Lie bracket maps to the commutator in its enveloping algebra, and internal Lie chains compute the ordered bar of that enveloping algebra.

58.6 Affine currents as chiral coefficients

The affine current algebra is a Beilinson–Drinfeld chiral Lie algebra: its simple pole is the Lie bracket and its double pole is the invariant cocycle. Its chiral enveloping construction produces the universal affine chiral coefficient. This construction takes place in the chiral pseudo-tensor category and is distinct from the internal enveloping algebra of Theorem 58.4, which begins with a Lie object for the pointwise tensor product.

To obtain a strictly noncommutative ordered chiral algebra from affine data one must supply a transverse associative object: for example an endomorphism algebra of affine boundary conditions, a quiver of affine modules, or a free tensor algebra generated by the affine factorisation coefficient. The resulting internal bar resolves that transverse multiplication. The chiral Chevalley complex resolves the affine chiral Lie operation. A comparison between them is an enveloping or Morita theorem and is never automatic. In either construction the level is not required to repair Jacobi; invariance of the cocycle is the identity that makes the central extension chiral-Lie.

58.7 Weyl and Clifford chiral algebras

Let M carry a perfect pairing $\omega : M \otimes M \to \mathbf{1}[d]$. The tensor algebra modulo

$$xy - (-1)^{|x||y|}yx = \hbar\,\omega(x, y) \tag{58.4}$$

produces a Weyl or Clifford object according to parity. In the chiral category, M may be a factorization D-module and ω may contain a higher normal symbol along the diagonal.

The relation is quadratic-linear-constant. Its Koszul dual is curved: the constant term becomes the curvature m_0 of the dual coalgebra. The curvature belongs to the algebraic presentation. It is not a Hodge class on moduli.

Example 58.5 (The $\beta\gamma$ system)**.** The chiral Weyl algebra generated by β and γ has singular relation

$$\beta(z)\gamma(w) \sim \frac{1}{z-w}.$$

With matrix Chan–Paton factors, the ordered product is matrix multiplication while the singular coefficient is the Weyl pairing. The two operations are visibly independent.

Example 58.6 (The *bc* system)**.** The odd analogue has Clifford relation. Its determinant line contributes the ghost conformal anomaly. The Weyl–Clifford Koszul exchange is a statement about quadratic relations; the cancellation of their conformal anomalies is a separate determinant-line calculation.

58.8 Chiral differential operators

For a smooth variety Y, a sheaf of chiral differential operators exists after its obstruction gerbe has been trivialized. It supplies a chiral factorisation coefficient whose local fields include functions and vector fields on Y. The vertex or chiral operation of that coefficient is not, by itself, the transverse E_1 multiplication on the boundary endomorphism object.

Strict ordered examples are obtained from associative endomorphism objects in a category of chiral-differential-operator modules, from quiver endomorphisms of boundary conditions, or from matrix and tensor-algebra extensions of a dualizable coefficient. When Y carries a group action, BRST or quantum Hamiltonian reduction acts on such an ordered endomorphism algebra provided the BRST operator is a derivation of the transverse product. The resulting comparison is governed by Chapter 52.

58.9 Boundary endomorphism algebras

Let $\mathcal{B}$ be a factorization category of boundary conditions on X. For objects b_i, set

$$A_{ij} = \mathrm{RHom}_{\mathcal{B}}(b_i, b_j).$$

Composition gives

$$A_{jk} \otimes A_{ij} \longrightarrow A_{ik}.$$

The direct sum

$$A_{\mathcal{B}} = \bigoplus_{i,j} A_{ij}$$

is an algebra over the semisimple idempotent object $S = \oplus_i \mathbf{1}e_i$. Its relative bar computes chains of boundary changes. This is the categorical form of the quiver example and the native algebra of open strings.

Theorem 58.7 (Morita reconstruction from a boundary generator)**.** *Let $b = \oplus_i b_i$ be a compact generator of a stable factorization category $\mathcal{B}$. Then*

$$A_b = \mathrm{RHom}_{\mathcal{B}}(b, b)$$

is an E_1 algebra in $\mathrm{FactD}(X)$ *and*

$$\mathcal{B} \simeq \mathrm{Mod}_{A_b}(\mathrm{FactD}(X))$$

under the usual compact-generation hypotheses.

Proof. Composition of boundary-changing morphisms gives the E_1 multiplication on A_b, and factorisation of the boundary category makes it a factorisable coefficient. The compact-generator Morita theorem gives an equivalence on every member of a factorising basis. Compatibility of the generator and the local equivalences with restriction identifies the equivalences on overlaps; factorisation descent then glues them globally. □

The theorem explains why noncommutative chiral algebras are primary in boundary physics. The category of boundary conditions comes first; its endomorphism algebra is a coordinate chart on that category.

58.10 Factorization quantum groups

A factorization quantum group is not merely an E_1 algebra. It combines an algebra object, a compatible coalgebra or tensor-category structure, and meromorphic exchange. In the ordered language these pieces occupy different levels:

$$\begin{array}{ccl} E_1 \text{ multiplication} & : & \text{fusion,} \\ \text{factorization coproduct} & : & \text{splitting of lines or representations,} \\ E_2 \text{ enhancement} & : & \text{exchange,} \\ \text{ribbon trace} & : & \text{closure and link invariants.} \end{array}$$

A Yangian possesses all these structures in a compatible completed sense. A generic ordered chiral algebra possesses only the first.

58.11 A noncommutative Koszul pair

Let V be dualizable and let $R \subset V \otimes V$ be a factorization subobject. Define

$$A = T_X(V)/(R), \qquad A^{!,\mathrm{quad}} = T_X(V^\vee[-1])/(R^\perp).$$

The canonical twisting morphism $A^{¡} \to A$ gives the Koszul complex

$$K(A) = A \otimes A^{¡}.$$

Theorem 58.8 (Quadratic recognition in the ordered chiral category)**.** *Assume positive weight, degreewise dualizability, and strong convergence of the weight filtration. The following are equivalent:*

(i) $A^{¡} \to \mathrm{Bar}^{\mathrm{ord}}_X(A)$ *is a quasi-isomorphism;*

(ii) $\mathrm{Cobar}_X(A^{¡}) \to A$ *is a quasi-isomorphism;*

(iii) the left and right quadratic Koszul complexes are acyclic off weight zero;

(iv) $\mathrm{Tor}^A(\mathbf{1}, \mathbf{1})$ *is concentrated on the quadratic diagonal.*

Proof. Apply the fundamental twisting-morphism theorem internally in $\mathrm{FactD}(X)$ to the canonical quadratic twisting morphism $A^{¡} \to A$. It identifies the bar comparison, the cobar comparison, and acyclicity of the left and right twisted tensor products. The positive weight filtration is bounded in each total weight and converges strongly, so diagonal concentration of Tor is equivalent to those acyclicity statements. □

The proof is the fundamental theorem of twisting morphisms internal to $\mathrm{FactD}(X)$. The chiral coefficient changes the tensor products and support conditions, but not the formal equivalence of the four tests. When these conditions hold and finite duals exist, $A^{!,\mathrm{quad}}$ identifies with the boundary endomorphism algebra $\mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$; before that comparison the two symbols denote different constructions.

58.12 What these examples establish

The examples exhibit three independent kinds of noncommutativity.

Quiver and matrix algebras have noncommutative transverse multiplication even when their chiral coefficient is free or commutative.

Internal enveloping algebras have noncommutative transverse multiplication generated by an internal Lie object. Affine current algebras supply chiral coefficients; a strictly ordered affine example requires the additional boundary, quiver, tensor, or endomorphism multiplication described above.

Yangians and line categories add meromorphic braid exchange to noncommutative fusion.

The ordered chiral language accommodates all three without identifying them. That separation is the condition under which their Koszul dualities can be stated correctly.

Chapter 59

Arithmetic traces and automorphic forms

59.1 Arithmetic enters through periods and traces

The chiral operation is defined over an algebraic or analytic coefficient field. Arithmetic appears when one takes periods of configuration spaces, traces of monodromy, dimensions of conformal blocks, theta sums over lattices, or regularized determinants. These operations are functorial shadows. Their arithmetic is controlled by the geometry of the chosen trace.

There are two basic sources. Ordered iterated integrals produce multiple zeta values and their elliptic analogues. Symmetric and modular traces produce modular forms, theta series, and characteristic numbers. The two sources can interact, but neither is determined by the other.

59.2 The KZ associator

The regularized holonomy of the KZ connection on $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ is the Drinfeld associator. Its coefficients are multiple zeta values. The weight filtration is the length of iterated integration. The pentagon equation expresses compatibility around the five degenerations forming the boundary of a real associahedral pentagon inside $\overline{\mathcal{M}}_{0,5}$; the complex Deligne–Mumford boundary itself has ten irreducible divisors.

This is a direct arithmetic consequence of ordered configuration geometry. Passing to the Grothendieck ring of the representation category discards the associator but retains fusion multiplicities. Thus ordered and symmetric traces have visibly different arithmetic content.

59.3 Verlinde numbers

For a modular tensor category with simple objects indexed by λ and modular S-matrix, the dimension of the genus-g conformal-block space is

$$\dim \mathbb{V}_g(\lambda_1, \ldots, \lambda_n) = \sum_{\mu} (S_{0\mu})^{2-2g-n} \prod_{i=1}^{n} S_{\lambda_i \mu}. \tag{59.1}$$

For $\widehat{\mathfrak{sl}}_2$ at level k with the vacuum insertion data,

$$\dim \mathbb{V}_g = \left(\frac{k+2}{2}\right)^{g-1} \sum_{j=1}^{k+1} \sin\left(\frac{\pi j}{k+2}\right)^{2-2g}. \tag{59.2}$$

The cosecant power sums are polynomials in $k+2$ with Bernoulli-number coefficients. This arithmetic follows from the modular tensor category and the geometry of flat connections. It is not a bar-cohomology dimension formula.

59.4 Hodge integrals

Faber and Pandharipande evaluate the one-point Hodge integrals

$$\int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2} \lambda_g. \tag{59.3}$$

Their generating function is

$$1 + \sum_{g \geq 1} \left(\int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2} \lambda_g \right) t^{2g} = \frac{t/2}{\sin(t/2)}. \tag{59.4}$$

This identity is a theorem of the tautological geometry of moduli [77]. It contributes to any field theory whose stable-graph trace reduces to these Hodge integrals. It is not the universal genus expansion of every chiral algebra.

59.5 Theta series

For an even lattice L, the degree-g Siegel theta series is

$$\Theta_L^{(g)}(\Omega) = \sum_{x_1,\ldots,x_g \in L} \exp\left(\pi i \sum_{a,b} (x_a, x_b) \Omega_{ab} \right). \tag{59.5}$$

The theta series $\Theta_L^{(g)}$ is a Siegel modular form when L is even unimodular. The genus-g partition function of a lattice theory contains (59.5) multiplied by an oscillator determinant.

Theta series are traces over zero modes. They forget the cocycle and many OPE coefficients. Nonisometric lattices may have equal theta series in low genus. The Schottky form illustrates the first place at which the two rank-sixteen even-unimodular lattice theta series differ on the full Siegel space, while their restrictions to the genus-four Jacobian locus agree [102, 109].

59.6 Symmetric products and the DMVV formula

Let $\phi_M(\tau, z)$ be the elliptic genus of a compact Calabi–Yau manifold M. Dijkgraaf–Moore–Verlinde–Verlinde prove the product formula

$$\sum_{N \geq 0} p^N \phi_{\mathrm{Sym}^N M}(\tau, z) = \prod_{n>0, m \geq 0, \ell \in \mathbb{Z}} \left(1 - p^n q^m y^\ell\right)^{-c(nm,\ell)}, \tag{59.6}$$

where $c(m,\ell)$ are the Fourier coefficients of ϕ_M [70]. The logarithm of the product is a sum of Hecke transforms. This is the plethystic exponential of connected single-particle data.

The formula has the same combinatorial exponential structure as a cofree commutative coalgebra and as the connected-graph identity (54.6). The equality of combinatorial forms does not identify the DMVV state space with a chiral Chevalley coalgebra. Such an identification requires a functor from the sigma-model or BPS category to chiral factorization data.

59.7 Borcherds products

A weak Jacobi form with integral Fourier coefficients can produce an automorphic product whose exponents are those coefficients. Gritsenko and Nikulin use such products to construct automorphic corrections of Lorentzian Kac–Moody algebras [88]. The Igusa cusp form Φ_{10} is the Borcherds–DMVV multiplicative lift of the K3 elliptic genus, and the DMVV second-quantized partition function is expressed by $1/\Phi_{10}$.

The weight of a Borcherds product is half the constant coefficient of its input in the standard normalization. This weight is an automorphic invariant. It is not the central charge, the rank of a lattice, or a universal Koszul conductor.

59.8 K3 and geometric quantum groups

Maulik and Okounkov construct Yangians from the geometry of Nakajima quiver varieties using stable envelopes and R-matrices [99]. Nakajima correspondences give a Heisenberg action on the cohomology of Hilbert schemes of points on a smooth surface; equivariant stable envelopes in symplectic resolutions give the larger quantum-group actions used here. Braverman–Finkelberg–Nakajima construct Coulomb-branch algebras that include shifted Yangians [54].

These are exact geometric constructions of quantum groups. A K3 surface also supplies a lattice of signature $(4, 20)$, elliptic genera, and Hilbert-scheme Fock spaces. None of these facts alone identifies a K3 Hall algebra with the ordered Koszul dual of a particular chiral algebra. The identification would require a functor preserving products, coproducts, pairings, completions, and R-matrices.

59.9 Moonshine

Moonshine associates modular functions to traces of Monster elements on $V^\natural$. Borcherds' proof uses a generalized Kac–Moody algebra and an automorphic denominator identity. The chiral algebra supplies the graded representation and its vertex operators; the denominator identity supplies an automorphic trace.

Chiral Koszul duality applies to the local chiral operation of $V^\natural$. Moonshine is additional equivariant and arithmetic structure. A trace remembers the conjugacy class and graded multiplicities but not the complete OPE.

59.10 Genus one and genus two

Genus one detects characters, modular transformations, and one-variable theta series. Genus two detects interactions between two cycles and introduces Siegel modular forms. The rank-sixteen lattice theta series first differ on the full degree-four Siegel space, while their difference vanishes on the genus-four Jacobian locus. Therefore the phrase "genus two detects" has mathematical content only after the function, the ambient moduli space, and the restriction locus have been named.

The same precision applies to L-functions. A modular form determines an L-function; a graph amplitude equals a special value only after an integral representation or a Rankin–Selberg unfolding has been proved. No operator-product coefficient is automatically an L-value.

59.11 Arithmetic duality

Dualizing a local system sends its period matrix to the contragredient one. Poincare duality pairs homology and cohomology periods. Functional equations of L-functions reflect duality of motives. These are distinct dualities, connected only when the displayed Hall, automorphic, and chiral realization functors commute with the bar and trace constructions.

The common language is a diagram of realizations:

$$\begin{aligned}\text{algebraic coefficient object} &\longrightarrow \text{de Rham local system} \longrightarrow \text{Betti monodromy},\\ \text{Betti monodromy} &\longrightarrow \text{period or trace}.\end{aligned}$$

Chiral Koszul duality acts at the first stage. Arithmetic enters at the final stages.

Chapter 60

Quantum field theory in the same language

60.1 Observables as a factorization algebra

A perturbative quantum field theory assigns a cochain complex of observables to each open set. Observables with disjoint supports multiply. Locality and descent make these complexes a factorization algebra [58, 59]. The differential combines the equations of motion, gauge symmetry, and quantum corrections.

A holomorphic theory on a complex curve produces a holomorphic factorization algebra. When the local observables are regular holonomic, translation equivariant, and finite in each conformal weight, their factorisation product determines the corresponding vertex or chiral algebra. A topological–holomorphic theory on $M \times X$ may produce a chiral algebra on a boundary parallel to X. The OPE is the short-distance expansion of the factorization product.

60.2 Renormalization and compactified configurations

A Feynman graph with vertices at points of a manifold defines a differential form or distribution on the configuration space of distinct vertices. Divergences occur at collision diagonals. Compactifying the configuration space resolves these diagonals into boundary faces indexed by forests. Renormalization extends the weight to the boundary by adding local counterterms.

The forest formula and the Fulton–MacPherson forest resolution have the same incidence category. The coefficient systems differ: a Feynman weight contains propagators, gauge-fixing data, and densities; the chiral Chevalley complex contains the diagonal operation. A comparison theorem identifies them when the boundary asymptotics of the propagator reproduce the chiral operation.

60.3 Classical and quantum master equations

Let $\mathcal{E}$ be the BV complex of fields with odd Poisson bracket $\{-,-\}$. A classical action satisfies

$$QS + \frac{1}{2}\{S, S\} = 0.$$

A quantization satisfies
$$QS + \frac{1}{2}\{S, S\} + \hbar\Delta S = 0$$
after renormalization. The anomaly is the cohomology class of the failure of this equation.

The modular equation (54.3) has the same form because both are generated by graph contractions. The identification is exact when the Feynman rules define an algebra over the same modular cooperad. The boundary strata of compactified configurations then prove the master equation by Stokes' theorem.

60.4 Chern–Simons theory and affine currents

Chern–Simons theory on a three-manifold with the standard chiral boundary condition induces a chiral Wess–Zumino–Witten boundary theory. The boundary currents form an affine Kac–Moody algebra. In the abelian theory, the boundary current OPE is proportional to the level and is computed directly from the Green kernel. In the nonabelian theory, one-loop effects produce the familiar shifted coupling in KZ and surgery formulas.

Gwilliam and Williams construct a one-loop exact quantization of a mixed holomorphic–topological presentation of Chern–Simons theory and analyze its boundary behavior [89]. The factorization algebra of boundary observables is the analytic counterpart of the affine chiral algebra.

Wilson lines form a braided tensor category. Their exchange is described by quantum-group R-matrices; their closure produces knot invariants. This is the physical realization of Theorems 49.2 and 50.2.

60.5 Four-dimensional Chern–Simons and integrability

In four-dimensional Chern–Simons theory, line operators are topological along a surface and holomorphic in a spectral parameter. Their OPE produces rational, trigonometric, or elliptic R-matrices depending on the curve. The Yang–Baxter equation follows from moving three lines without crossing a singular configuration. Transfer matrices commute because separated line networks are isotopic.

The Gaudin model arises from the semiclassical rational connection. Spin chains arise from choosing representations on parallel lines. Baxter relations are exact sequences or functional relations in the representation category. These structures are ordered because the sequence and positions of lines matter.

60.6 BRST, ghosts, and strings

Gauge fixing introduces ghosts and a BRST differential. The total central charge of matter and ghosts controls the conformal anomaly of the worldsheet theory. For the bosonic string, the (b, c) ghost system has central charge -26; BRST nilpotence requires matter central charge 26 together with the usual normal-ordering or intercept condition in the state-space formulation.

Polyakov's path integral produces determinant lines over moduli. The Weyl anomaly is the curvature of these determinant lines. The Belavin–Knizhnik theorem and the theory of Quillen metrics make this relation geometric. The cancellation of matter and ghost determinant lines is the global form of the central-charge condition.

This anomaly cancellation is not a statement that the matter VOA is Koszul dual to the ghost VOA. It is a tensor-product cancellation in the conformal-block and determinant-line theory.

60.7 Brown–Henneaux and Cardy

Three-dimensional gravity with negative cosmological constant has asymptotic Virasoro symmetry with

$$c = \frac{3\ell}{2G} \tag{60.1}$$

by the Brown–Henneaux theorem [56]. In a unitary modular-invariant two-dimensional CFT with a sparse light spectrum and vacuum dominance, the Cardy formula gives the high-energy density of states. Substituting (60.1) reproduces the Bekenstein–Hawking entropy of the BTZ black hole in the semiclassical regime.

The rigorous algebraic input is the boundary Virasoro algebra and modular covariance of its partition function. A derived center of the Virasoro algebra is not, by itself, a nonperturbative theory of three-dimensional gravity.

60.8 Entanglement in two-dimensional conformal field theory

For the vacuum of a unitary CFT on the line, the entanglement entropy of an interval of length L with ultraviolet cutoff ϵ is

$$S_A = \frac{c+\bar{c}}{6}\log\frac{L}{\epsilon} + \text{const.} \tag{60.2}$$

The replica derivation uses twist fields and conformal maps. The coefficient is fixed by the left- and right-moving stress-tensor anomalies; in a parity-invariant theory with $c = \bar{c}$ it reduces to $c/3$.

Equation (60.2) is a trace statement. It does not determine the OPE, braid monodromy, or representation category. Two CFTs with the same c may have different entanglement spectra and operator algebras.

60.9 Bulk–boundary maps

A boundary condition for a factorization algebra of bulk observables defines an action on boundary observables. By the universal property of the derived center, there is a map

$$\mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z^{\mathrm{der}}_{\mathrm{ch}}(A).$$

Costello and Gaiotto use this structure in twisted holography, where protected bulk sectors and boundary vertex algebras can sometimes be compared exactly [60]. The comparison is a theorem of the particular twisted theory, not a universal identity for all chiral algebras.

A decoupled bulk factor remains invisible to the boundary. Conversely, a boundary algebra may admit several bulk completions with different global observables. The center classifies central actions, not all global physics.

60.10 Celestial chiral algebras

In celestial and twistor constructions, scattering amplitudes can be encoded by correlators of chiral algebras. Costello–Paquette and collaborators construct examples in which gauge-theory amplitudes arise from current-algebra correlators and their deformations [61]. Associativity of the celestial OPE is then a nontrivial constraint on loop corrections.

The precise statement depends on the background geometry, asymptotic states, and the map from scattering observables to chiral fields. Soft theorems provide currents in the chiral algebra, but no universal sequence of OPE coefficients equals the gravitational soft expansion across all theories.

60.11 Calabi–Yau categories and topological strings

A Calabi–Yau A_∞ category has a cyclic pairing. Costello's construction associates to a smooth proper cyclic A_∞ category a two-dimensional topological conformal field theory whose closed states are Hochschild chains. The Connes operator and the cyclic pairing supply the BV structure.

A holomorphic field theory on a Calabi–Yau variety may produce factorization algebras and BPS operator algebras. Cohomological Hall algebras, shifted Yangians, and Borcherds products arise in specific geometries. Passing from such a category to a chiral algebra on a curve requires an explicit compactification, reduction, or boundary functor. The dimension shift and cyclic pairing alone do not define the functor.

60.12 The physical comparison diagram

The exact relation between the mathematical and physical objects is a commutative diagram of maps, not an equality of names:

$$\begin{tikzcd} \mathrm{coChev}^{\mathrm{ch}}(\mathfrak{g}) \arrow[r] \arrow[d] & \text{algebraic factorization coalgebra} \arrow[d] \\ \text{renormalized collision complex} \arrow[r] & \mathrm{Obs}_\partial \end{tikzcd} \tag{60.3}$$

The bulk-to-boundary action has the separately typed form

$$\mathrm{Obs}_{\mathrm{bulk}}|_\partial \longrightarrow Z^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Obs}_\partial). \tag{60.4}$$

The vertical maps are Riemann–Hilbert, analytification, or explicit field-theoretic comparison maps. When they are quasi-isomorphisms, the mathematical and physical master equations are the same equation in two models.

Chapter 61

The physical derivation of ordered chirality

61.1 A local experiment

Consider a quantum field theory on

$$X \times \mathbb{R}_t,$$

holomorphic along the complex curve X and topological along the oriented coordinate t. Let $\mathcal{O}_i(z_i, t_i)$ be local observables with

$$t_1 < t_2 < \cdots < t_n.$$

Topological invariance makes the correlator constant under motions of the t_i inside a fixed ordered chamber. Holomorphy permits the z_i to move except at collisions. Hence a correlator is locally constant in the ordered chamber of the t_i and meromorphic or distributional in the z_i.

The short-distance product has the form

$$\mathcal{O}_1(z_1, t_1) \cdots \mathcal{O}_n(z_n, t_n) \sim \sum_\alpha C_\alpha(z_1, \ldots, z_n)\mathcal{O}_\alpha(z, t), \tag{61.1}$$

where the coefficient C_α is a chiral distribution and the multiplication order is the order of the t_i. Formula (61.1) is the physical content of an E_1 algebra in factorization D-modules.

Principle 61.1 (Ordered OPE principle). Holomorphic collision determines the coefficient distribution. Topological ordering determines the associative word. The full product is their interchange law.

This principle is more primitive than a vertex-algebra expansion. A local coordinate on X converts the coefficient distribution into modes. Choosing an ordering prescription on $\mathbb{R}$ converts the topological factor into a time-ordered product.

61.2 Topological descent and the E_1 product

Let Q be the cohomological supercharge of the twist. Topological invariance in the real direction means that the translation generator is Q-exact:

$$P_t = [Q, G_t].$$

For a Q-closed observable $\mathcal{O}$,

$$\partial_t \mathcal{O} = [Q, G_t \mathcal{O}].$$

The difference between insertions at two values of t is therefore Q-exact. Ordered products are independent of the positions on cohomology.

The identity $P_t = [Q, G_t]$ is the first descent equation. A chain-level E_1 structure requires the coherent extension of this identity to families of interval embeddings: the descendant operators must satisfy the Cartan homotopy relations and their higher compatibility on compactified configuration cells. Integrating the coherent descendants gives the A_∞ operations. The boundary of an associahedron is the sum of its parenthesized degenerations, so Stokes' theorem gives the Stasheff identities.

Theorem 61.2 (Descent produces ordered multiplication)**.** *Let a factorization algebra of observables on $X \times \mathbb{R}$ be holomorphic in X and equipped with coherent topological descent in the real direction, so that its restriction to intervals is a locally constant factorization algebra. Then restriction to product disks determines an E_1 algebra in holomorphic factorization observables on X. A local operator G_t with $P_t = [Q, G_t]$ exhibits the first coherence; when it extends to the stated descent datum, the product on Q-cohomology is independent of the transverse positions.*

Proof. A locally constant factorisation algebra on the oriented line is equivalent to an E_1 algebra by evaluation on one interval and gluing of ordered subintervals. Applying this construction to every holomorphic disk in X gives an E_1 multiplication in the category of holomorphic factorisation observables. Coherent descent identifies isotopic interval configurations; the equation $P_t = [Q, G_t]$ is the infinitesimal homotopy, so the induced product on Q-cohomology is independent of the positions. □

The theorem is the physical proof of Theorem 39.2 for a cohomological field theory.

61.3 Why the OPE is a D-module

The coefficient in (61.1) is not a function at the collision locus. Ward identities imply differential equations in the insertion points. Contact terms are distributions supported on diagonals. A right D-module records both the differential equations away from collisions and the allowed supported terms at collisions.

For a conserved holomorphic current,

$$\bar{\partial} J = 0$$

away from other insertions. In a correlator,

$$\bar{\partial}_z \frac{1}{z - w} = \pi \delta^{(2)}(z - w)$$

in the distributional normalization. Thus the singular OPE and the contact Ward identity are the same distribution seen before and after applying $\bar{\partial}$. The D-module language retains this equality without choosing a Laurent contour at every step.

61.4 Boundary conditions and open strings

Let the theory occupy $X \times \mathbb{R}_{\geq 0}$ and let b be a boundary condition at $t = 0$. Boundary operators form the endomorphism algebra of b. If several boundary conditions b_i are present, a boundary-changing field lies in

$$A_{ij} = \mathrm{RHom}(b_i, b_j).$$

Composition has the order

$$A_{jk} \otimes A_{ij} \longrightarrow A_{ik}.$$

The boundary multiplication is the open-string product. The insertion point on X supplies its chiral dependence.

The strip with boundary conditions b_i and b_j has state complex A_{ij}. Gluing strips is composition. Closing a strip into an annulus gives Hochschild homology when a cyclic trace and the required dualizability are present. Thickening the Feynman graphs of a cyclic open-string interaction then gives ribbon surfaces. These hypotheses are exactly the transverse modular data of Chapter 53.

61.5 The interval bar as a propagator-independent amplitude

Place the augmentation boundary at the two ends of an interval. Factorization homology gives

$$\int_{([0,1],\partial)} A \simeq \mathbf{1} \otimes^L_A \mathbf{1}.$$

A cellular decomposition of the interval yields the bar complex. Its differential composes adjacent operator insertions. This construction uses no gauge-fixing propagator and no logarithmic form. It is topological and exact.

A Feynman expansion is another model of the same derived tensor product when the field theory admits an action and a gauge fixing. The comparison map sends a bar word to the corresponding time-ordered amplitude. Its proof is a homological perturbation or factorization-excision theorem, not an identification of the bar differential with a residue.

61.6 A second topological direction and exchange

Now place the theory on $X \times \mathbb{R}^2$. Two parallel lines can move around one another. The exchange path defines

$$R_{12}(z_1, z_2) : L_{z_1} \otimes L_{z_2} \longrightarrow L_{z_2} \otimes L_{z_1}.$$

Three-line isotopy gives the Yang–Baxter equation. The exchange depends meromorphically on $z_1 - z_2$ because a singularity occurs when the spectral points collide.

This derivation explains the codimension hierarchy:

$$\begin{array}{lcl} \mathbb{R}^1 & : & E_1 \text{ fusion,} \\ \mathbb{R}^2 & : & E_2 \text{ braid exchange,} \\ \mathbb{R}^3 & : & E_3 \text{ higher linking operations.} \end{array}$$

The chiral curve remains a holomorphic coefficient direction; a little-disks index counts it only after a topologization supplies null-homotopies for its real translations.

61.7 Four-dimensional Chern–Simons theory

The action

$$S[A] = \frac{1}{2\pi\hbar} \int_{\Sigma \times X} \omega \wedge \left(\frac{1}{2}(A, dA) + \frac{1}{6}(A, [A, A]) \right)$$

is topological on the oriented surface Σ and holomorphic on the curve X. A Wilson line lies on a curve in Σ and at a point of X. The point of X is its spectral parameter.

At tree level, exchange of two Wilson lines gives the Casimir kernel

$$r(z) = \frac{\Omega}{z}.$$

The configuration of three lines has two isotopic resolutions, which gives the classical Yang–Baxter equation. Quantization replaces r by an R-matrix. The one-loop anomaly of line gauge invariance forces the Yangian deformation in the rational theory. Thus the Yangian is not inferred from order; in the formal perturbative rational theory it is selected by the gauge-theory anomaly and Ward identities [63, 64].

For $X = \mathbb{C}^\times$ and for elliptic X, the propagator has trigonometric and elliptic periodicity. The quantum symmetry changes accordingly. The spectral curve is part of the physics.

61.8 Three-dimensional gauge theory and boundary chiral algebra

A holomorphic twist of a three-dimensional supersymmetric theory on $X \times \mathbb{R}_{\geq 0}$ may admit a boundary condition preserving the holomorphic supercharge. Boundary local operators form a chiral algebra on X. Boundary line junctions carry an ordered product inherited from the normal direction. Bulk lines act by modules.

In a Lagrangian theory, the boundary algebra is computed from boundary fields and BRST reduction. Gauge anomalies must cancel for the boundary condition to exist quantum mechanically. Different boundary conditions can give different E_1 chiral algebras with equivalent central bulk actions. This is the physical origin of Morita and Koszul dual boundary descriptions.

61.9 Drinfeld–Sokolov reduction as gauge fixing

Start from affine currents and couple them to ghosts imposing a moment-map constraint. The BRST charge is

$$Q = \oint \left(\sum_a c^a (J_a - \chi_a) - \frac{1}{2} \sum_{a,b,c} f_{ab}{}^c : c^a c^b b_c : + \cdots \right).$$

The equation $Q^2 = 0$ follows from the current algebra, invariance of the level cocycle, and cancellation of the gauge anomaly. The reduced chiral algebra is $H_Q(A)$.

Because Q is a derivation of the transverse product, it acts on the ordered bar. The total differential is

$$D = b_A + Q_{\mathrm{bar}}.$$

The spectral sequence obtained by first taking BRST cohomology compares the bar of the reduced algebra with the reduction of the bar. Degeneration is a theorem of the particular good grading and filtration. The higher W-operations are the homotopy-transferred remnants of affine multiplication and ghost contraction.

61.10 Polyakov's determinant and the chiral anomaly

For a free scalar on a Riemann surface,

$$Z[g] = (\det{}'\Delta_g)^{-1/2}$$

after the zero-mode measure has been factored out. Under $g \mapsto e^{2\sigma} g$, zeta regularization gives the Polyakov action

$$\log \frac{Z[e^{2\sigma} g]}{Z[g]} = \frac{1}{24\pi} \int \left(|\nabla\sigma|^2 + R\sigma\right) d\mathrm{vol}.$$

For a theory of central charge c, the right side is multiplied by c. The same coefficient appears in the fourth-order pole of the stress tensor because both measure the central extension of local conformal transformations.

On a family of curves, the determinant forms a line bundle with Quillen connection. Its curvature is computed by the family index theorem. This is the exact bridge from the local Virasoro cocycle to the Hodge class. It does not identify the central charge with affine level, ribbon genus, or a Koszul curvature.

61.11 Worldsheet sewing

Choose local coordinates $zw = q$ around two punctures. Sewing inserts

$$q^{L_0 - c/24}$$

between dual states. The factor $q^{-c/24}$ is the vacuum energy on the cylinder. The sum over a homogeneous basis is the inverse two-point pairing. Convergence for $|q| < 1$ is the analytic sewing theorem.

A nonseparating sewing increases the genus of the chiral curve. An annular trace of the transverse E_1 algebra increases ribbon genus. The two operations can occur in the same amplitude, giving the bidegree of Chapter 53.

61.12 The Chern–Simons/WZW comparison theorem and its hypotheses

For compact gauge group at integral level, and after the standard choice of polarization and framing data, the Chern–Simons/WZW and Reshetikhin–Turaev constructions give the fundamental example in which the quantized surface state spaces, boundary current algebra, Wilson-line labels, conformal blocks, and ribbon-category invariants are connected by comparison theorems.

The boundary current algebra is affine Kac–Moody. Canonical quantization on a surface gives the WZW conformal-block space. Wilson lines are labelled by admissible representations. The Chern–Simons/WZW and Drinfeld–Kohno comparison theorems identify their exchange with the corresponding quantum-group braiding. Closing lines with the ribbon trace gives link invariants. Under these comparison theorems, the framing dependence of the three-dimensional theory matches the ribbon twist and the projective anomaly data of the WZW conformal blocks.

Each statement has a different mathematical carrier:

$$\begin{array}{ccl} \text{current OPE} & : & \text{chiral } D\text{-module}, \\ \text{boundary composition} & : & E_1 \text{ algebra}, \\ \text{Wilson exchange} & : & E_2 \text{ line category}, \\ \text{surface states} & : & \text{conformal blocks}, \\ \text{framing phase} & : & \text{invertible anomaly theory}. \end{array}$$

The strength of the example lies in the comparison maps between these carriers, not in collapsing them into one object.

61.13 Large N and the double genus

For N coincident branes, boundary operators acquire matrix indices. A connected ribbon graph contributes N^{2-2h-b}. Independently, a closed-string worldsheet of genus g contributes g_s^{2g-2}. Open–closed duality can reorganize the first expansion into the second after summing boundaries and applying a dynamical transition.

Before that transition, the amplitude is naturally

$$\sum_{g,h,b} g_s^{2g-2} N^{2-2h-b} \mathcal{F}_{g,h,b}.$$

The bimodular formalism is the algebraic form of this double expansion. The relation between g_s and N is a physical duality parameter, not an identity in operad theory.

61.14 Holographic reconstruction

Let A be the ordered chiral algebra of a boundary condition. Bulk observables act centrally on A, giving

$$\beta : \mathrm{Obs}_{\mathrm{bulk}} \longrightarrow Z_{\mathrm{ord,ch}}(A).$$

If the boundary condition is complete, β is a quasi-isomorphism. Then the protected bulk local algebra is reconstructed from the boundary. Two boundary descriptions have the same centre when an explicit Morita context identifies their bimodule categories; a Koszul equivalence of augmentation-generated sectors does not by itself imply this stronger statement.

This is the precise algebraic statement behind twisted holography. It reconstructs a protected local sector. A metric, Hilbert space, nonperturbative spectrum, and integration cycle require additional physical structure.

61.15 The field-theoretic meaning of the dual

The augmentation $A \to \mathbf{1}$ is a distinguished boundary or defect. Its derived endomorphism algebra

$$A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$$

acts on deformations of that defect. The bar complex

$$\mathbf{1} \otimes^L_A \mathbf{1}$$

is the complex of interval amplitudes between two copies of the defect. This is why Koszul duality repeatedly appears in line-operator physics: it exchanges local bulk couplings near the defect with operators living on the defect.

The dual is a completion near the chosen augmentation. It need not recover sectors disconnected from that boundary condition. The physical meaning of double duality is therefore completion of the local theory around the chosen vacuum.

61.16 The physical theorem in one sentence

A holomorphic–topological quantum field theory whose protected observables are product-generated, algebraizable as factorisable D-modules, locally constant in the ordered direction, and anomaly-free produces an ordered chiral algebra: collision on the complex curve gives the chiral coefficient and ordered fusion gives the E_1 product. A second topological direction supplies exchange, a cyclic trace supplies ribbon loops, variation of the curve supplies conformal sewing, and each anomaly is the invertible theory obstructing the corresponding gluing.

Chapter 62

Higher-dimensional sources and reduction to a curve

62.1 Pushforward is the source of chirality

Let Y be a compact framed d-manifold and let $p : Y \times X \times \mathbb{R} \to X \times \mathbb{R}$. Suppose $\mathcal{F}$ is a factorization algebra that is locally constant in the Y and $\mathbb{R}$ directions and holomorphic in X. Define

$$(p_*^{\mathrm{fact}}\mathcal{F})(U \times I) := \int_{Y \times U \times I} \mathcal{F}. \tag{62.1}$$

Excision gives factorization maps on the target.

Theorem 62.1 (Topological pushforward to ordered chirality)**.** *Assume the local constancy, dualisability, and excision hypotheses required for factorisation homology, and assume that the pushforward is product-generated and its holomorphic coefficient admits the factorisable D-module realisation of Theorem 39.1. Then (62.1) is a rectangular holomorphic–topological factorisation algebra on $X \times \mathbb{R}$ and determines an ordered chiral algebra in $\mathrm{Alg}_{E_1}(\mathrm{FactD}(X))$.*

Proof. Fubini for factorization homology identifies integration over disjoint product opens with iterated integration over Y and over the target open. Local constancy in Y and $\mathbb{R}$ is preserved by the pushforward. Holomorphic differential equations in X commute with integration in the topological directions. Weiss descent follows from excision. Apply Theorem 39.2. □

The theorem gives the invariant shape of the higher-dimensional constructions treated here under these hypotheses. The source is a factorization algebra; the reduction is factorization homology; the output is an E_1 algebra in chiral coefficients.

62.2 Dimensional bookkeeping

If a theory is locally constant in m transverse directions, its local operators carry E_m structure. Integrating over a framed r-manifold with $r < m$ leaves E_{m-r} structure. A holomorphic curve remains a coefficient direction. Thus a theory on

$$Y^r \times X \times \mathbb{R}^{m-r}$$

can reduce to an E_{m-r} algebra in $\mathrm{FactD}(X)$.

This formula replaces a universal "chirality ladder." The output depends on the actual number of topological directions and on the tangential structure used in the pushforward.

62.3 Calabi–Yau categories

A smooth proper Calabi–Yau A_∞ category $\mathcal{C}$ has a nondegenerate cyclic trace. Its open sector is the family of morphism complexes $\mathrm{RHom}(E, F)$ with A_∞ composition. Its closed state complex is Hochschild homology $\mathrm{HH}_\bullet(\mathcal{C})$. The Connes operator and the cyclic pairing supply the circle and surface operations of the associated topological conformal field theory.

These data are topological. To obtain a chiral algebra on X, one needs a family $\mathcal{C}_x$ varying holomorphically with $x \in X$, or a holomorphic field theory whose boundary category is $\mathcal{C}$. The ordered chiral algebra is then the endomorphism object of a chosen boundary generator, or the pushforward (62.1).

Theorem 62.2 (Categorical source theorem)**.** *Let $\mathcal{C}$ be a factorization category over X with a compact generator b and with an ordered topological composition direction. Then*

$$A_b = \mathrm{RHom}_{\mathcal{C}}(b, b)$$

is an ordered chiral algebra and

$$\mathcal{C} \simeq \mathrm{Mod}_{A_b}(\mathrm{FactD}(X))$$

under compact Morita hypotheses. If $\mathcal{C}$ is Calabi–Yau, the induced trace on A_b supplies the transverse cyclic structure of Theorem 53.1.

Proof. Composition gives A_b its transverse E_1 multiplication, while factorisation of the category gives the coefficient system on X. The compact-generator Morita theorem applies locally and glues by factorisation descent, yielding the displayed equivalence. A Calabi–Yau trace is cyclic under composition; restricting it to endomorphisms of b gives the cyclic trace required for transverse sewing. □

This theorem is the correct passage from a Calabi–Yau category to noncommutative chirality. It requires a factorization family over the curve; the Calabi–Yau trace alone does not create one.

62.4 Naturality of obstruction classes

Let $\mathfrak{g}$ and $\mathfrak{h}$ be complete filtered L_∞ algebras controlling two deformation problems, and let

$$F : \mathfrak{g} \longrightarrow \mathfrak{h}$$

be a filtered L_∞ morphism. It sends Maurer–Cartan elements to Maurer–Cartan elements and gauge equivalences to gauge equivalences.

Proposition 62.3 (Transgression of obstructions)**.** *Let $\alpha_{\le r}$ be a Maurer–Cartan solution modulo filtration $F^{r+1}\mathfrak{g}$, with obstruction*

$$o_{r+1}(\alpha) \in H^2(\mathrm{gr}^{r+1}\,\mathfrak{g}, d_\alpha).$$

Then the obstruction to extending $F_*(\alpha_{\le r})$ is the image

$$F_* o_{r+1}(\alpha) \in H^2(\mathrm{gr}^{r+1}\,\mathfrak{h}, d_{F_*(\alpha)}).$$

Proof. Take the filtration-$r+1$ component of the L_∞ identities for F. The curvature of $F_*(\alpha_{\le r})$ is the image of the curvature of $\alpha_{\le r}$ modulo F^{r+2}. The linear term is the induced chain map between the twisted associated-graded complexes. Passing to cohomology sends the source obstruction class to the target obstruction class. □

The proof is the order-$r+1$ component of the L_∞ identities. This proposition is the rigorous content of every assertion that a Calabi–Yau, Hall, or gauge-theoretic obstruction becomes a chiral obstruction. The essential construction is the morphism F.

62.5 Cohomological Hall algebras

Let $\mathcal{M}$ be a moduli stack of objects in an abelian category of finite type, or a derived category whose extension stack is locally of finite type and admits the stated pushforwards. Correspondences of short exact sequences define a convolution product on Borel–Moore homology, equivariant cohomology, or K-theory for which the extension correspondence has defined pullback and proper pushforward. The resulting cohomological Hall algebra is an E_1 algebra. Its multiplication is noncommutative because extension flags are ordered.

A coproduct requires additional geometry, typically localization, a stability condition, or a choice of chamber. A Drinfeld double requires a nondegenerate Hopf pairing. An R-matrix requires a stable-envelope or line-exchange construction. These are separate structures.

Criterion 62.4 (From a Hall algebra to a quantum group)**.** *A Hall algebra determines a quasitriangular quantum group only after the construction of:*

(i) a compatible completed coproduct;

(ii) a nondegenerate pairing and a double;

(iii) a PBW theorem and a presentation of the relation ideal;

(iv) a universal or categorical R-matrix;

(v) convergence of the completion on the chosen representation category.

The ordered chiral pushforward of a Hall theory retains the E_1 multiplication. The remaining items must survive the pushforward to identify its line category with a quantum group.

62.6 Coulomb branches and shifted Yangians

For three-dimensional $\mathcal{N}=4$ gauge theories, the Braverman–Finkelberg–Nakajima construction defines Coulomb-branch algebras by convolution on affine-Grassmannian spaces. In type A quiver examples, quantized Coulomb branches are truncated shifted Yangians. The coweight shift records magnetic or boundary data.

In the holomorphic–topological twist, these algebras act on line defects. A boundary reduction to X gives a factorization family of modules with spectral parameter. The ordered chiral algebra records fusion of these lines. The BFN identification supplies the named shifted Yangian after the geometric convolution algebra and the Drinfeld presentation have been compared.

62.7 K3 geometry

Let S be a K3 surface. Its Mukai lattice

$$\widetilde{H}(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z})$$

has signature $(4, 20)$. Nakajima correspondences give a Heisenberg action on

$$\bigoplus_{n \geq 0} H^*(S^{[n]}).$$

The elliptic genera of the symmetric products satisfy the Dijkgraaf–Moore–Verlinde–Verlinde product formula. The reciprocal Igusa cusp form encodes the second-quantized K3 elliptic genus.

These are exact statements about cohomology, modular forms, and second quantization. They do not by themselves define an ordered chiral algebra or its Koszul dual.

Principle 62.5 (K3 separation principle). The Mukai lattice, Nakajima Heisenberg action, Hall multiplication, elliptic genus, Siegel modular form, and a possible line-defect quantum group are distinct realizations of K3 geometry. A common chiral object exists only after explicit functors identify their operations and pairings.

A valid construction may begin with a K3-associated field theory on $S \times X \times \mathbb{R}$, integrate over S or a cycle in S, and apply Theorem 62.1. Its comparison with a Hall algebra or a Borcherds product is then a theorem about that pushforward.

62.8 Borcherds products and scalar traces

A Borcherds lift takes a weakly holomorphic vector-valued modular form with integral principal part, or the corresponding Jacobi input after theta decomposition to an automorphic product. The weight is determined by the constant term of the input. The product can be interpreted as a denominator identity of a generalized Kac–Moody algebra in favorable cases.

The automorphic product is a scalar or section of a line bundle. It may arise as a trace, index, or partition function of a categorical or chiral object. Reconstruction in the opposite direction is not unique: different categories can have the same index. The ordered multiplication, braiding, and extension data are lost under the trace.

62.9 Omega backgrounds and BPS algebras

An Omega background makes rotations equivariantly exact and can reduce a higher-dimensional field theory to a noncommutative algebra of protected operators. In M-theory and gauge-theory examples, the resulting algebras include Yangians and their deformations. The deformation parameters are equivariant weights; line-defect spectral parameters are positions in a residual holomorphic curve.

The algebraic structure is obtained from the factorization OPE of protected operators. Koszul duality appears when local bulk observables are coupled to a defect and one computes derived endomorphisms of the trivial defect. The named BPS algebra follows from a separate comparison with geometric correspondences or representation-theoretic generators.

62.10 Shifted symplectic geometry

Derived moduli stacks of fields or objects often carry shifted symplectic forms. A boundary condition is represented by a Lagrangian morphism. The derived intersection of two Lagrangians has a shifted Poisson structure and may quantize to the algebra of boundary-changing operators.

Let $L_0, L_1 \to \mathcal{M}$ be Lagrangians in an n-shifted symplectic stack. Their derived intersection is $(n-1)$-shifted symplectic. Its $(n-1)$-shifted symplectic form determines a P_n-structure. Whenever an E_n-quantisation of that P_n-structure has been constructed and varies factorisably over a chiral curve, the resulting observables form an E_n algebra in factorisation D-modules.

This gives a geometric source of ordered chiral algebras when the ambient symplectic shift is $n = 1$, so that the Lagrangian intersection is 0-shifted symplectic and its quantisation is associative. The quantization and the factorization family are part of the theorem; the shifted symplectic form alone is classical data.

62.11 A source-to-chiral comparison

Suppose the higher-dimensional factorisation pushforward constructs

$$A_{\mathrm{geom}} \in \mathrm{Alg}_{E_1}(\mathrm{FactD}(X)),$$

and a Hall, BPS, or localization construction independently produces

$$A_{\mathrm{src}} \in \mathrm{Alg}_{E_1}(\mathrm{FactD}(X)).$$

The comparison datum is an E_1-algebra morphism

$$\phi : A_{\mathrm{src}} \longrightarrow A_{\mathrm{geom}}$$

compatible with the chiral coefficient. Applying the ordered bar gives the commutative square

$$\begin{tikzcd}
A_{\mathrm{src}} \arrow[r, "\phi"] \arrow[d, "\mathrm{Bar}^{\mathrm{ord}}_X"'] & A_{\mathrm{geom}} \arrow[d, "\mathrm{Bar}^{\mathrm{ord}}_X"] \\
\mathrm{Bar}^{\mathrm{ord}}_X(A_{\mathrm{src}}) \arrow[r, "\mathrm{Bar}(\phi)"'] & \mathrm{Bar}^{\mathrm{ord}}_X(A_{\mathrm{geom}}).
\end{tikzcd} \tag{62.2}$$

The upper objects are associative and the lower objects are coassociative; the functorial bar map is what relates the two types. The construction of A_{src} contains the Hall or BPS correspondences, while the construction of A_{geom} contains factorisation pushforward. Their equality is the theorem that ϕ is an equivalence.

If ϕ is an equivalence on a chosen complete sector, then (62.2) identifies the corresponding Hall or BPS resolution with the ordered chiral bar. A quantum-group identification still requires the coproduct, pairing, R-matrix, and PBW data of Criterion 62.4.

62.12 The higher-dimensional conclusion

Higher-dimensional geometry enters ordered chiral Koszul duality through factorization pushforward, boundary endomorphisms, and morphisms of deformation complexes. Cyclic traces produce surfaces; Hall correspondences produce associative multiplication; Omega backgrounds produce deformation parameters; shifted symplectic forms produce Poisson brackets. The chiral curve carries the spectral dependence of the resulting operators.

The language remains unchanged: an E_1 algebra in factorization D-modules is the target through which these sources can be compared without erasing their origin.

Part X

Frontier completions

Chapter 63

Formal geometry: from the disc to the curve

A chiral operation is intrinsic on a curve, whereas a vertex operator is written in a coordinate. The passage between the two is descent along the torsor of formal coordinates. Gelfand–Kazhdan descent carries formal-disc operations to the curve and places the stress tensor, the Virasoro cocycle, and the projective anomaly in one geometric construction.

63.1 The coordinate torsor

Let X be a smooth complex curve. For $x \in X$, write $\widehat{X}_x$ for the formal completion at x and $\widehat{\mathcal{O}}_x$ for its complete local ring. A formal coordinate is an isomorphism

$$\varphi : \mathbb{C}[[z]] \xrightarrow{\sim} \widehat{\mathcal{O}}_x$$

that carries the maximal ideal (z) onto the maximal ideal of x. The collection of all such coordinates forms a principal bundle

$$\mathrm{Coord}_X \longrightarrow X$$

for the pro-algebraic group $\mathrm{Aut}\,\mathbb{C}[[z]]$.

The Lie algebra of $\mathrm{Aut}\,\mathbb{C}[[z]]$ is

$$\mathrm{Der}_+\,\mathbb{C}[[z]] = z\mathbb{C}[[z]]\partial_z.$$

Allowing translations gives the Harish–Chandra pair

$$\big(\mathrm{Der}\,\mathbb{C}[[z]], \mathrm{Aut}\,\mathbb{C}[[z]]\big), \qquad \mathrm{Der}\,\mathbb{C}[[z]] = \mathbb{C}[[z]]\partial_z.$$

The coordinate torsor carries the Gelfand–Kazhdan form

$$\omega_{\mathrm{GK}} \in \Omega^1\big(\mathrm{Coord}_X; \mathrm{Der}\,\mathbb{C}[[z]]\big).$$

At a tangent vector to the coordinate torsor, ω_{GK} records the infinitesimal formal vector field that produces the variation of the coordinate. The Maurer–Cartan identity is

$$d\omega_{\mathrm{GK}} + \frac{1}{2}[\omega_{\mathrm{GK}}, \omega_{\mathrm{GK}}] = 0.$$

Theorem 63.1 (Gelfand–Kazhdan descent)**.** *Let M be a complete module for the Harish–Chandra pair $\big(\mathrm{Der}\,\mathbb{C}[[z]], \mathrm{Aut}\,\mathbb{C}[[z]]\big)$. The associated bundle*

$$\mathcal{M}_X = \mathrm{Coord}_X \times^{\mathrm{Aut}\,\mathbb{C}[[z]]} M$$

carries the flat connection

$$\nabla_{\mathrm{GK}} = d + \rho(\omega_{\mathrm{GK}}).$$

Horizontal sections of $\mathcal{M}_X$ are equivalent to equivariant functions $f : \mathrm{Coord}_X \to M$ satisfying

$$df + \rho(\omega_{\mathrm{GK}})f = 0.$$

The construction is functorial in M and symmetric monoidal.

Proof. The Maurer–Cartan identity and the representation identity for ρ give

$$\nabla^2_{\mathrm{GK}} = \rho\bigg(d\omega_{\mathrm{GK}} + \frac{1}{2}[\omega_{\mathrm{GK}}, \omega_{\mathrm{GK}}]\bigg) = 0.$$

The connection is basic for the right action of $\mathrm{Aut}\,\mathbb{C}[[z]]$ and therefore descends to the associated bundle. A section of the associated bundle is an equivariant function on the torsor; horizontality is precisely the displayed differential equation. Tensor products are preserved because ρ acts by the coproduct on a tensor product. □

63.2 Fields as descended distributions

Let V be a translation-equivariant vertex algebra. The formal distribution

$$Y(a,z)b \in V((z))$$

is obtained by evaluating a diagonal-supported section on the coordinate torsor in the coordinate z. The coordinate torsor turns the state space into a bundle $\mathcal{V}_X$, and the translation operator T supplies the action of ∂_z. The positive formal vector fields act by the conformal change-of-coordinate operators whenever V carries the corresponding quasi-conformal structure.

For the diagonal $\Delta \subset X^2$, set

$$j : X^2 \setminus \Delta \hookrightarrow X^2.$$

The complete singular operation is a morphism

$$\mu : j_* j^*(\mathcal{V}_X \boxtimes \mathcal{V}_X) \longrightarrow \Delta_! \mathcal{V}_X.$$

In a coordinate z near x, the normal coordinate to the diagonal is $u = z_1 - z_2$. The pole filtration

$$\mathcal{O}_{X^2}(\Delta) \subset \mathcal{O}_{X^2}(2\Delta) \subset \cdots$$

produces the expansion

$$\mu(a \boxtimes b) \sim \sum_{n \geq 0} u^{-n-1}(a_{(n)}b) + \sum_{m \geq 0} u^m (a_{(-m-1)}b).$$

The first sum is the singular chiral operation. The second sum is the regular factorisation product. The complete filtered distribution contains both.

Proposition 63.2 (Triangular covariance of modes). *Let $w = f(z)$ be a formal coordinate change with $f'(0) \neq 0$. For every N, the span of the normal symbols*

$$[u^{-1}], \ldots, [u^{-N}]$$

is preserved by the coordinate change. In the ordered basis of increasing pole order, the transformation matrix is upper triangular and its diagonal entries are

$$(f'(0))^{-1}, \ldots, (f'(0))^{-N}.$$

Consequently the pole filtration is intrinsic, while the individual coefficients $a_{(n)}b$ depend on a splitting of that filtration.

Proof. Write

$$f(z_1) - f(z_2) = f'(z_2)u + \frac{1}{2}f''(z_2)u^2 + \cdots .$$

Then

$$(f(z_1) - f(z_2))^{-N} = (f'(z_2))^{-N}u^{-N}\left(1 + c_1u + c_2u^2 + \cdots\right)^{-N}.$$

Expanding the final factor gives a linear combination of $u^{-N}, u^{-N+1}, \ldots$. Modulo regular terms only poles of order at most N occur, and the leading coefficient is $(f')^{-N}$. The argument is compatible with composition of coordinate changes, hence defines the intrinsic filtered action. ☐

The proposition gives the geometric meaning of a mode. A mode is a coordinate component of a filtered normal distribution. The filtration is the invariant object.

63.3 The stress tensor and projective connection

Suppose V contains a conformal vector with field

$$T(z) = \sum_{n \in \mathbb{Z}} L_n z^{-n-2}$$

and central charge c. The action of formal vector fields is projective because the Witt algebra acquires the Virasoro cocycle

$$\psi(f\partial_z, g\partial_z) = \frac{1}{12}\operatorname{Res}_{z=0}\big(f'''(z)g(z)\,dz\big).$$

The central extension is

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}.$$

Proposition 63.3 (Schwarzian transformation). *Under a coordinate change $w = f(z)$, the stress tensor transforms by*

$$T_z(z) = (f'(z))^2 T_w(f(z)) + \frac{c}{12}\{f, z\},$$

where

$$\{f, z\} = \frac{f'''(z)}{f'(z)} - \frac{3}{2}\left(\frac{f''(z)}{f'(z)}\right)^2$$

is the Schwarzian derivative. The affine space of splittings of this transformation law is the space of projective connections on X.

Proof. The coordinate change is implemented by exponentiating the positive Virasoro modes and the scaling mode. The adjoint action on T has a tensorial term of weight two. The central part is the transgression of the Virasoro cocycle. The unique local differential polynomial of order three satisfying the cocycle identity and vanishing on Möbius transformations is the Schwarzian derivative. Matching the infinitesimal variation

$$\delta_v T = vT' + 2v'T + \frac{c}{12}v'''$$

fixes the coefficient $c/12$. □

A projective connection trivializes the projective part of the formal-coordinate descent. It does not alter the internal vertex differential. It supplies a geometric splitting of the Virasoro extension over the curve.

63.4 Determinant lines and the Polyakov class

Let $\pi : \mathcal{C} \to S$ be a smooth proper family of curves and let $\bar{\partial}_E$ be a family of Cauchy–Riemann operators. The determinant line is

$$\lambda(E) = \det R\pi_* E = \bigotimes_i \big(\det R^i \pi_* E\big)^{(-1)^i}.$$

A Quillen metric and its compatible connection turn the local conformal anomaly into curvature of $\lambda(E)$.

Theorem 63.4 (Local-to-global anomaly)**.** *Let a chiral theory have projective formal-coordinate descent with Virasoro central charge c. Suppose its family of conformal blocks is finite rank and carries the standard projectively flat connection. The scalar curvature of that connection is the image of the Virasoro cocycle under the family index map. Its Hodge component equals*

$$\frac{c}{2}\,\mathrm{rk}(\mathbb{V})\,\lambda,$$

while marked-point and boundary components are determined by conformal weights and factorisation channels.

Proof. The infinitesimal formal-coordinate anomaly is the Gelfand–Fuchs class represented by the Virasoro cocycle. Descent along the universal curve transgresses that class to the Atiyah class of the determinant line. Grothendieck–Riemann–Roch evaluates the degree-two component on the Hodge bundle. Sewing at marked and nodal points inserts the eigenvalues of L_0 in the corresponding channels, giving the remaining ψ- and boundary terms. Each contribution belongs to the same projective connection because each is produced by the same descent datum on the universal family. □

The theorem identifies a precise chain of carriers:

$$\text{formal vector-field cocycle} \longrightarrow \text{projective descent} \longrightarrow \text{determinant line},$$
$$\text{determinant line} \longrightarrow \text{curvature on moduli}.$$

Affine level, BRST anomaly, framing anomaly, and curved Koszul data enter through other maps and remain distinct until a comparison theorem identifies them.

63.5 The formal-geometry square

The local and global forms of the theory fit into the commutative diagram

$$\begin{tikzcd}
\text{formal-disc chiral operations} \arrow[r, "\text{GK descent}"] \arrow[d, "\text{pole filtration}"'] & extchiralD\text{-module operations on } X \arrow[d, "\text{normal filtration}"] \\
\text{classical symbol operations} \arrow[r, "\text{GK descent}"'] & extPoisson/chiralsymboloperationsonX.
\end{tikzcd}$$

The horizontal maps are descent. The vertical maps are associated-graded constructions. Their commutativity follows because the coordinate action preserves the pole filtration.

Corollary 63.5 (Global classical limit)**.** *A complete filtered translation-equivariant chiral algebra on the formal disc whose coordinate action preserves the filtration descends to a filtered chiral algebra on X. Its associated graded is the descent of the classical operadic symbol. In particular, the vertex-to-Poisson-vertex degeneration is compatible with passage from the formal disc to the curve.*

Proof. Apply Theorem 63.1 in each filtration degree. Exactness of the finite graded quotients identifies descent of the associated graded with the associated graded of the descended bundle. Operadic compositions commute with descent because the construction is symmetric monoidal. □

The coordinate torsor therefore supplies the missing bridge between local fields and global chiral geometry. The field is the calculation. The D-module is the invariant statement. The stress tensor measures the obstruction to making the calculation independent of the coordinate without a projective connection.

Chapter 64

Associative chiral fields and rational nonlocality

The chiral product has two algebraic forms. A skew-symmetric chiral operation produces a vertex algebra. An ordered associative chiral operation produces a rational nonlocal field algebra. Braided exchange upgrades the latter to a quantum vertex algebra. The three structures share the same principal-part geometry and differ in their operadic symmetries.

64.1 Ordered rational distributions

Let V be a complete vector space with translation operator T. Set

$$\mathcal{R}_n = \mathbb{C}[z_i - z_j, (z_i - z_j)^{-1}]_{1 \leq i < j \leq n}.$$

An n-ary ordered chiral operation is a translation-equivariant map

$$\Phi : V^{\widehat{\otimes} n} \widehat{\otimes} \mathcal{R}_n \longrightarrow V[\lambda_1, \dots, \lambda_n] / \left\langle T + \sum_i \lambda_i \right\rangle$$

that is compatible with substitution on domains where nested collisions are defined. The order of the inputs is retained; no symmetric-group invariance is imposed.

Definition 64.1 (Associative chiral field algebra)**.** An associative chiral field algebra is a unital algebra over the non-symmetric operad of ordered rational chiral operations. Its binary field is written

$$Y(a, z)b \in V((z)),$$

and its defining identity is rational associativity: for every $a, b, c \in V$, there exists $N \geq 0$ such that

$$(z_1 - z_2)^N Y(a, z_1) Y(b, z_2) c = (z_1 - z_2)^N Y\big(Y(a, z_1 - z_2) b, z_2\big) c$$

as an element of the common expansion of a rational V-valued function.

The identity is operadic composition written in two nested coordinate systems. The factor $(z_1 - z_2)^N$ places both expansions in the same principal-part window.

Proposition 64.2 (Regular associative differential algebras)**.** *Let* $(B, m, 1, T)$ *be a unital* *associative algebra with a derivation T. The formula*

$$Y_B(a,z)b = (e^{zT}a)b$$

defines an associative chiral field algebra. The field is regular in z, the vacuum is 1, and translation covariance is

$$[T, Y_B(a,z)] = \partial_z Y_B(a,z).$$

Proof. The vacuum identities follow from $e^{zT}1 = 1$ and evaluation at $z=0$. Translation covariance follows from the derivation identity. For associativity,

$$\begin{aligned} Y_B(a,z_1)Y_B(b,z_2)c &= (e^{z_1T}a)(e^{z_2T}b)c, \\ Y_B(Y_B(a,z_1-z_2)b,z_2)c &= e^{z_2T}\big((e^{(z_1-z_2)T}a)b\big)c \\ &= (e^{z_1T}a)(e^{z_2T}b)c. \end{aligned}$$

No pole-clearing factor is needed. □

This construction is the regular atom of associative chirality. Tensoring it with a local vertex algebra gives an explicit doubly noncommutative field theory.

Proposition 64.3 (Tensor product with a vertex algebra)**.** *Let V be a vertex algebra and B an associative differential algebra. On $A = B \otimes V$, define*

$$Y_A(b\otimes v,z)(b'\otimes v') = (e^{zT_B}b)b' \otimes Y_V(v,z)v'.$$

Then A is an associative chiral field algebra. Its singular principal parts are those of V with coefficients in the ordered product of B, while its regular product is noncommutative whenever B is noncommutative.

Proof. Translation covariance is the tensor product of the two covariance identities. Rational associativity is the product of strict associativity in B and vertex associativity in V. The pole order is inherited from the vertex factor because the B-valued exponential is regular. □

64.2 Generators and relations

Let W be a complete translation module. The free associative chiral algebra is

$$\mathrm{Free}_{\mathsf{Ass}^{\mathrm{ch}}_X}(W) = \bigoplus_{n\geq 0} \mathsf{Ass}^{\mathrm{ch}}_X(n)\widehat{\otimes} W^{\widehat{\otimes} n}.$$

A presentation with relations $R \to \mathrm{Free}_{\mathsf{Ass}^{\mathrm{ch}}_X}(W)$ is the homotopy pushout

$$\begin{tikzcd} \mathrm{Free}_{\mathsf{Ass}^{\mathrm{ch}}_X}(R) \arrow[r] \arrow[d] & \mathrm{Free}_{\mathsf{Ass}^{\mathrm{ch}}_X}(W) \arrow[d] \\ \mathbf{1} \arrow[r] & A. \end{tikzcd}$$

The lower-left object enforces the relations by a derived quotient. The homotopy pushout retains the syzygies that an ordinary quotient discards.

For quadratic relations, the arity-three overlap space is the first test. Suppose generators $x_i(z)$ satisfy

$$x_i(z_1)x_j(z_2) = \sum_{k,l} R_{ij}^{kl}(z_1 - z_2)x_k(z_2)x_l(z_1).$$

Two reductions of a triple word agree exactly when

$$R_{12}(u)R_{13}(u+v)R_{23}(v) = R_{23}(v)R_{13}(u+v)R_{12}(u).$$

Theorem 64.4 (Confluence of Zamolodchikov–Faddeev normal ordering)**.** *Assume that $R(u)$ is invertible, satisfies the spectral Yang–Baxter equation, and obeys unitarity*

$$R_{21}(-u)R_{12}(u) = 1.$$

Then the quadratic exchange relations define a confluent rewriting system on words whose spectral parameters are pairwise distinct. Every word has a unique normally ordered form. Conversely, confluence in length three implies the Yang–Baxter equation, and reversibility of a two-letter reduction implies unitarity.

Proof. Choose the inversion number of the spectral ordering as a termination measure. Each adjacent interchange lowers the inversion number, so every reduction terminates. The only nontrivial critical overlap is a word of length three. Reducing the left pair, then the middle pair, then the right pair gives the left side of Yang–Baxter; the opposite reduction gives the right side. The diamond lemma gives confluence. Applying an interchange followed by its reverse gives the unitarity condition. The converse follows by evaluating the unique length-three diamond and the unique length-two reversal. □

64.3 Braided locality and quantum vertex algebras

Let $\mathcal{S}(z) : V \otimes V \to V \otimes V \otimes \mathbb{C}((z))$ be an exchange operator. Write

$$\mathcal{S}(z)(a \otimes b) = \sum_i a^i \otimes b^i \, f_i(z).$$

Braided locality is the equality, after multiplication by a sufficient power of $z_1 - z_2$,

$$Y(a, z_1)Y(b, z_2) = \sum_i f_i(z_1 - z_2)Y(b^i, z_2)Y(a^i, z_1).$$

Theorem 64.5 (Associative chiral field recognition)**.** *Let V be a complete associative chiral field algebra. Suppose $\mathcal{S}(z)$ satisfies translation covariance,*

$$[T \otimes 1, \mathcal{S}(z)] = -\partial_z \mathcal{S}(z), \qquad [1 \otimes T, \mathcal{S}(z)] = \partial_z \mathcal{S}(z),$$

unitarity, the spectral Yang–Baxter equation, and braided locality. Then $(V, Y, T, 1, \mathcal{S})$ is a quantum vertex algebra. The exchange operator is uniquely determined on the span of matrix coefficients of fields when those coefficients separate $V \otimes V$.

Proof. Associative chiral composition supplies rational associativity and the vacuum and translation axioms. Braided locality supplies the exchange axiom. Unitarity identifies the two orientations of a two-point exchange. Yang–Baxter identifies the two paths through a three-point configuration. The compatibility of $\mathcal{S}$ with Y follows by applying braided locality inside a threefold product and using associativity to compare the two parenthesizations. Separation of matrix coefficients gives uniqueness. □

The theorem separates three layers of structure:

$$\text{associative chiral composition} \quad + \quad \text{exchange transport} \quad = \quad \text{quantum vertex algebra.}$$

The first layer exists on an ordered formal disc. The second requires paths in a plane or a meromorphic flat connection.

64.4 The associative chiral bar

For an augmented associative chiral algebra C, its operadic bar is

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathrm{Ass}}(C) = \bigoplus_{T} \mathrm{or}(E(T)) \otimes \bigotimes_{v \in V(T)} s\overline{C}(\mathrm{in}(v)),$$

where T ranges over ordered rooted collision trees. The differential contracts one internal edge and applies the chiral multiplication at the resulting vertex.

Proposition 64.6 (Coderivation identity)**.** *Let m_X denote the associative chiral multiplication. The bar coderivation d_{m_X} satisfies*

$$d^2_{m_X} = \frac{1}{2} d_{[m_X, m_X]}.$$

Hence the bar differential is square-zero exactly when the associative chiral Maurer–Cartan equation $[m_X, m_X] = 0$ holds.

Proof. A double edge contraction is indexed by an ordered pair of internal edges. Disjoint contractions cancel after reversing their determinant-line order. Nested contractions assemble into the two parenthesizations of a threefold chiral product. Their difference is the operadic bracket $[m_X, m_X]$. Summing over trees gives the formula. □

The transverse interval bar and the associative chiral bar therefore resolve different multiplications. A mixed distributive law is the datum that allows their differentials to be totalized.

64.5 A finite exact model

Let $B_q = \mathbb{C}\langle x, y\rangle/(yx - qxy)$, with $q \in \mathbb{C}^\times$, and let V be a local vertex algebra. Set $A_q = B_q \otimes V$ with the regular nonlocal field from Theorem 64.3. The augmentation $x, y \mapsto 0$ gives the transverse Koszul resolution

$$0 \longrightarrow B_q e_{xy} \xrightarrow{d_2} B_q e_x \oplus B_q e_y \xrightarrow{d_1} B_q \longrightarrow \mathbb{C} \longrightarrow 0,$$

with

$$d_1(e_x) = x, \qquad d_1(e_y) = y, \qquad d_2(e_{xy}) = y e_x - q x e_y.$$

Tensoring with $\mathbb{C}$ over B_q gives

$$\mathrm{Tor}^{B_q}_*(\mathbb{C}, \mathbb{C}) \cong \mathbb{C} \oplus \mathbb{C}^2[1] \oplus \mathbb{C}[2].$$

Every term remains tensored with the complete chiral coefficient of V. Thus the transverse bar sees the noncommutative quadratic resolution, while the chiral coefficient retains all modes of V.

Corollary 64.7 (Exact double-bar series)**.** *If the associative chiral multiplication on V is free on a finite-dimensional space M and the mixed law is strict, then the bivariate homology series of the iterated bar of $A_q \otimes \mathrm{Free}_{\mathsf{Ass}^{\mathrm{ch}}_X}(M)$ is*

$$P(s,t) = (1 + 2s + s^2)(1 + (\dim M)t),$$

where s records transverse bar degree and t records associative chiral bar degree.

Proof. The quantum-plane resolution has Betti polynomial $1 + 2s + s^2$. The reduced bar of a free associative algebra contracts to its generator space in degree one, giving $1 + (\dim M)t$. Strict interchange identifies the iterated complex with the tensor product of the two resolutions, so the Künneth theorem multiplies the polynomials. □

The model exhibits the simplest exact form of double noncommutativity: one multiplication is the quantum-plane product, one is free associative chiral composition, and the vertex coefficient may independently carry singular local fields.

Chapter 65

The bi-Rees geometry of mixed Koszul duality

Two filtrations govern the local theory. The pole filtration measures approach to a chiral diagonal. The word filtration measures the length of transverse ordered fusion. Their simultaneous Rees object exhibits the classical limit of the collision operation and the Koszul limit of the ordered product in one flat family.

65.1 Two filtrations

Let A be a doubly associative chiral algebra. Write $P_\bullet A$ for the increasing pole filtration of its chiral coefficient and $W_\bullet A$ for an exhaustive increasing filtration preserved by the transverse multiplication. The bifiltered pieces are

$$F_{p,q}A = P_pA \cap W_qA.$$

Assume that the filtrations are complete, separated, and compatible:

$$m_X(F_{p,q}, F_{p',q'}) \subset F_{p+p',q+q'}, \qquad m_\perp(F_{p,q}, F_{p',q'}) \subset F_{p+p',q+q'}.$$

The bi-Rees object is

$$\mathcal{R}_{P,W}(A) = \bigoplus_{p,q\geq 0} F_{p,q}A\, u^p v^q.$$

The bi-Rees object is a module over $\mathbb{C}[u,v]$. The fibre at $(1,1)$ is A. The fibre at $(0,1)$ is the pole symbol algebra. The fibre at $(1,0)$ is the word symbol algebra. The fibre at $(0,0)$ is the simultaneous symbol.

Theorem 65.1 (Flat bi-Rees degeneration)**.** *Suppose each bifiltered quotient $F_{p,q}A/(F_{p-1,q}A + F_{p,q-1}A)$ is flat over the ground ring. Then $\mathcal{R}_{P,W}(A)$ is flat over $\mathbb{C}[u,v]$. Operadic composition extends over the family, and the four fibres are algebras over the corresponding specializations of the mixed operad.*

Proof. The bifiltration gives an iterated Rees module. Forming the P-Rees module first, the induced W-filtration has flat graded pieces by hypothesis; the ordinary Rees flatness criterion gives flatness over $\mathbb{C}[u,v]$. The compatibility inequalities make each multiplication homogeneous for the two Rees variables. Every mixed coherence equation is polynomial in the structure maps and therefore specializes to every fibre. ☐

65.2 Normal cones and the classical chiral limit

For a smooth collision divisor $D \subset Y$, deformation to the normal cone is

$$\mathrm{DNC}_D(Y) = \mathrm{Spec}_Y \left(\bigoplus_{m \geq 0} \mathcal{I}_D^{-m} u^m \right).$$

The generic fibre is Y and the special fibre contains the normal bundle $N_{D/Y}$. For a normal-crossing family of collision divisors, the multi-Rees construction produces the simultaneous normal cone.

Each internal edge of a collision tree selects one normal direction. If the edge carries pole order $r+1$, its associated symbol lies in

$$N_e^{\otimes(r+1)}.$$

Composition of nested collisions is substitution in the direct sum of normal directions. This gives the canonical associated-graded collision operad.

Proposition 65.2 (Canonical symbol composition). *Let T be a nested collision tree and D_T its boundary stratum in a Fulton–MacPherson compactification. There is a canonical splitting*

$$N_{D_T/Y} \cong \bigoplus_{e \in E(T)} N_e.$$

The associated-graded operation attached to T is the tensor product of the Laurent symmetric algebras of the $N_e^\vee$. Contracting a subtree agrees with substitution of the corresponding normal-cone operation.

Proof. Each edge represents one successive blow-up normal parameter. Nested-set coordinates identify a neighbourhood of D_T with a product of the stratum and one normal coordinate for each edge, up to terms of higher total normal order. Passing to the associated graded removes those higher terms and makes the splitting canonical. Contraction of a subtree forgets its internal scaling parameters after composing the corresponding relative configurations; the induced map on normal symbols is precisely substitution. □

The filtered lift depends on formal tubular data, while the associated graded is canonical. Two choices of lift differ by a filtration-positive automorphism. The pronilpotent group of such automorphisms acts by gauge transformations on the Maurer–Cartan space. Thus the filtered operation has a canonical gauge class.

65.3 The mixed quadratic datum

Let E be the generating collection of the two-coloured mixed operad. It has two binary generators

$$\mu_\perp, \qquad \mu_X,$$

and three families of quadratic relations:

$$R_\perp, \qquad R_X, \qquad R_{\mathrm{mix}}.$$

The first two encode associativity in their respective directions. The mixed relation is defined on the aligned-decomposition summand and compares

$$\mu_X \circ (\mu_\perp, \mu_\perp) \quad \text{with} \quad \mu_\perp \circ (\mu_X, \mu_X).$$

The quadratic dual cooperad is generated by $sE^\vee$ with corelations $s^2R^\perp$. Its cobar is the two-coloured homotopy operad of mixed ordered chiral algebras.

Theorem 65.3 (Mixed Koszul criterion)**.** *Assume:*

1. *the transverse associative operad and the associative chiral operad are Koszul on the chosen finite windows;*
2. *the aligned mixed relation defines a distributive law on associated gradeds;*
3. *the induced rewriting system on two-coloured trees is confluent in arity three.*

Then the mixed quadratic operad is Koszul on those windows. Its bar homology is concentrated on the diagonal of weight and homological degree, and its cobar reconstructs every connected algebra in the window.

Proof. Order two-coloured trees by moving transverse vertices below chiral vertices. The mixed relation is the rewriting rule. Termination follows from the number of inverted adjacent colour pairs. Critical overlaps occur in arity three and are resolved by hypothesis. The distributive-law theorem identifies the associated graded of the mixed operad with the composite of the two Koszul operads. The composite Koszul complex is the tensor product of the two diagonal complexes, hence diagonal. The comparison spectral sequence collapses. □

65.4 The double bar and its spectral sequence

For a mixed algebra A, form the bicomplex

$$B_{p,q}(A) = \overline{A}_\perp^{\otimes p} \widehat{\otimes} \overline{A}_X^{\star_{\mathrm{ch}} q}$$

tensored with the orientation line of the two edge sets. The differentials are

$$d_\perp : B_{p,q} \to B_{p-1,q}, \qquad d_X : B_{p,q} \to B_{p,q-1}.$$

The mixed boundary identity gives

$$d_\perp^2 = d_X^2 = 0, \qquad d_\perp d_X + d_X d_\perp = 0.$$

Theorem 65.4 (Double-bar Fubini theorem)**.** *Let A be a connected complete mixed algebra. Suppose both bar constructions preserve the completion and the mixed law is represented by a strong distributive law. Then there are natural equivalences*

$$\mathrm{Bar}^\perp \mathrm{Bar}^{\mathrm{ch}}_{\mathrm{Ass}}(A) \simeq \mathrm{Bar}^{\mathrm{ch}}_{\mathrm{Ass}} \mathrm{Bar}^\perp(A) \simeq \mathrm{Tot}\, B_{\bullet,\bullet}(A).$$

If the strong law is available only on aligned windows, the equivalence holds windowwise and passes to the inverse limit under the Mittag–Leffler condition.

Proof. The two iterated bars are geometric realizations of the bisimplicial bar object. The distributive law identifies the two ways of moving one face map past the other. Eilenberg–Zilber gives the equivalence with the total normalized complex. On finite windows every sum is finite. Surjective transition maps kill the derived inverse-limit obstruction and complete the equivalence. □

Two spectral sequences follow:

$$E^1_{p,q} = H^X_q(B_{p,\bullet}) \Longrightarrow H_{p+q}(\mathrm{Tot}\, B),$$

$$'E^1_{p,q} = H^{\perp}_p(B_{\bullet,q}) \Longrightarrow H_{p+q}(\mathrm{Tot}\, B).$$

Their agreement is a calculable test of the mixed law.

65.5 A strict rectangular model

Let Q be the quiver

$$0 \xrightarrow{x} 1 \xrightarrow{y} 2$$

with relation $yx = 0$. Let Q' be an independent copy with arrows u, v and relation $vu = 0$. Over the semisimple vertex algebra $S = \mathbb{C}^3$, take the tensor product of the two relative path algebras. Horizontal paths define $m_\perp$ and vertical paths define m_X. The strict rectangle relation identifies the two length-two routes around a square whenever both are defined.

The relative transverse bar has Betti polynomial

$$P_\perp(s) = 3 + 2s + s^2,$$

and the vertical bar has

$$P_X(t) = 3 + 2t + t^2.$$

Calculation 65.5 (Rectangular mixed bar)**.** For the strict double-quiver algebra, the total relative bar has bivariate Betti polynomial

$$P(s,t) = (3 + 2s + s^2)(3 + 2t + t^2).$$

The horizontal relation gives a class in bidegree $(2, 0)$, the vertical relation gives a class in $(0, 2)$, and their rectangular product gives a class in $(2, 2)$.

Proof. Each relative bar is the minimal two-step resolution of the corresponding radical-square-zero path algebra. Strict interchange makes the double complex the tensor product over S of the two resolutions. Künneth multiplication gives the displayed polynomial and identifies the product class. □

The example realizes double noncommutativity as a bigraded derived invariant whose two axis restrictions recover the separate one-dimensional bars.

65.6 The bi-Rees master object

Let $\mathfrak{g}_{\mathrm{mix}}(A)$ be the complete convolution Lie algebra of the mixed Koszul dual cooperad with the endomorphism operad of A. Decompose its Maurer–Cartan element as

$$\Theta = \Theta_\perp + \Theta_X + \Theta_{\mathrm{mix}}.$$

The equation

$$d\Theta + \frac{1}{2}[\Theta, \Theta] = 0$$

splits into the two associativity equations and the mixed equation. The bi-Rees variables record their filtration orders:

$$\Theta(u, v) = u\Theta_X + v\Theta_\perp + uv\Theta_{\mathrm{mix}} + ext higher filtered terms.$$

At $(u, v) = (0, 0)$ the equation becomes the simultaneous Poisson–Koszul symbol. At $(1, 1)$ it is the complete doubly noncommutative theory. The bi-Rees family is therefore the geometric interpolation between the classical collision shadow and the ordered quantum algebra.

Chapter 66

Modules, interfaces, and the category of boundary conditions

The algebra of local fields is a chart on a category of boundary conditions. Modules and bimodules carry the observables that change a boundary or cross an interface. Their bar constructions are pointed configuration spaces, and their composition is relative factorisation homology.

66.1 Pointed bars

Let A be an augmented transversely ordered chiral algebra, M a right A-module, and N a left A-module. The two-sided simplicial bar is

$$B_r(M, A, N) = M \otimes_! A^{\otimes_! r} \otimes_! N.$$

Its inner faces multiply adjacent A-factors, the first face uses the action on M, the last uses the action on N, and degeneracies insert the unit.

Theorem 66.1 (Relative interval amplitude)**.** *There is a natural equivalence*

$$|B_\bullet(M, A, N)| \simeq M \otimes^{\mathbb{L}}_A N \simeq \int_{([0,1];0,1)} (M, A, N).$$

The factorisation coefficient on X is preserved in every simplicial degree.

Proof. The simplicial bar is the standard resolution of the relative tensor product in the stable monoidal category $\mathrm{FactD}(X)$. Excision for factorisation homology decomposes the pointed interval into collars of the endpoints and finitely many interior intervals; its colimit is the same simplicial object. All structure maps are morphisms of factorisation coefficients. □

The geometric carrier is a compactification of ordered points in an interval with two marked boundary points. Collision with the left endpoint applies the right action on M; collision with the right endpoint applies the left action on N.

66.2 Interfaces and bimodules

Let A and B be ordered chiral algebras. An (A, B)-bimodule K is a field-theoretic interface from the A-boundary to the B-boundary. If L is a (B, C)-bimodule, fusion of interfaces is

$$K \otimes_B^{\mathbb{L}} L.$$

Associativity of derived tensor product is the algebraic image of gluing three adjacent strips.

Proposition 66.2 (Factorisation of interface fusion). *Suppose the actions of A and B on K are morphisms of factorisable coefficients. Then $K \otimes_B^{\mathbb{L}} L$ is an (A, C)-bimodule in* $\mathrm{FactD}(X)$, *and interface fusion is compatible with disjoint chiral factorisation on X.*

Proof. Compute the relative tensor product by the two-sided bar. Every term is factorisable because pointwise tensor preserves the factorisation equivalences. Every face map is factorisable by the module hypotheses. Geometric realization in a presentable category preserves the factorisation diagrams. □

The bimodules, their relative tensor products, and their morphisms form the Morita $(\infty, 2)$-category of ordered chiral algebras. The centre is the endomorphism algebra of the identity bimodule.

66.3 Boundary-changing fields

Let b_0, b_1 be boundary conditions in a factorisation category $\mathcal{C}$. The boundary-changing coefficient is

$$M_{01} = \mathrm{RHom}_{\mathcal{C}}(b_0, b_1).$$

Composition gives

$$M_{12} \otimes_! M_{01} \longrightarrow M_{02}.$$

When the b_i are compact dualisable local generators, the matrix of all M_{ij} is a factorisation-enriched category and any chosen generator $b = \bigoplus_i b_i$ produces the algebra

$$A_b = \mathrm{End}_{\mathcal{C}}(b).$$

Idempotents $e_i \in A_b$ recover the sectors:

$$e_j A_b e_i \simeq M_{ij}.$$

Theorem 66.3 (Relative Morita reconstruction). *Let $S = \bigoplus_i \mathbb{C}e_i$ and suppose $b = \bigoplus_i b_i$ is a compact dualisable generator on a factorising basis. Then*

$$\mathcal{C} \simeq \mathrm{Mod}_{A_b}(\mathrm{FactD}(X)),$$

and the pointed bar over the semisimple base S computes derived boundary-changing compositions. The result is invariant under replacing b by another compact generator.

Proof. Local compact-generator Morita theory gives the equivalence on each basis open. Factorisation compatibility glues the local equivalences. The algebra map $S \to A_b$ records the idempotent decomposition. Relative bar constructions compute tensor products without collapsing the boundary labels. A change of compact generator is implemented by an invertible bimodule and therefore preserves the Morita category. □

66.4 The Koszul transform of a module

Let $C = \mathrm{Bar}^{\perp}(A)$ and let $\tau : C \to A$ be the universal twisting cochain. For a complete right A-module M, define

$$\mathbb{K}_A(M) = M \otimes_\tau C.$$

The differential is the sum of the internal differentials and the action twisted by τ.

Theorem 66.4 (Module Koszul duality)**.** *On the augmentation-complete and conilpotent loci, the functors*

$$M \longmapsto M \otimes_\tau C, \qquad N \longmapsto N \otimes_\tau A$$

are inverse equivalences between complete A-modules and conilpotent C-comodules. The two-sided transform sends the diagonal A-bimodule to the diagonal C-bicomodule.

Proof. Filter by bar length. The associated graded unit and counit are the contracting bar resolutions of the augmentation. Convergence follows on finite positive-weight windows and then on complete Mittag–Leffler limits. The two-sided statement is the same argument in the enveloping algebra $A \otimes_! A^{\mathrm{op}}$. □

As a consequence,

$$\mathrm{HH}_\bullet(A) \simeq C_\Delta \square^{\mathbb{R}}_{C^e} C_\Delta, \qquad \mathrm{HH}^\bullet(A) \simeq \mathrm{Coext}_{C^e}(C_\Delta, C_\Delta).$$

The annulus and the derived centre are the trace and endomorphism of the same diagonal object.

66.5 A quiver calculation

Let Q be $1 \xrightarrow{a} 2 \xrightarrow{b} 3$ with relation $ba = 0$, and let $A = \mathbb{C}Q/(ba)$. Set $S = \mathbb{C}e_1 \oplus \mathbb{C}e_2 \oplus \mathbb{C}e_3$. The relative augmentation ideal is spanned by a, b. The minimal relative bar has

$$B_0 = S, \qquad B_1 = SaS \oplus SbS, \qquad B_2 = S(ba)S,$$

with zero differential after tensoring with S on both sides. Hence

$$\mathrm{Tor}_0^A(S, S) = S, \quad \mathrm{Tor}_1^A(S, S) = \mathbb{C}a \oplus \mathbb{C}b, \quad \mathrm{Tor}_2^A(S, S) = \mathbb{C}(ba).$$

For the simple modules $S_i = e_i S$, the only nonzero positive extensions are

$$\mathrm{Ext}^1_A(S_2, S_1) = \mathbb{C}a, \qquad \mathrm{Ext}^1_A(S_3, S_2) = \mathbb{C}b, \qquad \mathrm{Ext}^2_A(S_3, S_1) = \mathbb{C}(ba).$$

The degree-two class is the derived memory of the vanishing composite. Tensoring A with a chiral coefficient produces the corresponding family of boundary-changing fields on X.

66.6 Bulk action and its exact target

Let $\mathrm{Obs}_{\mathrm{bulk}}$ be the factorisation algebra of a bulk theory adjacent to the boundary algebra A. Bringing a bulk observable to the boundary gives an E_2-central action

$$\Phi_{\mathrm{oc}} : \mathrm{Obs}_{\mathrm{bulk}} \longrightarrow \mathrm{HH}^{\bullet}_{\mathrm{FactD}(X)}(A, A).$$

Proposition 66.5 (Universal property of the boundary centre)**.** *Every homotopy-coherent central action of an E_2-algebra B on A factors through the internal Hochschild cochains. A physical bulk is identified with the algebraic centre precisely when its open–closed map Φ_{oc} is a quasi-isomorphism.*

Proof. The higher Deligne construction identifies Hochschild cochains with endomorphisms of the identity A-bimodule. A central action is an action on that identity object and therefore determines a map to its endomorphism algebra. The final assertion is the definition of identification in the derived category. □

The centre is universal. The physical bulk is a realization equipped with a comparison map. This distinction preserves both the algebraic theorem and the dynamical content of the field theory.

Chapter 67

Critical level, opers, and the derived Langlands fibre

At the critical level, the Sugawara stress tensor ceases to exist as an ordinary field and the affine centre enlarges to functions on opers. The correct algebraic operation is derived central reduction. Its bar construction records the conormal geometry of the oper fibre.

67.1 The affine centre

Let $\mathfrak{g}$ be a simple complex Lie algebra and let

$$V_k(\mathfrak{g})$$

be the universal affine vertex algebra at level k. The critical level is

$$k = -h^\vee.$$

At this level the Segal–Sugawara generators become central after the standard renormalization.

External theorem 67.1 (Feigin–Frenkel centre)**.** *The centre of the completed critical affine algebra is canonically isomorphic to the algebra of functions on the ind-scheme of Langlands-dual opers on the punctured disc:*

$$Z\big(\widehat{U}_{-h^\vee}(\widehat{\mathfrak{g}})\big) \cong \mathcal{O}\big(\mathrm{Op}_{{}^L\mathfrak{g}}(D^\times)\big).$$

The regular centre corresponds to opers on the disc. [78]

The theorem changes the meaning of a central character. A character is a point

$$\chi \in \mathrm{Op}_{{}^L\mathfrak{g}}(D^\times).$$

The central reduction is

$$A_\chi = A \otimes^{\mathbb{L}}_Z \mathbb{C}_\chi.$$

Derived tensor is essential at singular oper points.

67.2 The derived central fibre

Let Z be a commutative algebra and $\chi : Z \to \mathbb{C}$ a point. The self-intersection of the point is

$$\mathbb{C}_\chi \otimes^{\mathbb{L}}_Z \mathbb{C}_\chi.$$

Theorem 67.2 (Smooth central fibre). *If Z is smooth of finite type at χ, then*

$$\mathbb{C}_\chi \otimes^{\mathbb{L}}_Z \mathbb{C}_\chi \simeq \mathrm{Sym}\big(T^*_\chi \operatorname{Spec} Z[1]\big)$$

as a graded-commutative algebra. Equivalently,

$$\mathrm{Tor}^Z_i(\mathbb{C}_\chi, \mathbb{C}_\chi) \cong \Lambda^i T^*_\chi \operatorname{Spec} Z.$$

Proof. Choose regular parameters $f_1, \dots, f_d$ generating the maximal ideal at χ. The Koszul complex $K_Z(f_1, \dots, f_d)$ resolves $\mathbb{C}_\chi$. Tensoring with $\mathbb{C}_\chi$ kills the Koszul differential and leaves the exterior algebra on the classes $df_i[1]$. The coordinate-free form is the symmetric algebra on the shifted cotangent space. □

For the oper space, finite jet truncations give finite-dimensional smooth windows on the regular locus. Their inverse limit gives the completed exterior algebra of the continuous cotangent complex.

Corollary 67.3 (Oper conormal bar). *At a smooth regular oper χ, the relative bar of the critical central reduction over the Feigin–Frenkel centre contains the completed coalgebra*

$$\widehat{\mathrm{Sym}}\big(T^*_\chi \mathrm{Op}_{{}^L\mathfrak{g}}(D)[1]\big).$$

The primitive degree-one classes are infinitesimal oper deformations.

Proof. Apply Theorem 67.2 in every finite jet window and pass to the inverse limit. Surjective jet truncations satisfy the Mittag–Leffler condition. □

67.3 Critical degeneration of exchange

The KZ connection is

$$\nabla_{\mathrm{KZ}} = d - \frac{1}{k + h^\vee} \sum_{i<j} \Omega_{ij}\, d\log(z_i - z_j).$$

The coefficient diverges at the critical level. The critical theory is therefore not obtained by substituting $k = -h^\vee$ into the noncritical monodromy formula. Rescaling the connection produces the Gaudin/Hitchin classical limit.

Set

$$\epsilon = k + h^\vee.$$

Then

$$\epsilon \nabla_{\mathrm{KZ}} = \epsilon d - \sum_{i<j} \Omega_{ij}\, d\log(z_i - z_j).$$

As $\epsilon \to 0$, flat sections have WKB form

$$\Psi \sim \exp(S/\epsilon)(\psi_0 + \epsilon\psi_1 + \cdots),$$

and the leading equation is the spectral equation for the commuting Gaudin Hamiltonians.

Proposition 67.4 (Critical WKB symbol). *The leading WKB symbol of the KZ connection is the Higgs field*

$$\Phi = \sum_{i<j} \Omega_{ij}\, d\log(z_i - z_j).$$

Flatness of KZ implies the classical integrability equation

$$\Phi \wedge \Phi = 0$$

modulo the infinitesimal braid relations. The corresponding spectral invariants are the classical Gaudin Hamiltonians.

Proof. Insert the WKB ansatz into $\nabla_{\mathrm{KZ}}\Psi = 0$ and collect the coefficient of ϵ^{-1}. The curvature equation for $d - \epsilon^{-1}\Phi$ has leading term $\epsilon^{-2}\Phi \wedge \Phi$. Since the full curvature vanishes, this term vanishes. Expanding it gives Arnold forms multiplied by commutators of the Casimir tensors; the infinitesimal braid relations cancel the coefficients. □

67.4 Drinfeld–Sokolov reduction at and away from criticality

Let $f \in \mathfrak{g}$ be principal nilpotent. Quantum Drinfeld–Sokolov reduction is the BRST complex

$$C_{\mathrm{DS}}(V_k(\mathfrak{g})) = V_k(\mathfrak{g}) \otimes \mathrm{Ghost}(\mathfrak{n}_+ \oplus \mathfrak{n}_+^*),$$

with differential Q_{DS}. Away from criticality, its degree-zero cohomology is the principal W-algebra $\mathcal{W}_k(\mathfrak{g})$ under the standard concentration theorem.

The Feigin–Frenkel dual level ℓ is determined by

$$r^\vee(k + h^\vee)(\ell + {}^L h^\vee) = 1,$$

where $r^\vee$ is the lacing number in the chosen convention. This multiplicative shifted-level relation is the correct duality parameter.

Theorem 67.5 (Bar–BRST comparison). *Let A be a complete ordered chiral algebra with a filtration-preserving square-zero BRST derivation Q. Suppose the bar filtration and BRST filtration are bounded below in every conformal-weight window and the associated spectral sequence converges strongly. Then*

$$E_1 \cong \mathrm{Bar}^\perp\big(H_Q(A)\big) \Longrightarrow H_Q\big(\mathrm{Bar}^\perp(A)\big).$$

If $H_Q(A)$ is concentrated in degree zero and the spectral sequence degenerates, then

$$\mathrm{Bar}^\perp\big(H_Q^0(A)\big) \simeq H_Q^0\big(\mathrm{Bar}^\perp(A)\big).$$

Proof. The derivation property gives $[Q_{\mathrm{bar}}, d_{\mathrm{bar}}] = 0$. Filter the total bicomplex by BRST degree. The first differential is Q acting factorwise, and the Künneth hypothesis identifies its homology with the bar chains of $H_Q(A)$. Strong convergence gives the abutment. Concentration and degeneration identify the degree-zero edge map with a quasi-isomorphism. □

At critical level, reduction is naturally performed over the oper centre. Derived central reduction and BRST reduction form a two-step derived intersection. Their order may be exchanged when the central action commutes with Q and the corresponding Tor spectral sequence converges.

67.5 The Langlands fibre as a boundary category

A regular oper χ defines a critical block of modules. The augmentation boundary in that block has endomorphism algebra

$$A^!_\chi = \mathrm{RHom}_{A_\chi}(\mathbf{1}, \mathbf{1}).$$

The cotangent directions of the oper fibre act as degree-one central operators in the boundary dual. The resulting category is a local form of the derived Langlands fibre: it records both the representation-theoretic reduction and the derived geometry of the central character.

Recognition theorem 67.6 (Derived oper fibre)**.** *Suppose a critical block admits a compact generator, the Feigin–Frenkel centre acts through a smooth regular oper χ, and localization identifies the block with a category of sheaves on the corresponding oper fibre. Then the Koszul dual of the augmentation boundary contains the exterior algebra of the oper cotangent space, and the localization equivalence identifies this action with the derived self-intersection of the oper point.*

Proof. The compact generator gives a Morita chart. Central reduction is derived tensor over Z. Theorem 67.2 computes the derived self-intersection. Localization transports the central action and the relative bar to the geometric fibre. □

Critical level is a change of geometric phase: quantum monodromy degenerates to a classical integrable system, while the centre expands into the geometry of opers. Quantum monodromy degenerates to a classical integrable system, while the centre expands into the geometry of opers.

Chapter 68

Heisenberg theory, determinants, and Eisenstein geometry

The free boson is the exact Gaussian model in which local OPEs, interval bars, determinants, theta functions, and modular differential equations can all be computed. Its simplicity separates the structures rather than collapsing them.

68.1 The current algebra

Let $\mathfrak{h}$ be an r-dimensional vector space with nondegenerate symmetric form $(\ ,\ )$. The Heisenberg current algebra is generated by fields $J^a(z)$ with

$$J^a(z)J^b(w) \sim \frac{(a,b)}{(z-w)^2}.$$

Equivalently,

$$[J^a_m, J^b_n] = m(a,b)\delta_{m+n,0}.$$

The singular coefficient lies in the second normal symbol of the diagonal. The simple-pole bracket vanishes. The stress tensor

$$T(z) = \frac{1}{2}\sum_{i=1}^{r} : J^{e_i}(z)J^{e_i}(z) :$$

has central charge $c = r$.

Proposition 68.1 (Fock factorisation)**.** *For momentum $\lambda \in \mathfrak{h}$, the Fock module decomposes as*

$$\mathcal{F}_\lambda \cong \mathrm{Sym}\left(\bigoplus_{n\geq 1}\mathfrak{h}\,t^{-n}\right)\otimes e^\lambda.$$

Its character is

$$\mathrm{ch}_{\mathcal{F}_\lambda}(q) = q^{(\lambda,\lambda)/2-r/24}\prod_{n\geq 1}(1-q^n)^{-r}.$$

Proof. The negative modes act freely and commute, while positive modes annihilate the highest-weight vector e^λ. Each oscillator J^a_{-n} contributes a symmetric generator of energy n. Multiplying the geometric series over n and over a basis of $\mathfrak{h}$ gives the product. The exponent contains the momentum energy and the vacuum Casimir shift. □

68.2 The Gaussian determinant

Let T_q be the one-particle operator with eigenvalue q^n on the nth oscillator. Bosonic second quantization gives

$$\mathrm{Tr}_{\mathrm{Sym}(H)}(\Gamma(T_q)) = \det(1-T_q)^{-1}.$$

For rank r,

$$\widehat{Z}_r(q) = \det(1-T_q)^{-r} = \prod_{n\geq 1}(1-q^n)^{-r}.$$

Since

$$\eta(\tau) = q^{1/24}\prod_{n\geq 1}(1-q^n),$$

we obtain

$$\widehat{Z}_r(q) = q^{r/24}\eta(\tau)^{-r}, \qquad Z_r(\tau) = q^{-r/24}\widehat{Z}_r(q) = \eta(\tau)^{-r}.$$

Corollary 68.2 (Colored partitions)**.** *Write*

$$\prod_{n\geq 1}(1-q^n)^{-r} = \sum_{m\geq 0} p_r(m)q^m.$$

Then

$$mp_r(m) = r\sum_{k=1}^{m}\sigma_1(k)p_r(m-k), \qquad p_r(0) = 1.$$

Proof. Take the logarithmic derivative:

$$q\frac{d}{dq}\log\widehat{Z}_r = r\sum_{n,k\geq 1} nq^{nk} = r\sum_{m\geq 1}\sigma_1(m)q^m.$$

Multiplication by $\widehat{Z}_r$ and comparison of coefficients gives the recurrence. □

68.3 Eisenstein series from the determinant

The logarithmic derivative of η is

$$q\frac{d}{dq}\log\eta(\tau) = \frac{1}{24}E_2(\tau),$$

where

$$E_2(\tau) = 1 - 24\sum_{n\geq 1}\sigma_1(n)q^n.$$

Hence

$$q\frac{d}{dq}\log Z_r(\tau) = -\frac{r}{24}E_2(\tau).$$

The quasi-modularity of E_2 is the infinitesimal form of the determinant-line anomaly.

The Ramanujan differential equations are

$$q\frac{dE_2}{dq} = \frac{E_2^2 - E_4}{12}, \qquad q\frac{dE_4}{dq} = \frac{E_2E_4 - E_6}{3}, \qquad q\frac{dE_6}{dq} = \frac{E_2E_6 - E_4^2}{2}.$$

The Eisenstein series E_2, E_4, E_6 close the differential ring of quasimodular forms.

Proposition 68.3 (Serre derivative of the free-boson block)**.** *Let $D_k = q\frac{d}{dq} - \frac{k}{12}E_2$ be the Serre derivative on weight k. The rank-r oscillator block $Z_r = \eta^{-r}$ has weight $-r/2$ and satisfies*

$$D_{-r/2}Z_r = 0.$$

Proof. The logarithmic derivative is $-(r/24)E_2$. Since $-k/12 = r/24$ for $k = -r/2$, the two terms cancel. □

The first-order modular differential equation is the exact analytic shadow of the Gaussian determinant.

68.4 Lattice zero modes

Let L be an even positive-definite lattice of rank r. The lattice vertex algebra has partition function

$$Z_{V_L}(\tau) = \frac{\Theta_L(\tau)}{\eta(\tau)^r}, \qquad \Theta_L(\tau) = \sum_{\lambda\in L} q^{(\lambda,\lambda)/2}.$$

The numerator is the zero-mode sum. The denominator is the oscillator determinant. Their modular transformations arise from different operations: Poisson summation for the lattice and the determinant anomaly for the oscillators.

Proposition 68.4 (Modular weight balance)**.** *If L is even unimodular, then Θ_L has weight $r/2$ and η^{-r} has weight $-r/2$. Their quotient is a modular function of weight zero, with the multiplier determined by the vacuum energy.*

Proof. Poisson summation gives

$$\Theta_L(-1/\tau) = (-i\tau)^{r/2}\Theta_{L^\vee}(\tau).$$

Unimodularity identifies $L^\vee = L$. The eta transformation is

$$\eta(-1/\tau) = (-i\tau)^{1/2}\eta(\tau).$$

The weights cancel in the quotient. □

Equal central charge fixes the oscillator exponent and leaves the theta series free to vary. The Leech and Moonshine theories therefore share $c = 24$ while carrying different state spaces and different genus data.

68.5 Higher genus Gaussian sewing

Let Σ_g be a compact Riemann surface with period matrix Ω. The normalized bidifferential $B(z, w)$ is characterized by a double pole with coefficient one on the diagonal and vanishing

A-periods. Wick's theorem gives current correlators as sums over pairings:

$$\left\langle \prod_{i=1}^{2n} J^{a_i}(z_i) \right\rangle = \sum_{\text{pairings } P} \prod_{\{i,j\} \in P} (a_i, a_j) B(z_i, z_j).$$

Odd correlators vanish.

The oscillator partition function is the inverse determinant of the scalar Laplacian or, holomorphically, the holomorphic determinant section fixed by the chosen spin structure and Quillen normalization. Lattice zero modes give the Siegel theta function

$$\Theta_L^{(g)}(\Omega) = \sum_{\lambda_1, \ldots, \lambda_g \in L} \exp\left(\pi i \sum_{a,b} (\lambda_a, \lambda_b) \Omega_{ab} \right).$$

Theorem 68.5 (Gaussian sewing factorisation)**.** *Under a separating plumbing degeneration with period matrix*

$$\Omega(q) = \begin{pmatrix} \Omega_1 & 0 \\ 0 & \Omega_2 \end{pmatrix} + O(q),$$

the lattice theta function satisfies

$$\Theta_L^{(g_1+g_2)}(\Omega(q)) = \Theta_L^{(g_1)}(\Omega_1) \Theta_L^{(g_2)}(\Omega_2) + O(q).$$

The oscillator determinant obeys the complementary gluing law, with the sewing exponent determined by the L_0 spectrum.

Proof. At $q = 0$ the quadratic form in the theta exponent splits into the direct sum of the two period forms, so the lattice sum factors. Plumbing perturbation produces an absolutely convergent power series for positive-definite L. The determinant gluing formula follows by decomposing modes into those supported on the two components and the propagating states through the neck. □

68.6 The transverse bar of the Gaussian boundary

The chiral Heisenberg coefficient alone does not select a transverse multiplication. A boundary condition supplies an associative algebra B of Chan–Paton factors. For $A = B \otimes V_{\mathfrak{h}}$,

$$\mathrm{Bar}^{\perp}(A) \simeq \mathrm{Bar}(B) \otimes V_{\mathfrak{h}}^{\mathrm{fact}}$$

when the mixed law is strict. If $B = \mathbb{C}$, the transverse bar is contractible beyond the augmentation. The nontrivial Heisenberg central extension remains in the chiral coefficient and in the determinant line, not in a fictitious transverse residue differential.

Calculation 68.6 (Matrix boundary)**.** For $B = \mathrm{Mat}_N(\mathbb{C})$ regarded relative to its unit boundary category, the cyclic ribbon amplitude of a connected surface of genus h with b boundary components carries the double-line factor

$$N^{2-2h-b}.$$

Proof. Each matrix propagator carries two index lines. A connected ribbon graph contributes one factor of N for each index face. With V vertices and E edges, the exponent is the number

F of faces. Normalizing each vertex and propagator so that the power is $V - E + F$ gives the Euler characteristic $2 - 2h - b$. $\square$

The complete Gaussian model consequently has two independent exact expansions: a chiral genus expansion governed by theta functions and determinants, and a transverse ribbon expansion governed by matrix index faces.

Chapter 69

Orbifolds, permutation defects, and entanglement

A symmetric orbifold organizes many copies of a chiral theory by cycle type. The same cycle decomposition governs twisted sectors, second-quantized partition functions, and replica defects. Hochschild theory supplies the algebraic form of the permutation twist.

69.1 Crossed products and twisted sectors

Let A be an algebra and let a finite group G act by automorphisms. The crossed product is

$$A \rtimes G = \bigoplus_{g \in G} A\, g, \qquad (ag)(bh) = a\, g(b)\, gh.$$

The orbifold category is the equivariant module category, or equivalently the module category of $A \rtimes G$ under the usual finiteness hypotheses.

For $g \in G$, let ${}_gA$ be the bimodule equal to A as a vector space, with left action twisted by g. The g-twisted Hochschild chains are

$$C_\bullet(A, {}_gA).$$

Theorem 69.1 (Orbifold Hochschild decomposition). *Over a field whose characteristic does not divide $|G|$,*

$$\mathrm{HH}_\bullet(A \rtimes G) \cong \bigoplus_{[g] \subset G} \mathrm{HH}_\bullet(A, {}_gA)^{C(g)},$$

where the sum ranges over conjugacy classes and $C(g)$ is the centralizer.

Proof. Decompose the Hochschild chains of $A \rtimes G$ by the total group element around the cyclic word. The Hochschild differential preserves conjugacy class. Conjugation identifies the summands inside one class, and averaging over $C(g)$ is exact because $|G|$ is invertible. The resulting complex is the $C(g)$-invariant twisted Hochschild complex. □

For $A^{\otimes n}$ and $G = \Sigma_n$, a permutation g decomposes into cycles. Each cycle produces one long string sector. The centralizer divides by cyclic rotations and permutations of equal-length cycles.

69.2 Second quantization and the DMVV product

Let

$$Z(\tau,z)=\sum_{m,\ell}c(m,\ell)q^m y^\ell$$

be a graded trace of the seed theory. The symmetric-product generating function is

$$\sum_{N\geq 0}p^N Z_{\mathrm{Sym}^N}(\tau,z)=\prod_{n\geq 1}\prod_{m,\ell}(1-p^n q^m y^\ell)^{-c(nm,\ell)}.$$

The product follows from cycle decomposition and bosonic exponentiation of connected cycles.

Proposition 69.2 (Plethystic derivation)**.** *Let T_n be the Hecke transform of the seed trace associated with one cycle of length n. Then*

$$\sum_{N\geq 0}p^N Z_{\mathrm{Sym}^N}=\exp\left(\sum_{n\geq 1}\frac{p^n}{n}T_n Z\right).$$

Expanding the exponential gives the product above.

Proof. A permutation is a set of disjoint cycles. A cycle of length n has automorphism group of order n and contributes $T_n Z$. The exponential formula for sets of connected components gives the plethystic exponential. Expanding $\exp(\sum_{k\geq 1}x^k/k)=(1-x)^{-1}$ coefficientwise gives the Euler product. □

The first coefficient is the seed:

$$[p]Z_{\mathrm{Sym}}=Z.$$

The second coefficient is

$$[p^2]Z_{\mathrm{Sym}}=\frac{1}{2}\big(Z^2+T_2 Z\big),$$

which separates two one-cycles from one two-cycle.

69.3 Replica defects

Let A be the operator algebra of one copy of a quantum field theory. The n-replica theory has algebra $A^{\otimes n}$ and cyclic permutation

$$\sigma=(1\ 2\ \cdots\ n).$$

The algebraic defect associated to the branch point is the twisted bimodule

$${}_\sigma A^{\otimes n}.$$

Closing an interval around the defect gives twisted Hochschild chains

$$\mathrm{HH}_\bullet\big(A^{\otimes n},{}_\sigma A^{\otimes n}\big).$$

The geometric cover and the algebraic twist are two realizations of the same monodromy.

Proposition 69.3 (Twist conformal weight)**.** *For a two-dimensional conformal field theory of central charge c, the cyclic replica twist has holomorphic weight*

$$h_n = \frac{c}{24}\left(n - \frac{1}{n}\right).$$

Proof. Uniformize the n-sheeted neighbourhood by $z = w^n$. The vacuum expectation of the stress tensor vanishes in the w-coordinate. The Schwarzian transformation gives

$$\langle T(z)\rangle = \frac{c}{12}\{w, z\} = \frac{c}{24}\left(1 - \frac{1}{n^2}\right) z^{-2}.$$

The n sheets contribute a factor of n, giving

$$\langle T(z)\rangle_{\text{replica}} = h_n z^{-2}.$$

□

The two-point function of twists on the line is

$$\langle \mathcal{T}_n(0)\widetilde{\mathcal{T}}_n(L)\rangle = C_n \left(\frac{L}{\epsilon}\right)^{-4h_n}$$

for a full nonchiral theory with equal left and right central charges.

Corollary 69.4 (Interval entropy)**.** *The Rényi entropy of one interval in the vacuum is*

$$S_n = \frac{c}{6}\left(1 + \frac{1}{n}\right)\log\frac{L}{\epsilon} + \textit{constant},$$

and the von Neumann entropy is

$$S = \frac{c}{3}\log\frac{L}{\epsilon} + \textit{constant}.$$

Proof. Insert the twist two-point function into $S_n = (1-n)^{-1}\log \operatorname{Tr}\rho^n$ and use the value of h_n. The limit $n \to 1$ gives the second formula. □

69.4 Permutation defects in an ordered chiral theory

For an ordered chiral algebra A, the replicated object

$$A^{\otimes_! n}$$

retains transverse order within each copy and carries the permutation action on the copy labels. A permutation defect is therefore a bimodule in the Morita category of ordered chiral algebras. Its local chiral coefficient is obtained by descent along the branched cover; its transverse composition is relative tensor product of defects.

Theorem 69.5 (Replica–bar compatibility)**.** *Assume the permutation action preserves the augmentation and the factorisation coefficient. Then the transverse bar is equivariant:*

$$\operatorname{Bar}^{\perp}(A^{\otimes_! n}) \simeq \operatorname{Bar}^{\perp}(A)^{\otimes_! n}.$$

For a cyclic twist σ, closing the bar interval around the defect yields the twisted Hochschild complex of $A^{\otimes_! n}$.

Proof. The bar is constructed from pointwise tensor powers and multiplication maps, all of which commute with the permutation action. The exponential law identifies the bar of the tensor product with the tensor product of the bars under the stated exactness hypotheses. Gluing the endpoints applies the cyclic identification and therefore inserts the twisted bimodule. □

This theorem gives a concrete interface between Koszul duality and replica geometry. The bar resolves fusion along the cut; twisted Hochschild chains close the cut around the branch point.

69.5 Connected cycles and large-N geometry

Taking the logarithm of the symmetric-product partition function isolates connected cycles:

$$\log Z_{\mathrm{Sym}} = \sum_{n\geq 1} \frac{p^n}{n} T_n Z.$$

The cycle length is a winding number. In holographic examples, long cycles give long strings; the statement is an interpretation of the exact combinatorial decomposition, not an additional algebraic identity.

The same connected/disconnected distinction appears in graph sewing. Partition functions exponentiate connected amplitudes, while Hochschild logarithms isolate primitive cyclic operations. The common structure is the exponential formula for disjoint unions.

Chapter 70

Two dimensional ladders: higher chirality and higher transverse order

A theory on

$$\mathbb{A}^d_{\mathrm{hol}} \times \mathbb{R}^n_{\mathrm{top}}$$

has two independent dimensions. The holomorphic dimension d changes the collision operad. The topological dimension n changes the internal little-disks multiplication. Their interaction is an interchange problem.

70.1 The bidegree of a protected theory

Let X be a smooth complex d-fold. A higher chiral coefficient assigns complexes to finite configurations in X and singular operations to their diagonals. A transverse E_n-structure assigns operations to configurations of little n-disks. The primitive object is therefore

$$A \in \mathrm{Alg}_{E_n}\big(\mathrm{FactD}_d(X)\big),$$

with the coefficient category replaced by the chosen higher chiral model.

When $d = 1$, formal-disc operations recover vertex algebraic structures. When $n = 1$, the transverse operation is ordered fusion. Neither index determines the other.

External theorem 70.1 (Polysimplicial higher chirality)**.** *On affine space $\mathbb{A}^d$, the derived algebra of functions on labelled configuration spaces admits a polysimplicial dg model. The resulting dg operad of chiral operations on the shifted canonical sheaf is quasi-isomorphic to the L_∞ operad. [24]*

The theorem gives a first exact higher-dimensional chiral algebra. It concerns the holomorphic collision direction. Little-disks algebras govern the independent transverse direction.

70.2 Iterated bar and Dunn additivity

For an augmented E_n-algebra A in a stable symmetric monoidal category, the n-fold bar is factorisation homology of the pointed n-disk:

$$\mathrm{Bar}^{(n)}(A) \simeq \int_{(D^n,\partial D^n)} A.$$

Dunn additivity gives

$$E_m \otimes E_n \simeq E_{m+n}.$$

Thus an E_m-algebra internal to E_n-algebras is an E_{m+n}-algebra under the standard presentability hypotheses.

Theorem 70.2 (Iterated bar Fubini)**.** *Let A be a connected complete E_{m+n}-algebra. Decompose the pointed disk as*

$$(D^{m+n}, \partial) \simeq (D^m, \partial) \times (D^n, \partial)$$

in factorisation homology. Then

$$\operatorname{Bar}^{(m+n)}(A) \simeq \operatorname{Bar}^{(m)} \operatorname{Bar}^{(n)}(A) \simeq \operatorname{Bar}^{(n)} \operatorname{Bar}^{(m)}(A).$$

Proof. Factorisation homology satisfies the exponential law on product manifolds. Integrating first over the n-disk gives an E_m-algebra, and integrating it over the m-disk gives the left iterated bar. Reversing the order gives the right iterated bar. Both compute the integral over the product disk. □

For higher chiral coefficients, this theorem applies in the transverse category after the coefficient functor has been fixed. It does not replace the higher collision operad.

70.3 Higher centres

The centre of an E_n-algebra is an E_{n+1}-algebra. In factorisation terms, it is the algebra of operators that may move around an n-dimensional defect.

Proposition 70.3 (Dimensional meaning of the centre)**.** *Let a boundary theory on $X \times \mathbb{R}^n$ have an E_n transverse algebra A. The derived centre $Z_{E_n}(A)$ is naturally E_{n+1} in the transverse directions. If the holomorphic coefficient topologizes to an independent E_m structure, the combined centre is E_{m+n+1} by Dunn additivity.*

Proof. The higher Deligne theorem identifies the endomorphisms of the identity E_n-module with an E_{n+1}-algebra. The independent topological coefficient structure tensors with this operation; Dunn additivity adds the dimensions. □

The dimensional count is therefore exact. A chiral coefficient that becomes topological in two real directions contributes E_2. A transverse ordered boundary contributes E_1. Taking its centre adds one further transverse direction and gives E_4 after full topologization, not E_3.

70.4 Higher Koszul duality

The E_n operad is Koszul self-dual up to suspension. For connected augmented algebras, the n-fold bar is the Koszul dual coalgebra. Completion is essential beyond finite positive-weight windows.

Theorem 70.4 (Completed E_n duality)**.** *In a stable presentable monoidal category, the E_n bar–cobar adjunction restricts to an equivalence between nilcomplete augmented E_n-algebras and conilcomplete coalgebras over the Koszul dual cooperad. On connected positive-weight objects, the equivalence is computed by finite weight windows and passes to complete Mittag–Leffler limits.*

Proof. The higher-categorical equivalence on the nilcomplete/conilcomplete locus is the general Koszul-duality theorem. In positive weight, an operation of total weight N involves finitely many bar lengths and finitely many little-disks cells. The classical finite-stage bar–cobar contraction applies. Surjective completion maps remove the derived inverse-limit term. $\square$

70.5 Higher holomorphic collision symbols

For a smooth complex d-fold, the diagonal has complex codimension d. The normal bundle of the small diagonal in X^2 is T_X. Singular kernels are distributions on the complement whose wavefront or D-module singular support lies over the conormal bundle.

The associated graded of the pole or order filtration is built from symmetric powers of the normal bundle and differential operators in d variables. In dimension one, residues and Laurent coefficients provide a one-dimensional mode index. In higher dimension, spherical harmonics or multi-indices replace that linear mode order.

Proposition 70.5 (Independence of the two ladders)**.** *There exist theories with fixed holomorphic dimension d and arbitrary transverse E_n degree, and theories with fixed transverse degree n and varying holomorphic dimension. Consequently no functor of d alone reconstructs the E_n multiplication, and no functor of n alone reconstructs the higher chiral collision operad.*

Proof. Tensor a fixed higher chiral coefficient on $\mathbb{A}^d$ with any augmented E_n-algebra to vary the transverse structure while keeping the collision coefficient fixed. Conversely, tensor a fixed transverse E_n-algebra with the higher chiral coefficient on $\mathbb{A}^d$ for varying d. The resulting objects have the required independence. $\square$

70.6 Product geometry and mixed compactifications

A local configuration in $\mathbb{A}^d \times \mathbb{R}^n$ has collision strata indexed simultaneously by holomorphic clusters and transverse little-disk clusters. A mixed compactification is indexed by forests with two edge colours. The codimension-two corners have three forms: two holomorphic contractions, two transverse contractions, and one of each.

The product orientation gives

$$d_{\mathrm{hol}}d_{\mathrm{top}} + d_{\mathrm{top}}d_{\mathrm{hol}} = 0.$$

The coefficient maps on mixed corners are the interchange law. A geometric model exists whenever the two compactifications form Cartesian squares on mixed strata and the coefficient pull-push maps satisfy Beck–Chevalley.

Criterion 70.6 (Higher rectangular realization)**.** *A factorisation theory on $\mathbb{A}^d \times \mathbb{R}^n$ realizes an E_n-algebra in higher chiral coefficients when:*

1. *its observables are algebraic or holomorphic in $\mathbb{A}^d$ and locally constant in $\mathbb{R}^n$;*
2. *product disks generate the local topology;*
3. *the factorisation maps satisfy descent on the product basis;*
4. *mixed collision squares satisfy base change;*
5. *the required completions commute with factorisation homology.*

Under these conditions, the mathematics and physics share one carrier: product configurations. The holomorphic compactification controls ultraviolet singularity. The little-disks compactification controls ordered or braided composition. Their mixed corners encode compatibility.

Chapter 71

Prime forms, plumbing, and the analytic sewing calculus

Positive genus introduces global zero modes, period matrices, determinant lines, and nodal sewing. These structures belong to the geometry of the chiral curve. A cyclic transverse algebra supplies a second, independent sewing theory. The analytic calculus begins with the prime form and ends with a bicoloured stable-graph expansion.

71.1 The prime form and normalized bidifferential

Let Σ be a compact Riemann surface of genus g, with symplectic homology basis (A_i, B_i) and normalized holomorphic one-forms ω_i satisfying

$$\oint_{A_j} \omega_i = \delta_{ij}.$$

The period matrix is

$$\Omega_{ij} = \oint_{B_j} \omega_i.$$

The period matrix Ω lies in the Siegel upper half-space $\mathfrak{H}_g$.

Choose an odd nonsingular theta characteristic δ. The prime form is

$$E(z,w) = \frac{\theta[\delta](\int_w^z \omega \mid \Omega)}{h_\delta(z)h_\delta(w)}.$$

The prime form $E(z,w)$ is a $(-1/2,-1/2)$-form, antisymmetric in z,w, with a simple zero on the diagonal. In a local coordinate,

$$E(z,w) = \frac{z-w}{\sqrt{dz}\sqrt{dw}}\left(1 + O((z-w)^2)\right).$$

The normalized bidifferential is

$$B(z,w) = d_z d_w \log E(z,w),$$

characterized by a double pole with coefficient one on the diagonal, symmetry, and vanishing A-periods in each variable.

Proposition 71.1 (Coordinate-free current propagator)**.** *For a free current algebra with pairing (,), the two-point function*

$$\langle J^a(z)J^b(w)\rangle = (a,b)B(z,w)$$

is globally defined and has local OPE coefficient $(a,b)/(z-w)^2$*. Every higher Gaussian current correlator is the sum over pairings of this bidifferential.*

Proof. The defining properties of B make it the unique normalized symmetric bidifferential with the required diagonal singularity. Wick's theorem follows from the Gaussian functional integral or, algebraically, from the fact that the state is determined by its two-point function and vanishing odd moments. □

71.2 Plumbing a node

Let Σ_1 and Σ_2 have marked coordinates x and y at points p and q. The separating plumbing family is defined by

$$xy = t, \qquad |t| \ll 1.$$

For a nonseparating node, the two punctures lie on one surface. The parameter t is a normal coordinate to the Deligne–Mumford boundary.

Let V be a graded chiral theory with invariant pairing and homogeneous basis $\{u_\alpha\}$ with dual basis $\{u^\alpha\}$. The local-coordinate sewing tensor is

$$\mathcal{P}(t) = \sum_\alpha t^{L_0} u_\alpha \otimes u^\alpha.$$

If $L_0 u_\alpha = h_\alpha u_\alpha$, then

$$\mathcal{P}(t) = \sum_\alpha t^{h_\alpha} u_\alpha \otimes u^\alpha.$$

The cylindrical vacuum factor $t^{-c/24}$ enters after a choice of cylindrical coordinate or anomaly-line trivialization.

Theorem 71.2 (Formal sewing identity)**.** *Suppose every conformal-weight subspace is finite dimensional and the pairing is nondegenerate. For any fixed total conformal weight* N*, the coefficient of* t^N *in the sewn block is a finite sum and is independent of the chosen homogeneous dual basis. The formal series satisfies factorisation on the nodal boundary.*

Proof. Only states of weight at most N contribute to the coefficient of t^N, and their total space is finite dimensional. The tensor $\sum u_\alpha \otimes u^\alpha$ is the coevaluation element and is basis independent. Inserting it contracts the two marked-point modules and gives the factorisation map by definition of sewing. □

Analytic convergence requires growth estimates. If the multiplicity d_h of weight h satisfies

$$d_h \le Ce^{\alpha\sqrt{h}}$$

and matrix coefficients grow at most exponentially in $\sqrt{h}$, then the sewn series converges for sufficiently small $|t|$.

Proposition 71.3 (A sufficient sewing bound)**.** *Assume*

$$|\langle \varphi_1, u\rangle\langle u^\vee, \varphi_2\rangle| \leq C_1 e^{\beta\sqrt{h}}$$

for states of weight h, and $d_h \leq C_2 e^{\alpha\sqrt{h}}$. Then

$$\sum_h d_h C_1 e^{\beta\sqrt{h}} |t|^h$$

converges for every $|t| < 1$.

Proof. For any $0 < r < 1$, the exponential decay r^h dominates $e^{(\alpha+\beta)\sqrt{h}}$. The root test gives radius one. ☐

71.3 Degeneration of the period matrix

For separating plumbing,

$$\Omega(t) = \begin{pmatrix} \Omega_1 & 0 \\ 0 & \Omega_2 \end{pmatrix} + O(t).$$

For nonseparating plumbing,

$$\Omega(t) = \begin{pmatrix} \Omega_0 & u \\ u^T & \frac{1}{2\pi i}\log t + c \end{pmatrix} + O(t).$$

The logarithm records the long cylinder created by the neck.

A genus-g Siegel theta series of an even lattice L is

$$\Theta_L^{(g)}(\Omega) = \sum_{\lambda \in L^g} \exp\left(\pi i \operatorname{tr}(G(\lambda)\Omega)\right),$$

where $G(\lambda)_{ij} = (\lambda_i, \lambda_j)$.

Proposition 71.4 (Separating theta factorisation)**.** *At a separating boundary,*

$$\Theta_L^{(g_1+g_2)}\left(\begin{smallmatrix} \Omega_1 & 0 \\ 0 & \Omega_2 \end{smallmatrix}\right) = \Theta_L^{(g_1)}(\Omega_1)\Theta_L^{(g_2)}(\Omega_2).$$

Proof. A vector in $L^{g_1+g_2}$ decomposes uniquely into a vector in L^{g_1} and one in L^{g_2}, and the quadratic exponent splits additively. ☐

External theorem 71.5 (Rank-sixteen theta comparison)**.** *The even unimodular lattices $E_8 \oplus E_8$ and D_{16}^+ have equal Siegel theta series in genera $g \leq 3$. Their degree-four difference is proportional to the Schottky form and vanishes on the genus-four Jacobian locus. [102, 109]*

The theorem is a precise example of the information gained and lost by genus. Genus two does not separate these lattices. Degree four detects a Siegel modular difference, while restriction to Jacobians imposes a further geometric relation.

71.4 Conformal blocks and projective flatness

Let $\mathbb{V}_{g,n}(V; M_1, \dots, M_n)$ be the bundle of conformal blocks for a strongly rational vertex operator algebra. Factorisation across a node is

$$\mathbb{V}_{g,n}\big|_{\Delta} \cong \bigoplus_{P \in \mathrm{Irr}(V)} \mathbb{V}_{g_1,n_1+1}(\dots, P) \otimes \mathbb{V}_{g_2,n_2+1}(\dots, P^{\vee})$$

for a separating node, with the analogous self-sewing formula for a nonseparating node.

External theorem 71.6 (Modular functor from conformal blocks)**.** *For a strongly rational vertex operator algebra, the conformal blocks form a modular functor. The module category acquires the resulting modular fusion structure, and the modular functor extends to a three-dimensional topological field theory with a factorisation-homological description. [17]*

This theorem supplies the coefficient maps for Deligne–Mumford sewing in an important class of theories. The graph topology becomes an algebraic operation because the conformal blocks provide the contraction maps.

71.5 The transverse ribbon theory

Let B be a smooth proper cyclic dg algebra with trace

$$\tau : B \to \mathbb{C}[-d]$$

and nondegenerate pairing $\langle a, b \rangle = \tau(ab)$. The cyclic bar complex assigns amplitudes to ribbon graphs. Contracting an internal edge uses the inverse pairing. A connected ribbon graph G determines a surface of genus h with b boundary components, and its amplitude is

$$W_B(G) = \mathrm{Contr}_G(m_B, \langle\ ,\ \rangle^{-1}).$$

The graph differential contracts edges and the master equation follows from the oriented boundary of the ribbon-cell compactification.

71.6 A constructed two-genus theory

Let $\mathcal{V}$ be a modular chiral theory and B a cyclic dg algebra. Define the bidegree coefficient system

$$\mathcal{E}_{g,h;n,b} = \mathcal{V}_{g,n} \boxtimes C_B(h, b).$$

Its parameter space is the product

$$\overline{\mathcal{M}}_{g,n} imes \mathcal{R}_{h,b},$$

where $\mathcal{R}_{h,b}$ is a ribbon-graph compactification.

Theorem 71.7 (Strict two-genus realization)**.** *The external tensor-product theory has two commuting sewing actions. Mixed codimension-two strata are Cartesian products of one Deligne–Mumford boundary and one ribbon boundary. The total master element*

$$\Theta = \Theta_X \boxtimes 1 + 1 \boxtimes \Theta_{\perp}$$

satisfies

$$d\Theta + \frac{1}{2}[\Theta,\Theta]_X + \frac{1}{2}[\Theta,\Theta]_\perp + \hbar_X \Delta_X \Theta + \hbar_\perp \Delta_\perp \Theta = 0.$$

The amplitudes factor:

$$\mathcal{F}_{g,h,b}(\mathcal{V} \boxtimes B) = \mathcal{F}_g^X(\mathcal{V})\, \mathcal{F}_{h,b}^\perp(B).$$

Proof. Each sewing operator acts on one tensor factor. Operators on distinct factors supercommute with the product orientation. Pure X-corners give the chiral modular equation; pure transverse corners give the ribbon master equation; mixed corners cancel in opposite orders. The tensor-product contraction of a product graph is the product of the two contractions. □

For $B = \mathrm{Mat}_N(\mathbb{C})$ with the double-line normalization,

$$\mathcal{F}_{h,b}^\perp(B) = N^{2-2h-b}.$$

For a rank-one Deligne–Mumford theory with amplitude λ^{1-g},

$$\mathcal{F}_{g,h,b} = \lambda^{1-g} N^{2-2h-b}.$$

The two exponents vary independently, proving independence of the two genus variables in a complete worked model.

71.7 Stable graphs and connected cumulants

Let Γ range over stable bicoloured graphs. The partition function is

$$Z = 1 + \sum_\Gamma \frac{\hbar_X^{g_X(\Gamma)} \hbar_\perp^{g_\perp(\Gamma)}}{|\operatorname{Aut}\Gamma|} W(\Gamma).$$

The connected potential is

$$\mathcal{F} = \log Z.$$

The logarithm removes disjoint unions and retains connected graphs.

Proposition 71.8 (Connected graph exponential)**.** *If the graph weight is multiplicative on disjoint unions,*

$$W(\Gamma_1 \sqcup \Gamma_2) = W(\Gamma_1) W(\Gamma_2),$$

then $Z = \exp(\mathcal{F})$ with $\mathcal{F}$ the sum over connected graphs.

Proof. Every graph is a finite set of connected graphs. The automorphism factor decomposes into automorphisms of components and factorials of repeated components. The exponential formula for labelled combinatorial species gives the identity. □

The coefficients of $\mathcal{F}$ are canonical graph cumulants after the trace and sewing data have been specified. They replace scalar restrictions to arbitrary primary lines.

Chapter 72

Integrability, exchange, and gauge-theoretic line operators

Exchange is motion in a plane. A flat meromorphic connection turns that motion into braid transport. Tannaka reconstruction turns a rigid tensor category of transported lines into a quantum group. Transfer matrices then convert the categorical exchange into commuting Hamiltonians.

72.1 The rational classical r-matrix

Let $\mathfrak{g}$ be a reductive Lie algebra with invariant tensor

$$\Omega = \sum_a t_a \otimes t^a.$$

The rational classical r-matrix is

$$r(u) = \frac{\Omega}{u}.$$

The rational tensor $r(u)$ satisfies the spectral classical Yang–Baxter equation

$$[r_{12}(u), r_{13}(u+v)] + [r_{12}(u), r_{23}(v)] + [r_{13}(u+v), r_{23}(v)] = 0.$$

Proof. Multiply by $uv(u+v)$. Invariance of Ω gives

$$[\Omega_{12}, \Omega_{13} + \Omega_{23}] = 0, \quad [\Omega_{13}, \Omega_{12} + \Omega_{23}] = 0.$$

The coefficients of u and v cancel by these identities. □

72.2 KZ flatness

On $\mathrm{Conf}_n(\mathbb{C})$, define

$$\omega_{ij} = d\log(z_i - z_j).$$

The KZ connection on $V_1 \otimes \cdots \otimes V_n$ is

$$\nabla_{\mathrm{KZ}} = d - \kappa \sum_{i<j} \Omega_{ij}\omega_{ij}, \qquad \kappa = (k + h^\vee)^{-1}.$$

Theorem 72.1 (KZ flatness). *The KZ connection is flat.*

Proof. Since $d\omega_{ij} = 0$, the curvature is the wedge square. Terms with four distinct indices commute. For each triple i, j, k, the coefficient is a sum of commutators of $\Omega_{ij}, \Omega_{ik}, \Omega_{jk}$ multiplied by

$$\omega_{ij} \wedge \omega_{ik}, \quad \omega_{ij} \wedge \omega_{jk}, \quad \omega_{ik} \wedge \omega_{jk}.$$

The infinitesimal braid relations identify the commutators, and Arnold's relation makes the form coefficient vanish. □

Parallel transport gives a braid-group representation. The associator is the regularized holonomy between different asymptotic parenthesizations of a three-point configuration.

72.3 Gaudin Hamiltonians

Define

$$H_i = \sum_{j \neq i} \frac{\Omega_{ij}}{z_i - z_j}.$$

The KZ equation is

$$(\partial_{z_i} - \kappa H_i)\, \Psi = 0.$$

Proposition 72.2 (Gaudin commutativity). *For distinct i, j,*

$$[H_i, H_j] = 0.$$

Proof. Expand the commutator. Terms with disjoint index sets vanish. The remaining terms are grouped by triples i, j, k. Their rational coefficients satisfy the partial-fractions identity

$$\frac{1}{(z_i - z_j)(z_i - z_k)} + \frac{1}{(z_j - z_k)(z_j - z_i)} + \frac{1}{(z_k - z_i)(z_k - z_j)} = 0,$$

and their algebraic coefficients cancel by the infinitesimal braid relations. □

72.4 The finite $\mathfrak{sl}_2$ Bethe calculation

Let V_m be highest-weight $\mathfrak{sl}_2$ modules at positions z_i, with highest vector v_0. For Bethe roots $w_1, \ldots, w_M$, set

$$B(w) = \sum_i \frac{f^{(i)}}{w - z_i}, \qquad \Psi(w_1, \ldots, w_M) = B(w_1) \cdots B(w_M) v_0.$$

The Bethe equations are

$$\sum_i \frac{m_i}{w_a - z_i} - 2 \sum_{b \neq a} \frac{1}{w_a - w_b} = 0.$$

Proposition 72.3 (Off-shell Gaudin identity). *The action of H_i on the Bethe vector is the sum of an eigenvalue term and unwanted vectors whose coefficients are the Bethe equations. On a solution, Ψ is a joint eigenvector.*

Proof. Commute H_i through the ordered product of $B(w_a)$. The relation

$$[e^{(i)}, B(w)] = \frac{h^{(i)}}{w - z_i}$$

and the analogous commutators reduce the result to the original Bethe vector plus vectors in which one $B(w_a)$ is replaced by $f^{(i)}$. Summing all contractions gives the displayed Bethe coefficient. Its vanishing removes every unwanted vector. □

72.5 RTT and transfer matrices

Let

$$R(u) = 1 - \frac{\hbar P}{u}$$

on $\mathbb{C}^N \otimes \mathbb{C}^N$. It satisfies the quantum Yang–Baxter equation and unitarity up to the scalar factor

$$R_{12}(u)R_{21}(-u) = 1 - \frac{\hbar^2}{u^2}.$$

The monodromy matrix $T(u)$ satisfies

$$R_{12}(u - v)T_1(u)T_2(v) = T_2(v)T_1(u)R_{12}(u - v).$$

Theorem 72.4 (Commuting transfer matrices)**.** *Let K commute with $R(u)$ in the auxiliary tensor square. Then*

$$t(u) = \mathrm{tr}_{\mathrm{aux}}(KT(u))$$

satisfies

$$[t(u), t(v)] = 0.$$

Proof. Write $t(u)t(v)$ as a trace over two auxiliary spaces. Insert $R_{12}(u - v)^{-1}R_{12}(u - v)$, use RTT to interchange $T_1(u)$ and $T_2(v)$, move R through K_1K_2, and use cyclicity of the auxiliary trace. The result is $t(v)t(u)$. □

The coefficients of $t(u)$ generate commuting quantum Hamiltonians. Their classical limit recovers the spectral invariants of the Gaudin/Hitchin system.

72.6 Meromorphic Tannaka reconstruction

Let $\mathcal{L}$ be a rigid meromorphic tensor category of line defects and

$$F : \mathcal{L} \to \mathsf{Vect}_{\mathrm{fd}}$$

a faithful exact tensor functor. The coend

$$\mathcal{O}(\mathcal{H}) = \int^{L \in \mathcal{L}} F(L)^\vee \otimes F(L)$$

is the coordinate coalgebra of matrix coefficients. Tensor product gives multiplication. Braiding gives a coquasitriangular form.

Theorem 72.5 (Exact Tannaka output)**.** *Under the hypotheses above, the coend is a coquasi-triangular bialgebra. Rigidity supplies an antipode. If the fibre functor is quasi-monoidal, the coend is a coquasi-bialgebra with coassociator equal to the tensor constraint. A named Yangian or quantum loop algebra is obtained only after a faithful presentation theorem identifies the reconstructed bialgebra.*

Proof. The universal dinatural matrix coefficient maps define the coalgebra. Tensor product induces multiplication by multiplying matrix coefficients. The hexagon identities give coquasitriangularity. Dual objects define the antipode. For a quasi-monoidal functor, the pentagon defect is the coassociator. The final statement is logical: a presentation is an additional isomorphism from a specified algebra to the universal coend. □

72.7 Yangian recognition

Let $Y_\hbar(\mathfrak{gl}_N)$ be the completed RTT algebra. Suppose a line category is tensor-generated by evaluation modules, RTT acts on all tensor products, PBW holds, and finite tensor products separate points.

Recognition theorem 72.6 (RTT Yangian)**.** *Under these hypotheses, the Tannaka coordinate bialgebra is the restricted dual of the completed RTT Yangian, and its continuous dual is $Y_\hbar(\mathfrak{gl}_N)$.*

Proof. RTT matrix coefficients give a surjective map from the coordinate algebra generated by evaluations to the coend. Tensor generation proves surjectivity on all objects. Separation of points proves injectivity of the continuous dual map. PBW identifies the completed topology and excludes hidden relations. □

72.8 Four-dimensional Chern–Simons theory

Let $M = \Sigma \times C$, with Σ an oriented real surface and C a complex curve carrying a meromorphic one-form ω. The action is

$$S(A) = \frac{1}{2\pi\hbar} \int_{\Sigma\times C} \omega \wedge \mathrm{Tr}\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A\right).$$

The theory is topological on Σ and holomorphic on C. Wilson lines supported on curves in Σ and located at points of C carry a spectral parameter. Crossing two lines produces an R-matrix.

External theorem 72.7 (CWY rational line algebra)**.** *For rational four-dimensional Chern–Simons theory, perturbative quantization of Wilson lines produces the Yangian deformation of the polynomial current algebra, and line crossing produces the rational R-matrix. [15, 16]*

The geometric chain is

$$\text{topological line fusion} \longrightarrow \text{meromorphic exchange} \longrightarrow \text{RTT relations},$$
$$\text{RTT relations} \longrightarrow \text{Yangian recognition}.$$

Every arrow has an independent hypothesis. Fusion exists on the ordered line. Exchange uses the plane. RTT uses the crossing calculation. Recognition uses PBW and faithful evaluation modules.

72.9 Koszul duality of line couplings

A line defect coupled to a bulk field theory is represented by a Maurer–Cartan element in a deformation complex

$$\mathfrak{g}_{\mathrm{line}} = \mathrm{Def}(A_{\mathrm{bulk}}, A_{\mathrm{line}}).$$

Bar duality replaces the line algebra by a coalgebra and the coupling by a twisting cochain. Under a perfect Koszul pairing, duality transposes the monodromy representation. Verdier duality sends a local system to its dual, so braid matrices transform by inverse transpose.

Proposition 72.8 (Dual monodromy). *For a finite-rank flat local system with monodromy $\rho(\gamma)$, the Verdier-dual local system has monodromy*

$$\rho^{\vee}(\gamma) = \rho(\gamma)^{-T}.$$

If a nondegenerate invariant pairing identifies the representation with its dual, this becomes inverse monodromy.

Proof. Parallel transport on the dual bundle is defined by preservation of the evaluation pairing. If v is transported by M and ϕ by N, invariance requires $N^T M = 1$, hence $N = M^{-T}$. □

The proposition is the exact geometric inversion supplied by duality. Identification with a parameter transformation such as $q \mapsto q^{-1}$ requires a presentation in which the monodromy matrix depends faithfully on q.

Chapter 73

Calabi–Yau sources, Hall multiplication, and Coulomb branches

A higher-dimensional source produces a chiral theory only through a geometric pushforward or a protected-sector functor. Cyclic Calabi–Yau data supplies traces and loop contractions. Hall correspondences supply ordered multiplication. Coulomb branches supply quantized convolution algebras. The comparison maps among these structures form the geometric source of the chiral theory.

73.1 Cyclic categories and traces

Let $\mathcal{C}$ be a smooth proper dg category with a d-Calabi–Yau structure. The structure is a nondegenerate class

$$\theta \in HC^-_d(\mathcal{C})$$

whose image induces pairings

$$\mathrm{RHom}_{\mathcal{C}}(E,F) tensor\, \mathrm{RHom}_{\mathcal{C}}(F,E) \longrightarrow \mathbb{C}[-d].$$

The pairing contracts two flags of a graph and therefore supplies the loop operation in a cyclic or modular envelope.

Proposition 73.1 (Cyclic edge contraction)**.** *Let m_k be the cyclic A_∞ operations of a finite-dimensional model of $\mathcal{C}$. Decorating vertices of a ribbon graph by the m_k and contracting internal edges by the inverse Calabi–Yau pairing produces a graph weight invariant under isomorphism of ribbon graphs. The sum over codimension-one edge contractions satisfies the classical master equation.*

Proof. Cyclicity makes the vertex weight independent of the chosen outgoing flag. Nondegeneracy defines edge contraction. The A_∞ relations are precisely the oriented sum over ways of contracting one internal edge in a tree. The ribbon version follows by retaining the cyclic order at each vertex. □

73.2 Factorisation pushforward

Let

$$f : M \longrightarrow X$$

be a map from a manifold or derived stack carrying a factorisation theory $\mathcal{F}$ to a complex curve. Define

$$(f_*\mathcal{F})(U) = \mathcal{F}(f^{-1}U).$$

Theorem 73.2 (Factorisation pushforward)**.** *Suppose preimages of disjoint opens are disjoint, the factorisation maps of $\mathcal{F}$ are compatible with the chosen topology, and f_* preserves the required homotopy colimits. Then $f_*\mathcal{F}$ is a factorisation algebra on X. If $\mathcal{F}$ is locally constant along the fibres and holomorphic transversely to them, the pushforward is a chiral or holomorphic factorisation coefficient on X.*

Proof. For disjoint $U_i \subset V$, the preimages $f^{-1}U_i$ are disjoint inside $f^{-1}V$, so the factorisation map of $\mathcal{F}$ gives the multiplication of $f_*\mathcal{F}$. Associativity and symmetry are inherited. Descent follows from preservation of the factorisation homology colimits appearing in the displayed pushforward. The final assertion is local in product charts. □

A cycle Σ in a Calabi–Yau space and a projection to a curve therefore define a chiral shadow only after an actual map and a factorisation pushforward have been specified.

73.3 The source-to-chiral deformation map

Let $\mathrm{Def}_{\mathrm{cyc}}(\mathcal{C})$ be the cyclic deformation complex of the Calabi–Yau category, and let $\mathfrak{g}_A^{\mathrm{mod}}$ be the modular deformation complex of the chiral image $A = f_*\mathcal{F}_{\mathcal{C}}$.

Criterion 73.3 (Cyclic-to-chiral comparison)**.** *A source-to-chiral map is a filtered L_∞ morphism*

$$\Phi_{\Sigma,X} : \mathrm{Def}_{\mathrm{cyc}}(\mathcal{C}) \longrightarrow \mathfrak{g}_A^{\mathrm{mod}}$$

with the following properties:

1. *cyclic multiplications map to local collision operations;*
2. *the Connes operator maps to the graph-loop contraction;*
3. *the Calabi–Yau trace maps to the chiral sewing trace;*
4. *source filtrations map to pole, weight, and graph filtrations;*
5. *Maurer–Cartan elements map to Maurer–Cartan elements.*

The criterion states the exact theorem shape for K3, BPS, and CoHA applications. A scalar equality is the image of such a map under a trace; it cannot replace the map.

73.4 Hall multiplication

Let $\mathcal{A}$ be a finitary abelian category over $\mathbb{F}_q$. The Hall algebra has basis $[M]$ indexed by isomorphism classes and product

$$[M] * [N] = \sum_{[E]} g^E_{MN}[E],$$

where g^E_{MN} counts subobjects $N' \subset E$ with $N' \cong N$ and $E/N' \cong M$.

Theorem 73.4 (Associativity of Hall multiplication)**.** *The Hall product is associative.*

Proof. The coefficient of $[E]$ in $([L] * [M]) * [N]$ counts flags

$$0 \subset N' \subset K \subset E$$

with successive quotients N, M, L. The coefficient in $[L] * ([M] * [N])$ counts the same flags grouped by the quotient K/N'. The identity map on the flag set gives equality. □

For the quiver $1 \to 2$, let S_1, S_2 be the simple representations and M the indecomposable of dimension vector $(1,1)$. Direct counting gives

$$[S_1] * [S_2] = [S_1 \oplus S_2] + [M], \qquad [S_2] * [S_1] = [S_1 \oplus S_2].$$

After the Euler-form twist, the generators satisfy the two A_2 quantum Serre relations.

Calculation 73.5 (A_2 Hall–Serre relation)**.** With $v^2 = q$ and the standard twisted Hall generators E_1, E_2,

$$E_1^2 E_2 - (v + v^{-1}) E_1 E_2 E_1 + E_2 E_1^2 = 0,$$
$$E_2^2 E_1 - (v + v^{-1}) E_2 E_1 E_2 + E_1 E_2^2 = 0.$$

Proof. Classify representations of dimension vectors $(2,1)$ and $(1,2)$ by the rank of the arrow. Count two-step flags with successive simple quotients in the three possible orders. The counts are Gaussian binomial coefficients $q+1$ and 1. Multiplication by the Euler-form twist converts $q+1$ into $v(v+v^{-1})$, yielding the displayed cancellations. □

The Hall product is ordered: the extension direction distinguishes $[S_1] * [S_2]$ from $[S_2] * [S_1]$. This is a concrete source of an E_1 multiplication.

73.5 Cohomological Hall algebras

For a derived moduli stack $\mathfrak{M}$ of objects in a Calabi–Yau category, let

$$\mathfrak{E} \xrightarrow{(p_1,p_2)} \mathfrak{M} \times \mathfrak{M}, \qquad \mathfrak{E} \xrightarrow{p_3} \mathfrak{M}$$

be the stack of exact triangles or extensions. The CoHA product is the pull-push

$$\alpha * \beta = (p_3)_! \big(p_1^* \alpha \cup p_2^* \beta \cup e(\mathcal{N}_{\mathrm{vir}})\big).$$

Proposition 73.6 (Associativity of CoHA correspondences)**.** *If the extension correspondences are proper in the pushforward direction and the two Cartesian squares of one-step and two-step extension stacks satisfy derived proper base change, the CoHA product is associative.*

Proof. Both triple products are pull-push operations over the derived stack of two-step filtrations. One factors the filtration by its first extension and the other by its second. Base change identifies the two composites, and multiplicativity of the virtual Euler class identifies their coefficients. □

73.6 Coulomb branches

Let G be a complex reductive group and N a representation. The Braverman–Finkelberg–Nakajima construction defines a convolution algebra from an equivariant Borel–Moore homology correspondence in the affine Grassmannian.

External theorem 73.7 (BFN Coulomb branch)**.** *The BFN convolution algebra is commutative in its classical form, and its spectrum defines the Coulomb branch of the three-dimensional $\mathcal{N} = 4$ gauge theory. Equivariant loop rotation produces a filtered quantization. [10]*

In quiver gauge theories, the quantized Coulomb branch is often recognized as a shifted Yangian or a truncated shifted Yangian. The recognition requires an explicit map from Yangian generators, PBW comparison, and equality of defining relations or faithful representations.

Recognition theorem 73.8 (Shifted Yangian from a Coulomb branch)**.** *Suppose a filtered homomorphism*

$$Y_\mu^\lambda \longrightarrow \mathcal{A}_{\mathrm{Coul}}$$

from a shifted Yangian to a quantized Coulomb branch is surjective, induces an isomorphism on associated graded coordinate rings, and both filtrations are separated and complete. Then the homomorphism is an isomorphism.

Proof. An associated-graded isomorphism makes the kernel and cokernel have zero associated graded. Separatedness kills the kernel and completeness plus exhaustiveness kills the cokernel. □

73.7 Bar Euler characters and denominator products

Let

$$\mathfrak{n}_+ = \bigoplus_{\alpha>0} \mathfrak{g}_\alpha$$

be a positively root-graded Lie superalgebra with finite-dimensional root spaces. Its enveloping algebra has Chevalley bar complex

$$\operatorname{Bar} U(\mathfrak{n}_+) \simeq C_*^{\mathrm{CE}}(\mathfrak{n}_+).$$

Theorem 73.9 (Denominator Euler character)**.** *The root-graded Euler supercharacter is*

$$\chi_\Gamma \operatorname{Bar} U(\mathfrak{n}_+) = \prod_{\alpha>0} (1 - e^{-\alpha})^{\operatorname{sdim} \mathfrak{g}_\alpha}.$$

Proof. As a graded vector space, Chevalley chains are the super exterior algebra $\Lambda^\bullet \mathfrak{n}_+$. Euler character is invariant under the differential. An even root vector contributes $1 - e^{-\alpha}$; an odd root vector contributes its inverse. Multiplication over root spaces gives the formula. □

For a Borcherds–Kac–Moody algebra whose root multiplicities are Fourier coefficients of a Jacobi form, the product becomes an automorphic denominator. The Igusa product is one such typed identity after the root lattice, Weyl vector, and multiplicities have been identified.

73.8 A K3 realization criterion

A K3 surface provides a Mukai lattice, moduli spaces of sheaves, Hall or CoHA products, and automorphic forms. These objects enter chiral Koszul duality through a chain of comparisons.

Criterion 73.10 (K3-to-chiral realization)**.** *A geometric realization of a Borcherds denominator as the bar Euler character of a chiral theory requires:*

1. *a Calabi–Yau category attached to the K3 surface;*
2. *a Hall or CoHA algebra with identified positive-root grading;*
3. *a factorisation or protected-sector functor to a curve;*
4. *an identification of the resulting ordered chiral enveloping algebra;*
5. *equality of root multiplicities with the Fourier coefficients entering the automorphic product;*
6. *compatibility of traces so that the bar Euler character maps to the automorphic character.*

When these conditions hold, the automorphic product is the trace of a constructed bar object. Matching the product without the functor establishes an arithmetic identity and leaves the geometric realization open.

Chapter 74

Chiral quantization and the mixed obstruction complex

A classical chiral algebra is a Poisson vertex object. A quantization is a filtered chiral operation whose associated graded is the classical one. The deformation problem is Maurer–Cartan theory in the filtered chiral deformation complex. A doubly noncommutative theory adds transverse Hochschild deformations and a mixed compatibility obstruction.

74.1 The chiral and classical operads

Let V be a translation module. The chiral operad $\mathcal{P}^{\mathrm{ch}}(V)$ is filtered by the number and order of diagonal poles. The classical operad $\mathcal{P}^{\mathrm{cl}}(V)$ is indexed by directed graphs and records the normal symbols of those poles.

External theorem 74.1 (Chiral–classical operad comparison)**.** *The associated graded of the filtered chiral operad is canonically isomorphic to the classical operad:*

$$\operatorname{gr}\mathcal{P}^{\mathrm{ch}}(V) \cong \mathcal{P}^{\mathrm{cl}}(V).$$

An odd Maurer–Cartan element in the chiral operad is a vertex algebra structure; an odd Maurer–Cartan element in the classical operad is a Poisson vertex algebra structure. [3]

The theorem makes quantization a lifting problem through a complete filtration.

74.2 Order-by-order lifting

Let $(\mathfrak{g}, F^{\bullet})$ be a complete filtered dg Lie algebra with

$$[F^p\mathfrak{g}, F^q\mathfrak{g}] \subset F^{p+q}\mathfrak{g}.$$

Let

$$\alpha_m \in (\mathfrak{g}/F^{m+1}\mathfrak{g})^1$$

solve the Maurer–Cartan equation modulo F^{m+1}.

Theorem 74.2 (Filtered obstruction theorem)**.** *The curvature*

$$\mathcal{F}(\alpha_m) = d\alpha_m + \frac{1}{2}[\alpha_m, \alpha_m]$$

defines a cocycle

$$o_{m+1}(\alpha_m) \in H^2\big(F^{m+1}\mathfrak{g}/F^{m+2}\mathfrak{g}, d_{\alpha_0}\big).$$

The class vanishes exactly when α_m lifts to a Maurer–Cartan element modulo F^{m+2}. When a lift exists, its gauge-equivalence classes form a torsor for

$$H^1\big(F^{m+1}\mathfrak{g}/F^{m+2}\mathfrak{g}, d_{\alpha_0}\big).$$

Infinitesimal automorphisms lie in H^0.

Proof. The Bianchi identity

$$d_{\alpha_m}\mathcal{F}(\alpha_m) = 0$$

shows that the leading filtration component of the curvature is a cocycle. Replacing α_m by $\alpha_m + \beta$, with $\beta \in F^{m+1}\mathfrak{g}^1/F^{m+2}$, changes the leading curvature by $d_{\alpha_0}\beta$. Thus a lift exists exactly when the class vanishes. Two lifts differ by a degree-one cocycle, and a filtration-$m+1$ gauge transformation changes that cocycle by a coboundary. □

74.3 Chiral Poisson cohomology

Let R be a vertex Poisson algebra and let X_0 be its classical Maurer–Cartan element. The controlling complex is

$$\big(W^{\mathrm{cl}}(\Pi R), [X_0, -]\big).$$

A chiral quantization is a formal series

$$X_\hbar = X_0 + \hbar X_1 + \hbar^2 X_2 + \cdots$$

in the completed chiral operad satisfying

$$[X_\hbar, X_\hbar] = 0.$$

External theorem 74.3 (Deformation theory of chiral quantizations)**.** *The obstruction to extending a chiral quantization through the next order lies in chiral Poisson cohomology $H^2_c(R)$, and the space of extensions is a torsor for $H^1_c(R)$. For arc algebras of affine symplectic varieties, the controlling groups are identified with shifted de Rham cohomology. Boundary Virasoro minimal models are rigid under the deformations considered in that theory. [11]*

The theorem supplies a genuine tangent-obstruction theory. A classical symbol does not determine its quantization when H^1_c or H^2_c is nonzero.

74.4 Collision Rees quantization

Let A be a filtered chiral algebra with pole filtration P. Its Rees object

$$\mathcal{R}_P(A) = \bigoplus_{m\geq 0} P_m A\,\hbar^m$$

is a family over the affine line. The generic fibre is the quantum chiral operation and the special fibre is the Poisson vertex symbol.

Proposition 74.4 (Geometric first-order bracket)**.** *Suppose the Rees multiplication is flat. For lifts a, b of classical elements $\bar{a}, \bar{b}$, the formula*

$$\{\bar{a}, \bar{b}\} = \hbar^{-1}[a, b]\big|_{\hbar=0}$$

is well defined and gives the classical chiral bracket. Its Jacobi identity is the leading term of the quantum Maurer–Cartan equation.

Proof. Flatness implies that a commutator vanishing at $\hbar = 0$ is divisible by $\hbar$. Changing a lift changes the quotient by a term vanishing at $\hbar = 0$. Associativity gives the commutator Jacobi identity; division by the lowest power of $\hbar$ and specialization give the classical Jacobi identity. □

74.5 The transverse deformation complex

Let A_0 be an associative algebra object in the chiral coefficient category. Transverse associative deformations are controlled by shifted Hochschild cochains

$$\mathfrak{g}_\perp(A_0) = C^{\bullet+1}_{\mathrm{Hoch}}(A_0, A_0)$$

with the Gerstenhaber bracket. Chiral deformations are controlled by

$$\mathfrak{g}_X(A_0) = W^{\mathrm{ch}}(\Pi A_0).$$

A mixed structure is governed by the homotopy fibre of the two restriction maps from the deformation complex of the mixed operad.

Theorem 74.5 (Mixed obstruction complex)**.** *Let $\mathfrak{g}_{\mathrm{mix}}(A_0)$ be the convolution dg Lie algebra of the cofibrant mixed operad. Restriction to the two colours defines*

$$\rho : \mathfrak{g}_{\mathrm{mix}}(A_0) \longrightarrow \mathfrak{g}_\perp(A_0) \oplus \mathfrak{g}_X(A_0).$$

Fix Maurer–Cartan elements $m_\perp$ and m_X. The space of compatible mixed lifts is the Maurer–Cartan space of the twisted homotopy fibre

$$\mathfrak{f}_A = \mathrm{hofib}_{(m_\perp, m_X)}(\rho).$$

Its H^2 contains the obstruction to a mixed law, H^1 acts transitively on lifts when the obstruction vanishes, and H^0 is the infinitesimal symmetry algebra.

Proof. A mixed algebra is a Maurer–Cartan element whose restrictions are the two prescribed structures. Homotopy fibres compute derived fibres of Maurer–Cartan functors for complete pronilpotent dg Lie algebras. The tangent-obstruction statement is the filtered obstruction theorem applied to the twisted fibre. □

The associated long exact sequence begins

$$H^1(\mathfrak{g}_{\mathrm{mix}}) \to H^1(\mathfrak{g}_\perp) \oplus H^1(\mathfrak{g}_X) \to H^2(\mathfrak{f}_A) \to H^2(\mathfrak{g}_{\mathrm{mix}}).$$

The connecting map measures the failure of independently chosen transverse and chiral deformations to satisfy the mixed relation.

Corollary 74.6 (Exact compatibility obstruction). *If $H^2(\mathfrak{f}_A) = 0$, every pair of infinitesimal transverse and chiral deformations in the image of $H^1(\mathfrak{g}_{\mathrm{mix}})$ extends to first order as a mixed deformation. If in addition $H^1(\mathfrak{f}_A) = 0$, the extension is unique up to gauge.*

Proof. The first assertion is the vanishing of the connecting obstruction. The second is the torsor statement in the homotopy fibre. □

74.6 Curved PBW quantization

Let

$$A_\hbar = \mathbb{C}\langle x, y\rangle/(yx - xy - \hbar).$$

For $\hbar \neq 0$, the map $x, y \mapsto 0$ is not an augmentation. The correct dual is curved. The quadratic symbol at $\hbar = 0$ is $\mathbb{C}[x, y]$, and the first-order bracket is $\{y, x\} = 1$.

The curved Koszul dual has exterior generators ξ_x, ξ_y and curvature

$$m_0 = \hbar\,\xi_x\xi_y.$$

The curved differential satisfies

$$d^2 = [m_0, -].$$

Relative and Rees formulations recover the bar object from the natural base algebra and filtration of the Weyl system.

74.7 Three exact deformation families

74.7.1 Affine currents

For a fixed Lie algebra $\mathfrak{g}$, the level k multiplies the invariant double-pole cocycle. The simple-pole bracket remains the Lie bracket. Varying k is an unobstructed one-parameter central deformation because the invariant form is a closed two-cocycle.

74.7.2 The beta–gamma system

A beta–gamma system quantizes the cotangent arc space. Coordinate changes act through the chiral differential-operator gerbe. The obstruction to global gluing is a characteristic class; choosing a trivialization produces a sheaf of chiral differential operators. The local free-field quantization and the global descent problem occupy different cohomological degrees.

74.7.3 Boundary Virasoro models

The rigidity theorem for boundary Virasoro minimal models states that the chiral Poisson group H^1_c controlling first-order deformations vanishes. The absence of infinitesimal deformations is a cohomological result, not a consequence of central charge alone.

74.8 BV quantization and the chiral obstruction map

Let a holomorphic–topological BV theory have local deformation complex $\mathrm{Def}_{\mathrm{BV}}$. Restriction to a protected boundary sector gives an L_∞ morphism

$$\Phi_\partial : \mathrm{Def}_{\mathrm{BV}}\, o\mathfrak{g}_{\mathrm{mix}}(A).$$

The quantum master obstruction maps to the chiral and mixed obstruction classes.

Proposition 74.7 (Functoriality of anomalies)**.** *An L_∞ morphism of complete deformation complexes sends Maurer–Cartan obstructions to Maurer–Cartan obstructions. Hence a vanishing BV anomaly gives a vanishing obstruction in every protected chiral realization. The converse requires the protected-sector map to detect the source cohomology class.*

Proof. An L_∞ morphism intertwines the Maurer–Cartan curvature series. Passing to the first nonzero filtration component sends the source obstruction cocycle to the target obstruction cocycle. Detection is precisely injectivity on the displayed obstruction class in H^2. □

Quantization is therefore one deformation problem seen through several faithful realizations. The classical symbol, the chiral lift, the transverse product, and the BV action are connected by morphisms of deformation complexes rather than by a universal scalar.

Part XI

Exact boundaries and the functorial atlas

Chapter 75

The constructive theory and its exact boundaries

A universal construction is characterized by its output and by the hypotheses that make the output functorial. Each constructive theorem is therefore paired with an exact boundary theorem. The pair states both the carrier that produces a structure and the data required to compare it with another carrier.

75.1 Four families of exact boundaries

The boundary theorems fall into four mathematical families.

1. *Carrier boundaries.* Diagonal residues, screen cohomology, interval composition, planar exchange, cyclic trace, and stable-curve sewing belong to distinct chain complexes on distinct parameter spaces.
2. *Monoidal boundaries.* Pointwise tensor, chiral convolution, transverse E_1-multiplication, associative chiral E_1-multiplication, and braided exchange are connected by explicit distributive laws, commutator maps, enveloping functors, or realization functors.
3. *Duality boundaries.* The interval bar, associative chiral bar, chiral Chevalley chains, quadratic dual, boundary dual, Verdier dual, and derived centre have distinct variances, targets, and completion conditions.
4. *Physical boundaries.* A quantum lift, a physical bulk, analytic sewing, a numerical anomaly, and a quantum group require, respectively, a vanishing obstruction class, an open–closed comparison, convergence estimates, an evaluation functional, and Tannaka or Hall reconstruction data.

Every prohibited identification is witnessed either by a type mismatch or by an explicit counterexample stated next to the corresponding construction.

75.2 Constructive replacements

Each boundary theorem identifies the datum that completes the corresponding construction.

Partial datum	Complete geometric datum
logarithmic residue	principal-part filtration and normal-jet object
incidence poset	incidence resolution with screen cohomology
transverse order	E_1-multiplication in $(\mathrm{FactD}(X), \otimes_!)$
associative chiral composition	algebra over $\mathsf{Ass}^{\mathrm{ch}}_X$ and its formal-disc field realization
two associative structures	algebra over the aligned mixed operad $\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$
quadratic presentation	derived presentation and acyclic Koszul comparison
uncompleted bar–cobar pair	nilcomplete algebra and conilcomplete coalgebra
Weyl or matrix coefficient	curved, Rees, relative, or Morita model
symmetrisation kernel	derived braid descent and Tannaka data
stable graph	cyclic or modular coefficient maps on every clutching stratum
boundary centre	open–closed map with a quasi-isomorphism criterion
one anomaly number	typed anomaly classes and explicit comparison morphisms

Chapter 76

The functorial atlas

76.1 The primitive datum

Let X be a smooth complex curve. A complete doubly associative chiral datum is a tuple

$$\mathbb{A} = (F, m_\perp, m_X, \lambda, \epsilon),$$

where $F \in \mathrm{FactD}(X)$,

$$m_\perp : F \otimes_! F \longrightarrow F, \qquad m_X : F \star_{\mathrm{ch}} F \longrightarrow F,$$

are augmented associative structures and λ is an algebra structure over the aligned mixed operad $\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$. The tuple includes the filtrations and completions used by the two bar constructions. Four independent extensions encode the next geometric motions:

$$\begin{aligned} \beta &= \text{exchange data}, & \tau &= \text{cyclic trace data}, \\ \sigma &= \text{stable-curve sewing data}, & Q &= \text{BRST differential}. \end{aligned}$$

Write $A_\perp = (F, m_\perp)$ and $A_X = (F, m_X)$. Each extension has its own source category and coherence equations.

76.2 The atlas theorem

Theorem 76.1 (Functorial atlas)**.** *Let $\mathbb{A}$ be a complete doubly associative chiral datum. Impose the positive-weight or nilcomplete hypotheses of Theorems 8.3 and 8.4, the exactness and continuity hypotheses of Theorem 9.2, and the dualisability hypotheses required by Hochschild and Tannaka constructions. Then the following constructions are functorial.*

1. *The oriented interval produces the transverse coalgebra*

$$\mathrm{Bar}^\perp_X(\mathbb{A}) = \mathbf{1} \otimes^{\mathbb{L}}_{A_\perp} \mathbf{1}.$$

2. *Associative chiral composition produces $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A_X)$; the chiral Lie operation produces $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(A_X)$.*
3. *The mixed law λ produces an iterated bar and a total bar whose two edge differentials anticommute.*
4. *Nilcompletion of an algebra and conilcompletion of its bar coalgebra make the operadic*

bar and cobar functors inverse.

5. *The augmentation boundary produces*

$$A^!_\perp = \mathrm{RHom}_{A_\perp}(\mathbf{1}, \mathbf{1}),$$

and the two-sided twisting transform identifies the specified complete bimodule category with the specified conilpotent bicomodule category.

6. *The identity bimodule produces the internal centre*

$$Z(A_\perp) = \mathrm{RHom}_{A_\perp \otimes A_\perp^{\mathrm{op}}}(A_\perp, A_\perp),$$

with its E_2-structure.

7. *An exchange extension β produces braid transport. Rigidity and a faithful exact tensor fibre functor produce the coquasitriangular coordinate bialgebra of tensor automorphisms.*
8. *A compatible BRST differential Q produces the bar–BRST bicomplex and the convergent comparison spectral sequence of Theorem 22.2.*
9. *A cyclic trace τ produces Hochschild chains and ribbon sewing. Stable-curve data σ produces chiral clutching. A mixed square between the two edge contractions produces the two-edge master equation.*
10. *Extendible renormalised BV weights satisfying the QME define the operadic morphism of Theorem 21.1.*
11. *A trace, index, period, monodromy, or Euler-character functional produces a scalar invariant of its declared source object.*

Each functor remains in the category of its source until a named realization or comparison functor changes the category.

Proof. The interval statement is Theorem 6.2. The three bar-type constructions and their mixed comparison are Theorems 5.2, 5.3 and 7.1. Reconstruction is Theorems 8.3 and 8.4. Boundary duality and two-sided transport are Theorems 9.1 and 9.2. The centre is the endomorphism object of the identity bimodule and therefore carries the higher Deligne E_2-action. Exchange and reconstruction follow from Theorems 16.1 and 23.1 together with the Tannaka coend. BRST compatibility is Theorem 22.2. The two sewing operations and their mixed totalization are Theorems 24.3 and 26.2. The field-theoretic realization is Theorem 21.1. Applying a functional to a functorial object gives the final construction. □

76.3 Three realization ladders

The local ladder passes from a diagonal-supported operation to its formal-coordinate expansion:

$$\text{chiral operation on } X \longrightarrow \text{principal-part filtration} \longrightarrow \text{formal-disc fields.}$$

The boundary ladder passes from ordered fusion to the universal central action:

$$\text{oriented interval} \longrightarrow \mathrm{Bar}^{\perp}_{X}(A_\perp) \longrightarrow A^!_\perp \longrightarrow Z(A_\perp).$$

The planar and sewing ladders begin with their own geometric extensions:

$$\text{plane} \longrightarrow \text{braid transport} \longrightarrow \text{Tannaka symmetry,}$$

$$\text{circle} \longrightarrow \mathrm{HH}_\bullet(A_\perp) \longrightarrow \text{ribbon sewing},$$

$$\text{stable curve} \longrightarrow \text{clutching} \longrightarrow \text{modular sewing}.$$

The mixed operad, Morita transport, and open–closed maps assemble these ladders into commutative diagrams whose vertices retain their native object types.

Chapter 77

The structure theorem and the inner music

Theorem 77.1 (Structure theorem for ordered and associative chiral fields)**.** *Let X be a smooth complex curve, and let $\mathbb{A} = (F, m_\perp, m_X, \lambda, \epsilon)$ be a complete algebra over $\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$. Then:*

1. *F carries the complete chiral collision operation, including all principal parts and regular products;*
2. *$m_\perp$ carries ordered fusion along a transverse line;*
3. *m_X carries associative chiral composition on the curve;*
4. *λ supplies the mixed locality cells on aligned decompositions;*
5. *the transverse bar, associative chiral bar, and chiral Chevalley chains are independently defined and are related by the operadic maps supplied by $\mathbb{A}$;*
6. *the completed bar–cobar, Koszul–Morita, centre, exchange, BRST, ribbon, and modular constructions are the functors of Theorem 76.1;*
7. *each scalar invariant is the value of a specified functional on one of these structured outputs.*

The construction is invariant under equivalences of complete $\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$-algebras and under Morita equivalences for every Morita-invariant output.

Proof. Items 1–4 are the definition and geometric realization of $\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$ in Theorems 1.7, 1.8, 2.1 and 5.3. Item 5 is Theorem 7.1. Item 6 is Theorem 76.1. Item 7 is functorial evaluation. Equivalence invariance follows from functoriality; Morita invariance follows from the identity-bimodule descriptions of Hochschild chains and cochains and from the two-sided Koszul transform. □

77.1 The invariant sentence

A noncommutative chiral field theory on a curve is built from a diagonal geometry of singular collision and two independent associative geometries: ordered fusion along a transverse line and associative chiral composition on the curve. The aligned mixed operad records their coherent coexistence. The interval, plane, circle, and stable curve generate bar duality, exchange, trace, and sewing through functors whose hypotheses express the corresponding physical motions.

77.2 The inner music

A field approaches another field along the diagonal, and the diagonal carries the singular product. Two experiments occur in succession along the interval, and the interval carries multiplication. One defect moves around another in the plane, and the plane carries exchange. Closing an interval forms a circle, and the circle carries trace. Degenerating a worldsheet forms a stable curve, and the stable curve carries sewing.

One boundary principle governs the coherence equations. A codimension-one face composes operations. A codimension-two corner compares two compositions. The two corner orientations supply opposite signs. Geometry determines the pattern of compositions; the coefficient maps supply the algebra. Associativity, the Borcherds identity, the Yang–Baxter equation, BRST descent, and the quantum master equation are distinct realizations of this principle on distinct parameter spaces.

Chiral Koszul duality passes from local collision operations to the coalgebra of compatible finite configurations and back. Ordered Koszul duality performs the corresponding passage for interval composition. The mixed theory preserves both memories. Every construction remains attached to the geometry that creates it, and every comparison is a map.

Part XII

The invariant synthesis

Chapter 78

The invariant language of the subject

The subject begins with a local experiment. Two observables have a position in a complex curve and an order in a transverse real direction. Collision on the curve creates a singular distribution. Composition on the line creates an associative product. Motion in a plane creates exchange. Closing the line creates a trace. Degeneration of the curve creates sewing. Each motion has one geometric carrier and one algebraic operation.

78.1 The carrier theorem

Theorem 78.1 (Carrier theorem)**.** *Let X be a smooth complex curve. A complete local theory of ordered chiral observables is specified by the following data:*

1. *a factorisable D-module F on $\operatorname{Ran} X$ carrying the complete filtered diagonal distribution;*
2. *an associative multiplication*
$$m_\perp : F \otimes_! F \to F$$
for transverse ordered fusion;
3. *an associative chiral multiplication*
$$m_X : F \star_{\mathrm{ch}} F \to F$$
when the fields themselves compose nonlocally in an ordered chiral sense;
4. *a coherent mixed law on the aligned-decomposition sector;*
5. *an exchange enhancement when planar transport is part of the theory;*
6. *a cyclic trace when intervals may be closed;*
7. *a modular sewing action when the complex curve varies.*

Every operation in the resulting theory is obtained from these data by bar construction, factorisation homology, Koszul duality, Hochschild theory, monodromy, or graph sewing.

Proof. The factorisable coefficient is the local-to-global carrier of collision. The two multiplications are the operations associated with the interval and with ordered chiral substitution. Their mixed law is the coefficient on mixed codimension-two corners. Exchange is parallel transport in a two-dimensional configuration space. Cyclic closure is factorisation homology of the circle. Modular sewing is contraction along Deligne–Mumford clutching maps. The listed functors are the universal constructions generated by these carriers. □

The theorem classifies the operations carried by a complete local theory and states where every datum lives and how it acts.

78.2 The theorem of three bars

The three fundamental resolutions are

$$\mathrm{Bar}^{\perp}(A), \qquad \mathrm{Bar}^{\mathrm{ch}}_{\mathrm{Ass}}(C), \qquad \mathrm{Bar}^{\mathrm{ch}}_{\mathrm{Lie}}(\mathfrak{g}) = \mathrm{Chev}^{\mathrm{ch}}(\mathfrak{g}).$$

Their differentials contract three different operations:

resolution	contracted operation	geometric carrier
$\mathrm{Bar}^{\perp}$	adjacent transverse multiplication	oriented interval
$\mathrm{Bar}^{\mathrm{ch}}_{\mathrm{Ass}}$	associative chiral composition	ordered collision trees
$\mathrm{Bar}^{\mathrm{ch}}_{\mathrm{Lie}}$	chiral Lie bracket	partition incidence on $\mathrm{Ran}\, X$.

A comparison exists when a functor maps one operation to another and preserves the corresponding bar construction. Similar notation carries no mathematical force.

Theorem 78.2 (Mixed-bar synthesis)**.** *For a complete doubly associative chiral algebra with a strong aligned distributive law, the transverse and associative chiral bars form a bicomplex. The two iterated bars and the total mixed bar are naturally equivalent. The chiral Chevalley complex enters through the commutator or enveloping comparison and remains a third functor.*

Proof. The mixed law makes the two face systems commute up to the determinant sign. Eilenberg–Zilber identifies the realizations of the bisimplicial object. The chiral Lie comparison is induced by the antisymmetrization or universal enveloping morphism and is therefore not part of the bisimplicial identity. □

78.3 Geometry produces the equations

Every coherence equation is the algebraic image of an oriented boundary.

For the interval, the boundary of an associahedron gives the Stasheff identities. For collision trees, the boundary of the Fulton–MacPherson resolution gives chiral associativity or Jacobi. For exchange paths, the fundamental groupoid of three-point configurations gives Yang–Baxter and associator equations. For ribbon graphs, edge contraction gives the cyclic master equation. For stable curves, clutching boundaries give modular sewing.

The common pattern is

$$\partial^2 = 0 \quad \Longrightarrow \quad \text{quadratic coherence equation,}$$

provided every face has already been decorated by a defined coefficient map.

Principle 78.3 (Coefficient-before-topology)*.* A compactification supplies the incidence and signs of an equation. An algebraic or physical theory supplies the operations decorating its faces. The master equation is the equality obtained after both structures have been constructed.

This principle is the precise relation between Stokes' theorem and algebra. Geometry determines the sum. The coefficient system determines what is being summed.

78.4 The complete deformation object

Let $\mathcal{P}_{\mathrm{full}}$ be a cofibrant coloured operad encoding the chiral coefficient, the two associative operations, exchange where present, cyclic contraction, and modular sewing. Its Koszul dual cooperad defines the complete convolution algebra

$$\mathfrak{g}_{\mathrm{full}}(A) = \mathrm{Hom}_{\mathbb{S}}\big(\mathcal{P}^{\mathrm{i}}_{\mathrm{full}}, \mathrm{End}_A\big).$$

The structure of the theory is a Maurer–Cartan element

$$\Theta_A \in \mathfrak{g}_{\mathrm{full}}(A)^1.$$

Its restrictions are

$$\Theta_\perp, \quad \Theta_X, \quad \Theta_{\mathrm{mix}}, \quad \Theta_{\mathrm{ex}}, \quad \Theta_{\mathrm{cyc}}, \quad \Theta_{\mathrm{mod}}.$$

The full equation

$$d\Theta_A + \frac{1}{2}[\Theta_A, \Theta_A] + \hbar_X \Delta_X \Theta_A + \hbar_\perp \Delta_\perp \Theta_A = 0$$

contains the lower equations as restrictions to coloured suboperads.

Theorem 78.4 (Unified obstruction theorem)**.** *Filter $\mathfrak{g}_{\mathrm{full}}(A)$ by total collision depth, transverse bar length, and graph complexity. A partial structure through filtration degree N extends to degree $N+1$ exactly when its curvature class vanishes in*

$$H^2\big(\mathrm{gr}^{N+1}\,\mathfrak{g}_{\mathrm{full}}(A), d_{\Theta_0}\big).$$

Extensions form a torsor for H^1, and infinitesimal symmetries are H^0. Restriction to any carrier sends the full obstruction to the obstruction of that realization.

Proof. Apply the filtered Maurer–Cartan obstruction theorem to the complete convolution algebra. Restriction to a coloured suboperad is a morphism of filtered dg Lie algebras and therefore preserves curvature and cohomology classes. □

The theorem replaces a collection of unrelated obstruction towers by one functorial deformation theory. Numerical shadows are obtained only after a trace, a projector, or a period map has been specified.

78.5 The Koszul–Morita movement

For the transverse multiplication, set

$$C = \mathrm{Bar}^\perp(A), \qquad A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1}).$$

The universal twisting cochain gives the equivalences

$$A\text{-modules}_{\mathrm{comp}} \simeq C\text{-comodules}_{\mathrm{conil}},$$

$$A\text{-bimodules}_{\mathrm{comp}} \simeq C\text{-bicomodules}_{\mathrm{conil}}.$$

The diagonal bimodule maps to the diagonal bicomodule. Tensor product maps to cotensor product. Hence

$$\mathrm{HH}_\bullet(A) \simeq \mathrm{coHH}_\bullet(C), \qquad \mathrm{HH}^\bullet(A) \simeq \mathrm{coHH}^\bullet(C).$$

The interval is the bar. The augmentation endpoint is the boundary dual. The annulus is Hochschild homology. The universal bulk action is Hochschild cohomology. These are four cuts through one Morita-invariant construction.

Quadratic duality, PBW duality, curved duality, Verdier duality, and double centralization compare to this movement through separate maps. Each carries its own hypotheses.

78.6 The passage from fusion to quantum symmetry

An ordered line gives multiplication. A plane gives exchange. A meromorphic connection turns exchange paths into matrices. A rigid tensor category and a fibre functor reconstruct the coordinate quantum group.

The complete chain is

$$\text{line defects} \xrightarrow{\text{fusion}} \text{monoidal category} \xrightarrow{\text{planar transport}} \text{braided category},$$
$$\text{braided category} \xrightarrow{\text{fibre functor}} \text{Tannaka coend} \xrightarrow{\text{recognition}} \text{named quantum group}.$$

KZ transport recognizes Drinfeld–Jimbo categories through Drinfeld–Kohno. Rational four-dimensional Chern–Simons theory recognizes Yangian line algebras after RTT, PBW, and faithfulness. Hall correspondences recognize quantum-group positive halves after finite-field flag counts and the Drinfeld-double construction.

The quantum group is therefore the algebraic reconstruction of a category of exchange. It is not a dimension difference and not a name for every ordered kernel.

78.7 The passage from local collision to modular sewing

A cyclic trace closes a transverse interval and produces ribbon genus $g_\perp$. A conformal-block sewing action glues algebraic curves and produces chiral genus g_X. The two genera occupy independent graph colours.

The strict product theory proves that both can coexist:

$$\mathcal{F}_{g_X, g_\perp, b} = \mathcal{F}^X_{g_X} \mathcal{F}^\perp_{g_\perp, b}.$$

Interactions between them are additional mixed vertices in the bicoloured graph complex. Their first obstruction is a class in the mixed graph deformation complex, not a presumed equality of the two genera.

Projective curvature of conformal blocks is a determinant-line class. A BV anomaly is a local functional cohomology class. A BRST anomaly is the square of a cohomological differential. A framing anomaly is an invertible three-dimensional field theory. Curved Koszul data is a nullary operation in a curved algebra. Family index and open–closed maps may relate these classes. Their types keep the relations exact.

78.8 Faithful realizations

The same operation can be calculated in several realizations:

realization	carrier
algebraic	D-module on diagonals
formal	vertex or nonlocal field
topological	E_n factorisation algebra
analytic	flat connection or conformal block
categorical	module, bimodule, or line category
physical	BV observable or defect amplitude
arithmetic	trace, index, theta series, or Euler product.

A faithful realization preserves the category of the object and its compositions. Agreement of faithful realizations is evidence because every comparison map has a source and target.

Theorem 78.5 (Scalar functoriality)**.** *Let $F : \mathcal{C} \to \mathcal{D}$ be a symmetric monoidal functor and let $\tau : \mathrm{End}_{\mathcal{D}}(\mathbf{1}) \to R$ be a trace. Any scalar invariant obtained as*

$$I(A) = \tau(F(\mathcal{O}(A)))$$

from a constructed operation $\mathcal{O}(A)$ is invariant under equivalences preserved by F and τ. Equality of scalars does not imply equality of the source operations unless the composite $\tau \circ F$ is conservative on the full subcategory generated by the displayed source operations.

Proof. Functoriality carries equivalent operations to equivalent endomorphisms, and the trace identifies equivalent endomorphisms. The converse is exactly conservativity of the scalarization functor. □

This theorem places characters, Verlinde dimensions, partition functions, central charges, theta coefficients, and automorphic products at the end of the construction. They are values of operations, not substitutes for the operations.

78.9 The final structure theorem

Theorem 78.6 (Chiral Koszul duality for ordered quantum fields)**.** *Let X be a smooth complex curve and let A be a connected complete doubly associative chiral algebra with:*

1. *a factorisable D-module coefficient carrying the complete diagonal distribution;*
2. *transverse and associative chiral E_1 structures;*
3. *an aligned mixed distributive law;*
4. *finite positive-weight windows with Mittag–Leffler completion;*
5. *perfectness whenever Verdier duality, cyclic sewing, or Tannaka duality is invoked.*

Then:

1. *the transverse and associative chiral bars form a mixed conilpotent coalgebra and satisfy the double-bar Fubini equivalence;*
2. *completed cobar reconstructs A on the nilcomplete/conilcomplete locus;*
3. *the augmentation boundary, module Koszul transform, diagonal bicomodule, Hochschild homology, and derived centre form a Koszul–Morita square;*

4. *the pole Rees construction gives the Poisson vertex symbol and the word Rees construction gives the ordered Koszul symbol;*
5. *exchange data reconstructs a coquasitriangular Tannaka object, with named quantum groups obtained by recognition theorems;*
6. *BRST derivations define compatible bar bicomplexes and descend under the stated convergence and concentration hypotheses;*
7. *cyclic and chiral sewing define independent loop operators and a bicoloured master equation;*
8. *every scalar output is the value of a specified trace, period, index, monodromy, or Euler-character functor on one of the constructed objects.*

Every assertion is natural under morphisms preserving the displayed data.

Proof. Statement (1) is Theorem 65.4. Statement (2) is completed bar–cobar duality on the stated locus. Statement (3) is module and bimodule Koszul duality together with the higher Deligne centre. Statement (4) is the flat bi-Rees theorem and the chiral–classical operad comparison. Statement (5) is meromorphic Tannaka reconstruction followed by a presentation theorem. Statement (6) is the bar–BRST spectral sequence. Statement (7) is the strict two-genus construction and its graphwise extension. Statement (8) is scalar functoriality. Naturality follows from the functoriality of every construction used in these steps. □

78.10 Coda

The diagonal carries the singular field. The interval carries the order of action. The plane carries exchange. The circle carries trace. The stable curve carries sewing. The coordinate torsor carries covariance. The bar carries resolution. The centre carries universal bulk action. The Rees family carries the classical limit. The deformation complex carries the possible theories and their obstructions.

The equations become simple when the carriers are separated. Associativity is a boundary of intervals. Borcherds identity is a boundary of collisions. Yang–Baxter is a boundary of exchange paths. The quantum master equation is a boundary of compactified graphs. Modular factorisation is a boundary of stable curves.

The inner unity of the subject is not the equality of these equations. It is the functorial passage from geometry to operation, from operation to resolution, from resolution to duality, and from duality to observable quantities. Chiral Koszul duality is that passage written in the language of the curve.

Chapter A

Suspensions, orientations, and low-arity signs

A.1 Cohomological suspension

For homogeneous a, $|sa| = |a| + 1$. Exchanging homogeneous factors u, v contributes the Koszul sign

$$u \otimes v \longmapsto (-1)^{|u||v|} v \otimes u.$$

The binary bar coderivation is fixed by

$$b_2(sa \otimes sb) = (-1)^{|a|} s(ab).$$

On a word,

$$b_2[a_1|\cdots|a_r] = \sum_{i=1}^{r-1} (-1)^{\sigma_i} [a_1|\cdots|a_i a_{i+1}|\cdots|a_r], \qquad \sigma_i = \sum_{j<i} (|a_j| + 1) + |a_i|.$$

A global conjugation by suspension changes the displayed convention and yields an isomorphic complex.

A.2 Four-letter bar word

For degree-zero letters,

$$b[a|b|c|d] = [ab|c|d] - [a|bc|d] + [a|b|cd].$$

A second application gives

$$\begin{aligned} b^2[a|b|c|d] = {} & [(ab)c|d] - [ab|cd] - [a(bc)|d] + [a|b(cd)] \\ & + [ab|cd] - [a|(bc)d]. \end{aligned}$$

The middle terms cancel, and associativity cancels the remaining pairs. The calculation is the cellular boundary identity for the two-dimensional associahedron in its word model.

A.3 Two edge colours

For a bicoloured graph G, set

$$\mathrm{or}(G) = \det E_X(G) \otimes \det E_\perp(G).$$

Contraction of an X-edge removes a basis vector from the first determinant. Contraction of a transverse edge removes one from the second. Reversing the order exchanges two odd normal directions, hence

$$d_X d_\perp + d_\perp d_X = 0.$$

The same determinant convention gives $\Delta_X \Delta_\perp + \Delta_\perp \Delta_X = 0$ in the two-edge master equation.

A.4 Arnold relation and KZ curvature

For three points, the logarithmic forms $\omega_{ij} = d\log(z_i - z_j)$ satisfy

$$\omega_{12} \wedge \omega_{23} + \omega_{23} \wedge \omega_{31} + \omega_{31} \wedge \omega_{12} = 0.$$

The arity-three KZ curvature is

$$F_\nabla = \kappa^2\big([\Omega_{12}, \Omega_{13}]\omega_{12} \wedge \omega_{13} + [\Omega_{12}, \Omega_{23}]\omega_{12} \wedge \omega_{23} \\ + [\Omega_{13}, \Omega_{23}]\omega_{13} \wedge \omega_{23}\big).$$

The infinitesimal braid identities express the commutators through one common coefficient. Arnold's relation then gives $F_\nabla = 0$.

A.5 Rational Yang–Baxter expansion

For $R_{ij}(u) = 1 - \hbar P_{ij}/u$, the linear terms in the Yang–Baxter equation agree. After multiplication by the common denominator, the quadratic terms reduce to

$$P_{12}P_{13} = P_{23}P_{12}, \qquad P_{13}P_{23} = P_{12}P_{13}, \qquad P_{12}P_{23} = P_{13}P_{12}.$$

The cubic terms agree because both products act by the longest permutation of three tensor factors.

Chapter B

Exact finite-window calculations

B.1 Quantum and Jordan planes

For the quantum plane $A_q = \Bbbk\langle x, y\rangle/(yx - qxy)$, multiplication in the PBW basis is

$$(x^a y^b)(x^c y^d) = q^{bc} x^{a+c} y^{b+d}.$$

For the Jordan plane $A_J = \Bbbk\langle x, y\rangle/(yx - xy - x^2)$, repeated use of $yx = xy + x^2$ gives a triangular PBW reduction. The normalized bar differential is obtained by multiplying adjacent PBW monomials.

Through total weight five, both bar complexes have

$$H_{1,1} \simeq \Bbbk^2, \qquad H_{2,2} \simeq \Bbbk,$$

and all other $H_{w,r}$ in this window vanish. The complete rank table is

algebra	w	r	$\dim C_r$	$\operatorname{rk} d_r$	$\operatorname{rk} d_{r+1}$	$\dim H_r$
$A_q, q = 2$	1	1	2	0	0	2
$A_q, q = 2$	2	1	3	0	3	0
$A_q, q = 2$	2	2	4	3	0	1
$A_q, q = 2$	3	1	4	0	4	0
$A_q, q = 2$	3	2	12	4	8	0
$A_q, q = 2$	3	3	8	8	0	0
$A_q, q = 2$	4	1	5	0	5	0
$A_q, q = 2$	4	2	25	5	20	0
$A_q, q = 2$	4	3	36	20	16	0
$A_q, q = 2$	4	4	16	16	0	0
$A_q, q = 2$	5	1	6	0	6	0
$A_q, q = 2$	5	2	44	6	38	0
$A_q, q = 2$	5	3	102	38	64	0
$A_q, q = 2$	5	4	96	64	32	0
$A_q, q = 2$	5	5	32	32	0	0
A_J	1	1	2	0	0	2
A_J	2	1	3	0	3	0
A_J	2	2	4	3	0	1
A_J	3	1	4	0	4	0

algebra	w	r	$\dim C_r$	$\operatorname{rk} d_r$	$\operatorname{rk} d_{r+1}$	$\dim H_r$
A_J	3	2	12	4	8	0
A_J	3	3	8	8	0	0
A_J	4	1	5	0	5	0
A_J	4	2	25	5	20	0
A_J	4	3	36	20	16	0
A_J	4	4	16	16	0	0
A_J	5	1	6	0	6	0
A_J	5	2	44	6	38	0
A_J	5	3	102	38	64	0
A_J	5	4	96	64	32	0
A_J	5	5	32	32	0	0

The degree-two generators are represented by

$$[y|x] - 2[x|y] \quad \text{for } A_{q=2}, \qquad [y|x] - [x|y] - [x|x] \quad \text{for } A_J.$$

Their bar boundaries are the defining relations.

B.2 The $\mathfrak{sl}_2$ Chevalley complex

With basis (e, f, h) and brackets $[e, f] = h$, $[h, e] = 2e$, $[h, f] = -2f$, the map

$$d_2 : \Lambda^2\mathfrak{sl}_2 \longrightarrow \mathfrak{sl}_2$$

has matrix, in the bases $(e \wedge f, e \wedge h, f \wedge h)$ and (e, f, h),

$$\begin{pmatrix} 0 & -2 & 0 \\ 0 & 0 & 2 \\ 1 & 0 & 0 \end{pmatrix}.$$

Its determinant is -4, so d_2 is an isomorphism. The top differential vanishes by Jacobi. Therefore

$$(\dim H_0, \dim H_1, \dim H_2, \dim H_3) = (1, 0, 0, 1).$$

B.3 A relative quiver bar

Let Q be the quiver $1 \xrightarrow{a} 2 \xrightarrow{b} 3$, let

$$S = \Bbbk e_1 \oplus \Bbbk e_2 \oplus \Bbbk e_3, \qquad A = \Bbbk Q/(ba),$$

and form $S \otimes^{\mathbb{L}}_A S$. The only composable bar word of length two is $[b|a]$, and

$$d[b|a] = [ba] = 0.$$

The resulting relative Tor dimensions are

$$\dim \operatorname{Tor}^A_0(S, S) = 3, \qquad \dim \operatorname{Tor}^A_1(S, S) = 2, \qquad \dim \operatorname{Tor}^A_2(S, S) = 1.$$

B.4 The A_2 Hall–Serre relation at $q = 2$

In the two-dimensional span of the isomorphism classes

$$X = 2S_1 + S_2, \qquad Y = S_1 + M,$$

the three products are

$$e_1^2 e_2 = (3, 3), \qquad e_1 e_2 e_1 = (3, 1), \qquad e_2 e_1^2 = (3, 0).$$

Hence

$$e_1^2 e_2 - 3e_1 e_2 e_1 + 2e_2 e_1^2 = (0, 0).$$

The arbitrary-q flag count in Theorem 18.1 gives the corresponding quantum Serre relation.

B.5 The stable trace bracket

For monomials $p_{a,b}$, define

$$\{p_{a,b}, p_{c,d}\} = (ad - bc)p_{a+c-1,b+d-1}.$$

The coefficient is the determinant of the exponent vectors. The Jacobiator is the image of the Schouten bracket of the standard symplectic Poisson bivector on the two-dimensional torus and therefore vanishes. Equivalently, expanding the three cyclic terms gives a common monomial with coefficient

$$(ad - bc)((a + c - 1)f - (b + d - 1)e) + \mathrm{cyc} = 0.$$

Chapter C

Homotopy-coherent presentation templates

C.1 Quadratic transverse presentation

For coefficient generators $V \in \mathrm{FactD}(X)$ and relations $R \to V \otimes_! V$, the strict algebra is

$$A = T_{\otimes_!}(V)/(R).$$

The derived presentation is the homotopy pushout

$$\begin{tikzcd} \mathrm{Free}_{\mathsf{E}_1}(R) \arrow[r, shift left, "\partial_0"] \arrow[r, shift right, "\partial_1"'] \arrow[d] & \mathrm{Free}_{\mathsf{E}_1}(V) \arrow[d] \\ \mathbf{1} \arrow[r] & A_{\mathrm{der}}. \end{tikzcd}$$

A cofibrant resolution attaches higher cells to relations among relations.

C.2 Associative chiral presentation

For formal-disc generators M and field relations R_X, use $\mathrm{Free}_{\mathsf{Ass}^{\mathrm{ch}}_X}(M)$. A finite-pole relation has the form

$$(z-w)^N \left(Y(a,z)Y(b,w) - \sum_i f_i(z-w)Y(b_i,w)Y(a_i,z) \right) = 0.$$

A quantum vertex relation includes the coefficients of an exchange operator $\mathcal{S}(z)$. The derived presentation records the relation, a null-homotopy, and all higher compatibilities under nested substitution.

C.3 Doubly associative presentation

A presentation of an algebra over $\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$ consists of

$$(V; R_\perp, R_X, R_{\mathrm{mix}}).$$

The first relation space governs transverse overlaps, the second governs chiral overlaps, and the third governs the aligned mixed squares. A cofibrant resolution adds transverse associahedral cells, chiral collision cells, mixed prisms, and higher fillers for the resulting spheres.

C.4 Relative presentation

For a semisimple base S, all free algebras, tensor products, and bars are formed in S-bimodules:

$$A = T_S(M)/(R), \qquad \mathrm{Bar}_S(A) = S \otimes^{\mathbb{L}}_A S.$$

A chiral quiver replaces each component $e_j M e_i$ by a factorisable coefficient and imposes the factorisation maps componentwise.

Chapter D

Historical coordinates of the language

Dirac's distribution theory makes singular support part of the definition of an operation. Wilson's operator product expansion organizes the short-distance coefficients of quantum fields. Borcherds and the vertex-algebra formalism encode their algebraic coherence [3, 35]. Beilinson and Drinfeld place the complete singular operation on the diagonal of a curve and globalize it over the Ran space [4]. Fulton and MacPherson resolve simultaneous collisions by a normal-crossing boundary, while Arnold's logarithmic forms compute the topology of the separated configuration [2, 27].

Stasheff's associahedra resolve ordered multiplication. Boardman–Vogt resolutions place associativity, chiral collision, and mixed coherence in a cellular operad [5, 6]. Francis–Gaitsgory pro-nilpotence turns chiral Chevalley chains and primitives into a Koszul equivalence [25]. Hochschild theory and the higher Deligne theorem identify the universal central action. Drinfeld and Kohno reconstruct quantum symmetry from braid transport, and Etingof–Kazhdan formulate the braided vertex-algebraic refinement [20, 21, 37].

Feynman graph expansion makes perturbation theory a sum over compactified configurations. Polyakov's determinant and anomaly line identify projective curvature as geometry on moduli. Witten's Chern–Simons theory relates line exchange and knot invariants [44]. Costello's BV renormalisation and factorisation observables supply chain-level maps from quantum field theory to algebraic operations [14]. Holomorphic–topological gauge theory realizes meromorphic line exchange [15, 16], while higher perturbative operations describe deformations, OPEs, and anomalies [28]. The common language is geometric: every operation is attached to the space of motions that produces it.

Chapter E

Comparison atlas

Geometry	Object	Operation	Required theorem or datum
partial diagonal	factorisable D-module	singular OPE and principal parts	diagonal extension, coordinate covariance, chiral identities
ordered interval	E_1-algebra for $\otimes_!$	transverse fusion	augmentation or relative base, interval excision
formal disc	$\mathsf{Ass}^{\mathrm{ch}}_X$-algebra	nonlocal field iteration	weak associativity, translation, completion
aligned rectangle	$\mathsf{Mix}^{\mathrm{ch}}_{X,\perp}$-algebra	mixed locality	aligned projector and mixed distributive law
transverse plane	E_2-lift or flat connection	exchange	braid paths, flatness, Yang–Baxter
circle	Hochschild chains	trace and closure	cyclic or Calabi–Yau structure
ribbon surface	cyclic A_∞-algebra	open sewing	perfect pairing and ribbon-graph action
stable curve	modular chiral object	conformal-block sewing	clutching maps, pairing, anomaly line, convergence
bicoloured graph	two-edge convolution algebra	mixed master equation	common corollas and mixed boundary square
augmentation boundary	module $\mathbf{1}$	boundary Koszul dual	compactness and completion
annulus	Hochschild theory	centre and trace pairing	two-sided Morita transform
quiver	relative S-algebra	boundary-changing composition	vertex-idempotent base and path relations
ZF spectral line	exchange algebra	ordered normal form	unitarity and Yang–Baxter
RTT category	coordinate bialgebra	matrix coefficients	rigidity, fibre functor, PBW recognition
Hall stack	correspondence algebra	extension product	proper pull–push, orientations, associativity

Geometry	**Object**	**Operation**	**Required theorem or datum**
derived commuting scheme	necklace algebra	Hamiltonian trace action	cyclic derivatives and derived quotient
BV configuration	graph-weight complex	quantum observables	renormalisation and quantum master equation
BRST complex	bicomplex	reduction	square-zero derivation, Kunneth, convergence
root grading	graded bar	denominator character	PBW and finite-decomposition completion
scalar target	field or coefficient ring	trace, period, index	specified functional on a structured object

Chapter F

Orientation lines and the collision differential

The square-zero identities are identities of oriented correspondences. One orientation convention yields both cancellations governing the collision complex and the corolla-normalized forest resolution. A coherent change of convention conjugates every differential by diagonal signs and does not change the resulting complex.

F.1 Cohomological suspension and unordered tensor products

Let (K, d_K) be a cochain complex. We write $sK = K[1]$ in the cohomological convention

$$(sK)^n = K^{n+1}, \qquad |sx| = |x| - 1, \qquad d_{sK}(sx) = -s(d_K x).$$

For a finite set S, the unordered tensor product is

$$\bigotimes_{s\in S}^{\text{unord}} K_s := \left(\bigoplus_{\alpha:\{1,\dots,|S|\} \xrightarrow{\sim} S} K_{\alpha(1)} \otimes \cdots \otimes K_{\alpha(|S|)} \right)_{\Sigma_{|S|}}, \tag{F.1}$$

where a permutation acts by the Koszul symmetry. An ordering of S identifies (F.1) with an ordinary tensor product, but no ordering is part of the object.

Let $\mu : \mathfrak{g} \otimes \mathfrak{g} \to \mathfrak{g}$ be a degree-zero graded Lie bracket. Its suspended form

$$\bar{\mu} : s\mathfrak{g} \otimes s\mathfrak{g} \longrightarrow s\mathfrak{g}$$

has cohomological degree $+1$ and is defined by

$$\bar{\mu}(sx, sy) = (-1)^{|x|} s[x, y]. \tag{F.2}$$

With this convention, graded antisymmetry of $[-,-]$ becomes graded symmetry of $\bar{\mu}$ with respect to the shifted degrees:

$$\bar{\mu}(u, v) = (-1)^{|u||v|} \bar{\mu}(v, u).$$

The Jacobi identity becomes

$$\bar{\mu}(\bar{\mu}(u,v),w) + (-1)^{|u|(|v|+|w|)}\bar{\mu}(\bar{\mu}(v,w),u) + (-1)^{|w|(|u|+|v|)}\bar{\mu}(\bar{\mu}(w,u),v) = 0. \tag{F.3}$$

The same formulas hold for a chiral Lie bracket after the tensor products are replaced by chiral products and the outputs are pushed forward to the common small diagonal.

F.2 Determinant lines of collision chains

Let I be finite and let $\Pi(I)$ be its partition lattice, ordered by refinement. A cover $\pi \lessdot \rho$ merges exactly two blocks. Write $e_{\pi,\rho}$ for the one-dimensional conormal direction of that merger. For a saturated chain

$$\gamma = (\pi_0 \lessdot \pi_1 \lessdot \cdots \lessdot \pi_r)$$

define

$$\mathrm{or}(\gamma) = \bigwedge^{r} \left(e_{\pi_0,\pi_1} \oplus \cdots \oplus e_{\pi_{r-1},\pi_r}\right).$$

Deleting the j-th edge gives the simplicial boundary sign $(-1)^{j-1}$. Interchanging two independent collision directions reverses the determinant orientation.

For a partition π, let

$$C_\pi(\mathfrak{g}) = \Delta_{\pi,!}\left(\bigotimes_{B\in\pi}^{\text{unord}} s\mathfrak{g}\right).$$

If $\pi \lessdot \rho$ merges blocks B and C, the map $d_{\pi,\rho} : C_\pi(\mathfrak{g}) \to C_\rho(\mathfrak{g})$ is the identity on the other blocks and $\bar{\mu}$ on the B, C factors, tensored with the oriented conormal line. The total collision differential is

$$d_{\text{coll}} = d_{\mathfrak{g}} + \sum_{\pi\lessdot\rho} d_{\pi,\rho}. \tag{F.4}$$

Lemma F.1 (Internal–collision cancellation)**.** *If $d_{\mathfrak{g}}$ is a derivation of the chiral bracket, then*

$$d_{\mathfrak{g}} d^{(1)}_{\text{coll}} + d^{(1)}_{\text{coll}} d_{\mathfrak{g}} = 0,$$

where $d^{(1)}_{\text{coll}}$ is the sum over covers.

Proof. On the two merged factors this is the suspended form of

$$d[x,y] = [dx,y] + (-1)^{|x|}[x,dy].$$

The identity $d(sx) = -s(dx)$ gives the sign required when the degree-$+1$ map $\bar{\mu}$ is interchanged with the unary differential. Untouched factors contribute the usual tensor-product signs. Because the operation is defined on the unordered tensor product, the calculation is independent of the temporary ordering used to write it. ☐

F.3 Rank-two intervals in the partition lattice

Every coefficient of $(d^{(1)}_{\text{coll}})^2$ is supported on a rank-two interval $[\pi,\sigma]$ of $\Pi(I)$. Such an interval has exactly one of two forms: two disjoint pairs of blocks are merged, or three blocks are merged into one.

Proposition F.2 (Disjoint-pair cancellation)**.** *Let B, C, D, E be four distinct blocks of π. The two composites that merge B with C and D with E in the two possible orders sum to zero.*

Proof. The two coefficient operations act on disjoint factors. The transposition of the two odd collision directions can be recorded either as the interchange sign of the two degree-one maps or as the reversal

$$e_{BC} \wedge e_{DE} = -e_{DE} \wedge e_{BC}$$

of the determinant line. These are two presentations of the same transposition, not two independent signs. In the total oriented convention the two paths induce opposite maps to the common codimension-two component. ☐

Proposition F.3 (Triple-collision coefficient)**.** *Let B, C, D be three blocks of π, and let σ be obtained by replacing them with $B \cup C \cup D$. The component $C_\pi(\mathfrak{g}) \to C_\sigma(\mathfrak{g})$ of $(d^{(1)}_{\mathrm{coll}})^2$ is one half of the suspended chiral Jacobiator.*

Proof. The three intermediate partitions are obtained by first merging B, C, first merging C, D, or first merging D, B. For a temporary ordering u, v, w of the suspended factors, the oriented sum is

$$\bar{\mu}(\bar{\mu}(u, v), w) + (-1)^{|u|(|v|+|w|)}\bar{\mu}(\bar{\mu}(v, w), u) + (-1)^{|w|(|u|+|v|)}\bar{\mu}(\bar{\mu}(w, u), v),$$

which is (F.3). The factor $1/2$ is the conventional conversion between the square of an odd coderivation and one half of its graded commutator with itself. ☐

Theorem F.4 (Square of the collision differential)**.** *For a graded chiral antisymmetric operation μ compatible with $d_{\mathfrak{g}}$,*

$$d^2_{\mathrm{coll}} = \frac{1}{2} d_{[\mu,\mu]}.$$

Consequently $d^2_{\mathrm{coll}} = 0$ if and only if μ satisfies the chiral Jacobi identity.

Proof. The internal square is zero and the mixed terms vanish by Theorem F.1. Disjoint mergers cancel by Theorem F.2; triple mergers give the Jacobiator by Theorem F.3. Conversely, the component from the discrete three-block partition to the one-block partition is precisely the Jacobiator. ☐

F.4 Signs in the corolla-normalized forest resolution

For a nonempty block B, let $K(B) = N_*(N\,\mathrm{Tree}(B))$ be the normalized nerve complex from Chapter 43. Its vertical differential ∂_B is the alternating simplicial boundary and satisfies $\partial_B^2 = 0$. The augmentation and terminal-corolla section

$$K(B) \xrightarrow{\epsilon_B} \mathbb{C} \xrightarrow{\iota_B} K(B)$$

are chain maps. For disjoint blocks B, C, set

$$m_{B,C} = \iota_{B\cup C}(\epsilon_B \otimes \epsilon_C) : K(B) \otimes K(C) \longrightarrow K(B \cup C). \tag{F.5}$$

The maps $m_{B,C}$ are symmetric and strictly associative, and commute with the vertical differentials.

The total complex

$$W_I^{\mathrm{FM}}(\mathfrak{g}) = \bigoplus_{\pi \in \Pi(I)} C_\pi(\mathfrak{g}) \otimes \bigotimes_{B \in \pi}^{\mathrm{unord}} K(B)$$

has vertical differential $\partial = \sum_B \partial_B$ and horizontal differential obtained by tensoring $d_{\pi,\rho}$ with $m_{B,C}$. The mixed commutator vanishes because (F.5) is a chain map. Strict associativity of the $m_{B,C}$ reduces the horizontal square to the partition-lattice calculation above. Hence the total square is zero exactly when the chiral Jacobi identity holds.

This construction is intentionally different from an assertion that a single internal-edge contraction merges disconnected roots. Vertical contractions resolve a rooted tree inside a fixed block. Horizontal collision maps merge blocks through the partition differential and insert the terminal corolla of the new block.

F.5 Residues and coorientations

Let Y be smooth and $D_1, \dots, D_r$ a normal-crossing divisor. A local logarithmic form can be written

$$\alpha = \sum_{J \subseteq \{1,\dots,r\}} \left(\bigwedge_{j \in J} \frac{dt_j}{t_j} \right) \wedge \alpha_J .$$

The iterated residue along an ordered tuple is the corresponding coefficient. Therefore

$$\mathrm{Res}_{D_i} \mathrm{Res}_{D_j} = -\, \mathrm{Res}_{D_j} \mathrm{Res}_{D_i} \tag{F.6}$$

when the residue maps include their conormal orientation lines. This concerns the logarithmic conormal factors. Higher OPE poles are first paired with normal jets, as in Chapter 44, and then subjected to the same residue orientation law.

F.6 Stable graphs

For a stable graph Γ, set $\mathrm{or}(\Gamma) = \det \mathbb{R}^{E(\Gamma)}$. A separating clutching inserts an edge joining two vertices; a nonseparating clutching inserts a loop. Ordering the newly inserted edges gives the graph differential. The determinant-line calculation proves that two successive node-forming operations anticommute. The automorphism group of Γ acts on $\mathrm{or}(\Gamma)$, and the modular convolution complex takes coinvariants with this twist.

In a BV presentation, joining different vertices gives the bracket and joining two flags of one vertex gives the loop operator Δ. With the standard modular-operad determinant convention, the codimension-two relations give

$$d^2 = 0, \qquad d\Delta + \Delta d = 0, \qquad \Delta^2 = 0,$$

together with the second-order identity expressing the bracket as the deviation of Δ from being a derivation. These equations are orientation identities of two node-forming parameters; no scalar curvature enters their proof.

Chapter G

Low-arity calculations

Low arity is where every abstract object can be seen without compression. The arity-two, arity-three, and arity-four calculations identify the Arnold and infinitesimal-braid cancellations and exhibit the first pole symbols, BRST descendants, and sewing contractions.

G.1 Two labels

Let $I = \{1, 2\}$. The partition lattice has two elements,

$$1|2 < 12.$$

For a chiral Lie algebra $\mathfrak{g}$, the labelled collision complex is

$$\Delta_{1|2,!}(s\mathfrak{g} \boxtimes s\mathfrak{g}) \xrightarrow{\bar{\mu}} \Delta_{12,!}(s\mathfrak{g}). \tag{G.1}$$

Since $\Delta_{1|2}$ is the identity of X^2 and Δ_{12} is the diagonal, the map is exactly the chiral bracket

$$j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}$$

written after suspension. There is no quadratic condition in arity two. Antisymmetry is the compatibility with the transposition of labels.

On a formal disc, a translation-invariant binary chiral operation is represented by

$$\{a_\lambda b\} + :ab:, \qquad \{a_\lambda b\} = \sum_{n \geq 0} \frac{\lambda^n}{n!} a_{(n)} b.$$

The summand $a_{(n)}b$ is the coefficient of the normal principal part of order $n+1$. Thus the arity-two chain already contains the full singular OPE and the regular product.

Example G.1 (Heisenberg)**.** For the rank-one Heisenberg chiral algebra with field J and

$$J(z)J(w) \sim \frac{k}{(z-w)^2},$$

the binary operation begins with the second normal symbol $k\, N^{\otimes 2}_{\Delta/X^2}$. The logarithmic residue of $d\log(z-w)$ does not produce k; the coefficient is supplied by the order-two principal-part component of the chiral operation.

Example G.2 (Affine currents)**.** For a finite-dimensional Lie algebra $\mathfrak{g}$ with invariant form

(,),

$$J^a(z)J^b(w) \sim \frac{k(a,b)}{(z-w)^2} + \frac{J^{[a,b]}(w)}{z-w}.$$

The first normal symbol is the Lie bracket and the second is the invariant form multiplied by the level. They have different tensorial weights and satisfy different identities: Jacobi for the first, invariance for the second.

G.2 Three labels and the Jacobiator

For $I = \{1,2,3\}$, the partition lattice is

$$1|2|3 < \{12|3, 13|2, 23|1\} < 123.$$

Suppressing the diagonal pushforwards, the collision complex has the form

$$(s\mathfrak{g})^{\otimes 3}_{\Sigma_0} \xrightarrow{d_1} \big((s\mathfrak{g})^{\otimes 2}\big)^{\oplus 3} \xrightarrow{d_2} s\mathfrak{g}. \tag{G.2}$$

The notation Σ_0 records fixed labels and the ordered chain complex before symmetric descent. For homogeneous $u, v, w \in s\mathfrak{g}$,

$$d_1(u \otimes v \otimes w) = \begin{pmatrix} \bar{\mu}(u,v) \otimes w \\ (-1)^{|v||w|}\bar{\mu}(u,w) \otimes v \\ (-1)^{|u|(|v|+|w|)}\bar{\mu}(v,w) \otimes u \end{pmatrix},$$

where the three rows correspond to 12|3, 13|2, and 23|1. With boundary orientations chosen cyclically,

$$d_2(x_{12|3}, x_{13|2}, x_{23|1}) = \bar{\mu}(x_{12}, x_3) - \bar{\mu}(x_{13}, x_2) + \bar{\mu}(x_{23}, x_1),$$

with the Koszul permutations implicit. Multiplication gives the suspended Jacobiator. This is the smallest calculation in which the square of the collision differential contains information.

G.3 Arnold's relation

On $\mathrm{Conf}_3(\mathbb{C})$ put

$$\omega_{ij} = d\log(z_i - z_j).$$

Set $x = z_1 - z_2$ and $y = z_2 - z_3$, so that $z_1 - z_3 = x + y$. Then

$$\omega_{12} = \frac{dx}{x}, \qquad \omega_{23} = \frac{dy}{y}, \qquad \omega_{13} = \frac{dx+dy}{x+y}.$$

A direct calculation gives

$$\begin{aligned} \omega_{12}\wedge\omega_{23} - \omega_{12}\wedge\omega_{13} - \omega_{13}\wedge\omega_{23} &= \left(\frac{1}{xy} - \frac{1}{x(x+y)} - \frac{1}{y(x+y)}\right) dx\wedge dy \\ &= 0. \end{aligned}$$

Equivalently,

$$\omega_{12}\wedge\omega_{23} + \omega_{23}\wedge\omega_{31} + \omega_{31}\wedge\omega_{12} = 0. \tag{G.3}$$

This relation lies in the coefficient algebra of configuration-space forms. It is not the Jacobi identity. A flat connection combines the two by tensoring ω_{ij} with endomorphisms that satisfy the infinitesimal braid relations.

G.4 The KZ curvature in arity three

Let V_1, V_2, V_3 be $\mathfrak{g}$-modules and let Ω_{ij} denote the invariant Casimir acting in the i-th and j-th factors. Put

$$A = \hbar(\Omega_{12}\omega_{12} + \Omega_{13}\omega_{13} + \Omega_{23}\omega_{23}), \qquad \nabla = d - A.$$

Since $d\omega_{ij} = 0$, the curvature is $F_\nabla = A \wedge A$. The terms with identical pairs vanish. After collecting coefficients,

$$\begin{aligned}\frac{1}{\hbar^2} F_\nabla = [\Omega_{12}, \Omega_{23}]\, \omega_{12} \wedge \omega_{23} + [\Omega_{12}, \Omega_{13}]\, \omega_{12} \wedge \omega_{13} \\ + [\Omega_{13}, \Omega_{23}]\, \omega_{13} \wedge \omega_{23}.\end{aligned} \tag{G.4}$$

Invariance of the Casimir gives

$$[\Omega_{12}, \Omega_{13} + \Omega_{23}] = 0, \qquad [\Omega_{13}, \Omega_{12} + \Omega_{23}] = 0.$$

Hence the three commutators in (G.4) are the same up to signs, and the curvature is that common commutator multiplied by (G.3). Therefore $F_\nabla = 0$.

This calculation displays the division of labor:

$$\text{Arnold relation} \quad \times \quad \text{invariance and Jacobi of } \mathfrak{g} \quad = \quad \text{flatness of KZ.}$$

The geometric and algebraic factors are independent inputs to the total coefficient.

G.5 The full vertex identity in arity three

For a vertex algebra, the coefficient operation is not exhausted by Ω_{ij}. Write

$$Y(a, z)Y(b, w) \sim \sum_{n \geq 0} \frac{Y(a_{(n)}b, w)}{(z - w)^{n+1}}.$$

The equality between the expansions in the regions $|z_1 - z_2| < |z_2 - z_3|$ and $|z_2 - z_3| < |z_1 - z_3|$ is the formal-distribution Jacobi identity. Multiplying by powers of the differences and comparing coefficients gives the Borcherds identity

$$\begin{aligned}\sum_{i \geq 0} \binom{m}{i} (a_{(n+i)}b)_{(m+k-i)}c = \sum_{i \geq 0} (-1)^i \binom{n}{i} \Big(a_{(m+n-i)} b_{(k+i)} c \\ - (-1)^n b_{(n+k-i)} a_{(m+i)} c \Big),\end{aligned} \tag{G.5}$$

with the usual parity modification in the super case. The binomial coefficients come from the expansion of translated formal variables. The chiral operad encodes (G.5) as a single Maurer–Cartan equation. Configuration-space forms can realize the domains and their boundary combinatorics, but they do not manufacture these mode coefficients.

G.6 Four labels

The partition lattice of four labels has ranks $0, 1, 2, 3$. The rank-one elements are the six partitions with one pair and two singletons. Rank two has two species:

$$12|34 \quad \text{and its two analogues,} \qquad 123|4 \quad \text{and its three analogues.}$$

Rank three is 1234. The two species of rank-two partitions separate the two mechanisms in the square of the differential:

- The interval from $1|2|3|4$ to $12|34$ contains two paths. Their cancellation is the disjoint-pair sign.
- The interval from $1|2|3|4$ to $123|4$ contains three paths. Their sum is the Jacobiator on the first three factors, tensored with the identity on the fourth.

The next differential, from rank two to rank three, tests compatibility of Jacobi with a fourth insertion. No new quadratic identity appears: the higher relation follows because the collision complex is the Chevalley complex of the chiral Lie algebra. In the Fulton–MacPherson resolution, however, the rank-three geometry has nontrivial screens. Their cohomology contributes additional form classes without changing the algebraic Chevalley differential.

G.7 The screen of three points

Consider three points on a complex tangent line modulo translation and dilation. Fix the first two at 0 and 1; the third gives a coordinate

$$u \in \mathbb{P}^1 \setminus \{0, 1, \infty\}.$$

A compactification is $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$, with boundary at $0, 1, \infty$. Thus the screen is not contractible. Its open cohomology is

$$H^0 \cong \mathbb{C}, \qquad H^1 \cong \mathbb{C}\, d\log u \oplus \mathbb{C}\, d\log(1-u).$$

The relation among the three boundary residues is the global residue theorem. The combinatorial forest category over the triple collision does have a terminal corolla and therefore contractible nerve. This explicit example proves why the incidence resolution and the geometric screen complex must be distinguished.

G.8 Virasoro pole symbols

The Virasoro OPE is

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \tag{G.6}$$

Let D be the pair diagonal. The three singular terms of (G.6) lie in

$$N_{D/X^2}^{\otimes 4}, \qquad N_{D/X^2}^{\otimes 2} \otimes \langle T \rangle, \qquad N_{D/X^2} \otimes \langle \partial T \rangle.$$

The regular term lies in filtration degree zero. A change of coordinate acts triangularly on this filtered object; the Schwarzian term records the extension by the fourth-order central symbol. No scalar extracted from one term determines the full filtered operation.

The lambda bracket is

$$[T_\lambda T] = (\partial + 2\lambda)T + \frac{c}{12}\lambda^3.$$

The Jacobi identity for this polynomial is equivalent to the Virasoro Lie conformal algebra identity. The normally ordered product is then part of the corresponding vertex algebra operation. This gives a direct formal-disc test of the chiral selection theorem.

G.9 A BRST bicomplex

Let (A, d_A) be a dg chiral algebra and let $Q : A \to A$ be a degree-one square-zero derivation anticommuting with d_A:

$$Q^2 = 0, \qquad d_A Q + Q d_A = 0, \qquad Q\mu = \mu(Q \otimes 1 + 1 \otimes Q).$$

On the collision complex, Q acts on every coefficient factor. Denote this extension by Q_{coll}. The derivation identity gives

$$Q_{\mathrm{coll}} d_{\mathrm{coll}} + d_{\mathrm{coll}} Q_{\mathrm{coll}} = 0.$$

Hence

$$D = d_A + d_{\mathrm{coll}} + Q_{\mathrm{coll}}$$

satisfies $D^2 = 0$. Filter by ghost number. If the spectral sequence is bounded below and its first page is concentrated in ghost degree zero, then

$$H_Q^0(C^{\mathrm{coll}}(A)) \cong C^{\mathrm{coll}}(H_Q^0(A)).$$

The displayed anticommutation is the low-arity mechanism behind bar–BRST interchange. In Drinfeld–Sokolov reduction the complex is infinite and filtered, so convergence must be proved from the good grading and PBW filtration; the algebraic identity above remains exact at every finite stage.

G.10 The first modular sewings

The stable pairs with smallest complexity are $(0, 3)$ and $(1, 1)$. The space $\overline{\mathcal{M}}_{0,3}$ is a point, so a cyclic three-point operation is a tensor

$$\theta_{0,3} \in (V^{\otimes 3})^{\Sigma_3}$$

with the cohomological degree and determinant twist displayed in the modular cooperad. Contracting two legs with the inverse pairing produces a genus-one one-point tensor

$$\Delta\theta_{0,3} \in V.$$

The complex compactification $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$ has three boundary divisors, and the global residue theorem relates their logarithmic residues. A real associahedral interval inside $\overline{\mathcal{M}}_{0,4}(\mathbb{R})$ has two endpoints; Stokes' theorem on that interval gives equality between the two adjacent parenthesizations. Analytic continuation, or the symmetric operadic action, relates the three complex channels. With coefficient tensors inserted, these statements give the associativity relation for the associative operad or the Jacobi relation for the Lie operad.

A basic codimension-two modular relation occurs when a loop contraction and a separating contraction meet. Its algebraic image is the BV second-order identity. The relation is unconditional for an algebra over the modular operad: all contractions are already defined. It does not assert that a given chiral algebra possesses a nondegenerate cyclic pairing or convergent sewing amplitudes.

G.11 The free boson determinant

Let q satisfy $|q| < 1$. On the one-particle space of a chiral boson, sewing a cylinder acts diagonally with eigenvalues q^n, $n \geq 1$. Its Fredholm determinant is

$$\det(1 - T_q) = \prod_{n \geq 1} (1 - q^n).$$

Second quantization gives the oscillator partition function

$$\mathrm{Tr}_{\mathrm{Fock}}\, q^{L_0 - 1/24} = q^{-1/24} \prod_{n \geq 1} (1 - q^n)^{-1} = \eta(\tau)^{-1}, \qquad q = e^{2\pi i \tau}.$$

The equality has three exact readings: a trace in the representation, a determinant of the sewing operator, and a Gaussian functional integral. The equality does not identify the central charge with an arbitrary Koszul curvature; it identifies three computations of the same free-field amplitude.

G.12 A diagnostic table

Low-arity datum	Mathematical identity	Physical reading
pair collision	chiral operation supported on Δ	local OPE/contact term
triple collision	chiral Jacobi or Borcherds identity	independence of nested short-distance expansion
Arnold form	relation in $H^2(\mathrm{Conf}_3)$	cancellation of configuration-space propagators
Casimir coefficient	infinitesimal braid relation	flat KZ transport
three-point screen	$\mathcal{M}_{0,4}$ geometry	relative configuration in a collision
loop contraction	BV operator Δ	one-loop sewing
anomaly cocycle	failure of the quantum master equation	obstruction to quantization
Fredholm determinant	trace-class sewing identity	Gaussian partition function

Every row compares objects only after the indicated coefficient map or realization has been constructed. This is the low-arity form of the governing rule: geometry supplies the carrier, algebra supplies the local operation, and physics supplies a representation by observables and amplitudes.

Chapter H

Low-arity calculations for the ordered theory

H.1 Arity zero and one

Let $A \to \mathbf{1}$ be an augmented ordered chiral algebra and write $\bar{A}$ for the augmentation ideal. The reduced bar begins

$$B_0 = \mathbf{1}, \qquad B_1 = \bar{A}[1].$$

The internal differential on B_1 is the suspension of the differential of A. There is no multiplication face in arity one.

The coaugmentation $\mathbf{1} \to B$ represents the empty ordered configuration. The projection $B \to \bar{A}[1]$ is the primitive quotient. The reduced coproduct vanishes on B_1.

H.2 Arity two

In arity two,

$$B_2 = \bar{A}[1] \otimes \bar{A}[1].$$

For homogeneous $a, b \in \bar{A}$, the multiplication face is

$$b_2[a|b] = (-1)^{|a|}[ab]$$

with the suspension convention of Chapter F. The internal differential is

$$b_1[a|b] = [da|b] + (-1)^{|a|-1}[a|db].$$

The identity $b_1 b_2 + b_2 b_1 = 0$ is the Leibniz rule for multiplication.

Geometrically, the configuration of two ordered intervals has one collision face. The chiral coefficient of ab may have a pole along $z_1 = z_2$. The bar face composes the two letters; it does not take a residue of that pole.

H.3 Arity three and associativity

There are two multiplication faces:

$$[a|b|c] \longmapsto [ab|c], \qquad [a|b|c] \longmapsto [a|bc].$$

Applying the multiplication differential twice gives

$$b_2^2[a|b|c] = \pm[(ab)c - a(bc)].$$

The two boundary points of the compactified one-dimensional moduli of three ordered intervals have opposite orientations. Associativity makes the coefficient operations equal. Hence $b_2^2 = 0$.

For an A_∞ algebra, the compactification is the interval K_4. Its interior operation is m_3, and the boundary identity is

$$dm_3 + m_2(m_2 \otimes 1) - m_2(1 \otimes m_2) + \text{terms containing } m_1 = 0.$$

The strict ordered chiral algebra is the case $m_{r\geq 3} = 0$.

H.4 The first mixed square

Take four product disks

$$U_i \times I_i \subset X \times \mathbb{R}, \qquad I_1 < I_2 < I_3 < I_4.$$

Suppose the first two chiral points collide while the last two transverse intervals become adjacent. There are two codimension-two paths:

$$\text{chiral collision then interval fusion}, \qquad \text{interval fusion then chiral collision}.$$

The coefficient maps agree by the interchange law (39.3). The conormal orientation lines occur in opposite orders, so the two boundary contributions cancel in the total differential.

This is the smallest calculation proving that the chiral resolution and ordered bar can be totalized without conflating their differentials.

H.5 Arity two: order and the additional exchange datum

The ordered chamber

$$\{t_1 < t_2\} \subset \mathrm{Conf}_2(\mathbb{R})$$

is contractible. A path remains in the same chamber and returns no monodromy. There is no path to the chamber $t_2 < t_1$ without crossing $t_1 = t_2$.

In two transverse dimensions,

$$\mathrm{Conf}_2(\mathbb{R}^2) \simeq S^1.$$

A half-rotation exchanges the two points; its square is the generator of the pure braid group $P_2 \cong \mathbb{Z}$. Thus the R-matrix first appears after the E_2 thickening.

H.6 The free algebra

Let $A = T_X(M)$. In bar degree two the map

$$M^{\otimes p} \mid M^{\otimes q} \longmapsto M^{\otimes(p+q)}$$

concatenates words. The contracting homotopy splits off the first letter of every word of total tensor length at least two. Consequently only the one-letter class survives. This gives the contraction of Theorem 58.1 explicitly.

H.7 The square-zero algebra

For $A = \mathbf{1} \oplus M$ with $M^2 = 0$, all multiplication faces vanish. Hence

$$H_\bullet \operatorname{Bar}_X^{\mathrm{ord}}(A) \cong \bigoplus_{r \geq 0} H_\bullet(M[1]^{\otimes r}).$$

The deconcatenation coproduct cuts a word at every position. Continuous duality gives the completed free algebra on $M^\vee[-1]$.

H.8 A quiver calculation

For the quiver $1 \xrightarrow{u} 2 \xrightarrow{v} 3$, the relative bar over $S = \mathbf{1}e_1 \oplus \mathbf{1}e_2 \oplus \mathbf{1}e_3$ has

$$B_1 = M_u \oplus M_v, \qquad B_2 = M_v \otimes M_u.$$

The multiplication face sends $[v|u]$ to the path vu. If no relation is imposed, the path belongs to the algebra and the two-term resolution is exact. If the relation $vu = 0$ is imposed, $[v|u]$ becomes the quadratic Koszul cogenerator. The calculation is internal to $\mathrm{FactD}(X)$, so M_u and M_v may carry singular chiral coefficients.

H.9 The enveloping algebra

Let $A = U_X(\mathfrak{g})$. Antisymmetrization sends

$$x_1 \wedge \cdots \wedge x_r \longmapsto \sum_{\sigma \in \Sigma_r} \operatorname{sgn}(\sigma)[x_{\sigma(1)}| \cdots |x_{\sigma(r)}].$$

The bar multiplication differential becomes the Chevalley differential

$$\sum_{i<j} (-1)^{\epsilon_{ij}} [x_i, x_j] \wedge x_1 \wedge \cdots \widehat{x_i} \cdots \widehat{x_j} \cdots \wedge x_r.$$

In arity three its square is the Jacobiator. PBW proves that antisymmetrization is a quasi-isomorphism.

H.10 BRST in arity two

Let Q be an odd derivation of A with $Q^2 = 0$. On $[a|b]$,

$$Q_{\mathrm{bar}}[a|b] = [Qa|b] + (-1)^{|a|-1}[a|Qb].$$

The mixed commutator with the bar face is

$$(Q_{\mathrm{bar}}b + bQ_{\mathrm{bar}})[a|b] = \pm\big(Q(ab) - (Qa)b - (-1)^{|a|}a(Qb)\big) = 0.$$

Thus $b + Q_{\mathrm{bar}}$ is square-zero. The comparison of its cohomology with the bar of $H_Q(A)$ is the spectral-sequence question, not this identity.

H.11 The annulus and the torus

The transverse annulus amplitude is

$$\mathrm{HH}_\bullet(A) = A \otimes^L_{A\otimes A^{\mathrm{op}}} A.$$

The annular trace closes an ordered interval and raises the ribbon genus. The genus-one chiral block is obtained by sewing two marked points of a chiral sphere and raises the curve genus. Their lowest chain models both contain a trace, but they act on different moduli spaces. In the bimodular complex they occupy bidegrees $(0, 1)$ and $(1, 0)$.

H.12 The first Tannaka coefficient

Let a meromorphic line category have exchange

$$R(z) = 1 + \hbar r(z) + O(\hbar^2).$$

The quantum Yang–Baxter equation at order $\hbar^2$ gives

$$[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = 0.$$

For $r(z) = \Omega/z$, the coefficient of each commutator is paired with the Arnold three-term relation. This is the first nontrivial compatibility between chiral collision geometry and transverse exchange. It requires the E_2 line category; it is absent from the bare E_1 bar.

Chapter I

The stratified Fulton–MacPherson spectral sequence

The Fulton–MacPherson space carries two different complexes that are easily conflated. The incidence complex remembers how boundary strata meet. The geometric complex also remembers the cohomology of the screens. A spectral sequence relates them and states exactly when the incidence complex is sufficient.

I.1 A filtered space with normal-crossing boundary

Let Y be a smooth complex variety and

$$D = \bigcup_{a \in A} D_a$$

a simple normal-crossing divisor. For $J \subseteq A$, put

$$D_J = \bigcap_{a \in J} D_a, \qquad D_J^\circ = D_J \setminus \bigcup_{b \notin J} D_b.$$

Let $j : Y \setminus D \hookrightarrow Y$. The increasing filtration by depth of support on the boundary gives a Cousin-type spectral sequence for $Rj_*\mathbb{C}$ or, algebraically, for the logarithmic de Rham complex. In one standard indexing,

$$E_1^{-p,q} = \bigoplus_{|J|=p} H^{q-2p}(D_J)(-p) \Longrightarrow H^{q-p}(Y \setminus D). \tag{I.1}$$

The first differential is the alternating sum of Gysin or residue maps. Its signs are the determinant signs of Chapter F. Deligne's mixed Hodge theory identifies (I.1) with the weight spectral sequence when Y is proper.

For coefficients in a complex $\mathcal{L}$ constructible along the boundary stratification, the same filtration gives

$$E_1^{-p,q}(\mathcal{L}) = \bigoplus_{|J|=p} \mathbb{H}^{q-2p}(D_J, \mathcal{L}|_{D_J})(-p) \Longrightarrow \mathbb{H}^{q-p}(Y \setminus D, \mathcal{L}). \tag{I.2}$$

Nothing in the construction requires $\mathcal{L}$ to be constant. In chiral applications it is assembled from external tensor powers of the coefficient D-module and from principal-part lines along the normal directions.

I.2 Boundary strata of Fulton–MacPherson space

Let X be a smooth curve and $Y_I = \mathrm{FM}_I(X)$. Boundary divisors are indexed by subsets $S \subseteq I$, $|S| \geq 2$, subject to the convention determined by the chosen compactification. A collection $\mathcal{N}$ of subsets labels a nonempty stratum exactly when it is nested: any two members are disjoint or one contains the other. The nested set is equivalently a rooted collision forest F.

The open stratum associated with F has the form of an iterated bundle whose base records the root configurations on X and whose fibers are screen configuration spaces. On a curve, the screen attached to a vertex with r incoming branches is the compactification of r points on an affine line modulo translations and dilations. Its interior is

$$\mathrm{Conf}_r(\mathbb{A}^1)/(\mathbb{G}_a \rtimes \mathbb{G}_m).$$

For $r = 2$ the screen is a point. For $r = 3$ its interior is $\mathcal{M}_{0,4}$ and its compactification is $\overline{\mathcal{M}}_{0,4}$. For larger r it has nontrivial cohomology in several degrees.

Consequently, a stratum is not determined by the forest alone. The forest gives its incidence type; the screens give its internal topology.

I.3 The forest resolution as the row of constants

Let $p : Y_I \to X^I$ be the blowdown. Consider the filtration of a geometric coefficient complex $\mathcal{K}_I$ by boundary depth. The E_1 page is a sum over forests,

$$E_1^{p,q} = \bigoplus_{\substack{F \in \mathrm{For}(I) \\ |E(F)| = p}} R^q(p_F)_*\big(\mathcal{K}_I|_{D_F^\circ}\big) \otimes \mathrm{or}(F), \tag{I.3}$$

where p_F is the projection of the stratum to the corresponding partial diagonal. The differential contracts one edge and applies the induced restriction or residue map.

The row $q = 0$ contains the locally constant functions on every screen. If the coefficient system has no higher screen cohomology and no nontrivial local monodromy, then after barycentric or nerve replacement of the face incidence category and blockwise terminal-corolla normalization, this row maps to the combinatorial forest resolution $W_I^{\mathrm{FM}}(\mathfrak{g})$ of Chapter 43. The augmentation to the partition collision complex is a quasi-isomorphism because every rooted-tree category has a terminal corolla.

Criterion I.1 (Incidence sufficiency)**.** *The incidence forest complex computes the proper pushforward of $\mathcal{K}_I$ if the following two conditions hold:*

(i) $R^q(p_F)_*(\mathcal{K}_I|_{D_F^\circ}) = 0$ *for every* $q > 0$ *and every forest* F*;*

(ii) *the degree-zero pushforwards identify with the coefficient system used in the forest complex, compatibly with the vertical incidence differential and the corolla-normalized horizontal collision merge.*

The criterion is a theorem of spectral sequences: under the two conditions, (I.3) is concentrated in one row. It is deliberately strong. Constant coefficients on complex screens generally do not satisfy it.

I.4 The first nonconstant row

For a triple-collision vertex, the screen interior is $\mathbb{P}^1 \setminus \{0, 1, \infty\}$. Its first cohomology has rank two. Therefore the $q = 1$ row of (I.3) already appears in arity three. Its logarithmic generators may be taken to be

$$d \log u, \qquad d \log(1-u).$$

Their residues at $0, 1, \infty$ satisfy one linear relation. This row carries the first period data of ordered configurations. Integrating its forms along paths gives logarithms; iterating at higher arity gives multiple polylogarithms and, for KZ transport, multiple zeta values.

The chiral Chevalley differential does not require these periods. It uses the diagonal coefficient operation and the incidence of partitions. Configuration-space integral realizations do require them. The spectral sequence locates the exact extra data.

I.5 Orlik–Solomon cohomology

For $X = \mathbb{A}^1$, the cohomology algebra of $\mathrm{Conf}_n(X)$ is the Orlik–Solomon algebra of the braid arrangement. It is generated in degree one by ω_{ij} with relations

$$\omega_{ij} = \omega_{ji}, \qquad \omega_{ij} \wedge \omega_{jk} + \omega_{jk} \wedge \omega_{ki} + \omega_{ki} \wedge \omega_{ij} = 0.$$

A no-broken-circuit basis gives the Poincaré polynomial

$$P_{\mathrm{Conf}_n(\mathbb{C})}(t) = \prod_{j=1}^{n-1} (1 + jt). \tag{I.4}$$

The top coefficient $(n-1)!$ is the dimension of the geometric screen cohomology; a chiral bar chain group additionally contains its algebraic coefficient object and all differentials. To form a geometric bar model one tensors or convolves (I.4) with the algebraic coefficient complex and includes all differentials. Dimensions of the tensor factors alone do not determine its homology.

I.6 Nested residues

On a normal-crossing chart with local boundary parameters t_e indexed by the edges of a collision forest, a logarithmic form has an expansion

$$\alpha = \sum_{J \subseteq E(F)} \left(\bigwedge_{e \in J} \frac{dt_e}{t_e} \right) \wedge \alpha_J.$$

The depth filtration counts $|J|$. The first differential of the residue spectral sequence extracts $\alpha_{\{e\}}$ and restricts it to $t_e = 0$. A second residue extracts $\alpha_{\{e,f\}}$. Because

$$\frac{dt_e}{t_e} \wedge \frac{dt_f}{t_f} = -\frac{dt_f}{t_f} \wedge \frac{dt_e}{t_e},$$

the codimension-two contribution cancels between the two orders. When the coefficient system itself changes under contraction, the remaining coefficient is the associator or Jacobiator of that system.

For higher poles, ordinary Poincare residue still acts only on the logarithmic conormal factor. The order-m_e coefficient is first paired with the dual m_e-jet of the normal line, producing a logarithmic coefficient; ordinary residue then removes dt_e/t_e. Invariantly this is the pairing

$$N_e^{\otimes m_e} \otimes N_e^{\vee \otimes m_e} \longrightarrow \mathcal{O}$$

followed by Poincare residue. Different pole orders remain in different associated-graded normal lines.

I.7 The Rees spectral sequence

Let $F_{\mathbf{m}}$ be the multi-pole filtration along the collision divisors. The multi-Rees object

$$\mathcal{R}(\mathcal{K}_I) = \bigoplus_{\mathbf{m} \geq 0} F_{\mathbf{m}} \mathcal{K}_I \, \mathbf{t}^{\mathbf{m}}$$

interpolates between the filtered chiral coefficient system and its normal-cone associated graded. Filtering the geometric complex simultaneously by boundary depth and pole order gives a bicomplex. One spectral sequence first resolves collision incidence and then pole order; the other first passes to normal symbols and then resolves the forest geometry. Under boundedness in each finite arity and pole window, the two converge to the same total cohomology.

This interchange explains the relation between chiral and Poisson vertex structures. The pole associated graded is classical; the extension data between pole orders is quantum. Fulton–MacPherson depth is independent of pole degree; it records how many nested collisions occur. The two filtrations must not be collapsed into one integer.

I.8 A comparison theorem for finite windows

Fix a finite set I, a maximum pole order M, and a bounded cohomological range. Let $\mathcal{K}_I^{\leq M}$ be the truncated principal-part coefficient system. Suppose there is a morphism

$$\Phi_I : W_I^{\mathrm{FM}}(\mathfrak{g}; \leq M) \longrightarrow Rp_* \mathcal{K}_I^{\leq M} \tag{I.5}$$

that is compatible with the vertical incidence differential and the corolla-normalized collision merge, and induces an isomorphism on every stratumwise cohomology group in the chosen range.

Theorem I.2 (Finite-window comparison)**.** *Under the preceding hypotheses,* (I.5) *is a quasi-isomorphism in the chosen cohomological range.*

Proof. Filter source and target by boundary depth. The map induces a morphism of the corresponding spectral sequences. By hypothesis it is an isomorphism on E_1 in the range. Since the filtration is finite for fixed I, the comparison theorem for spectral sequences gives an isomorphism on the associated graded of cohomology, hence on cohomology itself. □

The theorem is deliberately local in arity, pole order, and cohomological degree. Passage to an unbounded completed object requires control of inverse limits. Surjective transition maps imply the Mittag–Leffler condition and kill $\varprojlim^1$; absent such control, completion may create cohomology.

I.9 Renormalization and compactified configurations

Let M be a real manifold and consider a perturbative quantum field theory. A Feynman amplitude associated with a graph Γ is initially a differential form or distribution on the configuration space of its vertices away from the diagonals. Pulling it to a compactification separates the asymptotic regimes of every collision forest. Renormalization is an extension across the boundary satisfying locality and compatibility with disjoint unions.

The forest formula of perturbative renormalization has the same incidence category as the Fulton–MacPherson boundary. A divergent subgraph corresponds to a cluster of vertices; nested and disjoint divergent subgraphs correspond to nested and disjoint boundary divisors. Counterterms live on the associated diagonals. Stokes' theorem on the compactification converts the failure of a Ward identity into boundary integrals. The quantum master equation holds when the resulting local cocycle is exact or vanishes.

This physical realization uses more than the incidence complex:

(i) a propagator or parametrix;
(ii) a gauge-fixing operator;
(iii) an extension of singular distributions;
(iv) a renormalization prescription;
(v) a proof that the counterterms preserve the required symmetries.

When these data are fixed, the geometric boundary complex computes the anomaly. The combinatorial identity $\partial^2 = 0$ alone does not establish the absence of an anomaly.

I.10 Degeneration of curves

For a family of curves approaching a node, the compactification contains collision divisors of marked points and clutching divisors of the curve simultaneously: collisions of marked points in the fibers and degeneration of the curve. Logarithmic Fulton–MacPherson spaces resolve the first relative to the second. A local model of the node is

$$xy = q,$$

where q is the smoothing parameter. Collision parameters in the two branches and the nodal parameter q define independent conormal directions. The total residue complex therefore has a multigrading.

Separating degeneration corresponds to two components joined at the node. Nonseparating degeneration identifies two points on one component. The associated maps on conformal blocks are the two sewing operations of a modular operad. The screen cohomology of point collisions and the vanishing-cycle or nearby-cycle data of the node are distinct. A correct global comparison must include both.

I.11 Consequences

The spectral sequence yields four structural consequences.

(1) The partition collision complex is canonical and computes chiral Chevalley chains.
(2) The forest incidence complex is a canonical resolution of the partition complex.
(3) The geometric Fulton–MacPherson complex contains additional screen cohomology.
(4) An equality between the geometric and algebraic complexes is a comparison theorem, not a consequence of contractibility of the forest nerve.

From the physical side, the same four statements say that locality determines the pattern of counterterms, while the propagator and its periods determine their numerical amplitudes. The first is combinatorial and functorial. The second is analytic and theory-dependent. Chiral Koszul duality is the algebraic local-to-global theorem; configuration integrals are geometric realizations of its coefficient maps.

Chapter J

Formal neighborhoods, normal jets, and degeneration coordinates

The local geometry of a chiral operation is the geometry of a divisor and its formal neighborhood. Pole coefficients, delta derivatives, coordinate changes, and sewing parameters are four expressions of normal geometry.

J.1 The formal completion of a diagonal

Let X be a smooth curve and let $\Delta \subset X \times X$ be the diagonal. Its ideal sheaf $\mathcal{I}_\Delta$ satisfies

$$\mathcal{I}_\Delta/\mathcal{I}_\Delta^2 \cong \Omega^1_X, \qquad N_{\Delta/X^2} \cong T_X.$$

The m-th infinitesimal neighborhood is

$$\Delta^{(m)} = \operatorname{Spec}\bigl(\mathcal{O}_{X^2}/\mathcal{I}_\Delta^{m+1}\bigr).$$

The associated graded algebra of the formal completion is

$$\operatorname{gr}_{\mathcal{I}} \mathcal{O}_{\widehat{X^2}_\Delta} \cong \operatorname{Sym}_{\mathcal{O}_X}(\mathcal{I}_\Delta/\mathcal{I}_\Delta^2) \cong \operatorname{Sym}_{\mathcal{O}_X}(\Omega^1_X). \tag{J.1}$$

Dualizing gives the algebra of normal differential operators

$$\operatorname{Sym}_{\mathcal{O}_X}(T_X).$$

A coordinate difference $t = z_1 - z_2$ trivializes the conormal line and identifies the formal completion with $\mathcal{O}_X[[t]]$. A different coordinate changes t by a formal power series with invertible linear term. The formal neighborhood, not the chosen series, is invariant.

J.2 Principal parts

Let $D \subset Y$ be a smooth divisor with local equation $t = 0$. For $m \geq 0$, the sheaf $\mathcal{O}_Y(mD)$ consists locally of functions with poles of order at most m. There is an exact sequence

$$0 \longrightarrow \mathcal{O}_Y(mD) \longrightarrow \mathcal{O}_Y((m+1)D) \xrightarrow{\sigma_{m+1}} N_{D/Y}^{\otimes(m+1)} \longrightarrow 0. \tag{J.2}$$

To verify the target, write a section as

$$f = t^{-(m+1)}a_{-(m+1)} + t^{-m}a_{-m} + \cdots .$$

Modulo $\mathcal{O}_Y(mD)$ only the leading coefficient remains. If $t' = ut + O(t^2)$, then

$$(t')^{-(m+1)} = u^{-(m+1)}t^{-(m+1)} + O(t^{-m}),$$

so $a_{-(m+1)}$ transforms as the $(m+1)$-st power of the normal line. This proves (J.2) without choosing a global equation.

The pole filtration

$$F_m\mathcal{O}_Y(*D) = \mathcal{O}_Y(mD)$$

has associated graded

$$\mathrm{gr}^F_m \mathcal{O}_Y(*D) \cong N_{D/Y}^{\otimes m}, \qquad m \geq 1. \tag{J.3}$$

A complete OPE is a filtered morphism with coefficients in $\mathcal{O}_Y(*D)$. Its mode of order $m-1$ is the image under σ_m.

J.3 Differential operators and delta derivatives

Let $i : D \hookrightarrow Y$ be a smooth divisor. Distributions supported on D are generated by normal derivatives of the delta distribution. Algebraically, the Kashiwara equivalence identifies right D_Y-modules supported on D with right D_D-modules. Locally, a supported distribution has the form

$$T = \sum_{m=0}^{M} T_m\, \partial_t^m \delta(t). \tag{J.4}$$

The coefficient pairing of principal parts with normal jets is the formal residue identity

$$\mathrm{Res}_{t=0}\big(t^n t^{-m-1}\, dt\big) = \delta_{mn}. \tag{J.5}$$

The dual jet functional may be represented distributionally by a normalized derivative of $\delta(t)$. In complex analysis, (J.5) is Cauchy's coefficient formula; in D-module theory it is the pairing between local cohomology along the diagonal and normal differential operators.

For a vertex algebra, the singular commutator is

$$[Y(a,z), Y(b,w)] = \sum_{m\geq 0} Y(a_{(m)}b, w)\frac{1}{m!}\partial_w^m \delta(z-w). \tag{J.6}$$

Equation (J.6) is the distributional form of the chiral operation. It contains every singular mode. The normally ordered product is recovered from the regular extension, not from the supported commutator alone.

J.4 The Rees deformation

The Rees algebra of the ideal of a divisor is

$$\mathcal{R}(\mathcal{I}_D) = \bigoplus_{m\geq 0} \mathcal{I}_D^m u^{-m}.$$

Its relative spectrum over $\mathbb{A}^1_u$ is the deformation to the normal cone. In a local chart one may write

$$t = ux. \tag{J.7}$$

For $u \neq 0$, the fiber is isomorphic to the original neighborhood. For $u = 0$, the coordinate x is the normal coordinate. The corresponding construction for the pole filtration uses inverse powers and interpolates between meromorphic kernels and normal symbols.

For several normal-crossing divisors D_e with local equations t_e, the multi-Rees space has equations

$$t_e = u_e x_e.$$

Its deepest special fiber is the direct sum of the normal lines. On a Fulton–MacPherson stratum indexed by a forest F, the indices e are the internal edges of F. This proves the normal-bundle splitting

$$N_{D_F/\mathrm{FM}_I(X)} \cong \bigoplus_{e \in E(F)} N_e. \tag{J.8}$$

Operadic substitution on the associated graded is substitution of the normal coordinates x_e along contracted subtrees.

J.5 Coordinate changes and triangularity

Let $w = f(z)$ with $f'(z) \neq 0$. Near $z_1 = z_2 = z$, put $t = z_1 - z_2$. Taylor expansion gives

$$f(z+t) - f(z) = f'(z)t + \frac{1}{2}f''(z)t^2 + \frac{1}{6}f'''(z)t^3 + O(t^4), \tag{J.9}$$

$$\frac{1}{f(z+t) - f(z)} = \frac{1}{f'(z)t} - \frac{f''(z)}{2f'(z)^2} + O(t), \tag{J.10}$$

$$\frac{1}{(f(z+t) - f(z))^2} = \frac{1}{f'(z)^2t^2} - \frac{f''(z)}{f'(z)^3t} + O(1). \tag{J.11}$$

A pole of order m therefore transforms into a linear combination of poles of orders at most m, with leading factor $f'(z)^{-m}$. The associated graded is tensorial, while the filtered object is triangular.

This is exactly the transformation law of vertex modes. The top pole of an OPE between primary fields transforms by the tensor power prescribed by its conformal weights. Lower poles acquire corrections from derivatives of f. The complete chiral operation is coordinate-free because these corrections are the transition functions of the filtered principal-parts sheaf.

J.6 The Schwarzian derivative

The Schwarzian derivative is

$$\{f, z\} = \frac{f'''(z)}{f'(z)} - \frac{3}{2}\left(\frac{f''(z)}{f'(z)}\right)^2.$$

The Schwarzian derivative obeys the cocycle identity

$$\{f \circ g, z\} = \{f, g(z)\}g'(z)^2 + \{g, z\}. \tag{J.12}$$

A stress tensor of central charge c transforms as

$$T(z) = f'(z)^2 T(f(z)) + \frac{c}{12}\{f, z\}. \tag{J.13}$$

The first term is the tensorial transformation of a quadratic differential. The second is an affine cocycle. Thus stress tensors form a torsor for projective connections rather than the vector space of quadratic differentials.

Equation (J.13) follows from the fourth-order central term of the Virasoro OPE. The central term is a fourth normal symbol, while the Schwarzian is the extension class that mixes it into filtration degree zero under coordinate change. This is a precise example of why the associated graded and the filtered chiral operation contain different information.

J.7 The affine connection and the shifted level

For affine currents, the invariant form gives the second normal symbol. The KZ connection contains instead the Casimir divided by $k + h^\vee$:

$$\nabla = d - \frac{1}{k + h^\vee} \sum_{i<j} \Omega_{ij}\, d\log(z_i - z_j).$$

The level k and the shifted denominator $k + h^\vee$ arise in different calculations. The first is the coefficient of the central extension in the current OPE. The second is the coefficient of the Sugawara stress tensor and the KZ transport. Their relation is a theorem of the current-algebra representation, not a normalization identity that allows one to replace either by the other.

At $k = -h^\vee$, the Sugawara construction is singular. The current OPE and its level remain defined. The KZ connection in this normalization is singular. The Feigin–Frenkel centre appears. These facts are compatible because they concern different pieces of structure.

J.8 Prime form and normalized bidifferential

Let X be a compact Riemann surface of genus g, choose a symplectic basis of homology, and let Ω be the period matrix. The prime form $E(p, q)$ is a $(-1/2, -1/2)$-form with a simple zero on the diagonal. Locally,

$$E(p, q) = \frac{z(p) - z(q)}{\sqrt{dz(p)}\sqrt{dz(q)}}\big(1 + O((z(p) - z(q))^2)\big).$$

The normalized fundamental bidifferential is

$$B(p, q) = d_p d_q \log E(p, q) + \sum_{i,j=1}^{g} c_{ij}\, \omega_i(p)\omega_j(q), \tag{J.14}$$

where the coefficients c_{ij} are chosen, according to convention, so that the A-periods vanish. It has a double pole with biresidue one on the diagonal. Changing the symplectic basis acts by a modular transformation and changes the decomposition into prime-form and holomorphic pieces, while the normalized bidifferential transforms covariantly.

The prime form is not a global replacement for the coordinate difference as an ordinary function. It is a section of a line bundle with quasi-periodicity. A propagator constructed from it requires a choice of normalization, spin structure, or Green function fixed by the field content, spin structure, and normalization of the Green operator. Its local singular symbol is universal; its harmonic completion is global.

J.9 Plumbing coordinates

A nodal curve has local equation $xy = 0$. A smoothing is

$$xy = q, \qquad |q| < 1. \tag{J.15}$$

Removing discs around $x = 0$ and $y = 0$ and identifying the annuli by $y = q/x$ produces the smooth fiber. The parameter q is a normal coordinate to the Deligne–Mumford boundary divisor.

For a conformal field theory, sewing inserts a complete set of states:

$$\mathcal{A}_q = \sum_{\alpha} q^{L_0 - c/24} |\alpha\rangle\langle\alpha|, \tag{J.16}$$

with the dual basis defined by the two-point pairing. In a rational theory the sum is organized by finitely many simple modules and convergent graded traces. In a general vertex algebra the expression may be formal. The algebraic clutching map and the analytic convergence of (J.16) are separate assertions.

The factor $q^{-c/24}$ is the vacuum energy on the cylinder. Under a modular change of plumbing coordinate it contributes to the projective anomaly. The same central charge appears in the Virasoro OPE because both are representations of the same central extension of vector fields. The equality is specific and exact; it does not identify central charge with unrelated levels or arithmetic weights.

J.10 Separating and nonseparating nodes

A separating node joins curves of genera g_1 and g_2. Sewing gives a bilinear composition

$$\mathbb{V}_{g_1,n_1+1} \otimes \mathbb{V}_{g_2,n_2+1} \longrightarrow \mathbb{V}_{g_1+g_2,n_1+n_2}.$$

A nonseparating node identifies two marked points of one curve and contracts them with the inverse pairing:

$$\mathbb{V}_{g-1,n+2} \longrightarrow \mathbb{V}_{g,n}.$$

The separating and nonseparating clutching maps are the two modular-operad compositions. Their codimension-two compatibility is the geometric source of the bracket and loop terms in the quantum master equation.

The normal parameter q belongs to the base of the family of curves. Collision parameters t_e belong to relative configuration spaces in the fibers. In a degeneration with colliding insertions, both occur. A superconnection may mix their differentials, but an equation that equates a fiberwise endomorphism with dq/q or with a Hodge class is ill-typed unless the total complex and its base degrees have been defined.

J.11 Tangential base points

The spaces $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ and $\mathcal{M}_{0,n}$ are not simply connected. Iterated integrals between boundary points require tangential base points. A tangent vector at 0 specifies the asymptotic direction of approach and regularizes logarithmic divergences. The KZ associator is the regularized parallel transport from the tangent vector at 0 to the tangent vector at 1.

The choice is geometric: it is a nonzero vector in the normal line to a boundary divisor. Changing it conjugates the regularized monodromy by an exponential of the residue. This is the local origin of gauge dependence of associators. The braid category is invariant up to braided equivalence; a particular power-series associator is a representative.

J.12 Reality and positivity

Algebraic chiral operations are defined over a field and do not include a Hilbert-space structure. A unitary conformal field theory adds an antilinear involution and a positive Hermitian form compatible with the modes. The adjoint relation is

$$L_n^\dagger = L_{-n}$$

for the Virasoro modes in radial quantization, with analogous formulas for primary fields. Reflection positivity is an analytic inequality, not a consequence of the chiral Jacobi identity.

Similarly, a Euclidean Green function depends on a real structure and metric, while the singular chiral kernel depends only on complex geometry. The chiral algebra determines protected holomorphic quantities. Reconstruction of a unitary full conformal field theory requires left–right pairing, modular invariance, and positivity. These are additional structures.

J.13 Local geometry summarized

The formal neighborhood of a diagonal contains the jets. The pole filtration contains the singular modes. The Rees deformation contains the classical limit. The Fulton–MacPherson normal bundle contains nested collision scales. The plumbing parameter contains nodal sewing. A projective connection contains the stress-tensor anomaly. Each object is a line or filtered sheaf attached to a geometric normal direction. This is the local geometry in which the coordinate formulas of chiral algebra and the short-distance formulas of quantum field theory become intrinsic.

Chapter K

Field-theoretic derivations

Actions and Ward identities construct the physical factorisation structures. Every identification is stated after the boundary observables, pairings, and comparison maps have been specified.

K.1 Classical BV theory

A classical gauge theory in the Batalin–Vilkovisky formalism consists of a graded space of fields $\mathcal{E}$, an odd symplectic pairing of cohomological degree -1, and an action $S \in \mathcal{O}(\mathcal{E})$ of degree zero. The odd symplectic form induces a Poisson bracket $\{-,-\}$ of degree one. The classical master equation is

$$QS + \frac{1}{2}\{S, S\} = 0, \tag{K.1}$$

where Q is the linearized differential. Equivalently, the Hamiltonian vector field

$$d_S = Q + \{S, -\}$$

satisfies $d_S^2 = 0$.

The Taylor coefficients of S define cyclic multilinear operations. Equation (K.1) is their L_∞ relation. If the fields are local sections of bundles on a manifold, the operations are supported on the small diagonal. Thus the local L_∞ algebra of fields is the field-theoretic analogue of a diagonal-supported chiral operation.

Gauge fixing chooses a Lagrangian complement and a parametrix for the kinetic operator. This choice is needed to define Feynman amplitudes. The classical local operations and their master equation do not depend on the choice up to canonical homotopy.

K.2 Quantum master equation and anomaly

A regularized BV Laplacian Δ is a second-order operator of degree one. The quantum master equation is

$$(Q + \hbar\Delta)e^{S/\hbar} = 0, \tag{K.2}$$

or equivalently

$$QS + \frac{1}{2}\{S, S\} + \hbar\Delta S = 0. \tag{K.3}$$

In an interacting field theory the action is replaced by a scale-dependent effective interaction and Δ by a scale-dependent regularized operator. Renormalization constructs a compatible family of solutions when the obstruction vanishes [58, 59].

Suppose a solution has been constructed modulo $\hbar^N$. The coefficient of $\hbar^N$ in the failure of (K.3) is closed for the classical differential d_S. Its cohomology class is independent of changes of counterterm. This is the anomaly class. It vanishes exactly when the solution extends to the next order. This obstruction theory is the physical counterpart of the filtered Maurer–Cartan obstruction theory of Chapter 54.

The equality between the stable-graph master equation and the BV quantum master equation is exact for a cyclic local field theory: vertices are Taylor coefficients of S, internal edges are contractions by the propagator, and loops are contractions of two flags of one connected graph. The equality is an equality of graph expansions. It does not assert that an arbitrary chiral algebra is the boundary of a given field theory.

K.3 Factorization observables

For an open set U, let $\mathrm{Obs}^{\mathrm{cl}}(U)$ be the Chevalley–Eilenberg cochains of compactly supported fields in U. Extension by zero and disjoint union define structure maps

$$\mathrm{Obs}^{\mathrm{cl}}(U_1) \otimes \cdots \otimes \mathrm{Obs}^{\mathrm{cl}}(U_n) \longrightarrow \mathrm{Obs}^{\mathrm{cl}}(V)$$

for disjoint $U_i \subset V$. The cosheaf and descent properties make $\mathrm{Obs}^{\mathrm{cl}}$ a factorization algebra. A perturbative quantization deforms it to Obs^q over $\mathbb{C}[[\hbar]]$.

If translations in a specified set of n real directions are generated by Q-exact stress-tensor components, the factorization algebra is locally constant in those directions and determines an E_n algebra. If dependence on one complex direction is holomorphic, restriction to a holomorphic curve yields a chiral factorization algebra. Boundary conditions replace the bulk field complex by a relative complex and produce a boundary factorization algebra.

A comparison with an algebraic chiral algebra is a quasi-isomorphism

$$\Phi : U^{\mathrm{ch}}(\mathfrak{g}) \longrightarrow \mathrm{Obs}^q_\partial$$

compatible with factorization. Once Φ is given, the algebraic chiral bracket equals the renormalized boundary OPE in cohomology.

K.4 Chern–Simons theory and affine currents

Let M be an oriented three-manifold, G a compact Lie group with invariant form $(,)$, and $k \in \mathbb{Z}$. The Chern–Simons action is

$$S_{\mathrm{CS}}(A) = \frac{k}{4\pi} \int_M \left((A, dA) + \frac{2}{3}(A, A \wedge A) \right). \tag{K.4}$$

On a manifold with boundary, variation gives

$$\delta S_{\mathrm{CS}} = \frac{k}{2\pi} \int_M (\delta A, F_A) + \frac{k}{4\pi} \int_{\partial M} (A, \delta A).$$

A boundary condition chooses a Lagrangian subspace for the boundary symplectic form. In complex coordinates on the boundary, fixing one component produces a chiral boundary theory. The residual gauge symmetry has current algebra

$$J^a(z)J^b(w) \sim \frac{k(a,b)}{(z-w)^2} + \frac{f^{ab}{}_c J^c(w)}{z-w}. \tag{K.5}$$

The double pole is the boundary central extension determined by the coefficient of (K.4). The simple pole is the Lie bracket. The one-loop shift by $h^\vee$ enters the effective coupling and the Sugawara construction; it is not a change in the classical coefficient of the central extension.

Wilson lines are labelled by representations of G. Their expectation values depend on framing. Canonical quantization on a surface gives the space of conformal blocks of the affine algebra. Witten's path-integral derivation of knot invariants identifies the braiding and ribbon trace with quantum-group data [106, 117]. The Drinfeld–Kohno theorem supplies the algebraic comparison of braid categories.

K.5 One-loop exactness in a holomorphic gauge

A partially holomorphic gauge can make Chern–Simons theory a deformation of a topological–holomorphic BF theory. In the gauge constructed by Gwilliam and Williams, the perturbative quantization is one-loop exact and the bulk–boundary relation with chiral WZW theory is explicit [89]. One-loop exactness means that higher-loop counterterms vanish in that gauge after the local counterterm complex has vanishing cohomology in loop order greater than one. It does not mean that every boundary correlator is Gaussian; tree-level nonlinearity and representation-theoretic sums remain.

The exactness result supplies a rigorous model in which the boundary current OPE, the anomaly shift, and the factorization algebra of observables can be compared at chain level. It is one of the clearest physical realizations of chiral Koszul duality: the local Lie algebra of gauge transformations passes to Chevalley cochains, while boundary factorization organizes their states and operators.

K.6 Four-dimensional Chern–Simons theory

Let Σ be an oriented real surface and C a complex curve with meromorphic one-form ω. Four-dimensional Chern–Simons theory has action

$$S_{4d}(A) = \frac{1}{2\pi\hbar}\int_{\Sigma\times C} \omega \wedge \left((A, dA) + \frac{2}{3}(A, A\wedge A)\right). \tag{K.6}$$

The theory is topological along Σ and holomorphic along C. Wilson lines supported on curves in Σ and located at points of C possess a spectral parameter. Crossing two lines gives an R-matrix. The quantum Yang–Baxter equation follows from isotopy invariance of three line crossings. Rational, trigonometric, and elliptic kernels arise from the geometry of (C, ω) [63, 64].

The line-operator OPE gives a tensor category with spectral parameters. A Yangian or quantum loop algebra is recognized when the category is generated by evaluation representations and the R-matrix and relations agree with the corresponding presentation. The field theory provides the monodromy and operator products; the algebraic recognition identifies the named Hopf algebra.

K.7 The Polyakov anomaly

For a two-dimensional conformal field theory on a Riemann surface with metric g, the partition function is not generally invariant under a Weyl rescaling $g \mapsto e^{2\sigma} g$. The infinitesimal anomaly is

$$\delta_\sigma \log Z[g] = \frac{c}{24\pi} \int_X \sigma R_g \, d\mathrm{vol}_g + \text{boundary terms}. \tag{K.7}$$

Integrating gives the Polyakov action, up to local counterterms [108]. The coefficient c is the Virasoro central charge. On the moduli space of complex structures, the same anomaly is the curvature of a determinant or Hodge line.

For a free scalar, Gaussian integration gives

$$Z[g] \propto (\det{}' \Delta_g)^{-1/2}$$

with zero-mode factors. The variation of the zeta-regularized determinant reproduces (K.7) with $c = 1$. For free fermions, ghosts, and systems of several fields, determinants and Pfaffians give the corresponding central charges. The equality between OPE central charge and determinant-line curvature is a local index theorem for that field theory.

K.8 BRST quantization of the bosonic string

In conformal gauge, the bosonic string contains a matter CFT of central charge c_m and a (b, c) ghost system of central charge -26. The BRST current has the form

$$j_{\mathrm{BRST}} = c\left(T_m + \frac{1}{2} T_{gh}\right) + \text{derivative term}.$$

The BRST charge $Q = \oint j_{\mathrm{BRST}}$ satisfies

$$Q^2 \propto (c_m - 26) \sum_{n \in \mathbb{Z}} (n^3 - n) c_{-n} c_n. \tag{K.8}$$

Therefore $Q^2 = 0$ precisely when the total Virasoro central charge vanishes, apart from the intercept condition in the state-space formulation. This is anomaly cancellation. It is an exact equality of central extensions in the coupled matter–ghost chiral algebra.

The number 26 is not the value of a universal Koszul conductor. It is the negative central charge of the reparametrization ghost system of weights $(2, -1)$. Similar BRST systems in other reductions have different ghost contents and different anomaly polynomials.

K.9 Conformal blocks as states of quantization

Canonical quantization of Chern–Simons theory on a surface Σ produces a Hilbert space depending on the complex structure used in geometric quantization. Projective flatness identifies these spaces as the complex structure varies. The resulting projective bundle is identified with affine conformal blocks. Sewing surfaces corresponds to gluing three-manifolds and summing over intermediate representations.

For a modular tensor category with simple objects i, the dimension of the genus-g state space is

$$\dim \mathcal{H}_g = \sum_i S_{0i}^{2-2g}. \tag{K.9}$$

The resulting dimension formula is the Verlinde formula. The full modular functor contains the mapping-class-group representation and gluing maps, not only the dimensions [91, 116].

K.10 Cardy asymptotics

Let

$$Z(\tau, \bar{\tau}) = \mathrm{Tr}_{\mathcal{H}}\, q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - \bar{c}/24}$$

be a modular-invariant torus partition function with a discrete spectrum, nonnegative degeneracies, and a unique vacuum that dominates the low-temperature limit. Applying the modular transformation $\tau \mapsto -1/\tau$ and a saddle-point estimate gives the asymptotic density

$$\log \rho(h, \bar{h}) \sim 2\pi \sqrt{\frac{c_{\mathrm{eff}} h}{6}} + 2\pi \sqrt{\frac{\bar{c}_{\mathrm{eff}} \bar{h}}{6}}. \tag{K.10}$$

The assumptions are essential. Logarithmic theories, continuous spectra, noncompact targets, and additional light states can change the asymptotics.

For asymptotically AdS_3 gravity with Brown–Henneaux boundary conditions,

$$c = \frac{3\ell}{2G}.$$

When the gravitational spectrum satisfies the hypotheses leading to (K.10), the Cardy entropy agrees with the Bekenstein–Hawking entropy of the BTZ black hole. The equality identifies a regime of a gravitational theory with a regime of a boundary CFT; it is not a formal consequence of the Virasoro algebra alone [56].

K.11 Entanglement entropy

For the vacuum of a unitary two-dimensional CFT on the line, the replica method computes the entropy of an interval of length L with ultraviolet cutoff ϵ:

$$S_A = \frac{c + \bar{c}}{6} \log \frac{L}{\epsilon} + s_0. \tag{K.11}$$

The proof uses twist fields of dimensions

$$h_n = \frac{c}{24}\left(n - \frac{1}{n}\right), \qquad \bar{h}_n = \frac{\bar{c}}{24}\left(n - \frac{1}{n}\right),$$

and analytic continuation in the replica number. Equation (K.11) depends on the state, geometry, and full left–right CFT. A holomorphic chiral algebra supplies the central charge and twist-field constraints but not by itself the density matrix.

Relative entropy, modular Hamiltonians, and topological entanglement contain further information. They cannot be classified by the depth of a chiral obstruction tower without a theorem connecting that tower to replica correlators.

K.12 Boundary algebras in supersymmetric gauge theory

Topological or holomorphic twists of supersymmetric gauge theories often produce protected boundary operator algebras. Costello and Gaiotto construct vertex algebras associated with boundary conditions in three-dimensional $\mathcal{N} = 4$ gauge theories and relate their modules to bulk line defects [60]. Costello, Dimofte, and Gaiotto construct boundary chiral algebras for holomorphic twists of three-dimensional $\mathcal{N} = 2$ theories.

These constructions supply examples of the comparison map Φ. Mirror symmetry or duality of the bulk can induce equivalences of boundary chiral algebras or their module categories. The precise statement depends on the chosen twist and boundary condition. A boundary chiral algebra is a protected sector; it is not the complete untwisted quantum field theory.

K.13 Celestial and twistor chiral algebras

In four-dimensional gauge theories whose celestial or twistor transform produces a holomorphic factorisation algebra, scattering amplitudes can be organized by a chiral algebra on the celestial sphere or on twistor space. Costello and Paquette derive form-factor integrands from correlators of a chiral algebra obtained from a six-dimensional holomorphic theory [61]. The OPE associativity constrains loop corrections; anomalies of the chiral algebra correspond to obstructions in the bulk theory.

The exact object is a protected holomorphic observable algebra. Mellin transformation converts momentum variables to conformal weights when passing to the celestial basis. Soft theorems provide current insertions. Equality with a complete gravitational S-matrix requires control of infrared divergences, reality conditions, and the reconstruction transform; no such equality follows from chiral algebra alone.

K.14 Calabi–Yau categories and closed strings

A smooth and proper Calabi–Yau dg category has a nondegenerate cyclic trace. Its Hochschild chains carry the Connes operator, and its Hochschild cochains carry a Gerstenhaber structure. Costello constructs a topological conformal field theory from Calabi–Yau categorical data [62]. The cyclic trace permits the contraction of graph edges, exactly as in the modular envelope.

To obtain a chiral algebra on a curve from a higher-dimensional Calabi–Yau geometry one needs a field theory or factorization algebra, a compactification or defect, and a restriction map to the curve. The resulting morphism of deformation complexes may carry cyclic classes to chiral operations. The construction is geometry-dependent. Numerical invariants such as the Euler characteristic or a Borcherds-product weight do not determine the chiral operation without this morphism.

K.15 The physical synthesis

The equations derived above form one coherent structure:

$$\begin{aligned}\text{local action} &\longrightarrow \text{BV Maurer–Cartan element} \longrightarrow \text{factorization observables},\\ &\longrightarrow \text{boundary chiral operation.}\end{aligned}$$

A propagator resolves internal edges. Fulton–MacPherson geometry resolves collisions. The quantum master equation governs boundary strata of Feynman graphs. KZ transport governs exchange of current-algebra insertions. Conformal blocks govern sewing of states. Determinant lines govern projective anomalies. Each passage is exact when its comparison map has been constructed.

The resulting unity is physical in Feynman's sense and mathematical in Beilinson's sense. One begins with local interactions, follows their support under collision and gluing, and never equates objects merely because their formulas resemble one another.

Chapter L

Exact models seen from every side

A general language is convincing only when its principal objects agree in examples by independent calculations. Five models test the constructions in independent realizations. In each case the chiral operation, collision coalgebra, ordered transport, modular amplitude, and field-theoretic realization are kept distinct and then compared by explicit formulas.

L.1 The rank-one Heisenberg algebra

Let H_k be generated by a field J with OPE

$$J(z)J(w) \sim \frac{k}{(z-w)^2}. \tag{L.1}$$

Equivalently,

$$[J_\lambda J] = k\lambda.$$

The iterated lambda brackets vanish because $[J_\lambda J]$ is central; all three Jacobi terms are therefore zero. The full vertex algebra is the symmetric algebra on the negative modes J_{-n}, $n \geq 1$, applied to the vacuum.

Normal geometry.

The singular operation in (L.1) is the second normal symbol of the diagonal. Under $z \mapsto f(z)$, a weight-one current transforms as a one-form, and the coefficient k pairs the two normal lines. Let $\mathfrak{h}_{\mathrm{ch}}$ denote the abelian current chiral Lie algebra and let

$$0 \longrightarrow \omega_X \mathbf{K} \longrightarrow \widehat{\mathfrak{h}}_{\mathrm{ch}} \longrightarrow \mathfrak{h}_{\mathrm{ch}} \longrightarrow 0$$

be its central extension. The bracket of two current sections lands in the central line through the second normal symbol. Hence the Chevalley differential of $\widehat{\mathfrak{h}}_{\mathrm{ch}}$ is not zero when the central cogenerator is retained; the quotient $\mathfrak{h}_{\mathrm{ch}}$ is abelian.

Fock modules.

For a momentum α, the Fock module $\mathcal{F}_\alpha$ is generated by a vector $|\alpha\rangle$ such that

$$J_n|\alpha\rangle = 0 \quad (n > 0), \qquad J_0|\alpha\rangle = \alpha|\alpha\rangle.$$

The oscillator modes satisfy

$$[J_m, J_n] = km\,\delta_{m+n,0}.$$

After choosing a conformal vector

$$T = \frac{1}{2k} : JJ :$$

for $k \neq 0$, the central charge is one. The character is

$$\mathrm{ch}\mathcal{F}_\alpha(\tau) = \frac{q^{\alpha^2/(2k)-1/24}}{\eta(\tau)}. \tag{L.2}$$

The level k controls the momentum quadratic form, while the oscillator determinant has central charge one. This is an explicit demonstration that level and Virasoro anomaly are different invariants.

Sewing.

Let T_q be the one-particle sewing operator with eigenvalues q^n. Then

$$\det(1 - T_q) = \prod_{n \geq 1} (1 - q^n).$$

The bosonic Fock functor turns a one-particle determinant into its inverse, giving (L.2). On a compact Riemann surface of genus g, the nonzero-mode Gaussian integral gives a zeta-regularized determinant of the scalar Laplacian, while zero modes give a volume factor. A compactified boson with momentum lattice Λ multiplies the oscillator determinant by a Siegel theta series.

Field theory.

Abelian Chern–Simons theory with action

$$\frac{k}{4\pi} \int_M A \wedge dA$$

and a chiral boundary condition has boundary current algebra (L.1). A Wilson line of charge q acquires the braiding phase determined by the bilinear form $q_1 q_2 / k$ after the standard normalization of charges and level. The boundary algebra, line category, and Gaussian determinant therefore arise from different operations in one free field theory.

Koszul reading.

The quotient $\mathfrak{h}_{\mathrm{ch}}$ has a cofree commutative Chevalley coalgebra with zero bracket differential. For the central extension $\widehat{\mathfrak{h}}_{\mathrm{ch}}$, the Chevalley coderivation contracts two current cogenerators to the central cogenerator. In the level-k vacuum representation the central element acts as $k\mathbf{1}$; after reducing by the vacuum line, this contraction is represented as curvature in the reduced bar model. The completed cobar reconstructs the corresponding chiral enveloping algebra. Linear duality exchanges symmetric algebras and symmetric coalgebras under finite-weight completion.

L.2 Affine Kac–Moody currents

Let $\mathfrak{g}$ be simple and let $(,)$ be an invariant form. The universal affine vertex algebra at level k has currents with OPE

$$J^a(z)J^b(w) \sim \frac{k(a,b)}{(z-w)^2} + \frac{J^{[a,b]}(w)}{z-w}. \tag{L.3}$$

The lambda bracket is

$$[J^a_\lambda J^b] = J^{[a,b]} + k\lambda(a,b).$$

Its Jacobi identity has two coefficients. The coefficient independent of λ is Jacobi in $\mathfrak{g}$. The coefficient linear in the formal variables is invariance of $(,)$. The central term is not needed to repair Jacobi of the simple-pole bracket; it is a separate cocycle compatible with it.

Collision complex.

For the centrally extended chiral Lie algebra, the partition collision differential has two components: the simple-pole bracket merges currents to $J^{[a,b]}$, and the central cocycle merges them to the central line. The simple-pole component squares to zero by Jacobi; compatibility with the central component is exactly invariance of $(,)$. A geometric coefficient model that includes all principal parts carries both the first and second normal symbols.

Sugawara tensor.

For $k \neq -h^\vee$, the stress tensor is

$$T(z) = \frac{1}{2(k+h^\vee)} \sum_a :J^a J_a:(z), \tag{L.4}$$

with central charge

$$c = \frac{k \dim \mathfrak{g}}{k+h^\vee}. \tag{L.5}$$

The shift by $h^\vee$ comes from normal ordering. Equation (L.5) is rational in k; any linear expression in $k+h^\vee$ is not the affine conformal anomaly.

KZ transport.

For modules $V_1, \ldots, V_n$, conformal blocks satisfy the KZ equation

$$\left(\partial_{z_i} - \frac{1}{k+h^\vee} \sum_{j\neq i} \frac{\Omega_{ij}}{z_i - z_j}\right) F = 0.$$

Flatness is the calculation of Chapter G. The monodromy is a braid-group representation. Under the Drinfeld–Kohno comparison, it agrees with the braiding of the corresponding quantum-group category at a parameter fixed by the normalization of the invariant form and the exponentiation of $(k+h^\vee)^{-1}$.

Chern–Simons theory.

Three-dimensional Chern–Simons theory at integral level produces the same conformal blocks after quantization on a surface. Wilson-line braiding gives the same ribbon category. The equality is a bulk–boundary theorem involving the current algebra, not a consequence of the numerical level alone. Framing dependence corresponds to the ribbon twist and the gravitational Chern–Simons counterterm.

Critical level.

At $k = -h^\vee$, (L.4) is undefined. The affine algebra itself remains defined and develops the Feigin–Frenkel centre. The KZ equation in the displayed normalization is singular. The centre is identified with functions on opers for the Langlands dual group. These statements explain criticality without assigning the value zero to the current-algebra level.

Koszul and representation-theoretic dualities.

Chiral bar–cobar reconstruction applies in the pro-nilpotent Ran category. Quadratic PBW recognition begins from a chosen presentation. Kazhdan–Lusztig equivalence and Drinfeld–Kohno monodromy compare representation categories. Feigin–Frenkel duality compares principal W-algebras. Quadratic Koszul duality compares a presentation with its orthogonal relation space. A composite theorem requires the hypotheses and categories of every constituent map.

L.3 The Virasoro chiral algebra

The Virasoro algebra is the central extension of vector fields on the circle. Its OPE is

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \tag{L.6}$$

The lambda bracket is

$$[T_\lambda T] = (\partial + 2\lambda)T + \frac{c}{12}\lambda^3.$$

The Jacobi identity is a polynomial identity in two formal variables. It expresses the cocycle identity of the Gelfand–Fuchs central extension and the Lie bracket of vector fields.

Normal symbols.

The three singular terms of (L.6) occupy normal degrees four, two, and one. The normally ordered product occupies degree zero. A logarithmic residue detects only the degree-one component after a logarithmic form has been supplied. The chiral operation remembers all four pieces and their extensions.

Projective connections.

The transformation law

$$T(z) = f'(z)^2 T(f(z)) + \frac{c}{12}\{f, z\}$$

shows that the stress tensor is an affine projective connection. The Schwarzian cocycle is the coordinate form of the Virasoro central extension. On moduli, it gives a projective connection on conformal blocks. The Hodge coefficient is proportional to c, while insertions and boundary factorization contribute further classes.

Verma modules and null vectors.

A highest-weight module $M(c, h)$ is generated by v with

$$L_0 v = hv, \qquad L_n v = 0 \quad (n > 0).$$

The Kac determinant detects singular vectors. Passing from the universal Virasoro algebra to a minimal-model quotient changes the homological algebra because relations generated by null vectors are imposed. Koszulness of the universal object cannot be transferred to every simple quotient without a separate calculation.

The bosonic string.

The (b, c) reparametrization ghosts have central charge -26. Coupling them to a matter theory gives a nilpotent BRST charge when the total central charge and intercept anomaly vanish. The number 26 is therefore the cancellation value of a specific ghost system. The value 13 is the midpoint of the reflection $c \mapsto 26 - c$; it is not by itself a theorem of Virasoro self-duality.

Polyakov and determinant geometry.

The Weyl anomaly has coefficient c. For free fields it is computed from determinants. The curvature of the determinant line agrees with the central extension represented by (L.6). This is the rigorous bridge between local OPE and global moduli curvature. It does not imply that the Virasoro algebra determines a full nonchiral CFT or a gravitational path integral.

Three-dimensional gravity.

Brown–Henneaux boundary conditions produce two Virasoro algebras with

$$c = \bar{c} = \frac{3\ell}{2G}.$$

Under modular invariance, a sparse light spectrum, and vacuum dominance, Cardy asymptotics reproduce the BTZ entropy. The chiral algebra supplies the asymptotic symmetry and central extension. The spectrum and modular invariant are additional physical data.

L.4 Free fermions and ghost systems

A free complex fermion has fields ψ, ψ^* with

$$\psi(z)\psi^*(w) \sim \frac{1}{z - w}.$$

The associated Clifford algebra is quadratic. Wick's theorem expresses every correlator as a determinant. On a Riemann surface, a choice of spin structure enters the Szego kernel. The partition function is a determinant or Pfaffian of the Dirac operator.

A (b, c) system with conformal weights $(\lambda, 1-\lambda)$ has stress tensor

$$T = (1-\lambda){:}(\partial b)c{:} - \lambda{:}b\partial c{:}$$

and central charge

$$c_{bc} = 1 - 3(2\lambda - 1)^2. \tag{L.7}$$

A bosonic (β, γ) system with the same weights has the opposite determinant statistics and central charge

$$c_{\beta\gamma} = -c_{bc}.$$

The cancellation in a tensor product is a determinant identity and a Virasoro central-charge identity. The associated graded polynomial and exterior algebras are ordinary Koszul duals. The Weyl and Clifford algebras are PBW deformations of these two algebras and are related by the corresponding curved or inhomogeneous Koszul formalism when their constant quadratic terms are retained. These facts do not imply equality of their module categories as conformal theories.

Ghost number provides a second grading. BRST reduction uses the ghost differential and the vertex differential simultaneously. Spectral-sequence concentration can make the reduction exact. When concentration fails, higher BRST cohomology remains and cannot be discarded.

L.5 Lattice vertex algebras

Let Λ be an even lattice with bilinear form $(,)$ and let ε be a normalized two-cocycle on Λ. The lattice vertex algebra is

$$V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda].$$

For $\alpha, \beta \in \Lambda$, the vertex operators have leading product

$$Y(e^\alpha, z)Y(e^\beta, w) = (z-w)^{(\alpha,\beta)}\varepsilon(\alpha, \beta)Y(e^{\alpha+\beta}, w) + \cdots . \tag{L.8}$$

Evenness gives locality. The cocycle controls the ordered phase. Replacing ε by a cohomologous cocycle gives an isomorphic algebra; replacing it by its inverse reverses the commutator phase.

The genus-one character is

$$\mathrm{ch} V_\Lambda(\tau) = \frac{\Theta_\Lambda(\tau)}{\eta(\tau)^{\mathrm{rk}\,\Lambda}}. \tag{L.9}$$

For an even unimodular positive-definite lattice, the chiral central charge is $\mathrm{rk}\,\Lambda$ and the theory is holomorphic. At genus g, the zero-mode sum is the degree-g Siegel theta series

$$\Theta_\Lambda^{(g)}(\Omega) = \sum_{x_1,\ldots,x_g\in\Lambda} \exp\left(\pi i \sum_{i,j}(x_i, x_j)\Omega_{ij}\right).$$

The oscillator and lattice pieces are independent factors.

For the two even unimodular rank-sixteen lattices $E_8 \oplus E_8$ and D_{16}^+, the Siegel theta series agree in degrees $g = 1, 2, 3$. In degree four their difference is a nonzero scalar multiple of the Schottky form on $\mathfrak{H}_4$; its restriction to the genus-four Jacobian locus vanishes [102, 109]. Thus genus two does not separate the two lattice theories through their theta series, and even the degree-four difference disappears after restriction to smooth genus-four curves.

The Leech lattice VOA and the Moonshine VOA both have central charge 24. Their weight-one spaces differ: the Leech theory has a rank-twenty-four Heisenberg sector, while Moonshine has no weight-one states. Therefore central charge does not determine the chiral algebra, its OPE, or its automorphism group.

L.6 Principal W-algebras

Let $\mathfrak{g}$ be simple and f principal nilpotent. Quantum Drinfeld–Sokolov reduction forms a BRST complex from the affine algebra, charged ghosts, and neutral fields when required. Its degree-zero cohomology is the principal W-algebra $\mathcal{W}^k(\mathfrak{g}, f)$ under the standard hypotheses.

For the universal principal W-algebra at generic level, the standard strong generators have conformal weights $d_i + 1$, where d_i are the exponents of $\mathfrak{g}$; special levels may impose decoupling relations or collapse generators. Their OPEs contain nonlinear normally ordered polynomials. These terms are transferred operations of the BRST contraction; they are not captured by the simple-pole Lie bracket alone.

Feigin–Frenkel duality relates principal W-algebras for Langlands dual Lie algebras at levels satisfying

$$r^\vee(k + h^\vee)(\ell + {}^L h^\vee) = 1, \tag{L.10}$$

with the lacing convention included. The relation is multiplicative in shifted levels. It is not the affine reflection $k \mapsto -k - 2h^\vee$.

If a filtered BRST contraction is compatible with the chiral bar construction and its spectral sequence converges, then bar and BRST cohomology commute. This transports collision operations and Koszul comparisons through the reduction. The conclusion is a theorem about the specified contraction. It is not automatic for every nilpotent orbit or every completion.

L.7 Five comparisons in one table

Model	highest local pole	ordered datum	modular datum	physical realization
Heisenberg	double pole k	abelian phase	oscillator determinant and zero modes	abelian Chern–Simons boundary
affine	double pole plus Lie pole	KZ braid monodromy	affine conformal blocks	nonabelian Chern–Simons/WZW
Virasoro	fourth-order central pole	monodromy of Virasoro blocks	projective connection and determinant line	conformal anomaly; AdS_3 asymptotic symmetry
free ghosts	simple contraction	fermionic or bosonic exchange	determinant/Pfaffian	gauge fixing and BRST
lattice	integral power and cocycle	cocycle phase	Siegel theta series	compactified free boson
W-algebra	nonlinear higher-spin OPE	module-category braiding when constructed	W-conformal blocks	Hamiltonian reduction

The columns are not functions of the first column alone. The local pole controls the singular operation. Ordered transport requires modules and a connection. Modular data requires sewing. A physical realization requires an action and boundary condition. The exact models show that the same chiral operation can feed all four constructions without making them identical.

L.8 What the examples establish

The examples test the common language by independent constructions whose comparison maps are known.

For Heisenberg theory, the boundary quantization map identifies the current OPE with the abelian Chern–Simons boundary bracket, while Gaussian sewing identifies the oscillator trace with the determinant formula.

For affine theory, the Ward identities produce the KZ connection; Drinfeld–Kohno identifies its braid category with the corresponding quantum-group category, and the Chern–Simons/WZW comparison identifies the same braid data with Wilson-line exchange.

For Virasoro theory, the fourth normal symbol gives the central cocycle, the Ward identity gives the Schwarzian transformation, and the family index theorem identifies its moduli-space image with determinant-line curvature and the Weyl anomaly.

For ghost systems, the Clifford or Weyl quadratic relations determine the parity signs, the corresponding determinant ratio gives the central charge, and the BRST square measures the same total anomaly after the BRST comparison is made.

For lattice theories, the group-algebra cocycle determines ordered phases, Poisson summation gives the modular transformation of the theta series, and the compactified-boson path integral yields the same theta factor after zero-mode and oscillator separation.

Each agreement is mediated by an explicit construction. The evidentiary standard is: one operation, calculated in several faithful realizations, with every passage preserving the category of the object.

Chapter M

Exact distinctions, historical lineage, and the common language

A unifying language is useful only when it preserves distinctions. The logical relations among the objects follow from their sources, targets, and comparison maps; the historical and physical readings preserve those object types.

M.1 The primitive object

Fix a smooth complex curve X. The primitive datum is

$$A \in \mathrm{Alg}^{\mathrm{aug}}_{E_1}\big(\mathrm{FactD}(X), \otimes^{\mathrm{pt}}\big). \tag{M.1}$$

Thus A consists of a factorisable D-module together with a unital associative multiplication

$$\star : A \otimes^{\mathrm{pt}} A \longrightarrow A$$

and an augmentation $A \to \mathbf{1}$. The factorisable D-module carries the holomorphic collision data; the E_1 multiplication carries transverse ordered composition. These structures satisfy the interchange law because $\star$ is a morphism in $\mathrm{FactD}(X)$.

A diagonal-supported chiral Lie operation

$$\mu : j_* j^* (\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_! \mathfrak{g}$$

is a separate and important specialization of the chiral coefficient. It gives the Francis–Gaitsgory Chevalley branch, but it is not the source of the transverse E_1 multiplication and does not replace (M.1).

Construction	Output	Additional datum or hypothesis
ordered interval bar	conilpotent coassociative coalgebra in $\mathrm{FactD}(X)$	augmentation; completion when infinite sums occur
boundary endomorphisms	$A^! = \mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$	compactness and completion for double duality
quadratic comparison	presentation coalgebra $A^{¡} \to \mathrm{Bar}_X^{\mathrm{ord}}(A)$	chosen quadratic or filtered presentation
chiral Chevalley	commutative factorisation coalgebra on $\mathrm{Ran}\, X$	a chiral Lie coefficient and Ran pronilpotence
formal-disc realization	vertex, quantum-vertex, or nonlocal-vertex operations	coordinate, translation, and when present exchange data
Verdier dual	dual constructible or holonomic object	dualisability, continuity, and monoidal base change
E_2 enhancement	exchange paths and Yang–Baxter coherence	a transverse plane or equivalent flat exchange structure
Tannaka reconstruction	bialgebra/Hopf or coquasi/quasi-Hopf object	rigid tensor category and strong or quasi-monoidal fibre functor
two sewing structures	bicoloured stable-graph operations	transverse cyclic trace and chiral modular sewing
conformal blocks	sheaf or complex on moduli	modules, finiteness, projective connection, and sewing
internal Hochschild cochains	E_2 derived centre	homotopically correct diagonal bimodule resolution
physical observables	analytic holomorphic–topological factorisation algebra	fields, action, gauge fixing, renormalisation, and an algebraisation comparison

An equality between two rows requires a comparison morphism. The existence of a suggestive analogy does not supply that morphism.

M.2 Bar, Koszul dual, Verdier dual, and centre

Four neighboring words refer to four different constructions.

Bar.

For an augmented algebraic object A, the bar construction is a coalgebraic resolution. Its universal property is expressed by twisting morphisms. It exists under augmentation and conilpotence conditions whether or not A is quadratic or Koszul.

Quadratic Koszul dual.

A quadratic presentation $A = T(V)/(R)$ defines a quadratic coalgebra $A^{¡}$ and, when finite duals exist, an algebra $A^! = (A^{¡})^{\vee}$. The canonical map $A^{¡} \to B(A)$ is a quasi-isomorphism exactly on the Koszul locus. The quadratic dual depends on the presentation until that comparison has been proved invariant.

Verdier dual.

Verdier duality is a contravariant functor on sheaves or D-modules. Applied to a factorization coalgebra it may produce a factorization algebra when duality commutes with the pointwise and chiral tensor products, the diagonal pull-push maps, and the completed inverse limits used in the construction. This is a geometric duality, not the universal bar–cobar counit.

Derived centre.

The derived centre is Hochschild cochains of an associative object. For a chiral Lie algebra $\mathfrak{g}$, the typed definition passes through its chiral enveloping factorization algebra:

$$Z^{\mathrm{der}}_{\mathrm{ch}}(\mathfrak{g}) = \mathrm{HH}^{\bullet}_{\mathrm{Fact}}\big(U^{\mathrm{ch}}(\mathfrak{g}), U^{\mathrm{ch}}(\mathfrak{g})\big).$$

The derived centre carries an E_2 structure. It is neither the bar coalgebra nor a scalar trace of the bar coalgebra.

The four constructions can be related on favorable finite and perfect loci, but none is a synonym for another.

M.3 Four squares and four curvatures

The notation d^2 occurs in several theories. The symbols are equal only when the underlying operators are equal.

(i) An internal differential on a dg object satisfies $d^2 = 0$.

(ii) In a curved A_∞ algebra, m_1^2 is an inner operation determined by m_0 together with higher terms.

(iii) A connection satisfies $\nabla^2 = F_\nabla$, a two-form on the base with values in endomorphisms.

(iv) A BV quantum differential has square equal to the anomaly operator unless the quantum master equation holds.

Projective curvature belongs to the connection on a family of states; internal curved-algebra data belongs to the chiral differential. A central element has zero commutator, so the formula $d^2 = [m_0, -]$ cannot represent nonzero scalar projective curvature. An anomaly class can be cancelled by a counterterm when it is exact. A projective curvature can be cancelled by tensoring with a line of opposite Atiyah class when that line exists. These are different cancellation mechanisms.

M.4 Residue, principal part, and propagator

Three local objects are also distinct.

Residue.

Poincare residue extracts the coefficient of a logarithmic normal form. It is a map of line-bundle weight one.

Principal part.

The coefficient of a pole of order m is a section of $N_{D/Y}^{\otimes m}$. The direct sum over all m is the associated graded of the pole filtration. It contains the singular modes of a chiral operation.

Propagator.

A propagator is a Green kernel or parametrix selected by a kinetic operator and a gauge-fixing condition. It is analytic data. Its singular part may represent a principal part, but its regular and harmonic parts depend on global geometry.

The logarithmic form $d\log(z-w)$ is a canonical generator in the cohomology of a two-point configuration on an affine screen. It is not the universal propagator of every chiral theory and does not extract arbitrary OPE poles.

M.5 Ordering, descent, and braiding

Labelled configurations remember the pure braid group. Passing to unlabelled configurations is descent along a finite cover. Over a field of characteristic zero, homotopy invariants and homotopy coinvariants for the finite symmetric group are related by averaging, but the local system on the labelled configuration space can have nontrivial braid monodromy before descent.

A vector-space splitting of an averaging map is not a Drinfeld associator. An associator is a group-like series satisfying the pentagon and hexagon equations. It compares different parenthesized monodromies. The Grothendieck–Teichmuller group acts on associators, not on arbitrary linear splittings.

Verdier duality sends a local system with monodromy ρ to the dual local system with monodromy ρ^{-T}. An equality $R \mapsto R^{-1}$ follows when the chosen bilinear form identifies dual and original representations and when conventions convert inverse transpose to inverse. An equality of named quantum groups requires a Tannakian or presentation-theoretic recognition theorem.

M.6 Modular sewing and the master equation

The compactification $\overline{\mathcal{M}}_{g,n}$ has boundary divisors indexed by stable graphs with one edge. Codimension-two strata correspond to graphs with two edges. Their oriented incidence relation gives the differential in the Feynman transform. If a cyclic object assigns compatible operations to every clutching morphism, the total operation satisfies the quantum master equation

$$d\Theta + \frac{1}{2}[\Theta, \Theta] + \hbar\Delta\Theta = 0.$$

The equation follows from the compatibility maps. It does not construct them.

For conformal blocks, sewing is a theorem under the finiteness and representation-theoretic assumptions of the modular-functor construction. The connection is projectively flat. Its first Chern class contains a Hodge contribution proportional to the central charge, insertion contributions from conformal weights, and boundary contributions from factorization channels. Therefore no single scalar classifies the modular theory.

M.7 The mathematical history

The language assembled here was formed by a sequence of exact advances.

Historical note M.1 (Distributions). Dirac's delta function made an operation supported on a submanifold calculable. Schwartz's distribution theory made support and differentiation intrinsic. A chiral bracket is the holomorphic, D-module-valued descendant of this idea: the product is defined away from the diagonal and its singular commutator is supported on the diagonal.

Historical note M.2 (Operator products). Wilson separated short-distance singularity from long-distance state dependence. The operator product expansion is a local asymptotic expansion whose coefficients are local fields. Vertex algebra theory replaces convergence questions by a formal identity. Chiral algebra theory replaces a chosen coordinate by a geometric correspondence.

Historical note M.3 (Borcherds and Beilinson–Drinfeld). Borcherds recognized that locality, commutator formulas, and associativity are coefficient forms of one formal-distribution identity. Beilinson and Drinfeld placed that identity on a curve: poles are allowed away from diagonals, while operations land on diagonals. The formal variable becomes the normal direction to a geometric collision.

Historical note M.4 (Fulton–MacPherson and Kontsevich). Fulton and MacPherson resolved simultaneous collisions by adding screens that record relative configurations. Kontsevich used compactified configuration spaces to organize deformation quantization. In perturbative field theory the same spaces separate ultraviolet regions and make the forest structure of counterterms geometric.

Historical note M.5 (Ran space and chiral Koszul duality). The Ran space regards a finite subset of a curve as one variable. Factorization is multiplication over disjoint unions of finite subsets. Francis and Gaitsgory proved that, in the pro-nilpotent chiral monoidal category, chiral Lie algebras and conilpotent commutative factorization coalgebras are related by Chevalley chains and primitives. This is the categorical core of chiral Koszul duality.

Historical note M.6 (Braids and quantum groups). Knizhnik and Zamolodchikov produced a flat connection from current algebra. Kohno and Drinfeld identified its braid monodromy with the braided tensor structure of quantum groups. Witten's Chern–Simons theory gave the same braid representations as Wilson-line amplitudes. Costello–Witten–Yamazaki obtained rational, trigonometric, and elliptic integrable systems from four-dimensional gauge theory. These are exact comparisons of monodromy categories, not consequences of symmetrization alone.

Historical note M.7 (Polyakov, Segal, and modular geometry). Polyakov's path integral made the conformal anomaly a curvature on the space of metrics. Segal formulated conformal field theory by sewing Riemann surfaces. Tsuchiya–Ueno–Yamada constructed conformal blocks in families. The determinant and Hodge lines record the projective anomaly. Getzler–Kapranov's modular operads encode the topology of sewing independently of the analytic convergence of amplitudes.

Historical note M.8 (Feynman and BV). Feynman's graphs are the perturbative expansion of an action. The Batalin–Vilkovisky formalism turns gauge invariance into the classical and quantum master equations. Costello's renormalization theory constructs effective interactions on compactified configuration spaces and identifies anomalies as local cohomology classes. The stable-graph master equation of a cyclic chiral object has the same algebraic shape; an equality of theories requires a map identifying their fields, pairings, and vertices.

M.8 The physical dictionary

Let Obs be a factorization algebra of quantum observables on a manifold whose protected boundary geometry is $X^{\mathrm{an}} \times \mathbb{R}$. Suppose there is a quasi-isomorphism of $E_1 \otimes \mathsf{Ch}_X$ factorization objects

$$\Phi : A^{\mathrm{an}} \xrightarrow{\simeq} \mathrm{Obs}_\partial . \tag{M.2}$$

For an enveloping example one may take $A = U^{\mathrm{ch}}(\mathfrak{g})$ with its supplied transverse product. The following identifications are then exact consequences of Φ.

Chiral geometry	Quantum-field-theoretic realization
D-module of fields	cochain complex of boundary local observables
chiral operation	renormalized boundary OPE
partial diagonal	coincidence locus of insertions
principal-part filtration	scaling hierarchy of contact singularities
partition collision complex	local short-distance factorization complex
Fulton–MacPherson screen	relative ultraviolet configuration
chiral Chevalley coalgebra	coalgebra of factorized insertions
ordered local system	exchange transport of defects or lines
cyclic pairing	two-point function used to sew legs
modular graph operation	perturbative amplitude with fixed topology
BV anomaly class	obstruction to preserving the quantum master equation
Hochschild centre map	bulk observable acting centrally on the boundary

Equation (M.2) promotes the two mathematically parallel columns to two models of the same operations.

M.9 Exact boundary theorems

The following implications are false without additional hypotheses.

No-go theorem M.9 (No quantum group from dimension loss). The dimension of the kernel of $V^{\otimes n} \to \mathrm{Sym}^n V$ does not determine an r-matrix, associator, or quantum group. It contains no multiplication, coproduct, monodromy, or coherence equation.

No-go theorem M.10 (No modular object from topology alone). The equality $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$ does not assign coefficient operations to boundary strata. A modular master element exists only after cyclic contractions and clutching-compatible coefficient maps have been given.

No-go theorem M.11 (No bulk reconstruction from a boundary centre alone). The derived centre is terminal among algebraic central actions on the boundary. It does not detect a decoupled bulk sector. Therefore a physical bulk is not determined by the boundary chiral algebra unless a bulk-to-centre comparison is proved to be an equivalence.

No-go theorem M.12 (No classification by central charge). Central charge determines the Virasoro anomaly coefficient. It does not determine the full chiral operation, representation category, braid monodromy, or higher-genus conformal blocks. The Leech lattice vertex algebra and the Moonshine vertex algebra both have central charge 24, while their weight-one spaces have dimensions 24 and 0.

No-go theorem M.13 (No identification of anomalies of different types). A current-algebra level, a Virasoro central charge, a determinant-line curvature, a BRST ghost-number anomaly, and a Borcherds-product weight are invariants in different theories. They may be related by a theorem in a particular family. They are not one universal scalar.

M.10 The common language as a commutative diagram

The subject can be expressed by the following diagram. Solid arrows are universal for the stated input; arrows labelled by extra structure exist only when that structure has been supplied.

$$\begin{tikzcd} A \in \mathrm{Alg}^{\mathrm{aug}}_{E_1}(\mathrm{FactD}(X)) \arrow[rr, bend left, "\mathrm{Bar}^{\mathrm{ord}}_X"] \arrow[d, "\mathrm{RHom}_A(\mathbf{1},-)"'] & & \mathrm{Coalg}^{\mathrm{conil}}_{\mathrm{coAss}}(\mathrm{FactD}(X)) \arrow[ll, bend left, "\mathrm{Cobar}^{\mathrm{ord}}_X"] \\ A^! = \mathrm{RHom}_A(\mathbf{1},\mathbf{1}) & & Z_{\mathrm{ord,ch}}(A) \arrow[ll, "\text{boundary action}"] \end{tikzcd} \tag{M.3}$$

The analytic and exchange realizations form the companion diagram

$$\begin{tikzcd} A \arrow[r, "\mathrm{an}"] \arrow[d, "\mathrm{HH}^\bullet"'] & \mathrm{Obs}_\partial(X \times \mathbb{R}) \arrow[d, "\text{bulk action}"] & \\ Z_{\mathrm{ord,ch}}(A) \arrow[r, "E_2/E_3\ \text{extension}"'] & \mathcal{L}_A \arrow[r, "\text{Tannaka}"'] & \mathcal{H}_A. \end{tikzcd}$$

A chiral Lie coefficient $\mathfrak{g}$ supplies the independent branch

$$\mathfrak{g} \xrightarrow{\mathrm{coChev}^{\mathrm{ch}}} \mathrm{CoFact}_{\mathrm{conil}}(\mathrm{Ran}\, X),$$

and a monoidal comparison theorem relates this branch to the ordered interval bar when its hypotheses hold. A transverse cyclic trace and a chiral modular sewing system add the bicoloured Feynman transform and its master element Θ^{bi}_A. A field-theoretic realization adds propagators, renormalisation, and states. The common source is the ordered chiral algebra together with the extra structures displayed on the displayed comparison arrows.

M.11 The final principle

The mathematical phenomenon is the coexistence of holomorphic collision and transverse associative composition. The physical phenomenon is the short-distance product of observables together with their ordered fusion along a topological direction. The geometric carrier is the operadic product of chiral collision spaces and oriented intervals. The algebraic laws are the chiral identities of the coefficient, associativity of the E_1 multiplication, and their interchange. Chiral Koszul duality passes to factorised coalgebraic states in the Ran direction; ordered bar–cobar duality passes to interval states in the transverse direction. Ordering remembers composition, exchange data remembers transport, cyclicity permits loops, moduli record sewing, and the derived centre records universal central action.

These clauses form one statement because every noun has a category and every verb is a functor or a correspondence. The unity of the subject is not the equality of all its invariants. It is the fact that each invariant is obtained from the same ordered chiral object, with explicitly named additional structures, by an exact geometric construction.

Chapter N

Historical lineage of the language

N.1 Distributions and local fields

Dirac's delta distribution made support a mathematical datum rather than a limiting metaphor [71]. Quantum field theory inherited this lesson: products of fields are defined away from coincidence and acquire contact terms on diagonals. The chiral D-module operation is the algebraic form of that distributional fact. It records the singular product and its support before a coordinate expansion is chosen.

Wilson's operator product expansion separated short-distance coefficients from local operators [118]. Borcherds and the vertex-algebra tradition encoded the consistency of those coefficients in formal distributions. Beilinson and Drinfeld replaced the formal coordinate by a curve and the formal delta function by a morphism of D-modules supported on the diagonal. The Beilinson–Drinfeld object serves as the chiral coefficient of a genuinely associative ordered product.

N.2 Time ordering and the associative direction

Feynman's time-ordered product is not commutative multiplication [83]. It depends on an ordering of insertion times, and its perturbative expansion is indexed by graphs. Stasheff's associahedra later supplied the homotopy geometry of all parenthesizations [111, 112]. Factorization homology identifies the same geometry with the bar construction on an interval.

The ordered chiral algebra combines these two historical lines. Dirac and Beilinson govern support on the complex diagonal. Feynman and Stasheff govern composition along the ordered real direction.

N.3 Configuration spaces and renormalization

Arnold computed the cohomology of configuration spaces and isolated the three-term relation that underlies the flatness of logarithmic connections [48]. Fulton and MacPherson compactified configurations by replacing collisions with screens [84]. Axelrod–Singer and Kontsevich used related compactifications to make perturbative gauge-theory and deformation-quantization integrals finite and compatible with boundary strata [53, 97].

The compactification has two roles that must remain distinct. Its face category resolves collision incidence. Its screen spaces carry nontrivial cohomology and propagator integrals. The first is combinatorial; the second contains the analytic weight of Feynman graphs.

N.4 Chiral algebras and Koszul duality

Koszul, Priddy, Quillen, and the operadic tradition developed the passage between algebraic operations and coalgebraic resolutions. Beilinson–Drinfeld placed local Lie operations on the Ran space. Francis–Gaitsgory proved the corresponding chiral Koszul duality in a pro-nilpotent monoidal category.

The associative extension considered here uses the same Ran geometry with one additional ordered operadic direction. Its bar is the interval bar internal to factorization D-modules. Its dual is the derived endomorphism algebra of the augmentation boundary. This is the form of Koszul duality that appears in defect physics.

N.5 Anomalies and geometry

Polyakov identified the Weyl anomaly with a local action for the conformal factor [108]. Quillen and the family index theorem identified the same anomaly with curvature of a determinant line [105]. Belavin and Knizhnik related determinant geometry to string measures on moduli [55]. These are different realizations of a central extension of conformal transformations.

The construction preserves the chain of maps: local stress-tensor cocycle, Ward identity, Atiyah algebra of conformal blocks, determinant-line curvature. It does not replace the chain by a universal scalar.

N.6 Gauge theory, knots, and quantum groups

Witten's Chern–Simons theory connected three-dimensional gauge fields, two-dimensional current algebras, conformal blocks, braid groups, and knot invariants [117]. Drinfeld and Jimbo supplied the quantum groups [73, 92]; Kohno identified their braid representations with KZ monodromy [95]. The decisive structure is the category of lines and its exchange, not the dimension of an invariant subspace.

Costello, Witten, and Yamazaki added a holomorphic spectral curve to topological gauge theory [63, 64]. Wilson-line OPEs then produced rational, trigonometric, and elliptic quantum groups. Gauge-theoretic constructions of Gaiotto, Moore, and Neitzke and of Nekrasov placed line defects, framed BPS states, Omega backgrounds, and BPS/CFT Ward identities in the same protected sector [85, 100]; Coulomb-branch convolution gives a complementary algebraic realization [54]. The spectral curve carries chiral dependence; transverse topology carries fusion and exchange.

N.7 Factorization algebras and BV theory

Costello and Gwilliam formulated perturbative quantum field theory as a factorization algebra of observables [58, 59]. The BV classical and quantum master equations became local cohomological identities. Boundary and defect observables became factorization modules. This made the passage from an action functional to a chiral or topological operator algebra a functor with a defined source and target.

The present construction is compatible with that formalism. An ordered chiral algebra is the algebraic local observable structure on a holomorphic–topological product. A BV theory is one realization of it, together with fields, pairing, integration, and renormalization.

N.8 Open–closed theories and centres

Hochschild cochains have long governed deformations of associative algebras. Deligne's conjecture and the Swiss-cheese operad identified their E_2 structure and their universal action on an E_1 algebra. In open–closed field theory, the centre is the universal closed algebra acting on the open sector.

This universal property is exact but limited. It reconstructs the protected bulk from a boundary only when the bulk-to-centre map is an equivalence. Witten's open-string field theory and Costello's Calabi–Yau categories supply settings in which Hochschild chains and cyclic operations are physical closed states [62, 119]. Twisted holography supplies settings in which a boundary algebra determines a protected bulk sector.

N.9 The resulting synthesis

The historical constructions meet in one typed sentence. A noncommutative chiral algebra is an E_1 algebra in factorisation D-modules. Its chiral coefficient is a diagonal-supported distribution, and its multiplication is ordered fusion. Interval factorisation homology gives its bar. Hochschild cochains give the universal central action. Tannaka reconstruction applied to a braided line category gives its quantum symmetry. A cyclic trace produces ribbon loops, conformal blocks produce worldsheet sewing, and transgression compares the resulting invertible anomaly theories.

Every clause has an independent historical origin. Their unity is the geometry of support, order, exchange, and sewing.

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